User Manual **MN150016EN**

January 2026
Supersedes April 2025

PXQ Event Analysis System User Manual



Powering Business Worldwide

DISCLAIMER OF WARRANTIES AND LIMITATION OF LIABILITY

The information, recommendations, descriptions and safety notations in this document are based on Eaton Corporation's ("Eaton") experience and judgment and may not cover all contingencies. If further information is required, an Eaton sales office should be consulted. Sale of the product shown in this literature is subject to the terms and conditions outlined in appropriate Eaton selling policies or other contractual agreement between Eaton and the purchaser.

THERE ARE NO UNDERSTANDINGS, AGREEMENTS, WARRANTIES, EXPRESSED OR IMPLIED, INCLUDING WARRANTIES OF FITNESS FOR A PARTICULAR PURPOSE OR MERCHANTABILITY, OTHER THAN THOSE SPECIFICALLY SET OUT IN ANY EXISTING CONTRACT BETWEEN THE PARTIES. ANY SUCH CONTRACT STATES THE ENTIRE OBLIGATION OF EATON. THE CONTENTS OF THIS DOCUMENT SHALL NOT BECOME PART OF OR MODIFY ANY CONTRACT BETWEEN THE PARTIES.

In no event will Eaton be responsible to the purchaser or user in contract, in tort (including negligence), strict liability or other-wise for any special, indirect, incidental or consequential damage or loss whatsoever, including but not limited to damage or loss of use of equipment, plant or power system, cost of capital, loss of power, additional expenses in the use of existing power facilities, or claims against the purchaser or user by its customers resulting from the use of the information, recommendations and descriptions contained herein. The information contained in this manual is subject to change without notice.

CONTENTS

DISCLAIMER OF WARRANTIES AND LIMITATION OF LIABILITY II

Safety Precautions 1

Congratulations! 1

1. INTRODUCTION 2

1.1 Safety 2

1.2 Product Overview 2

1.3 Component Overview 2

1.4 Catalog Information 2

1.5 Support 2

2. INSTALLATION 3

2.1 Mounting 3

2.2 Field Installation (Per UL2808) 5

2.2.4 Mounting the PXQ 6

2.4 Metering specifications 12

2.6 Power supply input specifications 14

2.7 I/O Specifications 14

2.8 Safety and environmental specifications 15

3. CONFIGURING & COMMISSIONING 18

3.1 Establishing a connection to PXQ's Onboard Webserver via USB* 18

3.2 Establishing a connection to PXQ's Onboard Webserver via Ethernet 19

3.3 Logging Into PXQ Web Server 20

3.4 Basic Configuration 20

3.5 Additional Settings 20

4. WEB-SERVER SCREENS 21

4.1 Main Navigation 21

4.2 Overview 26

4.3 Energy 30

4.4 Metering 34

4.5 Timeline 40

4.6 Calendar 41

4.7 I/O 42

5. OPTIMIZED SCREENS FOR LOWER RESOLUTIONS 43

5.1 Simplified Overview 43

6. SETTINGS 45

6.1 General Settings 45

6.2 Triggers 48

6.3 Network 59

6.4 Users 72

6.5 Notifications 78

6.6 I/O 80

6.7 Diagnostics 92

7. TROUBLESHOOTING & MAINTENANCE 95
7.1 Factory Reset 95
7.2 Update 95
7.3 Reboot 97
7.4 Com Reset 97
7.5 Maintenance and Care 97
7.6 Battery 98
7.7 LED Status 98
7.8 Save and Restore Configurations 99

8. COMMUNICATIONS 100
8.1 Modbus TCP 100
8.2 BACnet IP 139

DISCLAIMER OF WARRANTIES AND LIMITATION OF LIABILITY

Please read this manual carefully before installation, operation, and maintenance of the PXQ.

The following symbols in this manual and on the PXQ are used to provide warning of danger or risk during the installation and operation of the system.

Safety Precautions

All safety codes, safety standards, and/or regulations must be strictly observed in the installation, operation, and maintenance of this device.

⚠ WARNING

The warnings and cautions included as part of the procedural steps in this document are for personnel safety and protection of equipment from damage. An example of a typical warning call-out is shown above. This will help to ensure that personnel are alert to warnings that may appear throughout the document. In addition, cautions are all upper case and boldfaced as shown below.

⚠ WARNING

Completely read and understand the material presented in this document before attempting installation, operation, or application of the equipment. Only qualified persons should be permitted to perform any work associated with the equipment. The wiring, installation and application use instructions presented in this document must be followed precisely. Failure to do so could cause permanent equipment damage, bodily injury, or death.

⚠ WARNING

Do not attempt to install or perform maintenance on equipment while it is energized. Death, severe personal injury, or substantial property damage can result from contact with energized equipment. Always verify that no voltage is present before proceeding with the task, and always follow generally accepted safety procedures.

Eaton is not liable for the misapplication or misinstallation of its products.

Congratulations!

You have purchased a Power Xpert Quality event analysis system, a state-of-the-art power quality monitoring instrument that provides intuitive visualization of complex power quality events, reliable and accurate analytics, and actionable data. This helps to minimize facility downtime, loss of productivity, and improve efficiency.

When you open the package, you will find the following items.

	QTY
1. PXQ System Core	1
2. PXQ Meter Module	1
3. Terminal Blocks	3
4. Installation Clips	4
5. Additional documentation (Quick Setup Guide)	2

To avoid complications, please read this manual carefully before installation and operation of the PXQ system.

1. Introduction

1.1 Safety

SHOCK HAZARDS: ALL MAINTENANCE PROCEDURES MUST BE PERFORMED ONLY BY QUALIFIED PERSONNEL. Failure to observe this warning could result in serious injury, death and/or equipment damage.

TROUBLESHOOTING PROCEDURES MAY INVOLVE WORKING IN EQUIPMENT AREAS WITH EXPOSED LIVE PARTS WHERE THE HAZARD OF A FATAL ELECTRIC SHOCK IS PRESENT. Personnel must exercise extreme caution to avoid injury or death.

ALWAYS DISCONNECT AND LOCK OUT THE CURRENT SOURCE AND CONTROL POWER SUPPLY BEFORE TOUCHING THE COMPONENTS ON THE REAR OF THE PXQ.

1.2 Product Overview

State-of-the-art power quality and connected data insights is the new standard for power distribution assemblies, along with the transition to a more sustainable, efficient facility. The implementation of advanced digital technology enables a rich experience for end-users operating today and tomorrow's distribution equipment. Eaton is now paving the way for improved manufacturing efficiencies by helping customers quickly diagnose power quality events to avoid costly business interruptions and achieve a greener, healthier system.

Eaton's Power Xpert Quality (PXQ) event analysis system sets the benchmark for intuitive visualization of complex power quality events by providing reliable and accurate analytics, actionable data to minimize facility downtime, loss of productivity, and reduce energy consumption all viewed through a data simplified and at-a-glance user interface.

1.3 Component Overview

The PXQ is a modular, software enhanced device that can be configured to suit the application. Each unit consists of a system core and installed hardware modules.

The System Core and Meter Module are offered together as a kit with the selected software package ease of selection.

- System Core (PXQ-STX-SC) – where X corresponds to the installed software tier
- Meter Module (PXQ-MM)
- Software Package (Available in future release)

1.4 Catalog Information

Catalog Number	Description
PXQ-ST1-1A1	Eaton Power Xpert Quality analysis system kit, Software Tier 1, System Core and Meter Module Kit
PXQ-ST2-1A1	Eaton Power Xpert Quality analysis system kit, Software Tier 2, System Core and Meter Module Kit
PXQ-ENH	PXQ enhanced capture module (10MHz event capture)
PXQ-SER	PXQ sequence of events recorder module (sequence of events, status monitoring and WAGES dashboard)

1.5 Support

For technical assistance or referral to a local authorized distributor, contact Eaton:

Email: mrsupport@eaton.com

1-844-435-8982

1-877-836-2273 option 2 Sub option 5,1,1

Additional documentation can be found at www.eaton.com/meters

2. Installation

The PXQ is designed to be installed, operated and maintained by adequately trained personnel. The instructions do not include all details, variations or combinations of the equipment, its storage, delivery, installation, checkout, safe operation of maintenance. Compliance with local, state, and international regulations, as well as with industry standard safety practices for this class of equipment is imperative.

This chapter describes mounting, wiring, startup and miscellaneous details associated with the PXQ. Every section should be reviewed prior to installing this device.

⚠ WARNING

SHOCK HAZARD: VERIFY THAT ANY INCOMING AC POWER OR FOREIGN POWER SOURCES ARE TURNED OFF AND LOCKED OUT BEFORE PERFORMING ANY WORK ON THE POWER XPERT QUALITY OR ASSOCIATED EQUIPMENT. Failure to do so can result in death, injury, or damage to equipment.

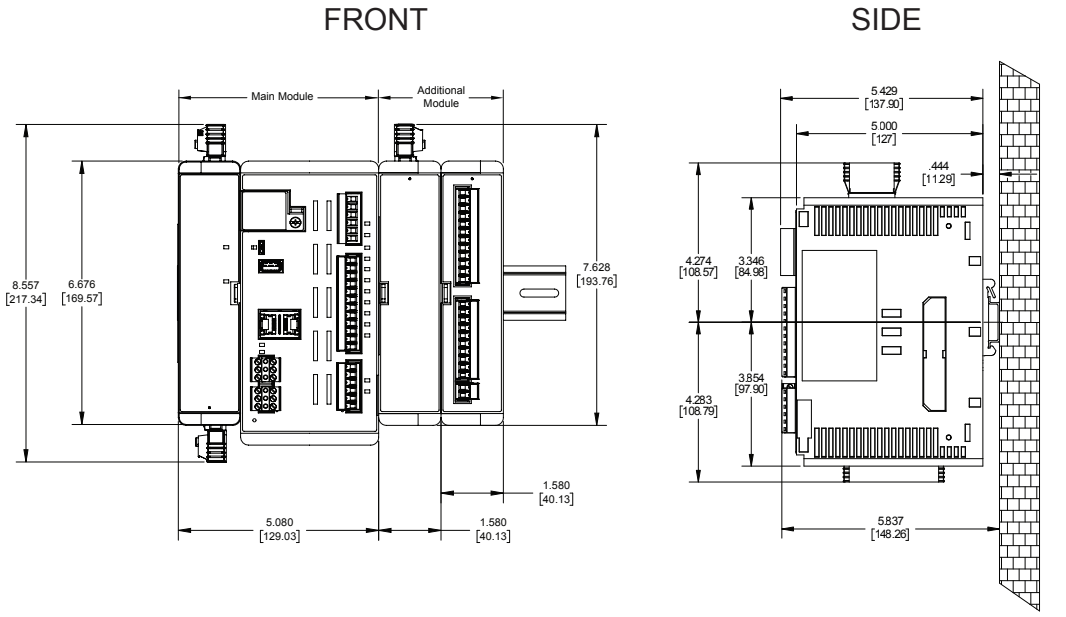
2.1 Mounting

It is recommended that the PXQ is mounted in an electrical switchgear enclosure that is suitable for its environment. The PXQ is designed with flexibility in mind. The PXQ can also be mounted remotely on a flat surface or panel elsewhere in the enclosure.

- To maintain proper ventilation, the PXQ must remain vertical at all times with 3 inches of clearance above and below the chassis. Allow for a minimum of 2 inches of clearance in front of the face plate for wiring clearance from the connectors.
- The PXQ should be installed in a Pollution Degree II environment.
- The PXQ is DIN Rail mountable with each module having its own DIN Rail connectors. For additional mechanical security, a DIN Rail stopper such as IDEC BNL6 can be attached to the DIN Rail on each side of the PXQ.
- The PXQ is DIN Rail mountable with each module having its own DIN Rail connector/s.
- Before mounting on the panel or door, check the required three-dimensional clearances for the PXQ case, particularly behind the panel (see the dimension drawings in this chapter). If mounting on a swinging door, check the swinging clearance of rear projections and wired connections. Remember to allow for extra space for all wiring, intermediate terminal strips, shorting blocks or any other required components.
- Attach the PXQ to the flat panel using DIN Rail previously mounted.

2.1.1 Dimensions

Eaton PXQ's unique modular design enables a customized solution to fit your application and grow along with you.



THERE WILL BE .4" GAP BEHIND SYSTEM

2.1.1.1 Base Unit Characteristics

- Module height (typical): 6.7″ (170.18mm)
- Height (including connectors): 8.557″ (217.34mm)
- Module width (each): 1.58″ (40.13mm)
- Module depth: 4.5″ (114.3mm)

2.2 Field Installation (Per UL2808)

2.2.1 When performing a field installation of a PXQ:

1. Devices that are field-installed are not suitable for Class 2 wiring methods and should not be connected to Class 2 equipment.
2. Secure the PXQ and route the conductors to avoid direct contact with live terminals or bus.

2.2.2 Use only copper conductors when connecting equipment with field-wiring terminals.

2.2.3 When field installing or removing current sensors and/ or PXQ:

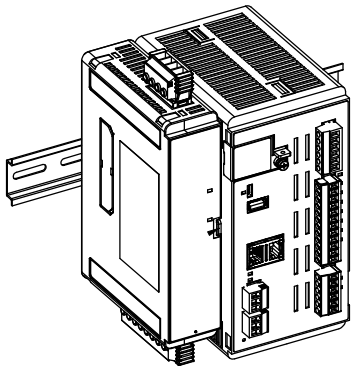
1. The Energy Meter may not be field installed in the wiring space of enclosures for switches or overcurrent devices if the total area of the PXQ, conductors, splices, taps, and equipment exceeds 75% of the cross-sectional area of that space.
2. Restrict installation of PXQ in areas where it would block ventilation openings.
3. Restrict installation of PXQ in areas of breaker arc venting.
4. The PXQ shall not be mounted within 50.8 mm (2 in.) of any live parts including primary conductors, primary terminals, primary lugs. This requirement excludes insulated cables.
5. PXQ attached to the enclosure shall not contact the panel's interior insulation.
6. Mounting provisions shall not be attached to any live part.
7. Voltage sensing and power supply connections to the primary voltage shall have overcurrent protection.
8. Do not install PXQ equipment in any area where breaker arc venting exhaust gases could be re-directed because of submetering equipment installation.

2.2.4 Mounting the PXQ

PXQ mounts on standard 35mm DIN rail. Allow sufficient wire bending radius when mounting.

To mount the unit:

1. Seat the DIN clip against the channel of the DIN rail.
2. Press in until the PXQ clips into place securely on the rail.

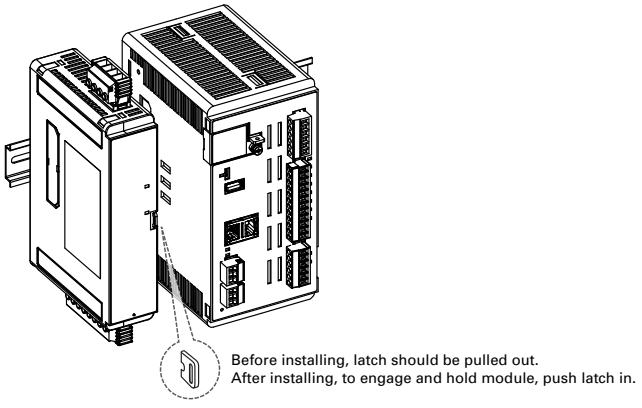


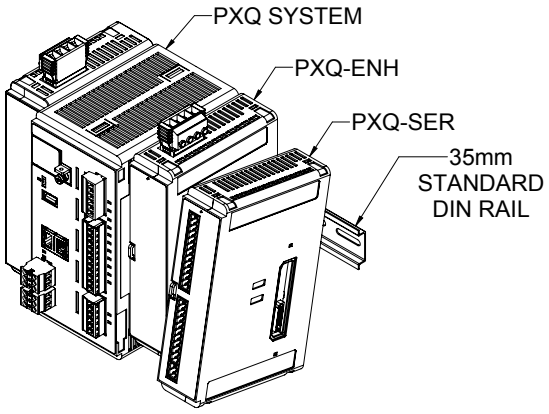
If mounting the PXQ is difficult, modules can be removed from the system core and installed one at a time. Ensure latch (see image below) is at pulled position before attaching meter module to system core.

Note: Do not attempt to install or uninstall meter or associated modules while it is energized.
When installing both PXQ-ENH & PXQ-SER, PXQ-ENH must be connected directly to PXQ System Core then PXQ-SER connected directly to PXQ-ENH.

To install a module:

1. Clip the module to the DIN rail.
2. Slide until it is flush with the rest of the PXQ assembly.
3. Press the latch inwards to engage and hold the module in place.





To remove a module:

1. Pull latch to disengage.
2. Module should pull away from the assembly.
3. If mounted on the DIN rail, slide module away from the assembly and unclip from DIN rail.

2.3 Wiring

Wiring of the PXQ must follow a suitable wiring plan drawing. The phase wiring plan refers to the drawings made for the specific application. It describes all electrical connections between the PXQ and external equipment. A network wiring diagram can also be helpful for networked systems. Specific wiring diagrams are useful when creating the overall wiring plan drawing. Wiring diagrams for each system configuration are addressed below.

⚠ WARNING

SHOCK HAZARDS: IF THIS DEVICE IS BEING USED ON A SINGLE PHASE SYSTEM, WIRE TO PHASE A AND NEUTRAL.

The following general considerations should be complied with during the wiring of the PXQ.

All wiring must conform to applicable Federal, State and Local codes.

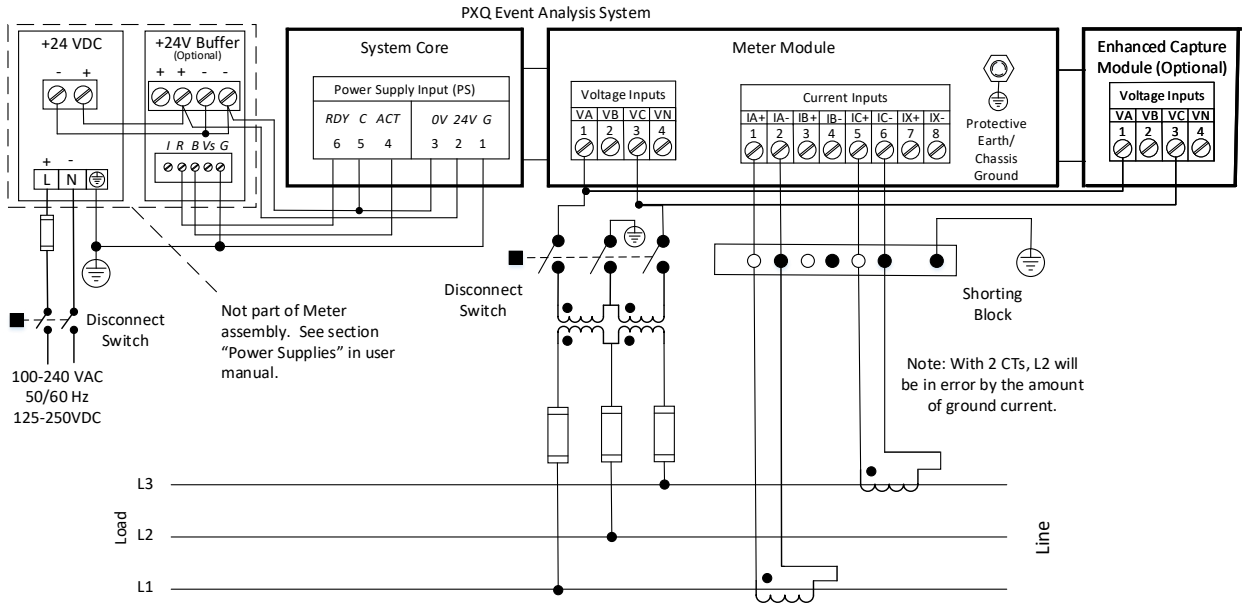
The wires to the terminal blocks must not be larger than AWG No. 10 (CT, VT, VX). Larger wire will not connect properly to the terminal block.

Wiring diagram contacts are shown in their de-energized position.

A good low impedance ground is essential for proper functioning.

Please use UL certified external CT's for installation.

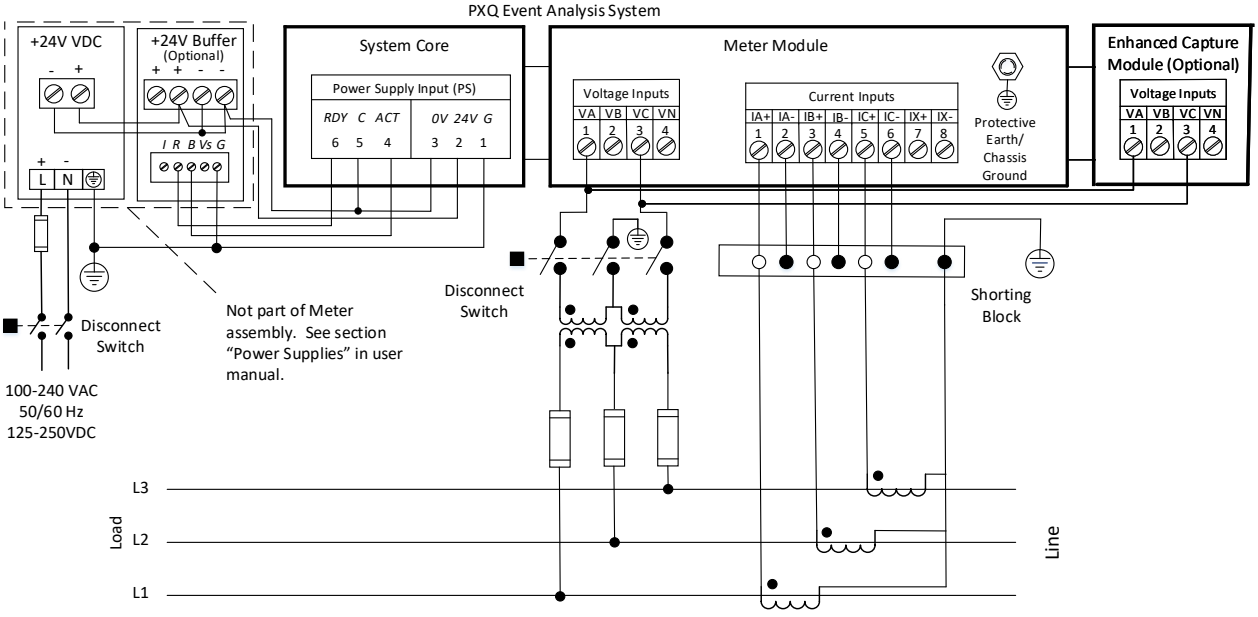
2.3.1 CT & VT Wiring



Note: For voltage connections, fuses should be sized in accordance with best practices to protect the instrumentation wire.

Figure 2.1 : 3-phase, 3-Wire Delta (up to 480 Volts) 2 CT. External Voltage and External Current Transformers.

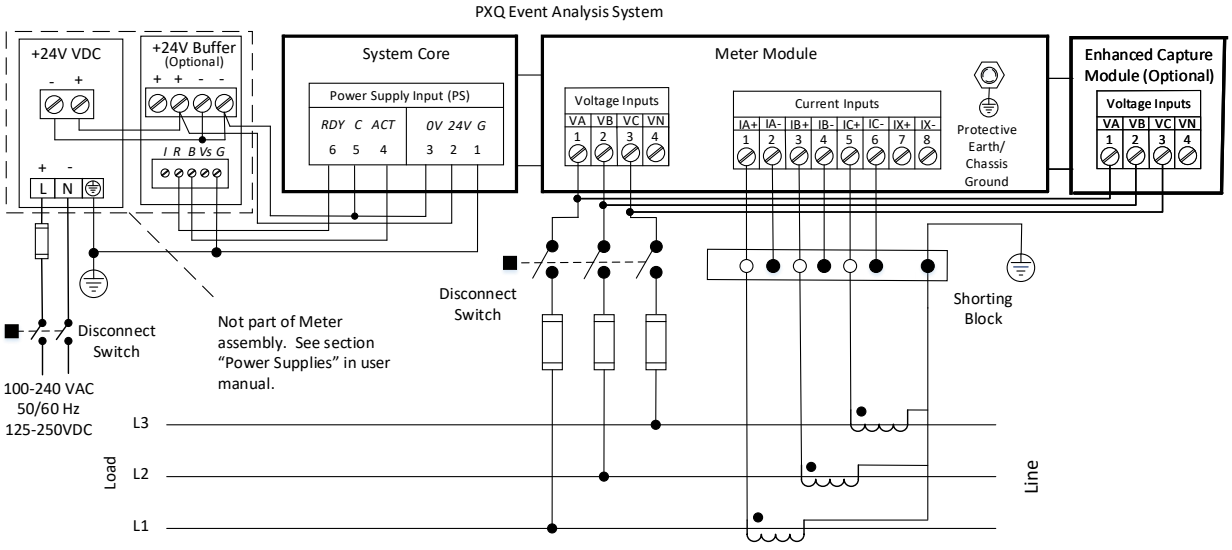
Figure 2.1: 3 Phase, 3 Wire Delta (2CT). Up to 480V, External Voltage and External Current Transformers.



Note: For voltage connections, fuses should be sized in accordance with best practices to protect the instrumentation wire.

Figure 2.2: 3-phase, 3-Wire Delta (up to 480 Volts) External Voltage and External Current Transformers.

Figure 2.2: 3 Phase, 3 Wire Delta (3CT). Up to 480V, External Voltage and External Current Transformers.

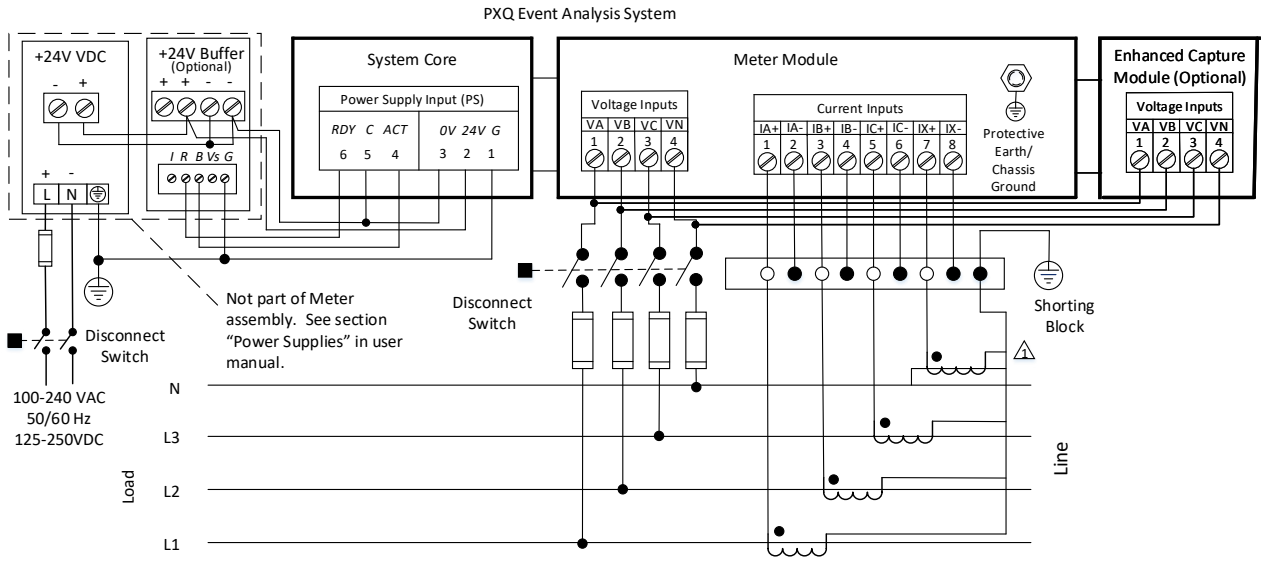


Note: For voltage connections, fuses should be sized in accordance with best practices to protect the instrumentation wire.

Figure 2.3: 3-phase, 3-Wire Delta (up to 480 Volts) Direct voltage and External Current Transformers

Figure 2.3: 3 Phase, 3 Wire Delta (3CT). Up to 480V, Direct Voltage and External Current Transformers.

Note: 3 Phase, 3 Wire Delta (2CT). Up to 480V, Direct Voltage and 2 External Current Transformers.



Note: For voltage connections, fuses should be sized in accordance with best practices to protect the instrumentation wire.

1 Optional Ground CT can be replaced by a zero-sequence CT

Figure 2.4: 3-phase, 4-Wire Y (up to 600 Volts) Direct Voltage and External Current Transformers

Figure 2.4: 3-phase, 4-Wire Y (3 CT). Up to 600V, Direct Voltage and External Current Transformers.

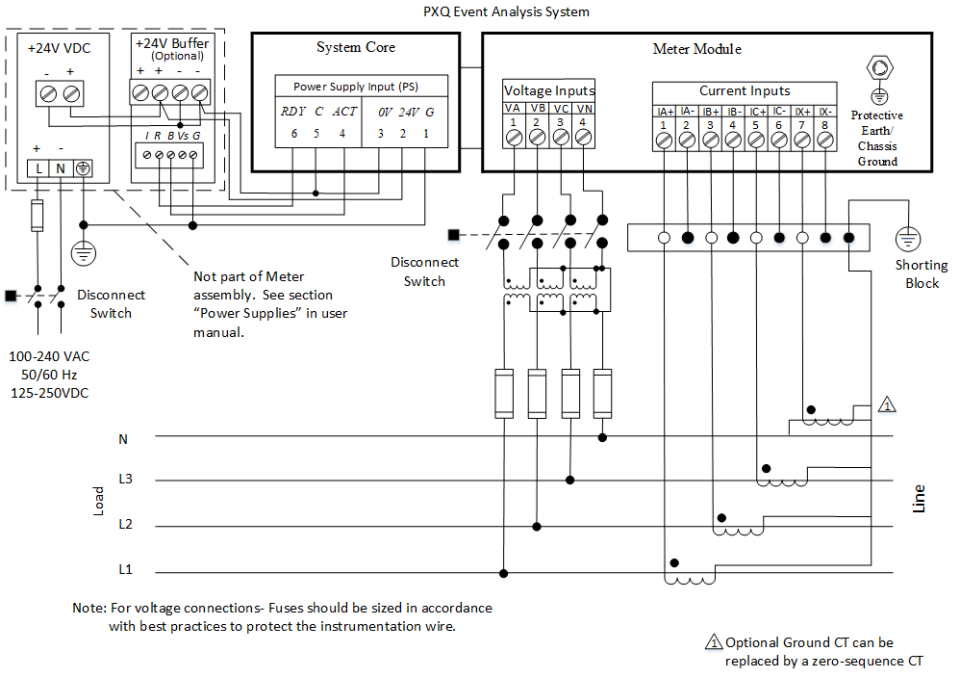


Figure 2.5: 3-phase, 4-Wire Y (3 CT). Up to 600V, External Voltage and External Current Transformers.

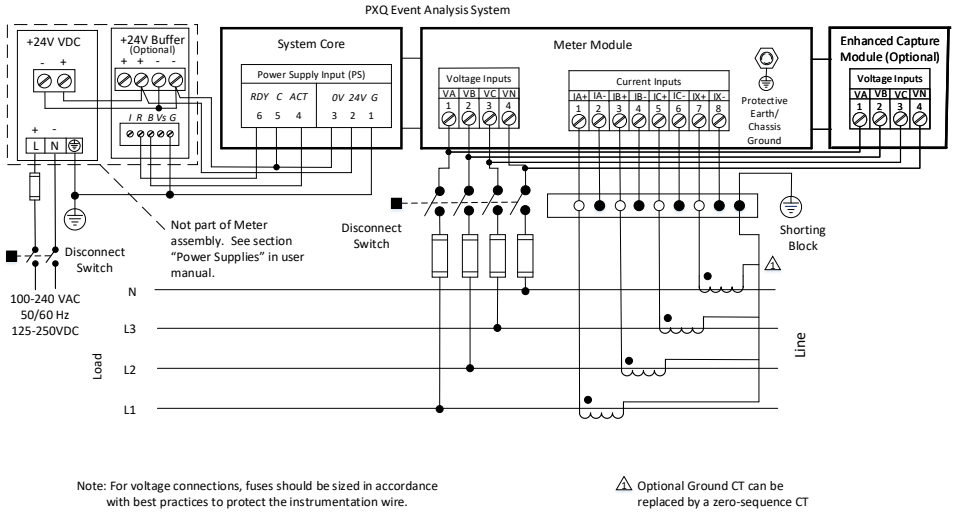


Figure 2.6: 3 Phase, 4 Wire, Center Tap Delta (3 CT). 1 3 2 N Up to 600 Volts, Direct Voltage and External Current Transformers.

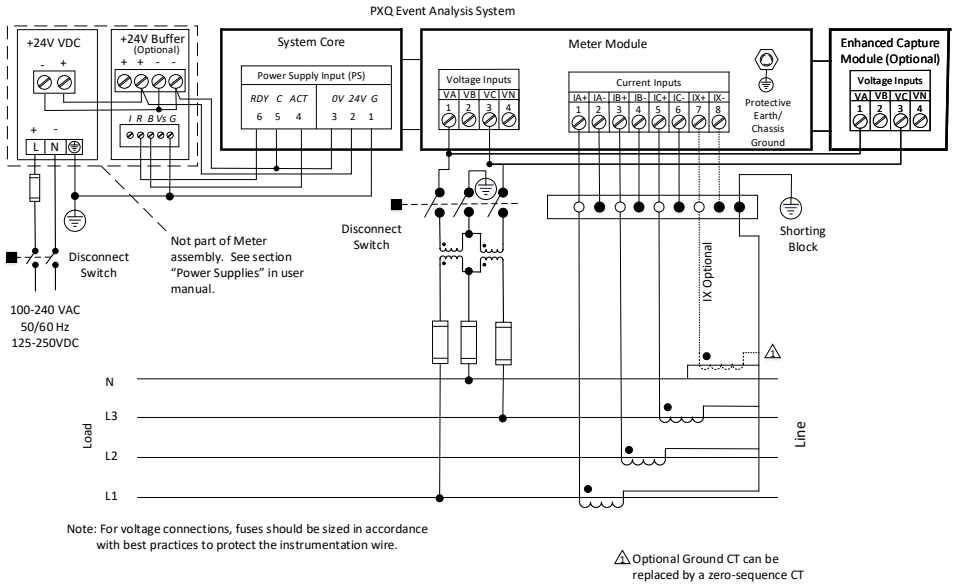


Figure 2.7: 3 Phase, 4 Wire Y (3CT, 2V) 2 ½ Element

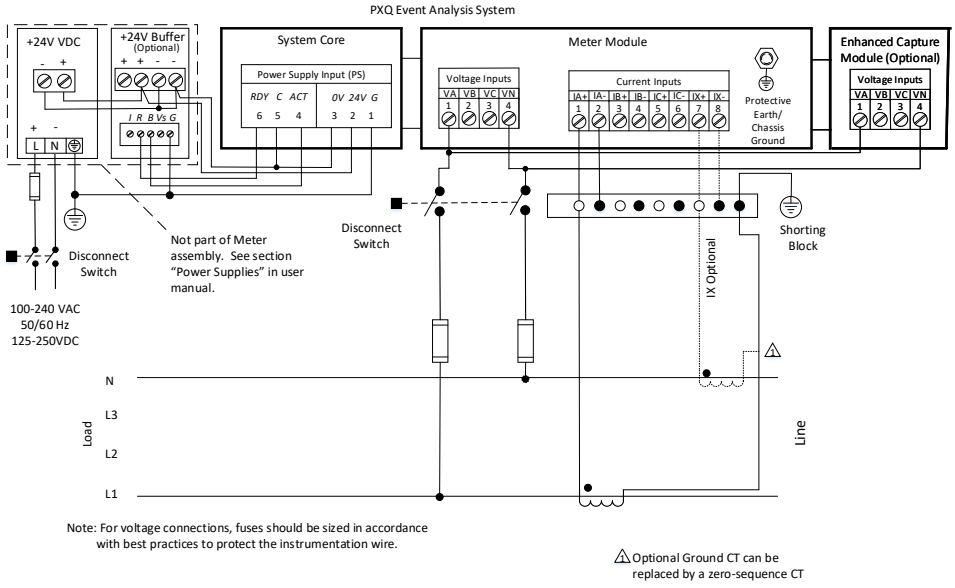


Figure 2.8: Single Phase 2 Wire (1CT).

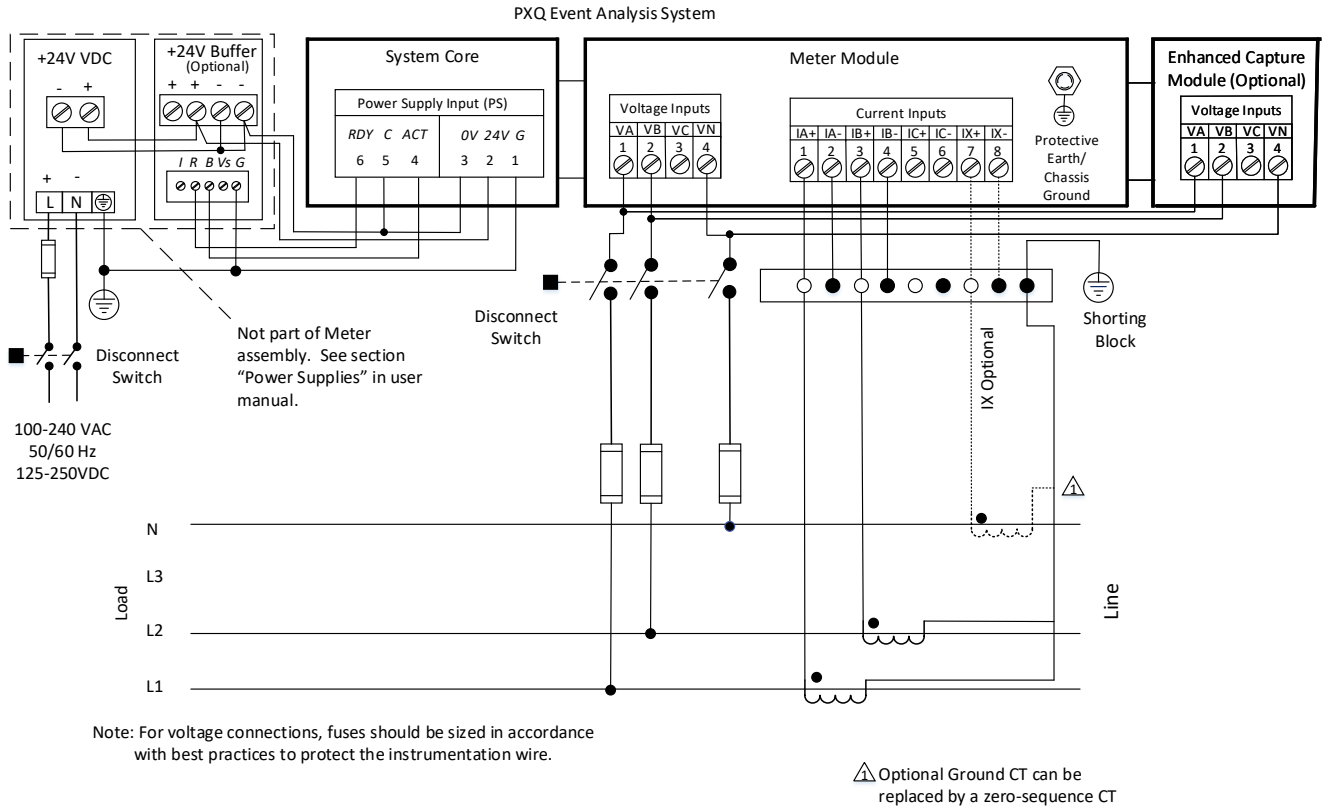


Figure 2.9: Single Phase 3 Wire (2CT).

2.3.2 Control Power Voltage Input

The PXQ event analysis system requires control power from a nominal 24-48 Vdc, 50VA supply (such as Eaton PSG60E). Accessory modules, such as PXQ enhanced capture module or sequence of events recorder modules, are also powered by this supply through an interface connector when attached to PXQ system core.

2.4 Metering specifications

Table 1. Metering standards compliance

Description	
Power Quality	61000-4-30, Class A IEC 62586 IEC 62586-2
Harmonics and spectral components	IEC 61000-4-7
Flicker	IEC 61000-4-15 IEEE 1453
Power Monitoring Devices (PMD)	IEC 61557-12/SD/K70.0.2 IEC 61557-12/SS/K70.0.2

Table 2. Metering accuracy

Description	
Current (A)	IEC 61557-12 Class 0.1
Voltage (V)	IEC 61557-12 Class 0.1

Active energy (Wh)	IEC 62053-22 Class 0.1 S ANSI C12.20:2015 Class 0.1 ANSI C12.1:2022 Class 0.1
Reactive energy (VARh)	IEC 62053-24 Class 0.5 S
Apparent energy (kVAh)	IEC 61557-12 Class 0.2
Active power (W)	IEC 61557-12 Class 0.1
Reactive power (Vars)	IEC 61557-12 Class 1
Apparent power (VA)	IEC 61557-12 Class 0.2

Table 3. Compatible power systems

Description	
Supported wiring configurations	Single Phase 2 Wire (1 CT) Single Phase 3 Wire (2 CT) 3 Phase, 3 Wire Delta (3 CT) 3 Phase, 3 Wire Delta (2 CT) 3 Phase, 4 Wire Y (3 CT) 3 Phase, 4 Wire, Center Tap Delta (3 CT) 3 Phase, 4 Wire 2 1/2 Element Y (3 CT, 2V)
Nominal V L-L (PT Primary)	100-150,000 V L-L
Nominal Application Current (CT Primary) Phases or Neutral/Ground CT	5-12,500 A
Power Monitoring Devices (PMD)	IEC 61557-12/SD/K70.0.2 IEC 61557-12/SS/K70.0.2

2.5 Current and voltage input specifications

Table 4. Meter current input (Meter Module IA, IB, IC, IX)

Description	
Installation category	CAT-III
CT rating	5A nominal
Sampling rate	Up to 1024 samples per cycle, true RMS
Accuracy	IEC 61557-12 Class 0.1
Metering range at specified accuracy	0.05 A to 20 A (Continuous)
Maximum current rating	20A (continuous)
Frequency range at specfied accuracy	42.5-69 Hz (50/60 Hz nominal)
Overload current withstand rating	500A rms for 1.0 second (non repeating)
Maximum voltage (dielectric withstand) rating	400V continuous 3500 Vac for 1 minute
Impedance	<0.010 Ohm (per phase)
Burden	<0.05 VA at 5A (per phase)
Wiring	10-18 AWG to removable 8-position terminal plug

Table 5. Meter standard voltage input (Meter Module VA, VB, VC, VN)

Description	
Installation category	CAT-III
Sampling rate	Up to 1024 samples per cycle, true RMS
Accuracy	IEC 61557-12 Class 0.1
Metering range at specified accuracy	30 to 700 Vac rms L-G
Frequency range at specfied accuracy	42.5-69 Hz (50/60 Hz nominal)

Maximum allowable voltage rating	347 Vac rms L-N 600 Vac rms L-L
Overload voltage withstand rating	1700 Vac rms (sustained)
Maximum voltage (dielectric withstand) rating	3500 Vac for 1 minute
Impedance	5 MOhm
Wiring	10-18 AWG to removable 4-position terminal plug

Table 6. Meter high speed transient voltage input (Enhanced Capture Module VA, VB, VC, VN)

Description	
Installation category	CAT-III
Sampling rate	Configurable, 1,000,000 samples per second or 10,000,000 samples per second
Accuracy	+/- 1V
Metering range at specified accuracy	1V-10kV
Overload voltage withstand rating	1700 Vac rms (sustained)
Maximum voltage rating	UL: 347 VAC RMS L:N /600 VAC RMS L:L IEC: 400VAC RMS L:N/690VAC RMS L:L
Frequency range at specfied accuracy	42.5-69 Hz (50/60 Hz nominal)
Impedance	5 MOhm
Wiring	10-18 AWG to removable 4-position terminal plug

2.6 Power supply input specifications

Table 7. Control power supply input (System core PS)

Description	
Voltage	24-48 Vdc +/-10%
Maximum current draw under load	50 VA PXQ SC + PXQ MM + PXQ ENH + 2-PXQ SER
Typical minimum ride-through time	With no optional accessories: 200 ms (12 cycles at 60 Hz)
Maximum ride-through time	With capacitive ride through: Up to 15 Seconds

2.7 I/O Specifications

Table 8. Digital inputs (System Core IN, D1-D8; SER Module IN, D1-D16)

Description	
Number of inputs	Included on System Core: 8 (built-in) PXQ-SER: 16 additional per module, maximum of 32 additional with 2 modules
Application	WAGES (pulse counting), sequence of events analysis, status change notifications
Event timestamp accuracy	+/- 0.1 ms when connected to PTP source, +/- 1 ms otherwise
Pickup (voltage on state)	>17.5 Vdc
Dropout (voltage off state)	<16.0 Vdc
Maximum voltage rating	60 Vdc
Minimum pulse width	10 ms
Maximum pulse rate	50 pulses per second (10 ms on, 10 ms off)

Nominal current draw (per input)	On: 2.5 mA Off: 0.5 mA
Voltage output (for external device activation)	Nominal 24 Vdc Suitable for driving external devices' dry contact for digital input pickup
Max number of simultaneous status changes	16
Wiring	12-18 AWG to removable 12-position terminal plug 0.25 mm2 to 2.5 mm2 wire ferrule recommended

Solid state outputs (System Core DO1/DO2 and SER DO)

Description	
Number of outputs	2
Type	Form C (NO and NC bidirectional)
Maximum pulse rate	20 pulses per second (25 ms on, 25 ms off)
Pickup/Dropout time	< 25 ms
Maximum voltage rating	60 Vdc
Isolation (Circuit to Ground or DO1 to DO2)	Up to 2kV for 1 minute
Rated Current/Load Current	170 mA
Wiring	12-18 AWG to removable 6-position terminal plug 0.25 mm2 to 2.5 mm2 wire ferrule recommended

2.8 Safety and environmental specifications

Table 9. Safety

Description	
Safety	IEC/EN61010-1 UL 61010-1 File E530482

Table 10. Environmental conditions

The PXQ and current sensors can be housed in a NEMA or UL enclosure that keeps the internal environment within the PXQ's environmental specification ranges and provides suitable fire and mechanical protection in the end-product installation.

Description	
Operating temperature	-20 to +70° C (-4 to +158° F)
Storage temperature	-40 to 85° C
Humidity rating	5-95% relative humidity, noncondensing
Installation category / Overvoltage category	III
Operating altitude	0 to 9843 ft (0 to 3000m)
IP degree of protection rating	IEC 60529 IP 20 minimum, IP30 preferred, NEMA 1

Table 11. Electromagnetic compatibilty

Description	
EMC standards	IEC62052-11, IEC61326-1, IEC61000-6-5
Conducted and radiated emissions*	EN55011 and EN55032 Class B, FCC Part 15 Class B, ICES-003 Class B
Emission limits for harmonics and flicker	Harmonics fluctuations: EN 61000-3-2 Voltage fluctuations: EN 61000-3-3
Immunity	Electrostatic discharge: IEC61000-4-2 Radiated fields: IEC61000-4-3 Fast transients: IEC 61000-4-4 Surges: IEC 61000-4-5 Conducted disturbances: IEC 61000-4-6, IEC61000-4-16 Power frequency magnetic fields: IEC61000-4-8 Conducted disturbances, 2-150 KHz: CLC/TR 50579 Voltage dips and interruptions: IEC 61000-4-11 Ring waves: IEC 61000-4-12
Surge withstand capability	IEEE/ANSI C37.90.1

Note: *Applies to date codes starting 2025. PXQ with date codes 2024 and prior comply with Class B without PXQ-ENH. Compliance is Class A with PXQ-ENH installed.

2.8 Communications Specifications

Table 12. Serial port specifications (System Core COM2)

Description	
Type	RS-485 2-wire, half duplex (D+, D-, Shield/Common)
Number of ports	1
Industrial Protocols	Modbus RTU
Max distance	Up to 4000 ft
Baud Rate	9600 to 115,200 baud Default: 9600
Data Bits	8
Parity	None, Even or Odd. Default: None
Stop Bits	1 or 2 Default: 1
Address range	1-247 Default: 2
Wiring	12-18 AWG to removable 6-position terminal plug 0.25 mm2 to 2.5 mm2 wire ferrule recommended

Note: COM 1 Reserved for future functionality.

Table 13. Ethernet port specifications (System Core E1, E2)

Description	
Connections	10/100/1000 BASE-TX (full duplex)
Number of ports	2
Connector type	RJ45
Cable type	Cat5/5E/6 Shielded cable recommended for network connections

Table 14. Communications protocols

Description	
Industrial communication	Modbus TCP/RTU, BACnet/IP
Network communications	SNMP v1, SNMP v3, HTTP, HTTPS, SMTP
Network addressing	IPv4, IPv6, DHCP
Time synchronization	NTP/SNTP, PTP
File transfer	sFTP (secure FTP)
Authentication	LDAP for OpenLDAP systems
Cryptography	TSL 1.2, SSL

3. Configuring & Commissioning

After the unit is installed, commissioning can begin.

Initial configuration is required to be performed using the PXQ USB Micro B port “Configuration”.

- PXQ USB Micro B port “Configuration” default IP address 192.168.200.101
- PXQ E1 port defaults to DHCP.
- PXQ E2 port defaults to 172.22.17.254.

A USB-A to USB Micro B cable is required.

The USB driver should automatically install with the proper PC permissions otherwise visit www.eaton.com/pxq to download the PX USB Configuration file (px-usb-signed-driver.zip).

Follow the steps below install the USB driver.

3.1 Establishing a connection to PXQ’s Onboard Webserver via USB*

*(Required when connecting forthe first time)

PXQ can communicate to your PC via an off-the-shelf USB A to USB Micro B cable.

To establish a connection to PXQ via USB, use the USB-A to USB Micro B cable to connect the PXQ System Core Configuration port to your computer’s USB port. With the appropriate permissions on your PC, the driver for this should install automatically, allowing you to connect to PXQ by directing your browser to http://192.168.200.101.

If it does not, please take the following steps.

1. Visit www.eaton.com/pxq to download the PX USB Configuration file (px-usb-signed-driver.zip), unzip the file to obtain the PX USB Signed Driver folder.
2. Connect PXQ USB Micro B port to PC USB Type A port.
3. Open **Device Manager** by opening Windows **Start** and typing Device in the search box. Select **Device Manager** when it appears in the list.
4. Look under **Network adapters** and **Ports (COM&LPT)**. You should see a USB Serial Device with a COM port associated: **USB Serial Device (COM#)**.
    - If you do not see this, you may need to look under **Universal Serial Bus controllers** and look for an Unknown Device.
    - If you’re having trouble finding the device, try unplugging the USB connection to the PXQ and reconnecting to verify which device comes up.
5. Right click the device (typically **USB Serial Device (COM#)**), to display the menu and select **Update Driver Software**.
6. When prompted: **How do you want to search for driver software?**, select **Browse my computer for driver software** and then select **Next**.
7. Follow the prompts and navigate to indicate the folder that contains the powerxpert.inf driver that you downloaded in Step 1.
8. The driver will begin to install. Wait for a message that says: “The best driver is already installed” or “Windows has successfully updated your driver software.”
9. Close the dialogue.
10. Launch web browser and then navigate to http://192.168.200.101. A PXQ Login screen is displayed.

Note: The connection will appear in Device Manager as e.g. Linux USB Ethernet/RNDIS Gadget #2.

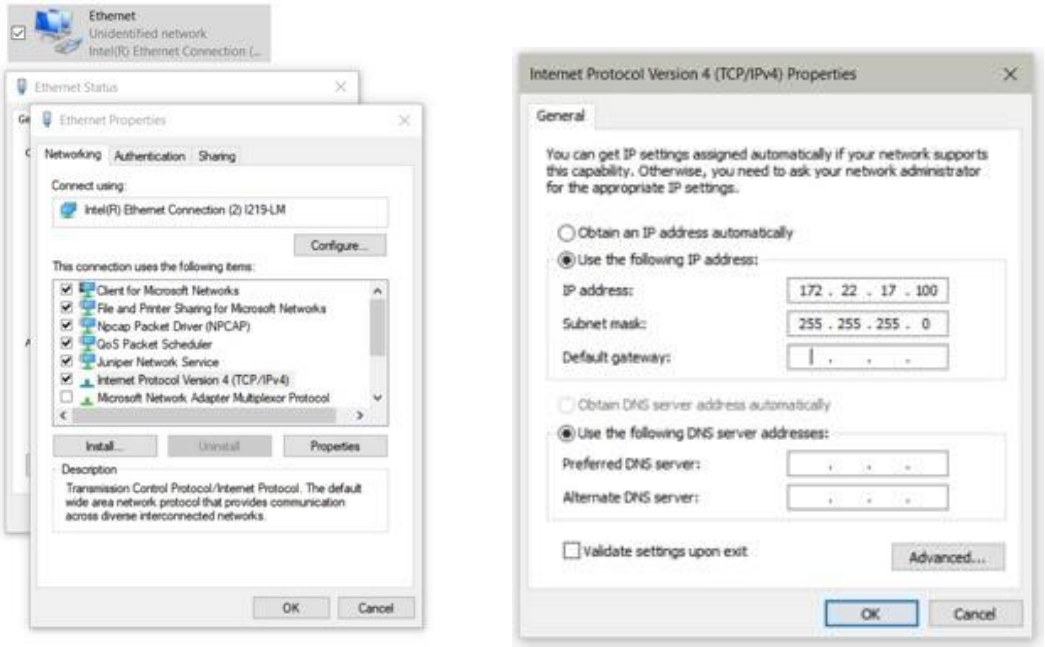
3.2 Establishing a connection to PXQ's Onboard Webserver via Ethernet

If a fixed IP address has been set for either PXQ Ethernet ports (E1 or E2), one may connect directly to the PXQ via Ethernet by setting the computer's IP address to the same subnet.

For this example, we will assume that you are connecting to the PXQ E2 port using its default IP address of 172.22.17.254.

Start by setting the Computer's IP address to a compatible IP address, e.g. set computer IP address to 172.22.17.100 and subnet mask to 255.255.255.0.

1. Go to Windows Control Panel - click Window's Start, search for Control Panel, click on Control Panel app.
2. Go to Network and Sharing Center "Change adapter settings" (there are 2 options);
   - In Control Panel > Network and Internet > View network status and tasks, click "Change adapter settings".
   - In Control Panel > All Control Panel Items > Network and Sharing Center, click "Change adapter settings".
3. Double click on Ethernet "Unidentified Network" or compatible.
4. In the "Ethernet Properties" dialog, double click on "Internet Protocol Version (TCP/IPv4)"
5. In the "Internet Protocol Version 4 (TCP/IPv4) Properties" dialog, select "Use the following IP address:";
   - Enter IP address: 172.22.17.100
   - Enter Subnet mask: 255.255.255.0
   - Leave all other fields blank, click OK
   - Click OK on Ethernet Properties dialog.
   - Close down Network Connections



Launch web browser and navigate to: http://172.22.17.254

A PXQ Login screen is displayed - default login is username: admin, and password: admin.

Note: Once the commissioning is completed and computer disconnected from the PXQ, change the computer IP address back to its default settings, e.g. go to Internet Protocol Version 4 (TCP/IPV4) Properties and select "Obtain an IP address automatically".

3.3 Logging Into PXQ Web Server

A user signs into the PXQ onboard web server by providing Username and corresponding password. Users can be created locally and/or granted access via an organization's directory server. LDAP (Lightweight Directory Access Protocol) compatible directory servers, such as servers supporting Active Directory, are supported.

The PXQ screen appears if the User has not previously signed in. The User provides login credentials by entering appropriate Username and Password. The login credentials are then validated by the web server after the User clicks the "Login" button.

Note: Note: The default administrative Username is admin. The default administrative Password is admin. For security purposes, please change the Password as soon as possible after login.

3.4 Basic Configuration

While logged in as the "admin" user, Navigate to Settings > General > Quick Setup.

The Quick Setup section provides the minimum parameters that must be set to define the PXQ configuration.

This includes:

- File Options
- PXQ name (Displayed in the title bar)
- Wiring Configuration
- Frequency (50Hz or 60Hz)
- Nominal L-L Voltage
- PT Ratio
- CT Ratio (for phases)
- Neutral/Ground CT Ratio
- Selection of calculated current (In or Ig)
- PT Compensation
- CT Compensation

3.5 Additional Settings

- The user may also set the system for the PXQ under General > Clock. The settings for NTP or PTP synchronized time server can also be found under this heading.
- To enable communications, navigate to Network > Access Control. Specific ports must be enabled if desired.
- To change the default IP address for the Ethernet port Eth1, Network > Ethernet/LAN Setup > Eth1 (or Eth2). From there, DHCP can be enabled or static IPv4 or IPv6 addresses can be assigned.
- To add users or edit the default user profiles, navigate to Users > User Configuration. Other settings, like the Password Policy, Role configurations and LDAP settings can be found under the Users section also.

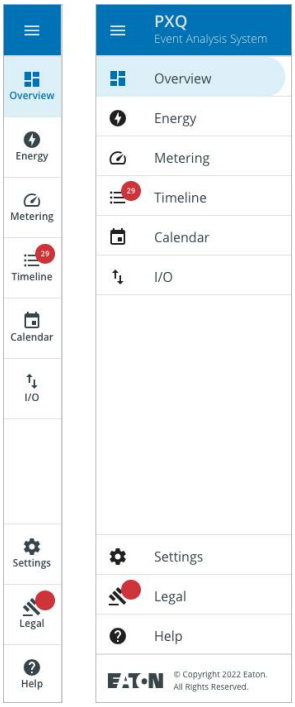
For additional details on PXQ settings, please reference PXQ User Manual, Section 5.

Note: Password Policy setting for Password Expires is defaulted to 90 days (Settings > Users > Password Policy), setting to 0 means it never expires.

4. Web-Server Screens

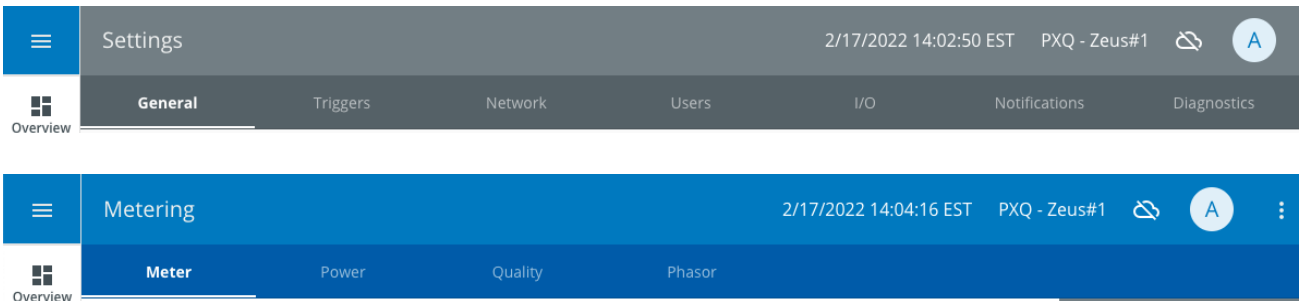
4.1 Main Navigation

The PXQ application is navigated with a navigation bar on the left side of the screen. This bar can be expanded or collapsed by selecting the three horizontal lines next to the application name. Depending on the screen resolution, additional icons may require scrolling.

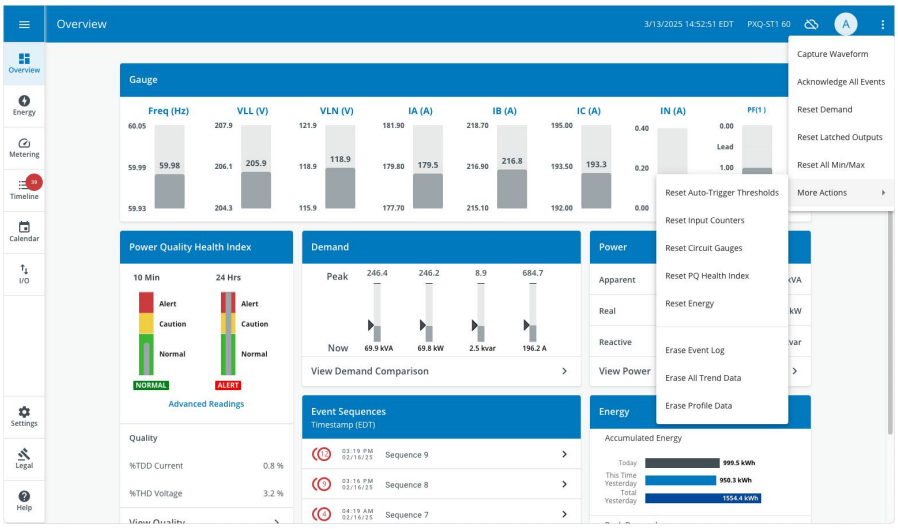


4.1.1 Secondary Navigation

Some areas such as Settings and Metering will have tabs across the top for navigating within that section.



4.1.2 Actions



Select the three dots next to the user's icon at the top right to access a list of actions that may be performed on the PXQ. This list may differ, depending on the user's permission settings.

- Capture Waveforms
- Acknowledge All Events
- Reset Demand
- Reset Latched Outputs
- Reset All Min/Max
- Reset Auto-Trigger Thresholds
- Reset Input Counters
- Reset Circuit Gauges
- Reset PQ Health Index
- Reset Energy
- Erase Event Log
- Erase All Trend Data
- Erase Profile Data

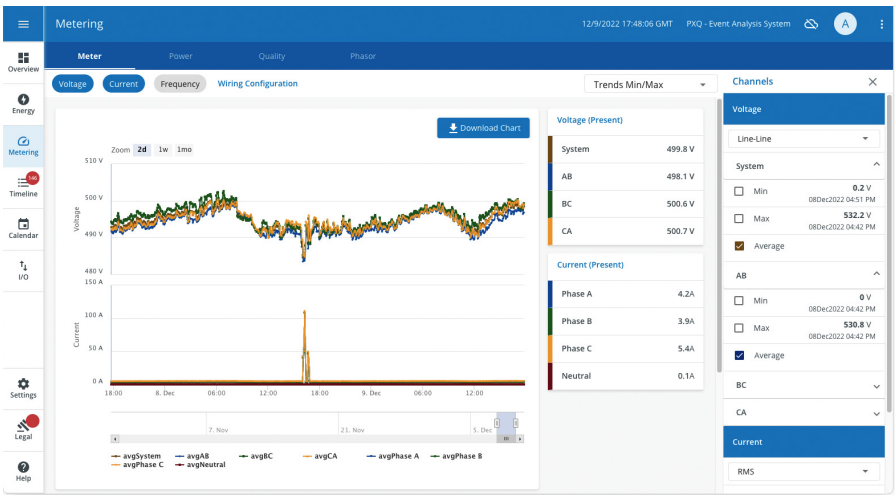
Note: It may take up to 5 minutes for resets or erasures of data (trend, event, etc) data to reflect in the interface.

4.1.3 User Info

Select the user icon at the top right to access features for Login History, Change Password and Log Out.



4.1.4 Interactive Charts



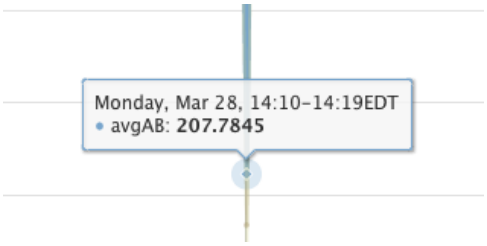
PXQ presents interactive charts that make it easy to explore data trends and anomalies in power quality experienced by the monitored circuit.



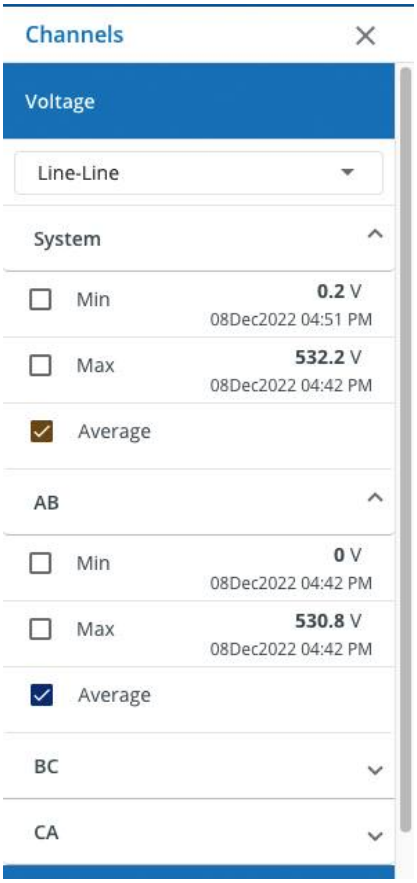
The **Zoom** level of the data can be chosen with the buttons indicating the zoom level. The user can zoom in on data to view it at a higher resolution with a 'rubber-band' technique, dragging from corner to corner of the area of interest.



Data can be further navigated by sliding the window on the **Navigator**. The window can be expanded to view a longer duration by manipulating the handles on either side.

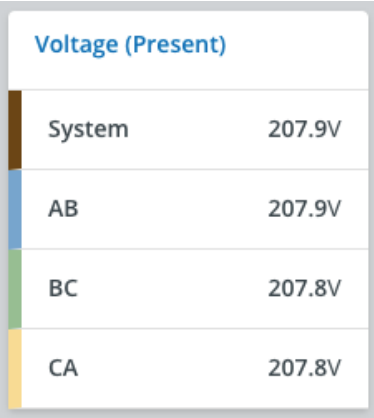


Hovering over the data brings up a **Tool-tip** that describes the parameter, the value and the time-stamp.



Plotted parameters can be selected from the **Channels Sidebar** on the right. Where multiple selections are supported, parameters can be hidden by deselecting them in the legend below each graph or removed by deselecting them in the Channels pane.

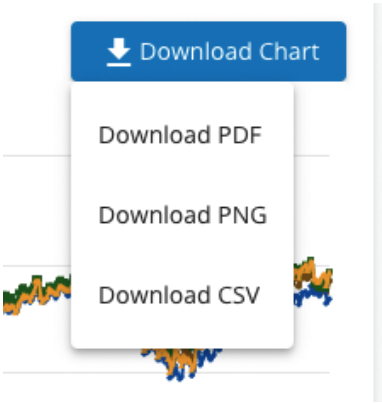
The Channels Sidebar is grouped by phase. The latest value for each parameter is displayed next to each parameter with a timestamp where applicable.



The **Present Values** of each selected parameter are also displayed in a card next to the grap.



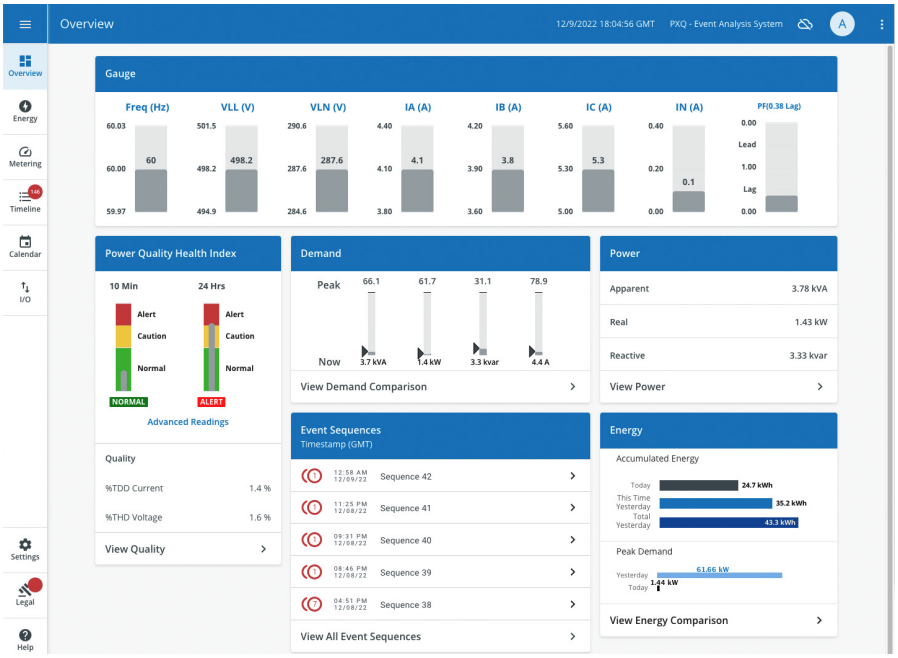
Where available, additional charts can be displayed or hidden with the **Chip Selectors** in the hite action bar. Up to two charts can be displayed at once.



The chart data can be **downloaded** as CSV (data only), or as an image in PDF or PNG format with the use of the download icon in the upper right of each chart.

4.2 Overview

From this screen the user can gain additional information about Overview, Energy, Metering, Timeline, Calendar, I/O, and Settings. These selectable options are displayed in a menu bar at the left of the screen, as illustrated below. When one of the options is selected, the main section of the screen will update to display information about that option.



Each card summarizes a data reading and provides the means to access more detailed information by clicking on the appropriate icons. For example, in the Power Quality Index section below, there are buttons labeled 10 Min and 24 Hrs. By clicking on these options, the user will be able to access information that occurred during those time frames.

Note: The setup screens require a password to edit settings.

4.2.1 Gauge

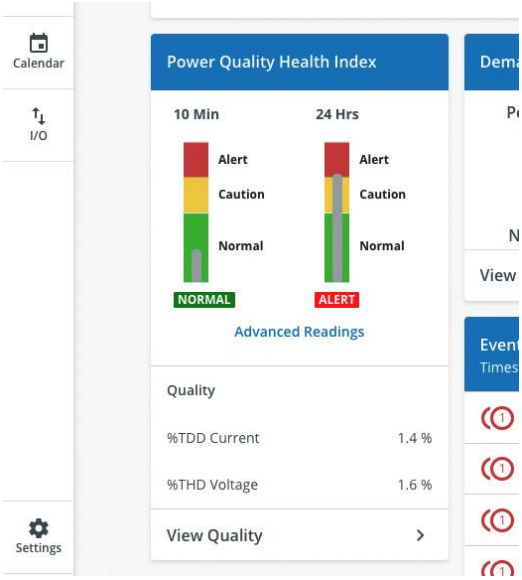


The Gauge card provides a unique at-a-glance view of key electrical parameters. PXQ learns about the electrical distribution system is characterized using statistical analysis. When the system is operating under normal conditions, the indicator is displayed at the middle of the gauge. Watching as the indicator rises or falls gives a visual on the present state of the system, relative to normal.

Each gauge lists an upper and lower statistical range as well as the average. The numerical spread of these values can be used to determine overall system variance. If the range is wide, operating conditions tend to fluctuate; if the range is narrow, the system is stable. The skew of the average within the range (closer to the upper or lower value) shows whether system variations occur evenly or if they are mostly increases or mostly decreases.

The present value is listed above each indicator.

4.2.2 Power Quality Health Index



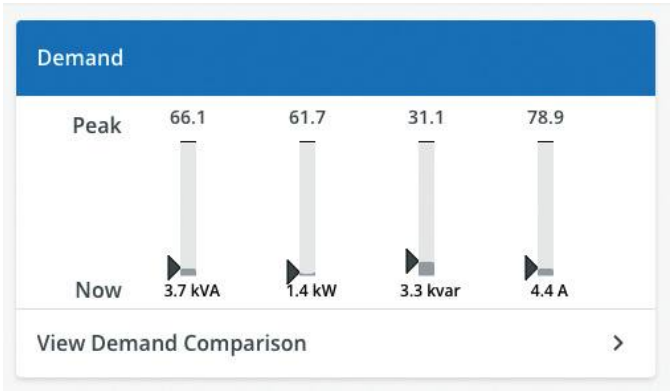
PXQ calculates power quality information for the last 10 minutes and the last 24 hours to provide an assessment of the circuit's health. This all-in-one number accounts for the impact of events and disturbances on the monitored circuit. The color-coded indicator makes it easy to determine the values at a glance.

Expanding either Latest 10 Minutes or Latest 24 Hours provides detailed information about Sags, Swells, dv/dt, TDDi, THDv and Flicker for the time period.

Under Quality, the %TDD-Current, %THD-Voltage and Flicker values are displayed along with a shortcut to the Quality screen in the Metering section.

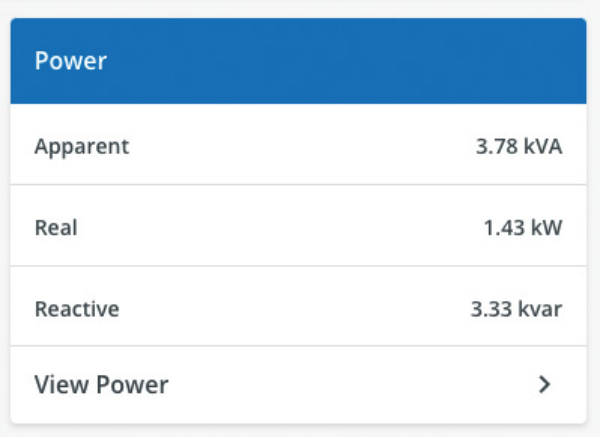
4.2.3 Demand (View Demand Comparison)

The Demand Comparison feature enables graphical comparisons of average power consumption between 2 date ranges. This graphical comparison has selections for comparing average power consumption for today vs yesterday, this week vs last week or this month vs last month. For additional comparisons, the user can select any 2 days, any 2 weeks or any 2 months. The selected view can be exported as a .png, .pdf or a .csv file. The ability to compare power consumption from 2 date ranges can be used to identify opportunities to reduce wasted energy.



4.2.4 Power (View Power)

The power card shows the real, apparent and reactive power being read by the PXQ. A shortcut is provided for quick navigation to the Power screen in the Metering section.

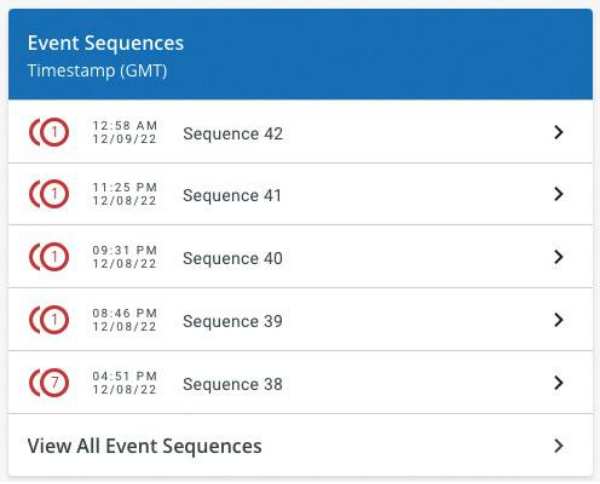


4.2.5 Event Sequence (View All Event Sequences)

The Event Sequences card displays the latest five sequences captured by PXQ along with the timestamp of each.



The icon on the left indicates how many events were part of each sequence.



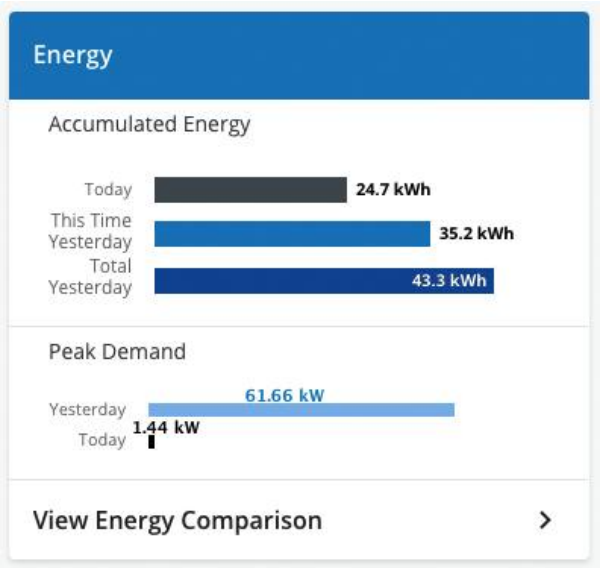
Selecting any event brings up the detailed view in the Timeline screen.

A shortcut, **View All Event Sequences**, is provided for quick navigation to all the events in the Timeline screen.

4.2.6 Energy Usage Summary (View Energy Comparison)

The Energy Usage Summary card displays a comparison of between the energy usage today and the usage of the previous day (yesterday). Accumulated energy and peak demand are displayed, providing an easy visual comparison.

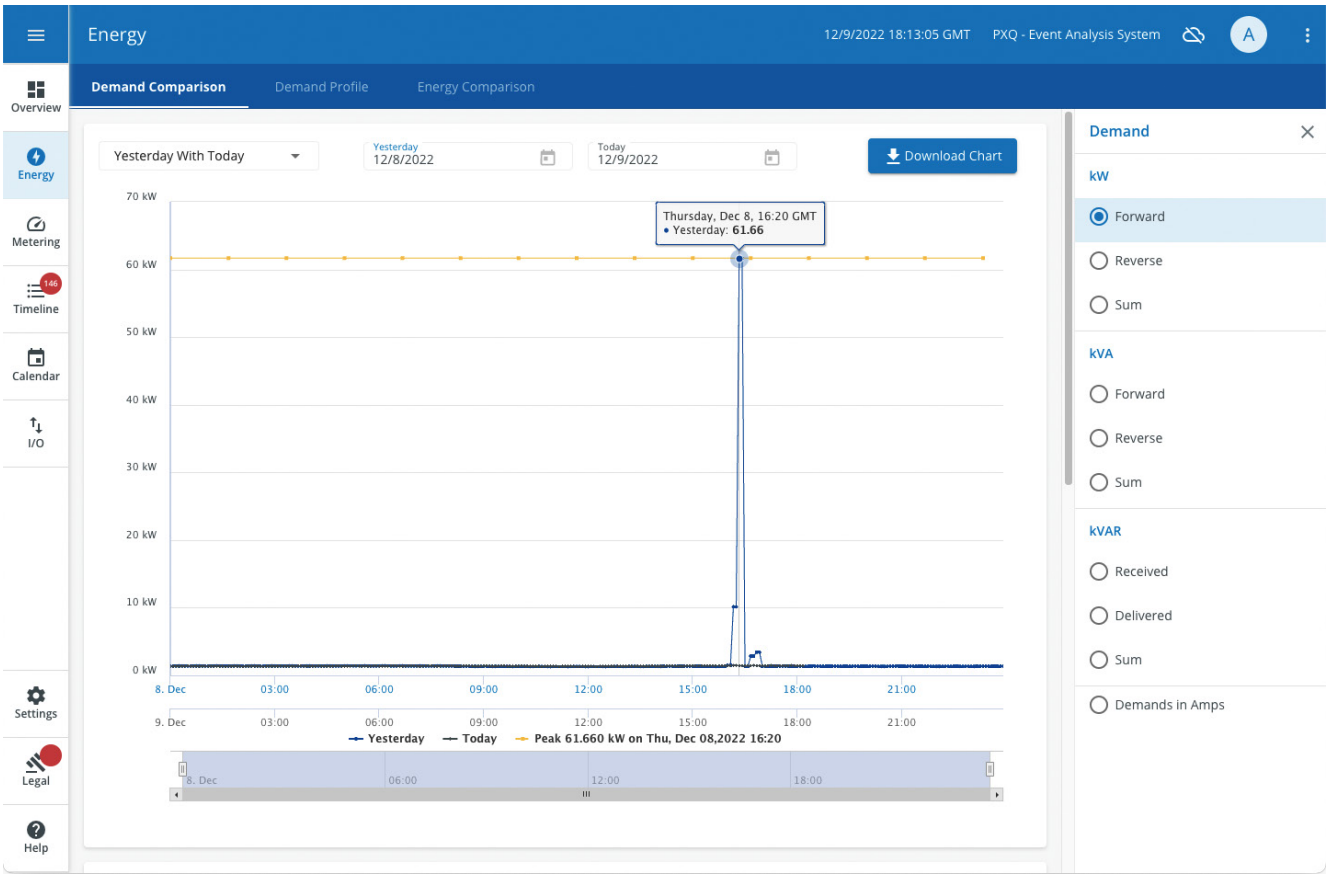
A shortcut, View Energy Comparison, is provided for quick navigation to Energy Comparison screen.



4.3 Energy

The Energy section provides views of the energy usage and demand.

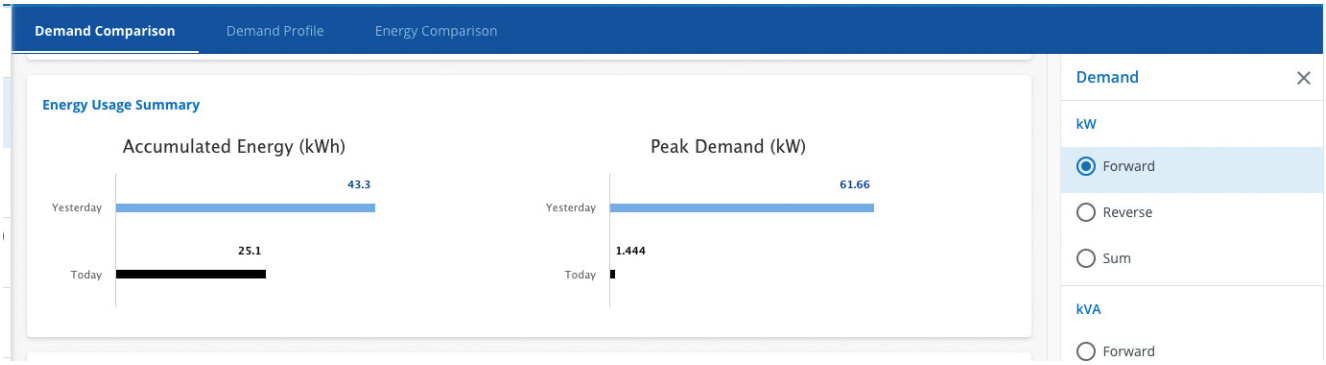
4.3.1 Demand Comparison



The **Demand Comparison** page allows the user to easily view their demand between two given periods. Preset comparisons can be selected from a dropdown. Custom comparison periods can be set by selecting the calendar controls at the top of each chart.

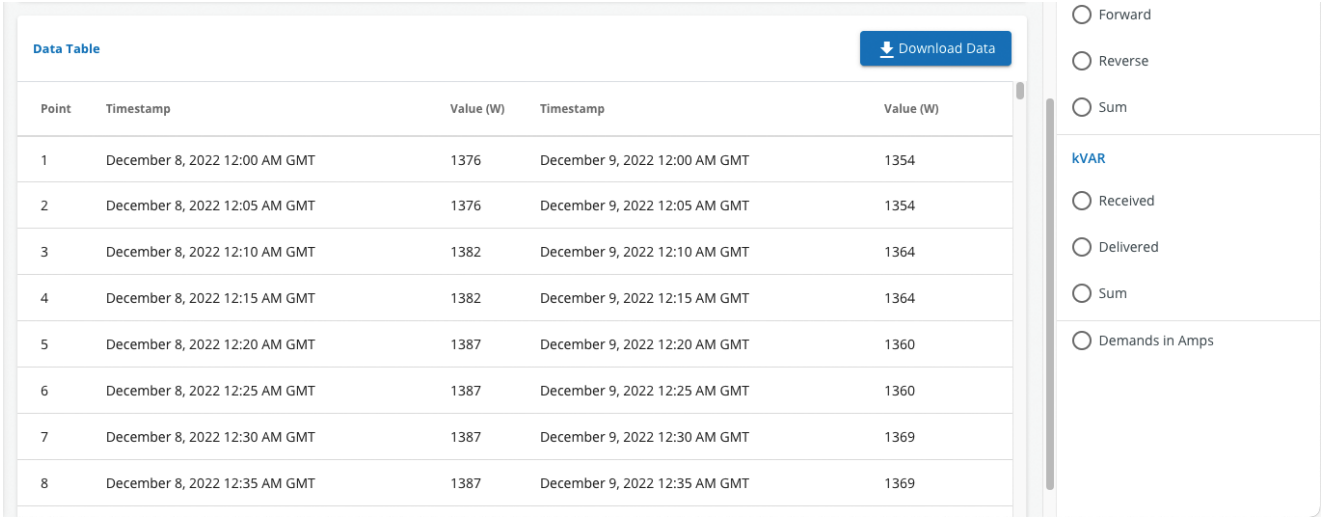
- Yesterday with Today
- Last Week with This Week
- Last Month with This Month
- Two Different Days – enables calendar input for date selection
- Two Different Weeks – enables calendar input for date selection
- Two Different Months – enables calendar input for date selection

The peak demand since last reset is represented as a dashed line.



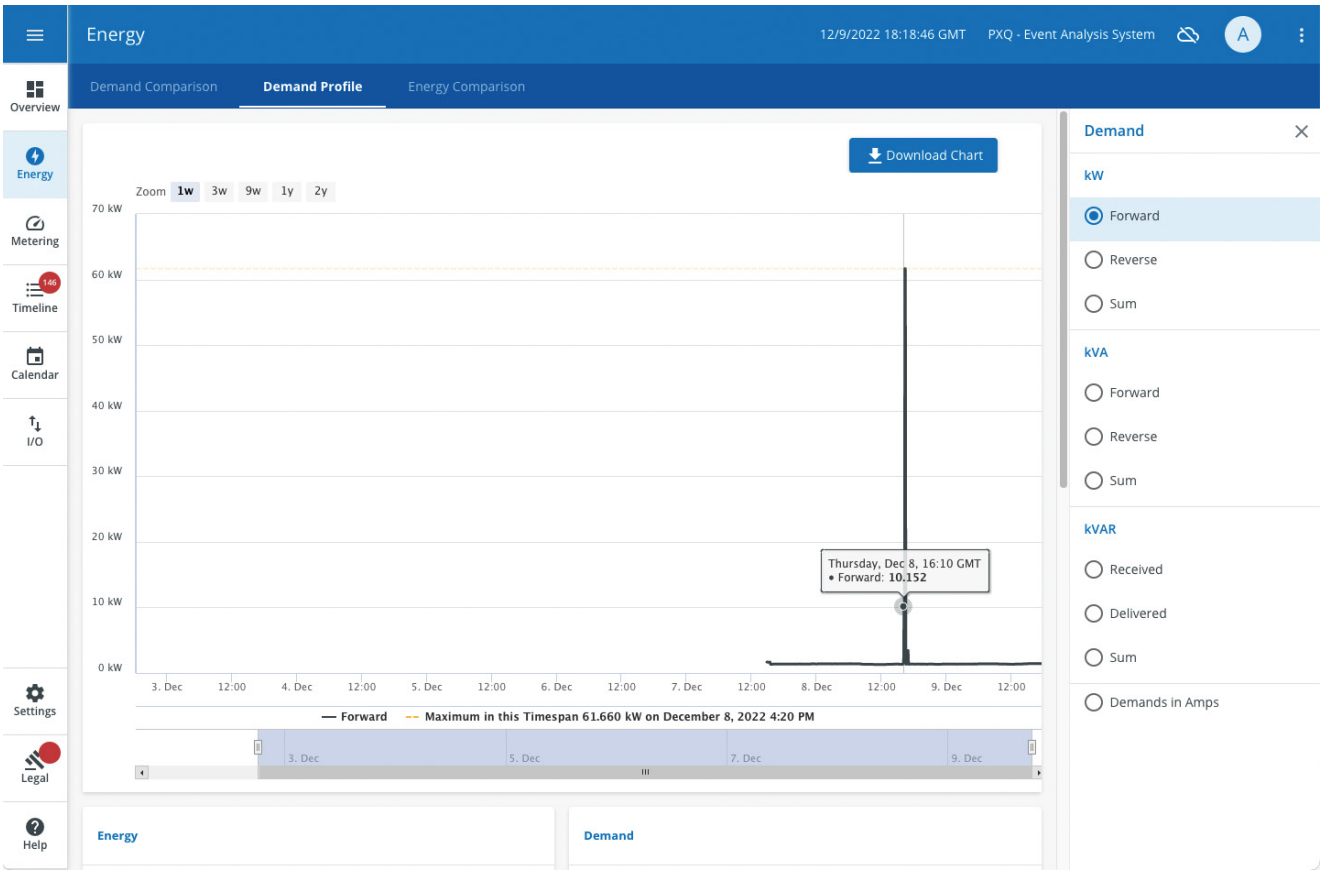
The **Energy Usage Summary** displays Accumulated Energy and Peak Demand for each of the two periods selected so they can be compared.

A selection between kW (forward, reverse or sum), kVA (forward or reverse), and kVAR (Received, Delivered, Sum, each of four quadrants, or total demand in Amps) is available in the side pane.



The **Data Table** for both traces is available here.

4.3.2 Demand Profile

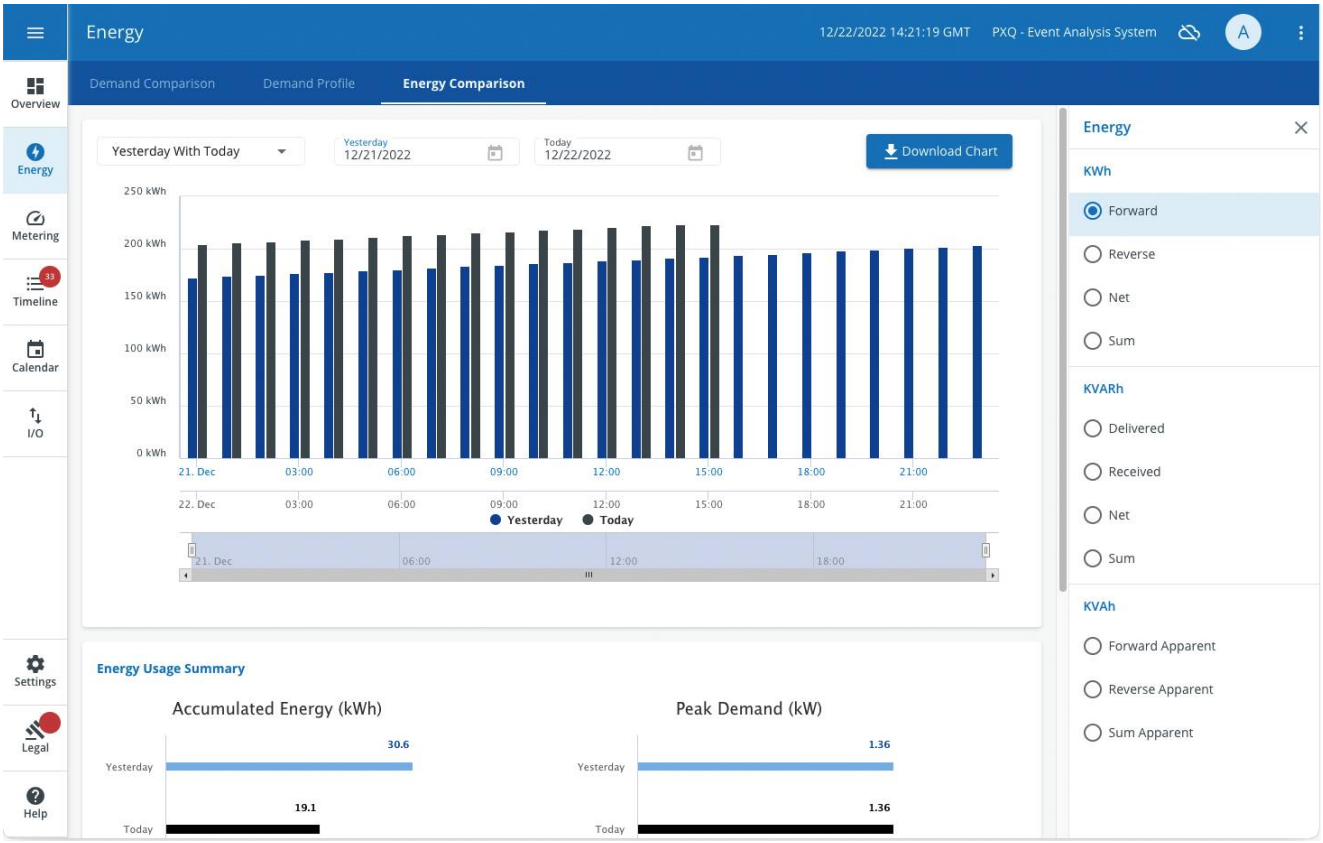


The **Demand Profile** page shows the user's demand history, similar to the comparison page.

A selection between kW (forward, reverse or sum), kVA (forward or reverse), and kVAR (Received, Delivered, Sum, each of four quadrants, or total demand in Amps) is available in the side pane.

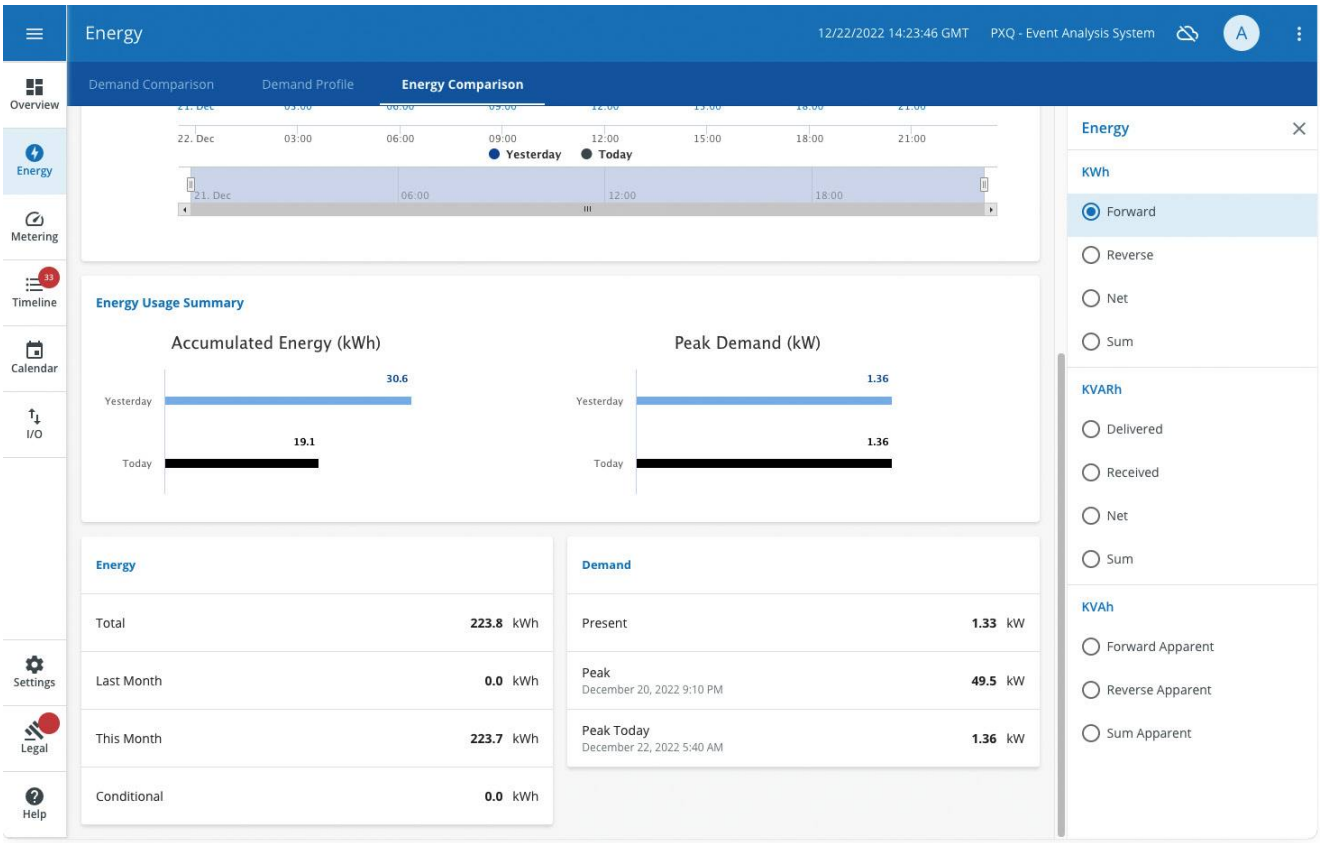
Zoom levels for 1 week, 3 weeks, 9 weeks, 1 year and 2 years are provided for easy access.

4.3.3 Energy Comparison



The **Energy Comparison** page allows the user to easily view their demand between two given periods. Preset comparisons can be selected from a dropdown. Custom comparison periods can be set by selecting the calendar controls at the top of each chart.

- Yesterday with Today (hourly usage)
- Last Week with This Week (daily usage)
- Last Month with This Month (daily usage)
- Two Different Days (hourly usage) – enables calendar input for date selection
- Two Different Weeks (daily usage) – enables calendar input for date selection
- Two Different Months (daily usage)– enables calendar input for date selection



The **Energy Usage Summary** displays Accumulated Energy and Peak Demand for each of the two periods selected so they can be compared.

A selection between kWh (forward, reverse, net or sum), kVARh (each of four quadrants, net or sum), and kVAh (forward, reverse, net or sum) is available in the side pane.

Energy usage for Last Month, This Month, Total and Conditional are displayed in the **Energy** card below the graph.

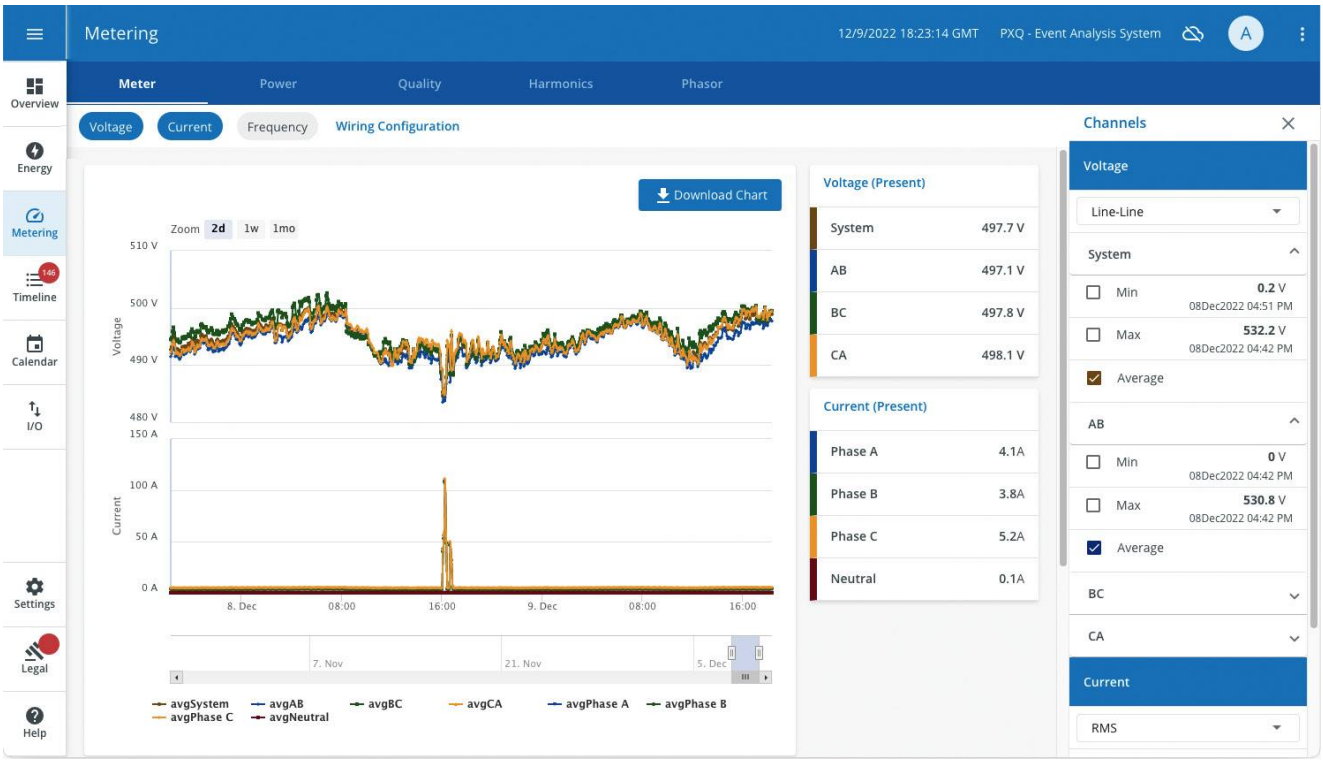
The present demand, peak demand (since last reset) and peak demand today are displayed in the **Demand** card below the graph.

4.4 Metering

4.4.1 General Navigation

The Metering section provides detailed views for users to investigate trends in their historical data. Data is plotted on interactive charts that can be customized by selecting different parameters or changing the time window.

On the right side of the action bar, a drop-down is presented to let the user choose trended data or the associated harmonics or waveform.



4.4.1.1 Trended Data (Meter, Power, Quality)

Trended Data is plotted at the following resolutions:

- 2-day view (5-minute interval data for 42 days)
- 1-week view (15-minute interval data for 180 days)
- 1-month view (5-hour interval data for 730 days)

Multiple channels are plotted at one time displaying Min, Max, Average values that are user selectable. Use Voltage and Current buttons to either display voltage and current, voltage only, or current only. Use Frequency button to display frequency.

Download Chart button provides options to download chart as PDF (screen capture), PNG (screen capture), or CSV file of data. Selecting Download CSV generates a csv file for each channel trended.

4.4.1.2 Harmonics

PXQ provides a view of harmonics up to the 85th harmonic.

Under Metering > Quality, Total Harmonic Distortion (THD) per phase can be displayed as one of four harmonic spectrum parameters: Total, Odd, Even or InterHarmonic along with the percentage. Only one harmonic spectrum parameter can be plotted at one time.

Under Metering > Harmonics, Harmonic data is presented three (4) ways:

Per phase Voltage and Current Distortion waveforms are displayed at the following views:

- 1 cycle
- 4 cycles
- 16 cycles
- 64 cycles

Harmonic Voltage and Current Data are plotted as a bar chart at the following resolutions:

- Interharmonics (frequencies falling between fundamental frequencies)
- Harmonics1-27 (fundamental frequencies 60Hz to 1620Hz)
- Harmonics 1-85 (fundamental frequencies 60Hz to 4860Hz)

Harmonic Data Table displays per phase Voltage & Angel and Current & Angel for fundamental frequencies.

Download Chart button provides options to download chart as PDF (screen capture), PNG (screen capture), or CSV file of data. Selecting Download CSV generates a csv file for each channel trended.

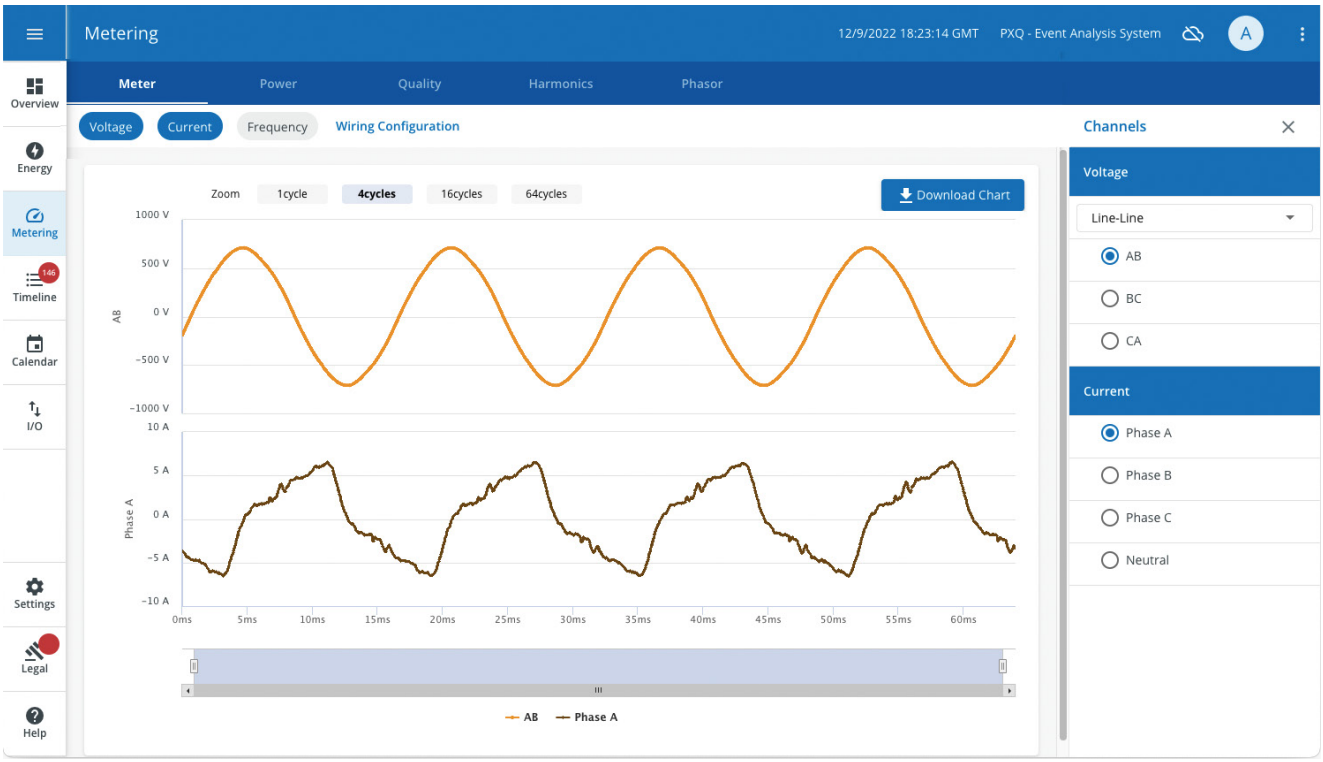
4.4.1.3 Waveform

The latest instantaneous waveform capture for Current and Voltage is displayed under Harmonics.. Currents and Voltages plots can be viewed simultaneously (one above the other) or togged on and off via the chips on the Action Bar. Zoom levels (1, 4, 16 and 64 cycles) are available at the top of the plot area and control both plots. Additional channels may be added or removed in the side bar. Similar to other plot areas, the waveform plot can be exported as PDF, PNG or CSV via the arrow icon in the upper right of the plot area.

4.4.2 Meter

The Meter page displays Voltages (Line-Line, Line-Neutral or Symmetric), Current (RMS or Symmetric) and Frequencies. Average, min and max are given for each parameter, along with a system-wide reading.

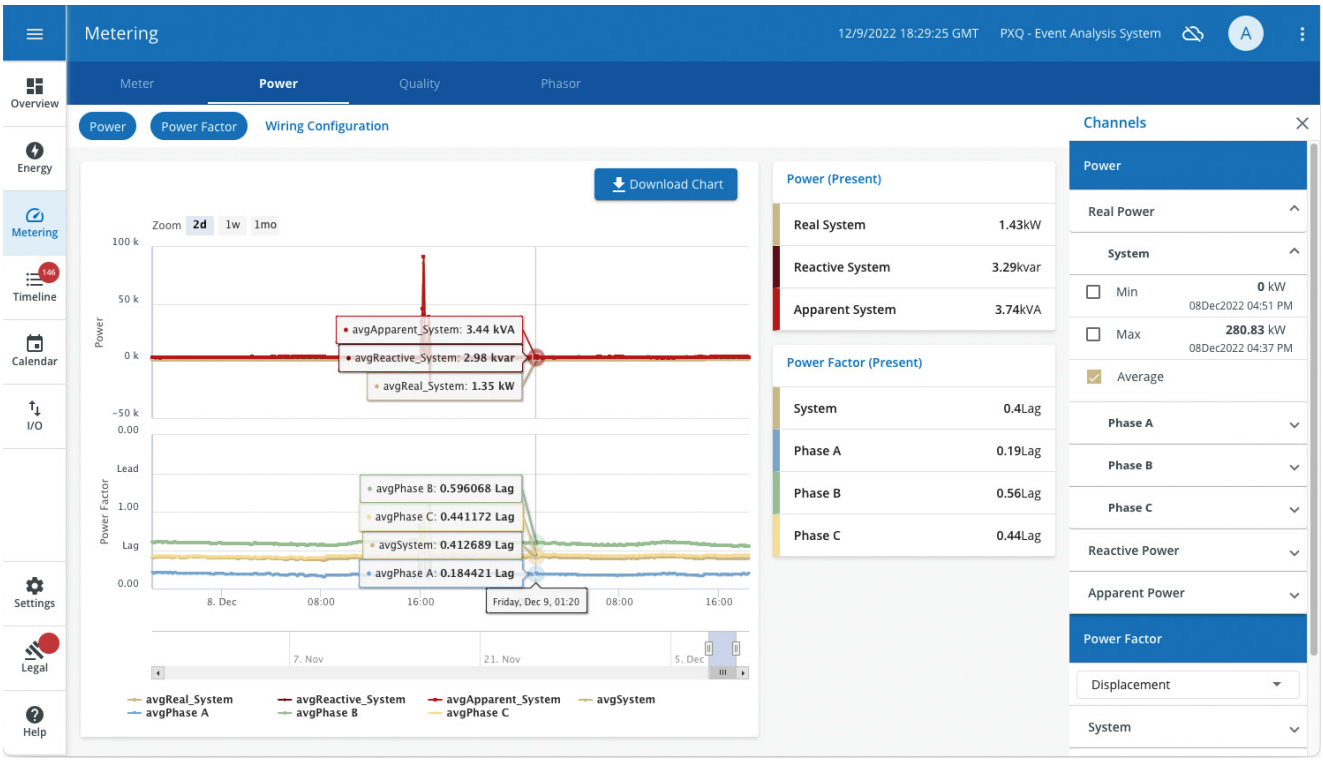
Either the Voltage and/or Current plot or the Frequency plot can be viewed.



4.4.3 Power

For the Power page, Real, Reactive and Apparent power are given for the system and all three phases A, B, and C. Each is described with the average for the period as well as the min and max (with a given timestamp).

Power factor (choice between displacement and apparent) for each phase is given as well as for the system.

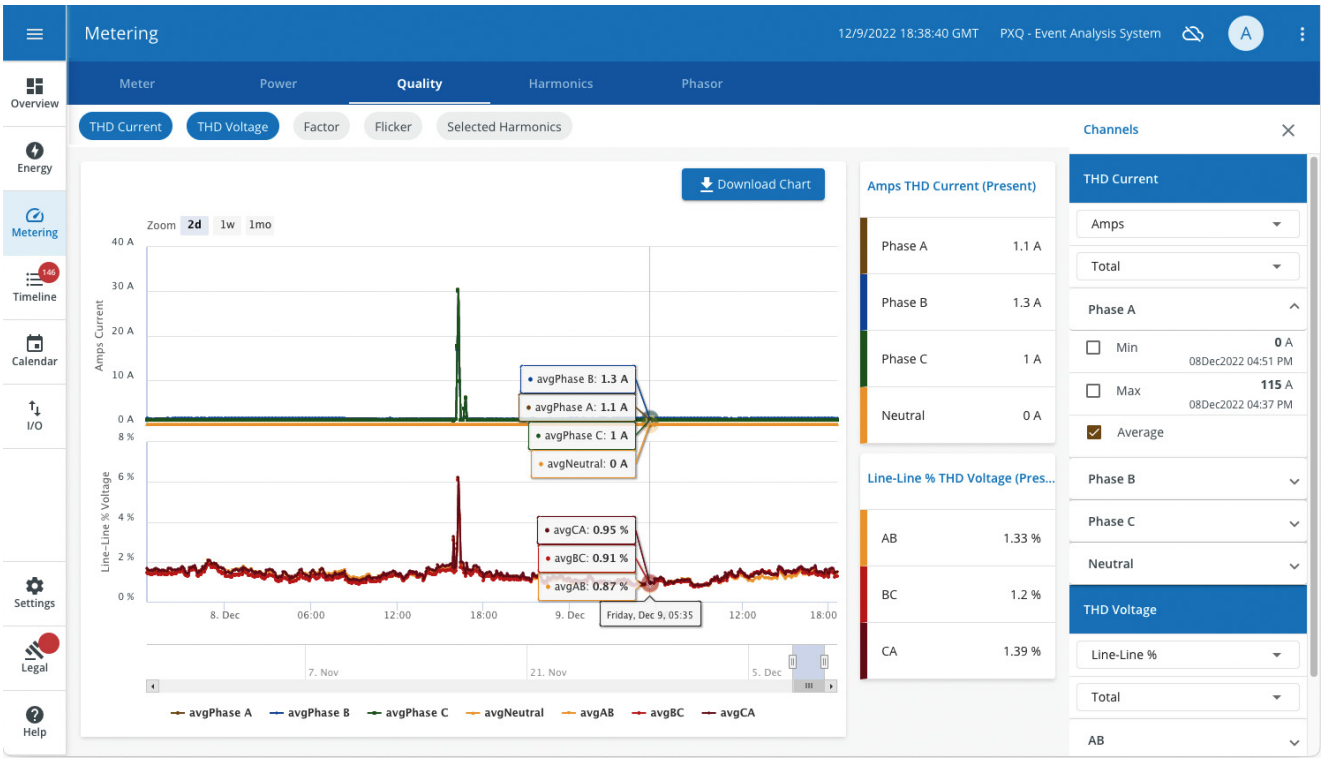


4.4.4 Quality

The quality page provides visibility to THD Current (expressed as Amps or Percent TDD) and THD Voltage (expressed as Line-Line %, Line-Line Volts, Line-Neutral % or Line-Neutral Volts). Both support the option of Even, Odd, Interharmonic or Total.

Alternatively, K or Crest factor can be viewed for all three phases.

Parameters are expressed as minimum, maximum or average.



4.4.4.1 Flicker

Flicker refers to the human perception of artificial light intensity fluctuations. All artificial light sources flicker. Flicker Fusion Threshold or Critical Flicker Fusion Frequency (CFF) is the frequency at which a flickering light source appears steady rather than flickering to a human observer. Humans can perceive flicker at frequencies up to 60-100 Hz. Long term exposure to flicker can cause discomfort, health issues, as well manufacturing issues.

Flicker perceptibility is caused by RMS voltage fluctuations and quantified using metrics such as Instantaneous Perceptibility, PST (Short-Term Flicker Severity), and PLT (Long-Term Flicker Severity) in perceptibility units P.

PXQ Flicker measurements and calculations are based on IEC61000-4-15, providing functional and design specifications for flicker measurements with the following recommended limits:

Pst ≤ 1.0: The value of the short-term flicker severity index should not exceed 1.0.

Plt ≤ 0.65: The value of the long-term flicker severity index should not exceed 0.65.

PXQ Metering > Quality > Flicker displays line-to-line and line-to-neutral Perceptibility values and trending on the following:

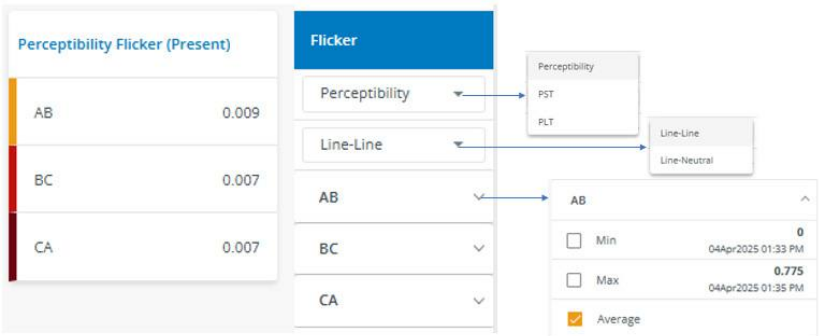
Perceptibility Flicker (Present) = Instantaneous Flicker Perceptibility (Min, Max, Average for 200ms window)

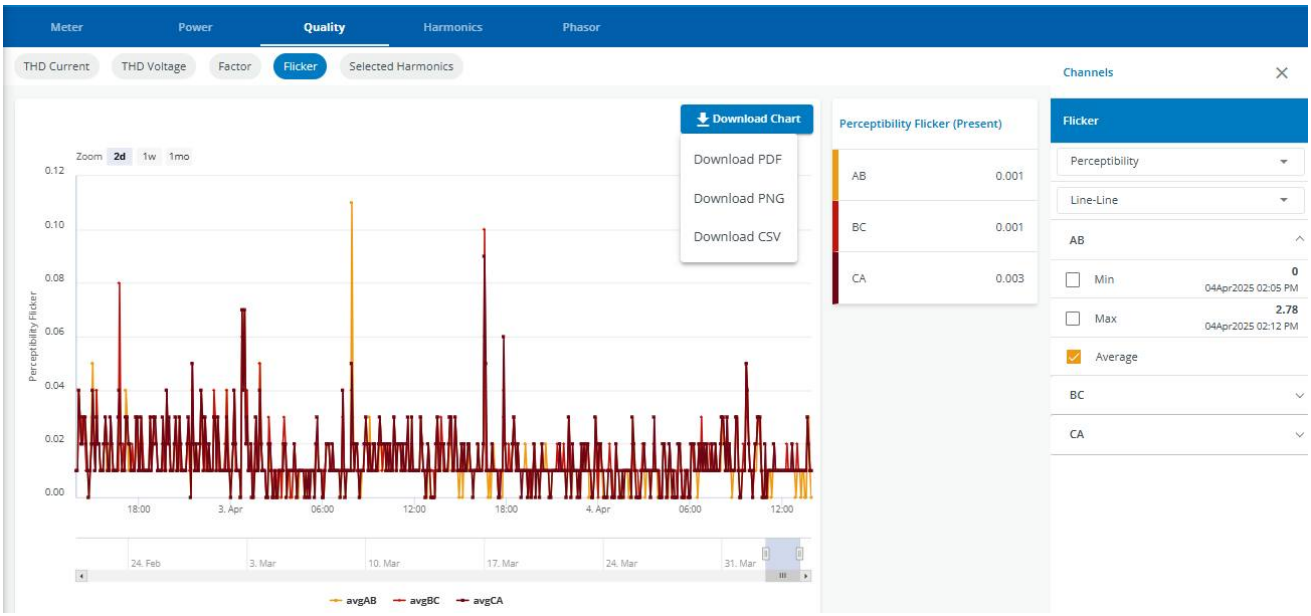
PST = Short Term Flicker Perceptibility (Min, Max, Average for 10-minute window)

PLT = Long Term Flicker Perceptibility (Min, Max, Average for 2-Hour window)

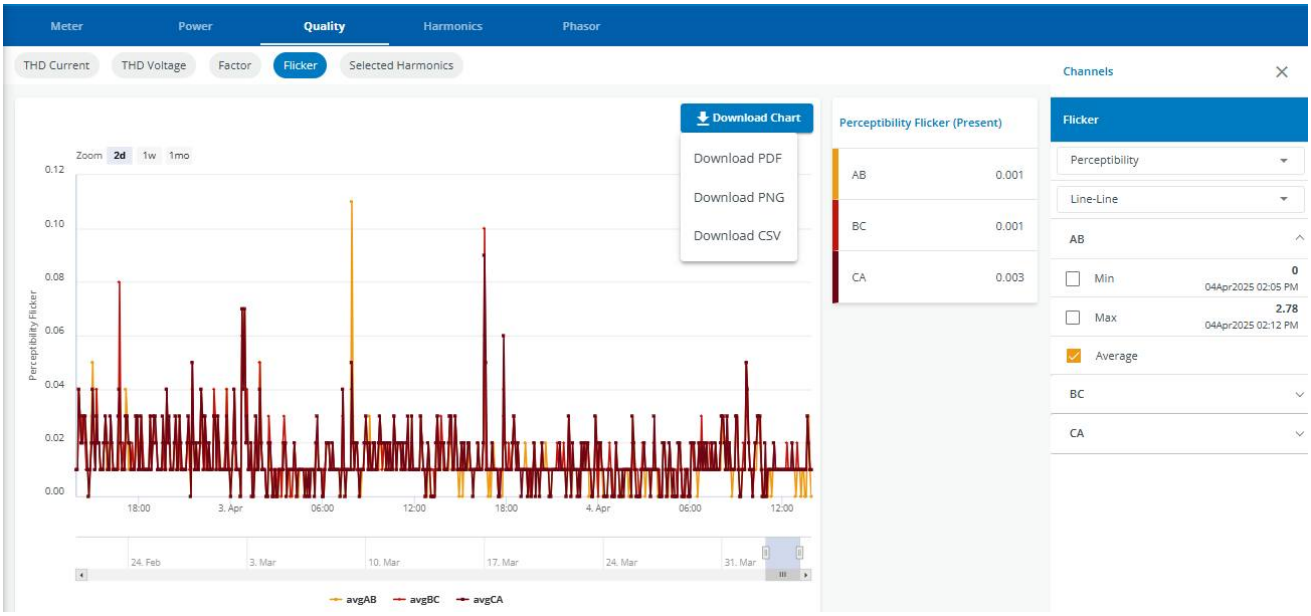
There are no settings for Flicker, Perceptibility calculations are based on IEC61000-4-15 algorithms.







Multiple Min, Max, Average boxes can be selected or deselected to have them displayed on the trend graph.



Selecting “Download PDF” or “Download PNG” provides a screen shot of the graph.

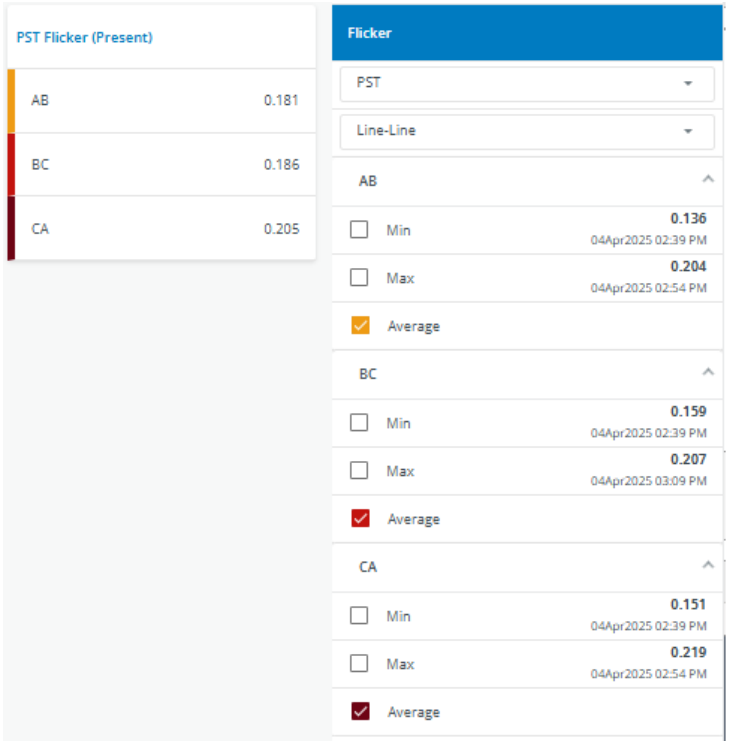
Selecting “Download CSV” will create a CSV file for each box selected.

e.g. Three (3) CSV were created based on these selections:

PXQ_ST1_66_2SL1_Mobile_Rack_admin_Trends_Quality_Flicker_BC_04-04-2025_08_25_45.csv

PXQ_ST1_66_2SL1_Mobile_Rack_admin_Trends_Quality_Flicker_CA_04-04-2025_08_25_45.csv

PXQ_ST1_66_2SL1_Mobile_Rack_admin_Trends_Quality_Flicker_AB_04-04-2025_08_25_45.csv

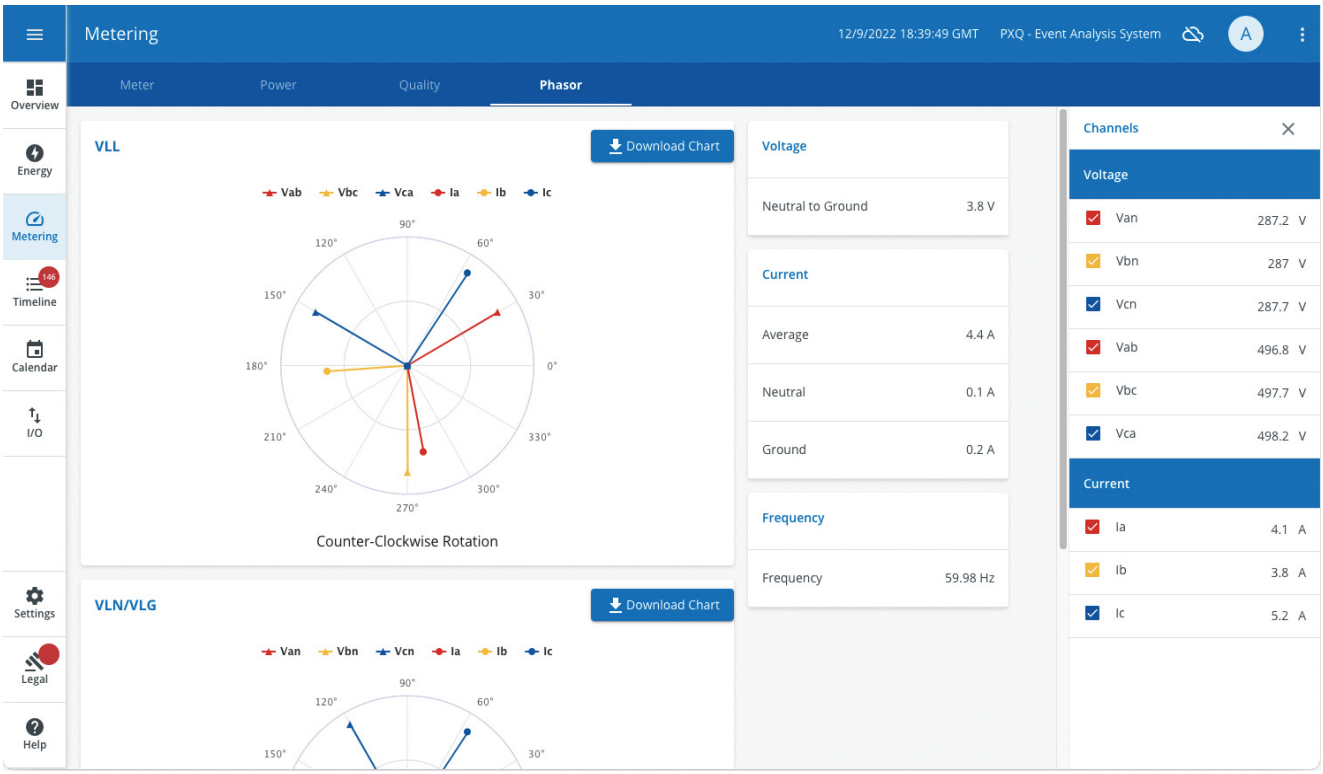


BC				
Point	Timestamp (GMT)	Avg	Min	Max
1	3/19/2025 13:55	0.18	0.18	0.18
2	3/19/2025 14:00	0.18	0.18	0.18
3	3/19/2025 14:05	0.19	0.18	0.2
4	3/19/2025 14:10	0.2	0.2	0.2
5	3/19/2025 14:15	0.2	0.2	0.2
6	3/19/2025 14:20	0.215	0.2	0.229
7	3/19/2025 14:25	0.229	0.229	0.229
8	3/19/2025 14:30	0.229	0.229	0.229
9	3/19/2025 14:35	0.219	0.208	0.229
10	3/19/2025 14:40	0.208	0.208	0.208
11	3/19/2025 14:45	0.208	0.208	0.208
12	3/19/2025 14:50	0.21	0.208	0.213
13	3/19/2025 14:55	0.213	0.213	0.213
14	3/19/2025 15:00	0.213	0.213	0.213
15	3/19/2025 15:05	0.223	0.213	0.233
16	3/19/2025 15:10	0.233	0.233	0.233
17	3/19/2025 15:15	0.233	0.233	0.233
18	3/19/2025 15:20	0.217	0.201	0.233

4.4.5 Phasor

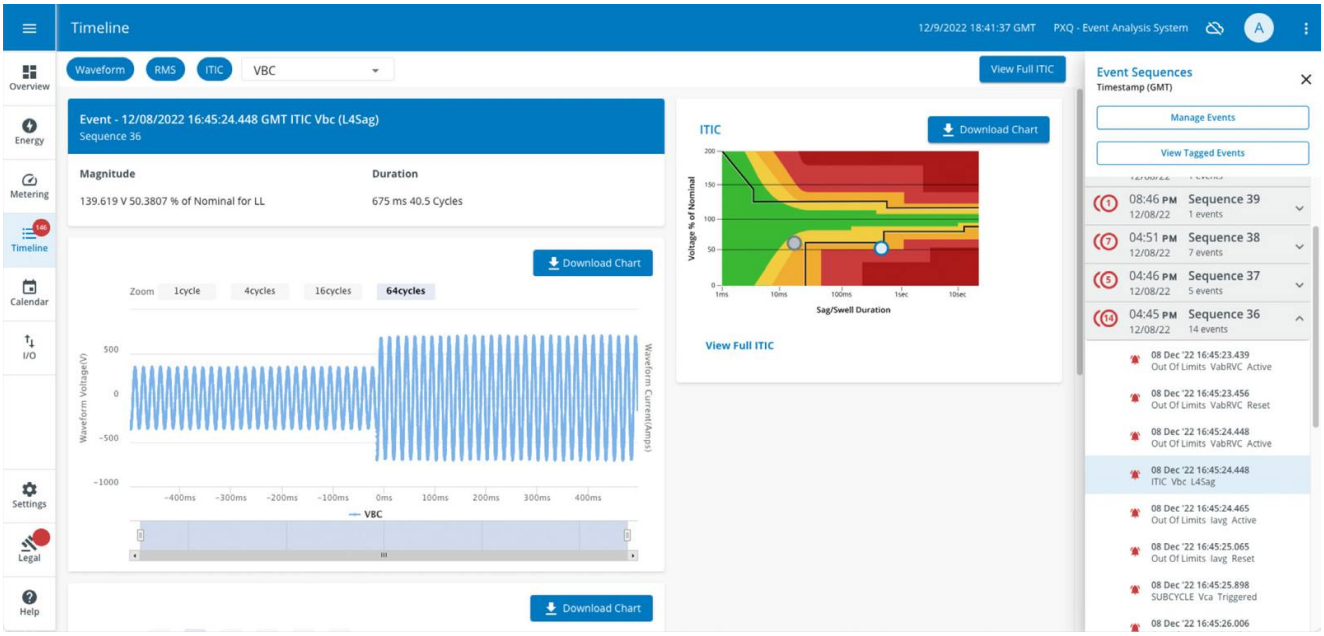
The phasor page provides plots for VLL and VLN/VLG. Corresponding current phasors are plotted as well. Similar to the trended data, vectors can be turned on or off via the legend or on the channels tab pane.

Present values for voltage, current and frequency are provided in cards next to the charts.



4.5 Timeline

PXQ integrated Timeline view provides powerful contextual insights into power quality events and disturbances. For a selected event, waveform and RMS views are displayed, alongside the event's position on the ITIC or Semi-F47 graph. Visibility of the waveform, RMS and ITIC charts can be toggled with the chips in the white action bar at the top of the screen.



4.5.1.1 Event Sequences

PXQ automatically groups events into sequences for easy analysis [Mid-Tier Only]. The number of unacknowledged events is given in the icon next to the sequence. If all events are acknowledged, no icon is displayed. The total number of events that comprise the sequence and the timestamp are given on each group.

The user can tag each event sequence with a custom name using the Create Tag button. This tag is the displayed on the sequence for easy recognition when collapsed.

For each event in a sequence, the acknowledgement status is indicated by the following:

Unacknowledged Events are red

Acknowledged Events are gray

The timestamp and ITIC classification are also provided. Options to manage all event sequences, as well as **View Tagged Events**, are available from the **Choose an action** dropdown.

Additionally you can also acknowledge all events from the Meter Actions menu (three dots at top right of all pages).

4.5.1.2 ITIC

All events in a sequence are plotted on the ITIC curve. This industry-standard curve displays events with respect to voltage magnitude and event duration, color coded by severity.

The selected event is marked with a bullseye. Other events in the sequence are each indicated on the ITIC curve by a faded bullseye.

View Full ITIC brings up a modal, displaying a counter of all ITIC events recorded by PXQ. Selecting the total of any counter allows the user to view all events of that type. To return to the prior screen without making a selection, close the modal.

Alternatively, the ITIC Curve can be displayed as a Semi-F47 curve. This can be changed in the Settings section.

4.5.1.3 Details

Details on the selected event are provided in a summary card. The timestamp and type are provided in the title of the card.

Magnitude is expressed in both relative (% of Nominal) and absolute (voltage) readings. Event duration is expressed as both seconds and cycles.

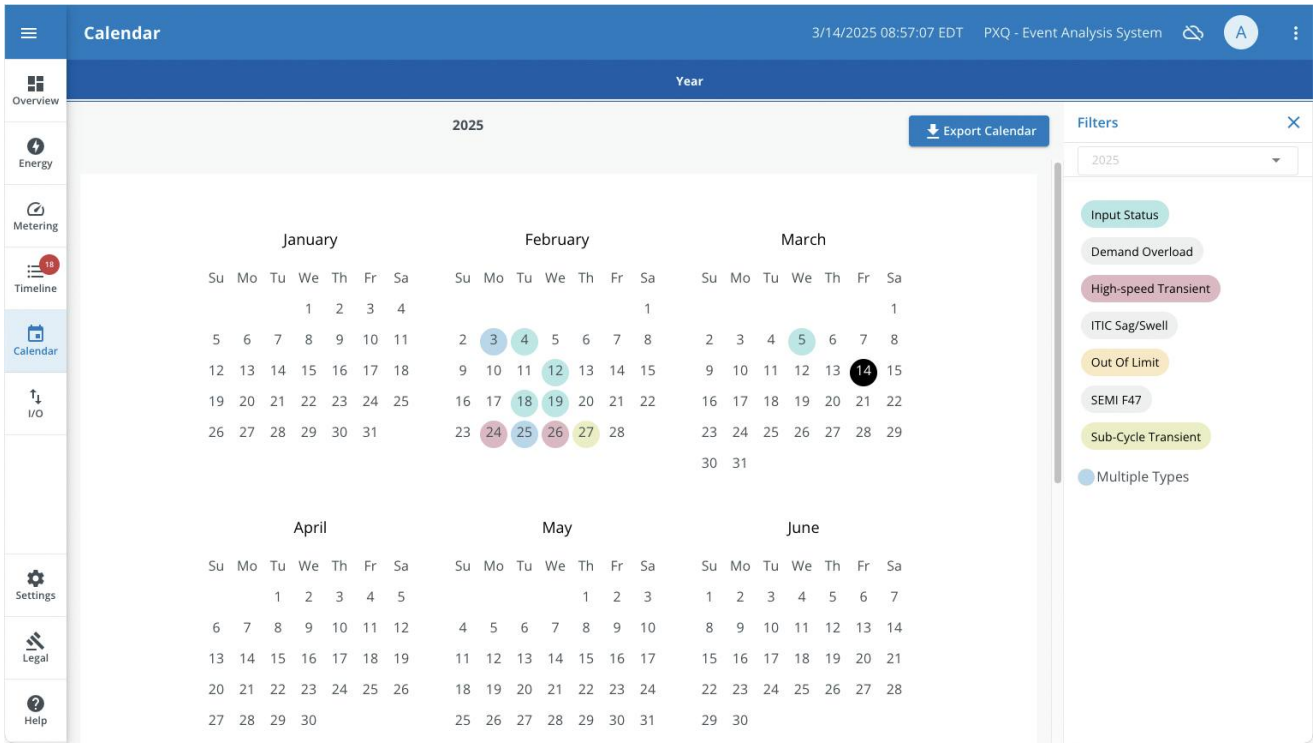
4.5.1.4 Charts

The integrated Waveform and RMS charts can be used to determine causality of the event. The charts share a single navigator control. The zoom levels are expressed in cycles on the waveform graph, and in seconds on the RMS view. Changing the zoom on one graph will automatically impact the other.

Additional channels for each graph can be added via the Select Channel dropdown in the action bar. These channels can be hidden by selecting the trace in the legend of either graph.

4.6 Calendar

The Calendar view provides an easy way to determine patterns in electrical disruptions and power quality events in the monitored circuit. The year can be selected with a drop down in the side pane.

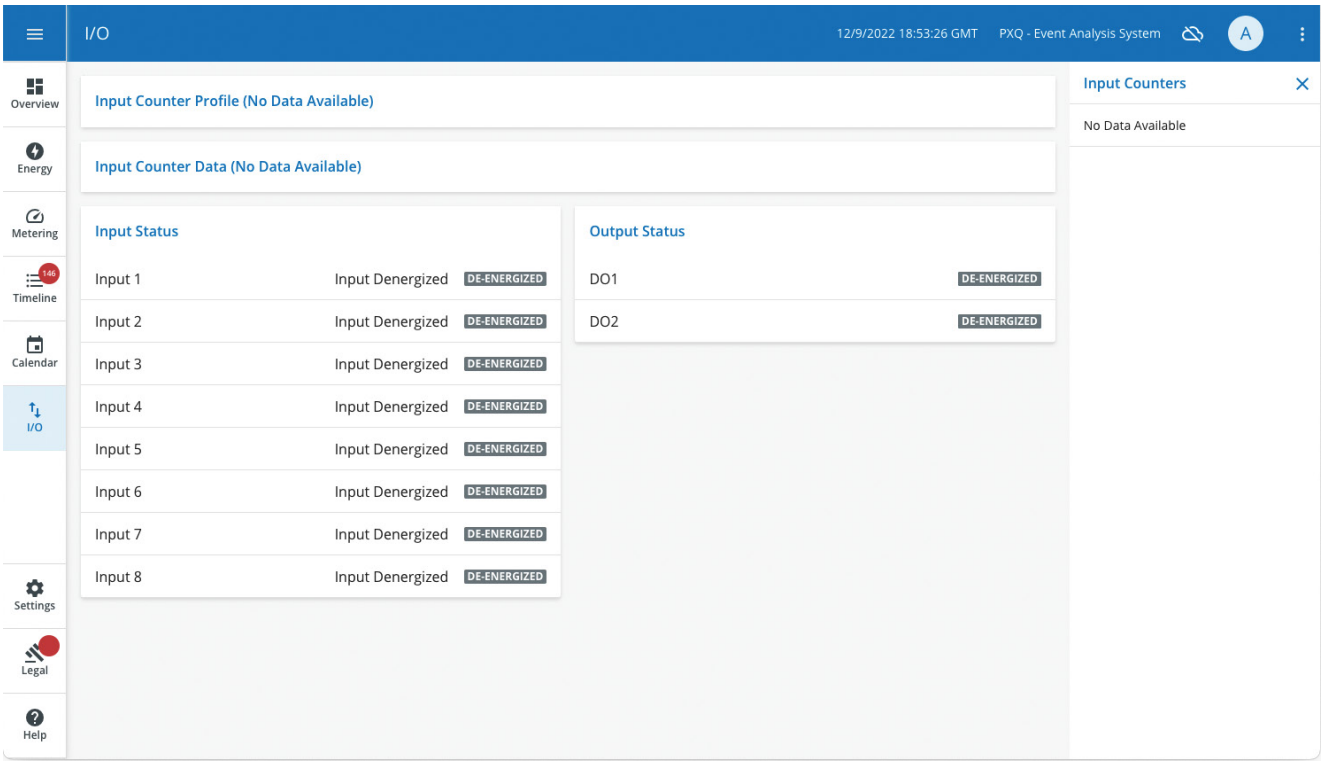


Different types of power quality events are color coded and can be selected or deselected in the side pane. The events monitored are:

- Input Status Change
- Out of Limits Trigger
- Demand Overload
- ITIC Sag/Swell
- Sub-Cycle Transient
- High-speed Transient
- Multiple event types.

Events are highlighted in the calendar in the relevant color. Selecting one of the marked dates brings up the events of that day in the Timeline view.

4.7 I/O



The I/O section shows the status and history of the on-board inputs and outputs.

Inputs configured as Pulse Counter (WAGES) can be viewed on the right-hand pane. The selected input is displayed in an interactive graph. Zoom levels for 1 day, 2 days, 1 week, 1 month, and 3 years is provided.

The readings card summarizes data from the displayed graph. Total accumulation, peak, and the demand for the previous and current intervals is displayed. The accumulation is expressed in user-defined units.

Inputs configured to display status are shown in the Status Input card. User-defined names for each input and status are used.

The status of each output is provided in the Status output card. User-defined names for each output and the corresponding statuses are used.

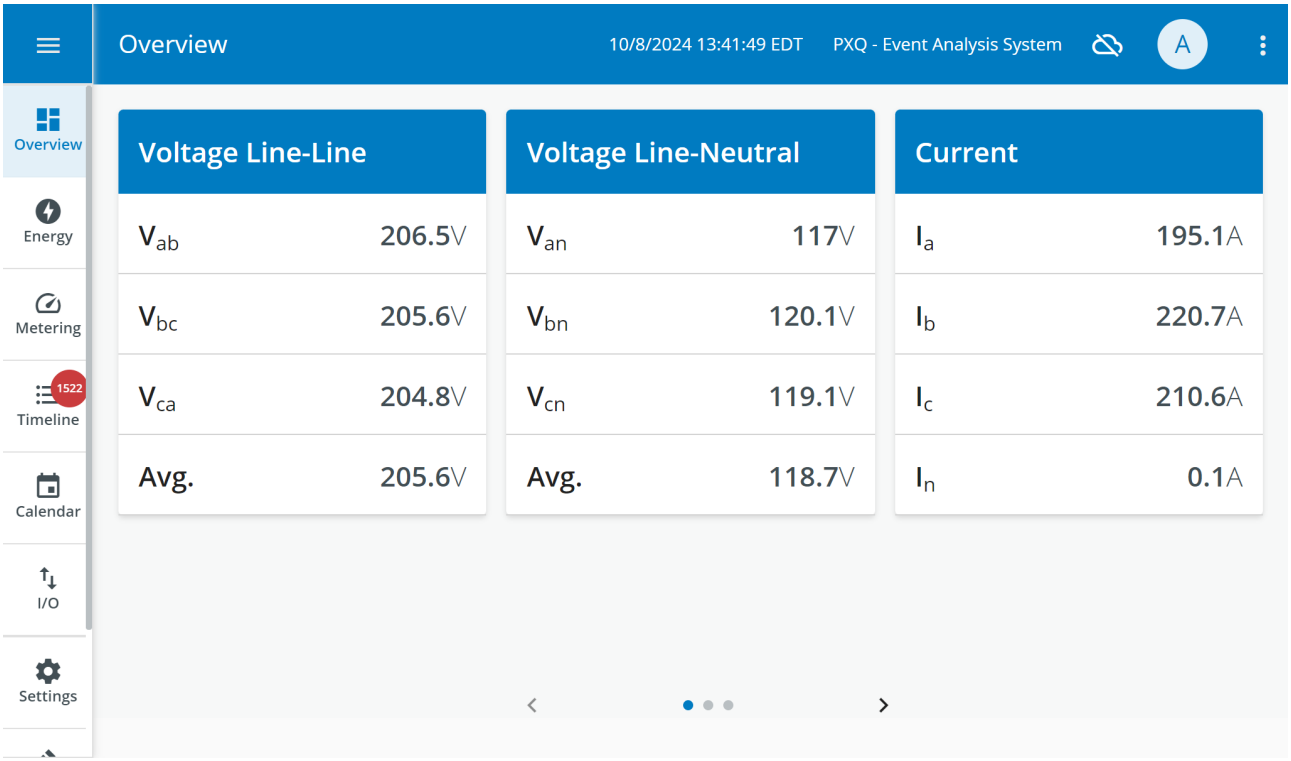
Inputs and Outputs can be configured in the Settings section of PXQ.

5. Optimized Screens for Lower resolutions

The PXQ includes select screens that are touch-optimized for lower resolution displays, such as HMIs and tablets. These screens are displayed at resolutions of 600 x 1024 (standard 7" diagonal) and below.

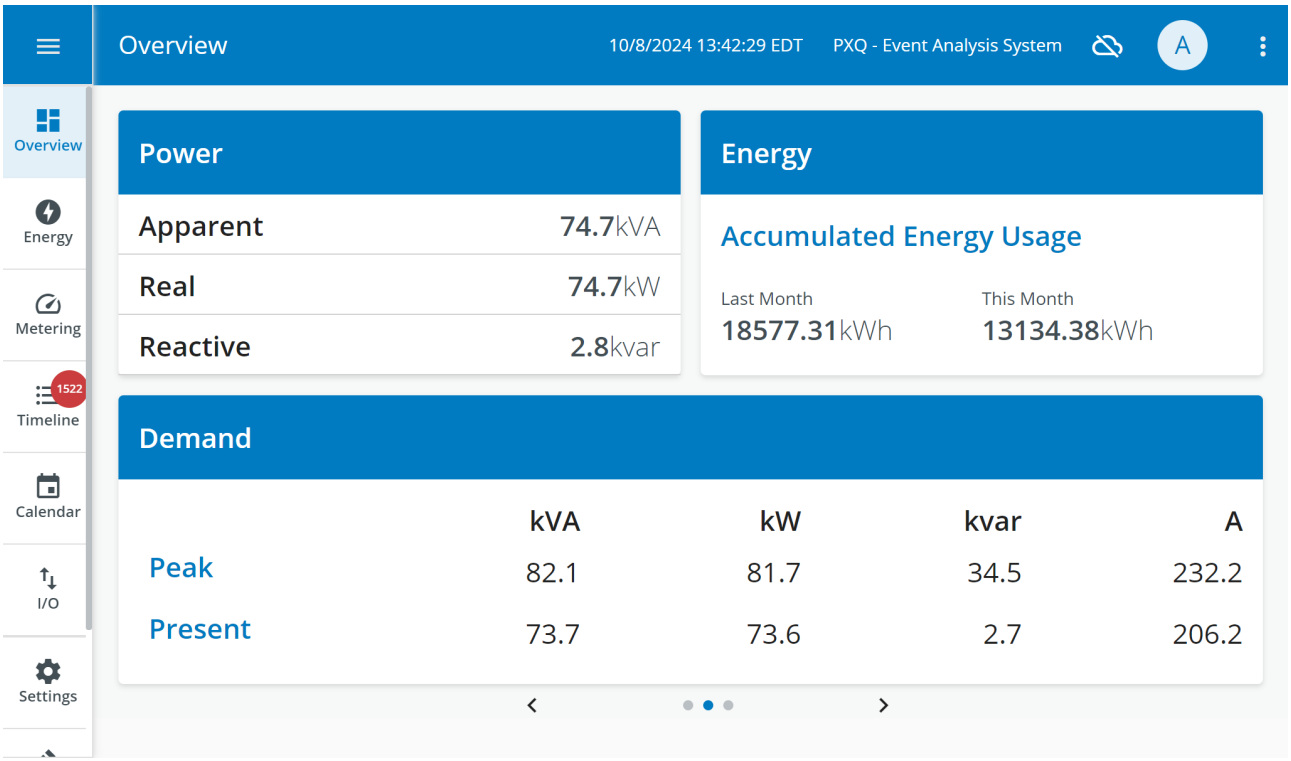
5.1 Simplified Overview

The overview is now simplified into three screens arranged as a carousel. Use the user-control at the bottom to switch between screens.



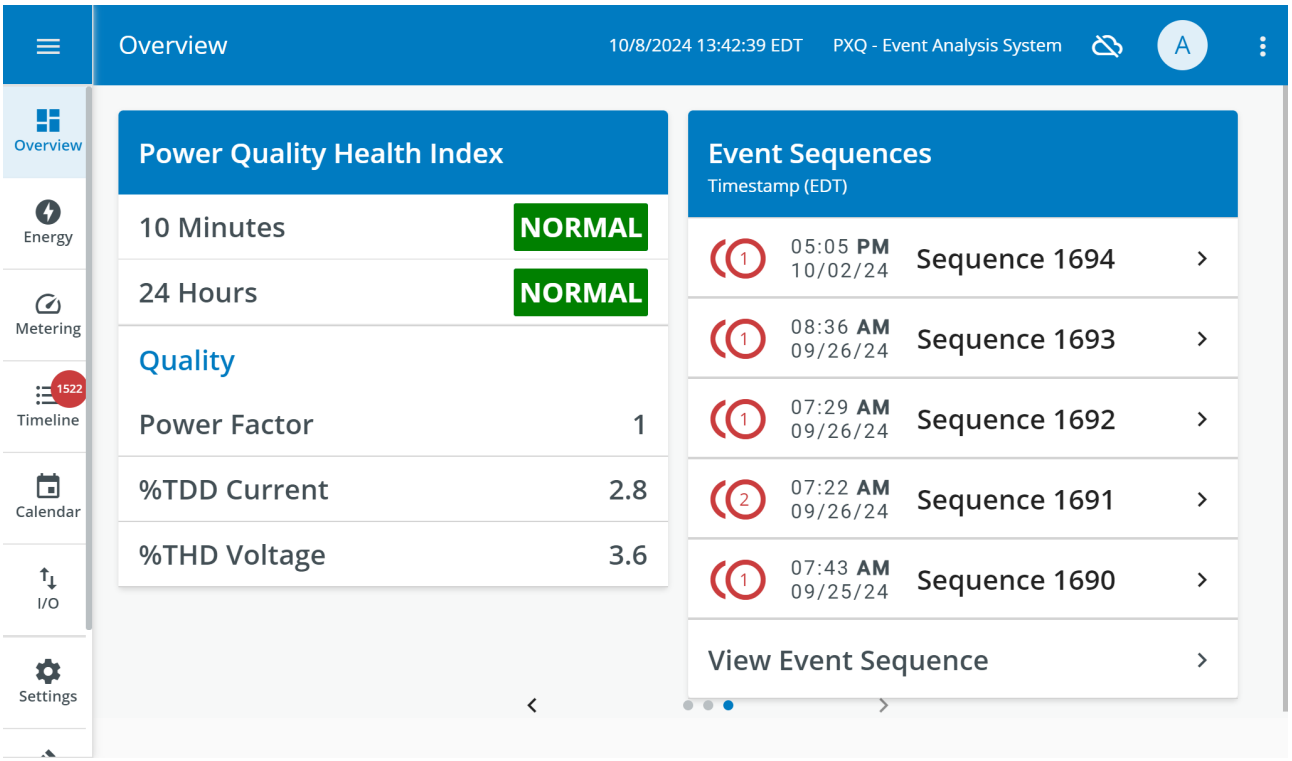
Screen 1: Displays phase and line voltages, and phase current.

5. Optimized Screens for Lower resolutions



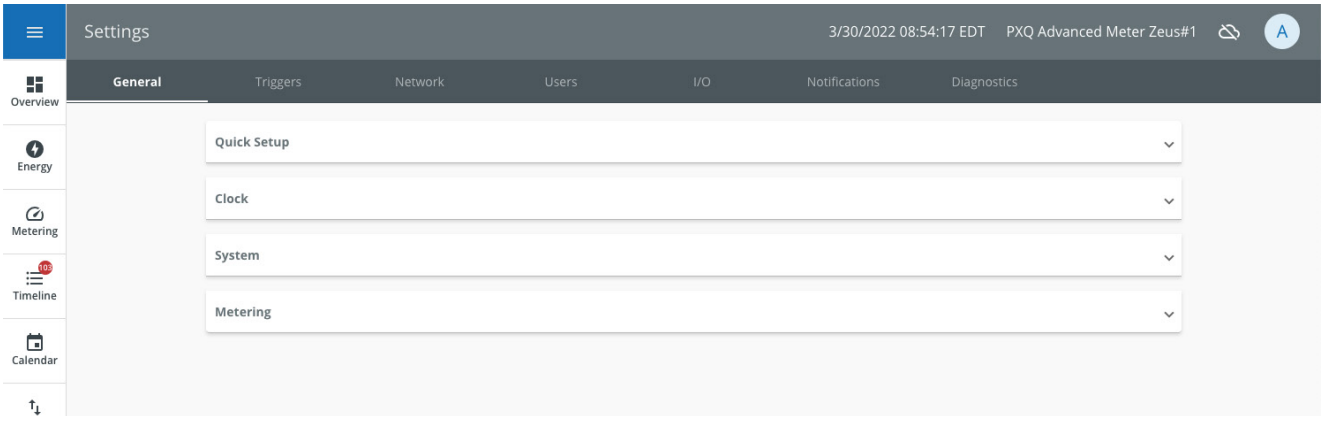
Screen 2: Shows real, reactive, and apparent power, along with accumulated energy for last month and this month. It also displays peak and present demands in kVA, kW, kvar, and A.

5. Optimized Screens for Lower resolutions



Screen 3: Provides Power Quality Health values for the last 10 minutes and the last 24 hours. Below this, power quality measurements are displayed. To the right, a list of the 5 most recent event sequences is shown. Selecting any of these will take you to the Event Analysis Timeline page for more details.

6. Settings



PXQ Configuration can be performed by users assigned to roles with the appropriate configuration or reset permissions. The count of unsaved changes is displayed until the user saves or cancels the edits.

6.1 General Settings

6.1.1 Quick Setup

Quick Setup provides the minimum parameters that must be set to define the PXQ configuration.

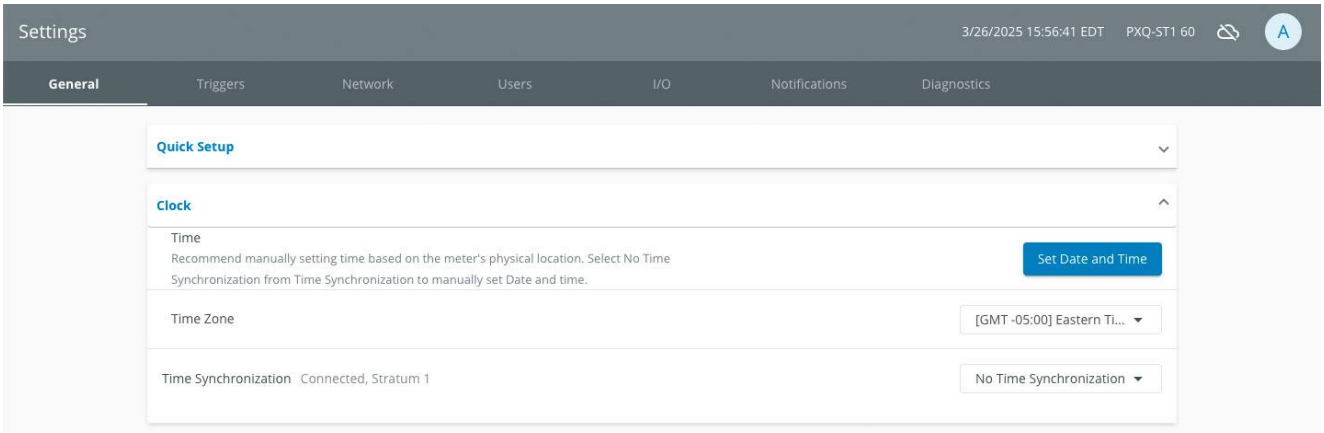
This includes:

- File Options
- PXQ name (Displayed in the title bar)
- Wiring Configuration:
  - Single Phase, 2 Wire (1CT)
  - Single Phase, 3 Wire (2CT)
  - 3 Phase, 3 Wire Delta (3CT)
  - 3 Phase, 3 Wire Delta (2CT)
  - 3 Phase, 4 Wire (Y 3CT)
  - 3 Phase, 4 Wire (CtrTap Delta 3CT)
  - 3 Phase, 4 Wire Y (3CT, 2V) 1 1/2 Element
- Frequency (50Hz or 60Hz)
- Nominal L-L Voltage
- PT Ratio
- CT Ratio (for phases)
- Neutral/Ground CT Ratio
- Selection of calculated current (In or Ig)
- PT Compensation
- CT Compensation

6.1.2 Clock

The user can manually set a date and time or specify a time zone and specify up to 3 NTP or PTP servers for time synchronization.

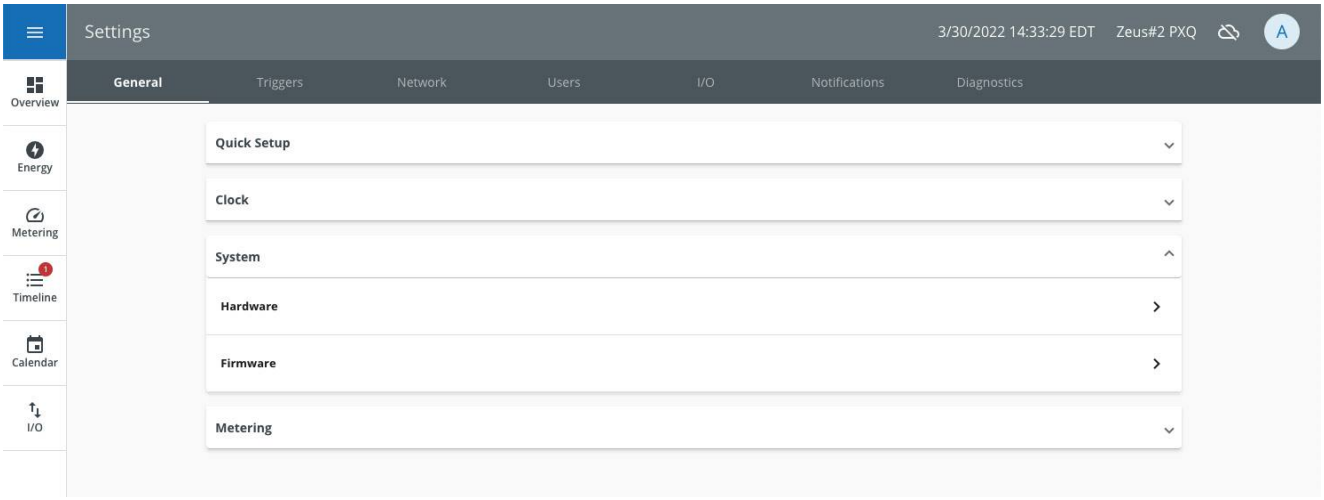
Note: When using **No Time Synchronization** under **Time Synchronization**, users may experience a pause in the time displayed during extended periods of inactivity. The system will correct this within one minute of the user re-engaging the system.



When manually setting PXQ time:

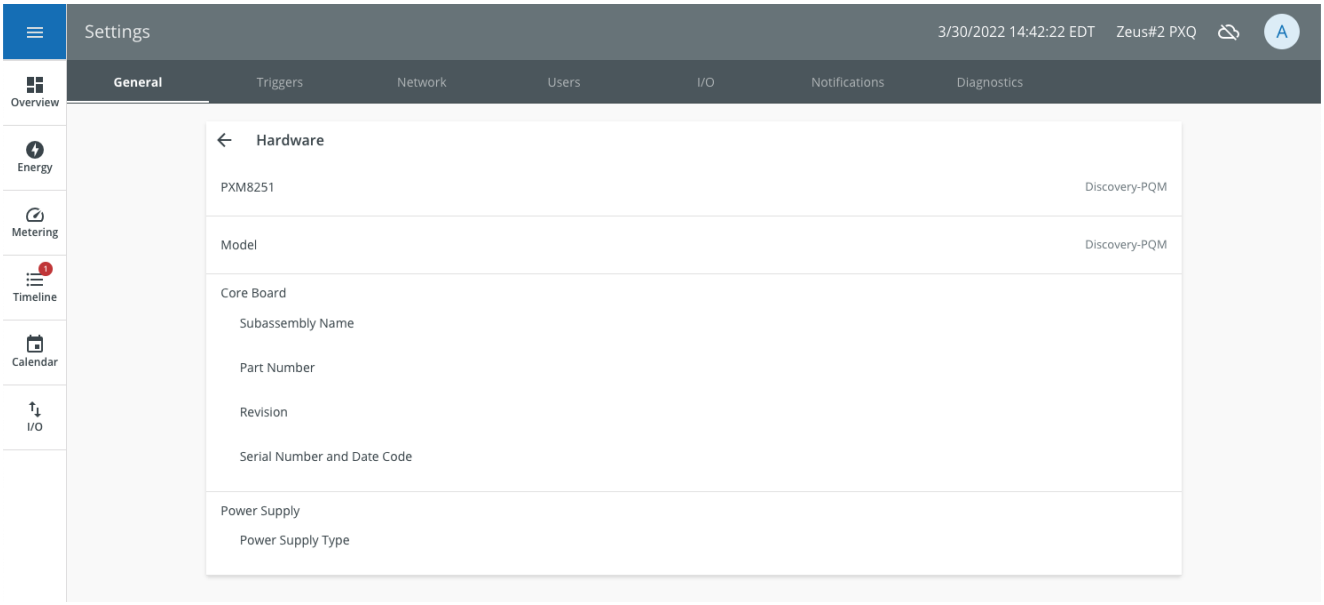
1. Set Time Zone to [GMT 00:00] Greenwich Mean Time
2. Set Date and Time
3. Change Time Zone to correct time zone.

6.1.3 System



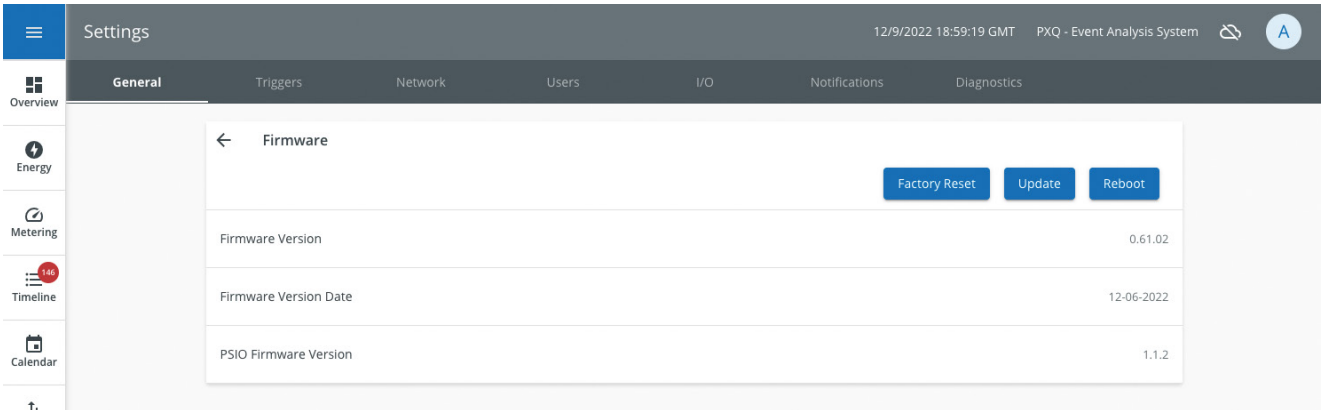
6.1.3.1 Hardware

For troubleshooting assistance, information about each module can be found here. This includes each subassembly, the model number, part number, hardware revision, serial number and date code.

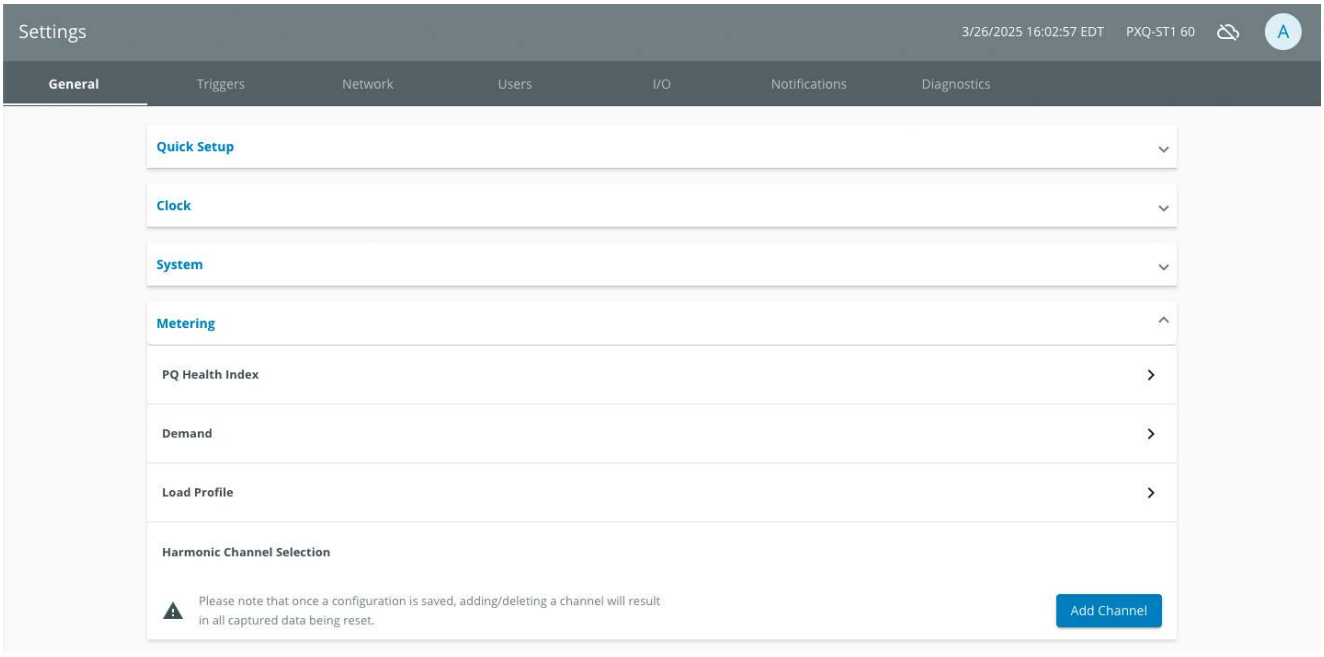


6.1.3.2 Firmware

PXQ's firmware version, UI version and release date cana be found here. Users can also perform a Factory Reset, Update, and Reboot.



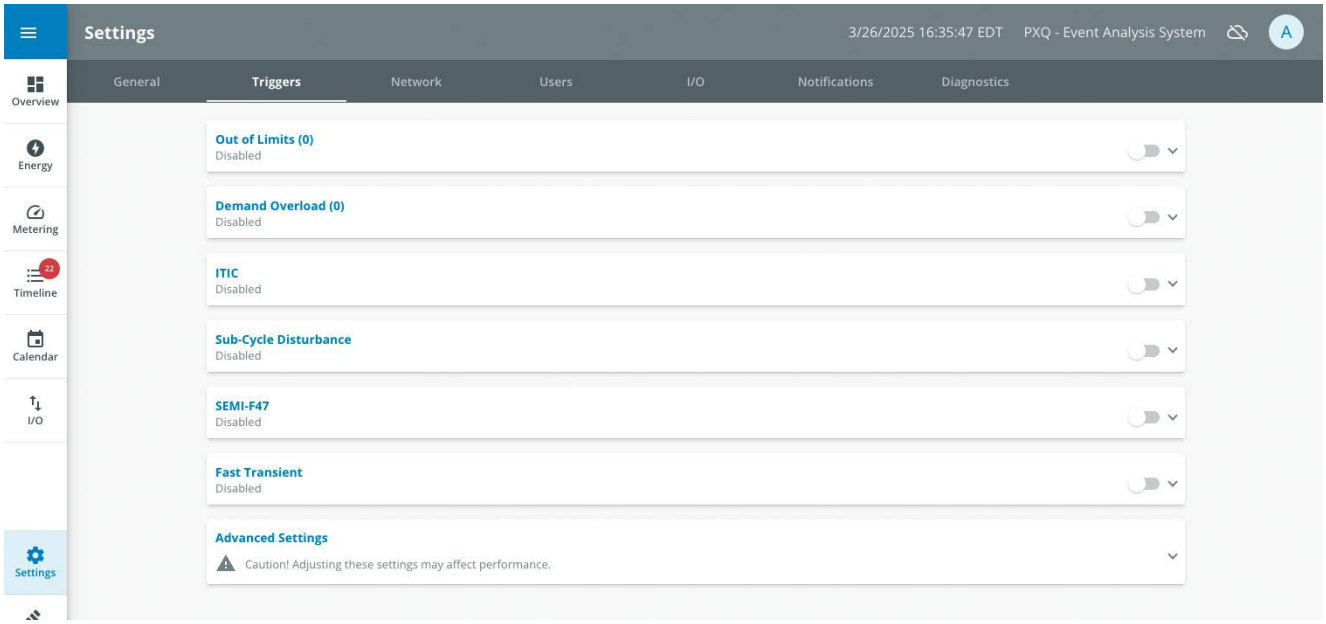
6.1.4 Metering



6.2 Triggers

In PXQ Settings > Triggers, various triggers can be set for parameters monitored by PXQ.

All Triggers are disabled by default therefore each trigger type needs to be enabled before using.



Capture Waveform and Send E-Mail are options that can be enabled or disabled for any trigger except for Demand Overload which only offers Send E-Mail.

PXQ Overview can be a resource for determining trigger settings.

Send E-Mail requires proper setup of the PXQ for emails to be sent for events.

1. Create User (PXQ Settings > Users > User Configuration > Add User)

Required

- User Name,
- Role Name,
- Email Address,
- Enable Receive Waveform Attachments

2. Configure Email (PXQ Settings > Notifications > Email)

Required

- Server IP or Hostname (must be valid email server, default=mail.etn.com)
- Username (e.g. PXQ)
- From (default = pxq@eaton.com)

Note: Note: Password and Require TLS are site specific and may not be required.

3.

4. Test Email configuration using PXQ Settings > Diagnostics > Tools > SMTP Test > Send Test.
   - "info: Successfully connected to the SMTP server."

5. Add Email Recipients (PXQ Settings > Notifications > Email Recipients > Add Recipient)

6.2.1 Out of Limits

Out of Limits triggers is based on instantaneous values of PXQ monitored parameters as shown in the table below.

PXQ Settings > Triggers > Out of Limits > Add Trigger requires five (5) settings with the option to enable Capture Waveform and/or Send E-Mail should an Out of Limits trigger occur. The table below provides an overview of Out of Limits settings.

Category	Type	Sub Type	Limit Type	Time Delay (Range: 0.01-999.99 seconds)	Capture Waveform	Send-Email
Voltage	Line-Line	System/AB/BC/CA	Upper/Lower	Default = 0	Enable **Disable**	Enable **Disable**
	Line-Neutral	System/AN/BN/CN/NG	Upper/Lower			
	Symmetric Components	V0/V1/V2	Upper			
	Volt Unbalance	-	Upper			
	Rapid Voltage Change	SystemLL/ AB/ BC/ CA/ SystemLN	Upper			
Current	RMS Current	Average/ Phase A/ Phase B/ Phase C/ Neutral	Upper	Default = 0	Enable **Disable**	Enable **Disable**
	Symmetrical Comp	Zero Sequence (I0)/ Positive Sequence (I1)/Negative Sequence (I2)				
	Current Unbalance	-				
Frequency	-	-	Upper/Lower	Default = 0	Enable **Disable**	Enable **Disable**
Power	Real Power Reactive Power Apparent Power	System/ Phase A/ Phase B/ Phase C	Upper	Default = 0	Enable **Disable**	Enable **Disable**
Power Factor	Apparent Displacement	System/ Phase A/ Phase B/ Phase C	Lead/Lagging	Default = 0	Enable **Disable**	Enable **Disable**
THD Voltage	Line-Line	System/ AB/ BC/ CA	Upper/Lower			
	Line-Neutral	System/ AN/ BN/ CN				
THD Current	RMS Current	System/ Phase A/ Phase B/ Phase C	Upper	Default = 0	Enable **Disable**	Enable **Disable**
Flicker Perceptibility (PST	Line-Line	System/ AB/ BC/ CA	Upper/Lower	Default = 0	Enable **Disable**	Enable **Disable**
	Line-Neutral	System/ AN/ BN/ CN				

Out of Limits events can be viewed on PXQ Timeline.

6.2.2 Demand Overload

Demand is the power consumption over a period of time (Fixed or Sliding) or in response to a designated Digital Input (Demand Sync) as configured in Settings > General > Metering > Demand.

Demand data for Real Power (kW), Reactive Power (kVAR), Apparent Power (kVA), and Current (Amps) is displayed on PXQ WebUI > Energy > Demand Comparison.

Demand Overload triggers can be set for Real Power, Reactive Power, Apparent Power, and Current demand values that cross a specified (upper) Trigger Threshold. The setup is like that of Out of Limits.

Settings > General > Metering > Demand > Add Trigger requires two (2) settings with the option to enable Send E-Mail should a Demand Overload trigger occur. The table below provides an overview of Demand Overload settings.

Type	Trigger Threshold	Send E-mail
Real Power Forward	Range: 0 - 180000 Watts	Enable / Disable
Real Power Reverse	Range: 0 - 180000 Watts	Enable / Disable
Real Power Net	Range: 0 - 180000 Watts	Enable / Disable
Real Power Sum	Range: 0 - 180000 Watts	Enable / Disable
Reactive Power Delivered	Range: 0 - 180000 VARs	Enable / Disable
Reactive Power Received	Range: 0 - 180000 VARs	Enable / Disable
Reactive Power Net	Range: 0 - 180000 VARs	Enable / Disable
Reactive Power Sum	Range: 0 - 180000 VARs	Enable / Disable
VA	Range: 0 - 180000 VA	Enable / Disable
Current	Range: 0 - 180000 Amps	Enable / Disable

Demand Overload events can be viewed on PXQ Timeline.

6.2.3 Sub-Synchronous Oscillation (SSO) Detection Setup

Sub-Synchronous Oscillations (SSO) are oscillatory phenomena that occur in power systems at frequencies below the system's synchronous frequency (50Hz or 60Hz). Electrical SSO are caused by dynamic, nonlinear, and poorly damped power loads. Rapid and significant changes in power demand causes oscillations of voltage and current that interact with the natural frequencies of the power system. SSOs can cause equipment damage, particularly to generator shafts and transformers, and can cause widespread brownouts and blackouts.

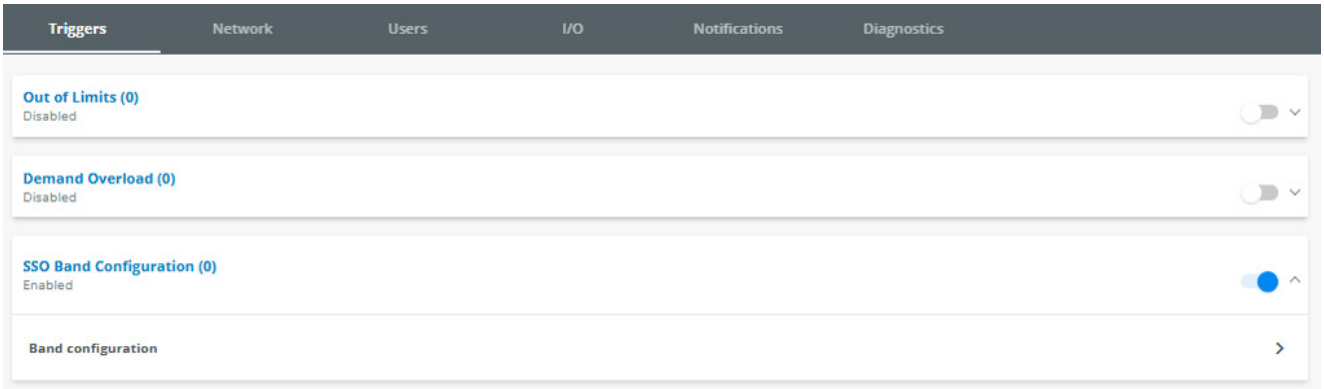
PXQ provides SSO detection and alarming, providing operators with an early warning of instability allowing operators to take proactive measures to prevent costly outages and equipment damage. In addition PXQ SSO detection supports data streaming via MQTT of half-cycle and FFT data.

- Timestamp with millisecond resolution
- 3-Phase Average RMS Voltage Line-Line
- 3-Phase Average RMS current
- 3-Phase Total/System Real Power (Watt)
- 3-Phase Total/System Reactive Power (VAR)
- 3-Phase Total/System Apparent Power (VA)
- 3-Phase Total Power Factor
- Frequency (Hz)

In general, SSO indicates an oscillation with a frequency of 5 to 55 Hz (for 60Hz fundamental frequency), and is differentiated by low frequency oscillation, which occurs at around 10 Hz to 12 Hz from a data center perspective.

PXQ provides 50 Bands of frequency analysis, each having their own setpoints, for SSO detection and alarming. By default, SSO detection is disabled, and all Band settings are blank.

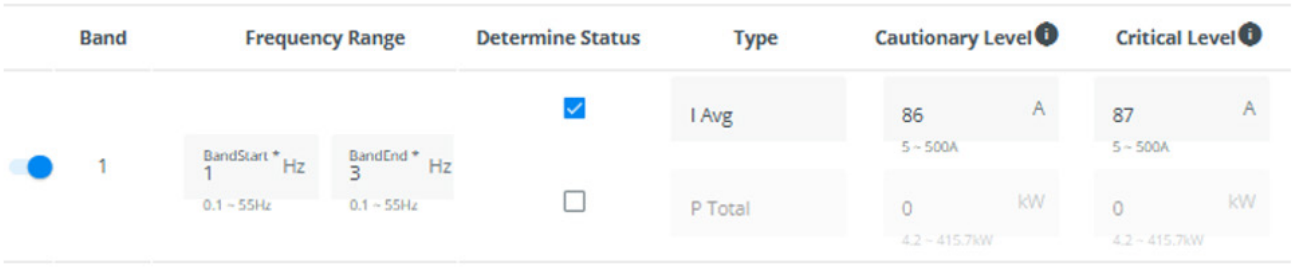
SSO detection is enabled via PXQ WebUI, PXQ WebUI > Settings > Triggers > SSO Band Configuration (0), (0 = indicates that none of the 50 Bands have been configured).



Once enabled, Band Configuration will appear as a menu item where additional configuration is required.

Bands are configured via PXQ WebUI, PXQ WebUI > Settings > Triggers > SSO Band Configuration (0). All settings are blank by default however each setting provides possible numerical values that are based on settings configured on PXQ WebUI > Settings > General > Quick Setup:

- Wiring Configuration
- Frequency
- Nominal L-L Voltage
- CT Ratio



Each of the 50 Bands of frequency analysis has the following setpoints:

BAND	Frequency Range		Determine Status (disabled by default)	Type	Cautionary Level (range)	Critical Level (range)
	BandStart (range)	BandEnd (range)				
1	0.1-55Hz (60Hz)	0.1-55Hz (60Hz)	Enable/Disable	I Avg	1%-100% (Primary CT Rating)	1%-100% (Primary CT Rating)
	0.1-45Hz (50Hz)	0.1-45Hz (50Hz)	Enable/Disable	P Total	1%-100% Total/System Watt	1%-100% Total/System Watt

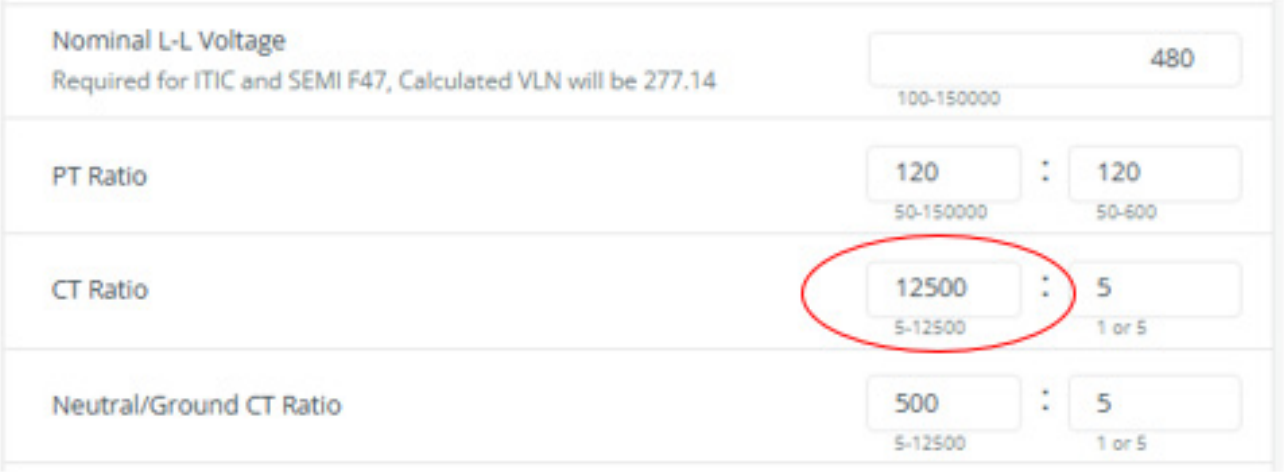
Each Band is treated independent of the other therefore Band Settings can overlap each other.

Band Setpoints:

- Frequency Range: BandStart must be smaller than BandEnd.

Based on PXQ WebUI > Settings > General > Quick Setup > Frequency setting.

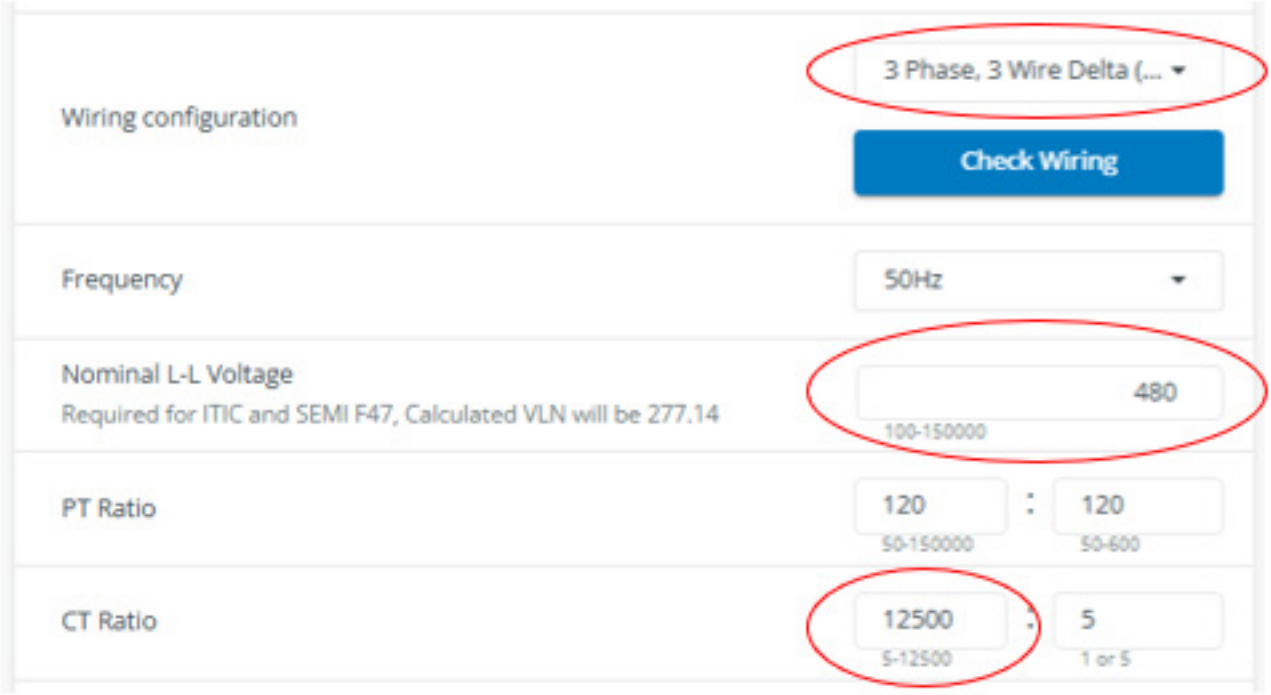
- Type: user can select either “I Avg” or “P Total” or Both
- I Avg Cautionary Level and Critical Level range is 1% to 100% of CT Primary Rating that is configured on PXQ WebUI > Settings > General > Quick Setup > CT Ratio.



e.g. Cautionary/Critical Level range would be 125 - 12500A

- P Total Cautionary Level & Critical Level range is 1% to 100% of Total/System Watts that is based on Nominal L-L Voltage and CT Primary Rating (PXQ WebUI > Settings > General > Quick Setup).

3 Phase, 3 Wire Delta (3CT) 3 Phase, 3 Wire Delta (2CT) 3 Phase, 4 Wire Y (3CT) 3 Phase, 4 Wire (CtrTapDelta 3CT)	3 x Nominal L-N Voltage x CT Primary Rating
Single Phase 2 Wire (1CT) Single Phase 3 Wire (2CT)	2 x Nominal L-N Voltage x CT Primary Rating



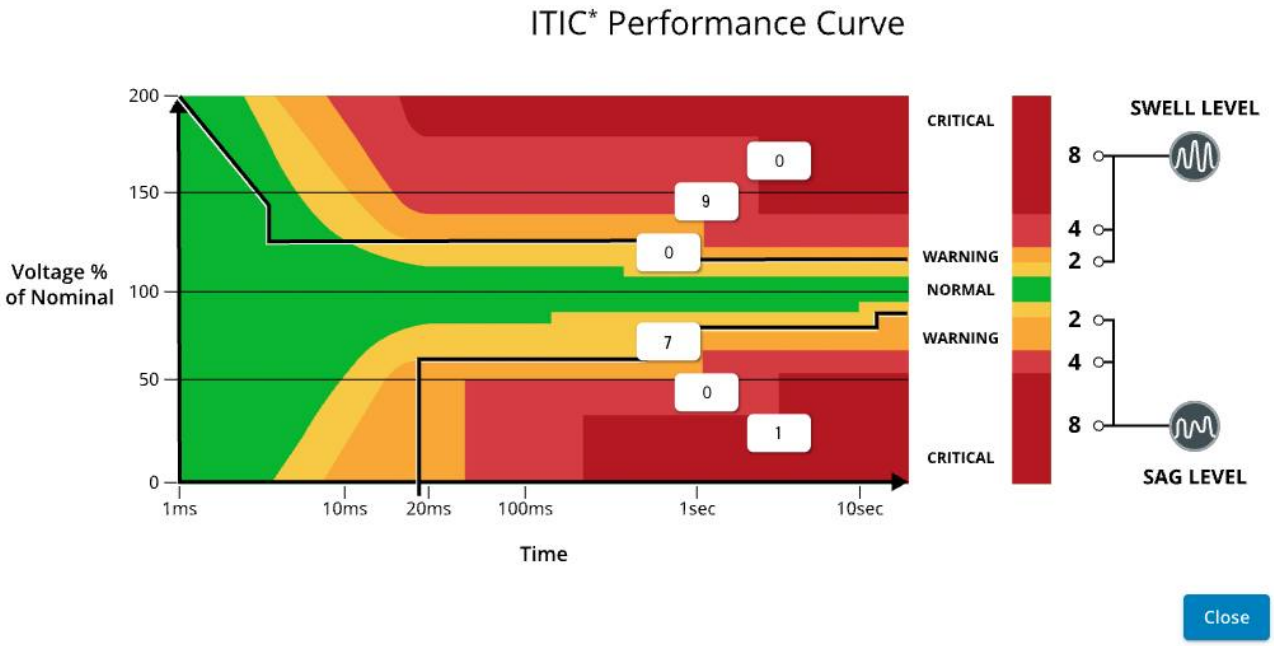
e.g. 3 x (480/Sqrt(3)) x 12500 = 10392305 therefore Cautionary/Critical Level range is 103.9 – 10392.8kW

Cautionary and Critical values exceeded within the Frequency Range are logged as Events and viewed on the PXQ WebUI > Timeline.

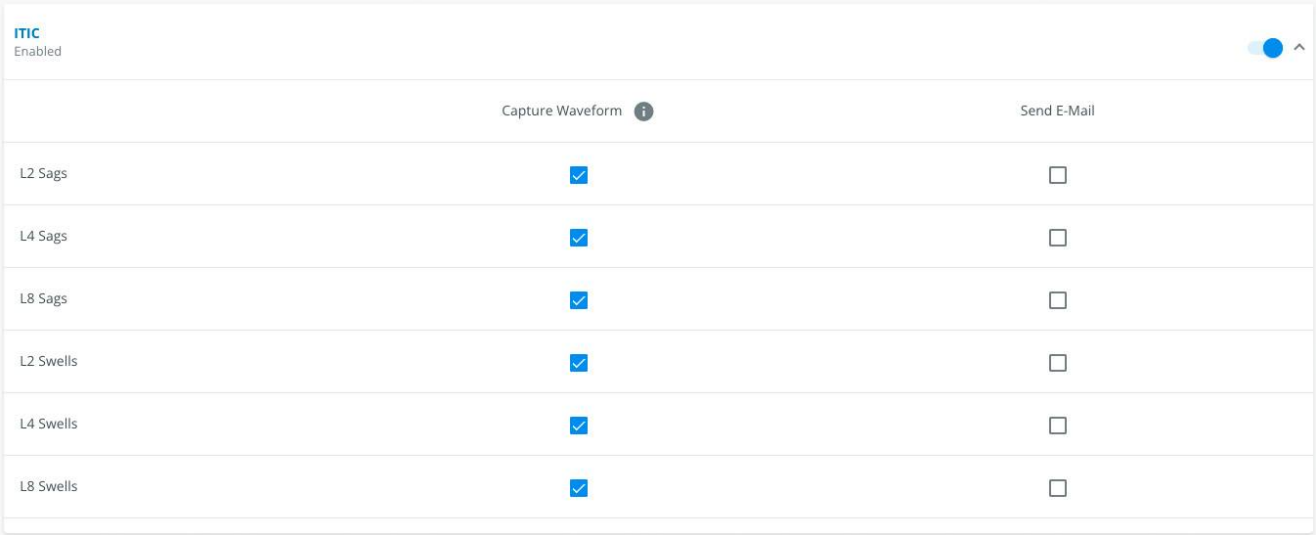
- magnitude greater than critical threshold is logged as a critically high SSO event.
- magnitude greater than cautionary threshold but less than critical threshold is logged as a cautionary high SSO event.
- If a band that is already in cautionary high state, moves to a critically high state in a subsequent evaluation than it is logged as a separate event.
- If a band is in critical state and moves to cautionary state, no event is generated.
- There are no repeat events for Bands that remain in a cautionary or critical state.
- If a band is in critical/cautionary state, it needs to drop to a reset state (-10% of Cautionary Level) before a new event can be generated. No event is generated for reset.
- If multiple events occur in same FFT values spread across different bands, First come First detect is used. Band B1 will be logged 1st then B2 and so on.
- Suspend Triggers On Overrun:
  - "Suspend Triggers On Overrun" is a global setting (PXQ WebUI > Settings > Triggers > Advance Settings). If enabled, the meter can suspend triggers for 10 minutes, allowing the user to get access the meter via the web interface and correct the setting.
  - PXQ SSO Events will be suspended if more than 20 events detected in 5 minute timeframe – this is not configurable by user.
  - For suspend trigger same logic as in Edge to be used which uses timer functionality to maintain the event count and number of events
  - Once events are suspended, then Do not log any events into Edge and update the suspended flag in above mentioned OOB , which UI can use for any user notification
  - Suspended Triggers is logged in syslog (PXQ WebUI > Settings > Diagnostics > Log > System Log)

6.2.3 ITIC

The ITIC Curve is published by the Information Technology Industry Council (ITIC) formerly known as Computer & Business Equipment Manufacturer's Association (CBEMA). It helps identify range and number of voltage sags and swells that a facility or load is subject to. The curve has defined regions identified as Level 2 (L2), Level 4 (L4), and Level 8 (L8) for both voltage sag and swell.



PXQ Settings > Triggers > ITIC by default is disabled. If enabled, there are no settings but there are options to enable Capture Waveform and/or Send E-Mail should a ITIC event occur.



ITIC events can be viewed on PXQ Timeline.

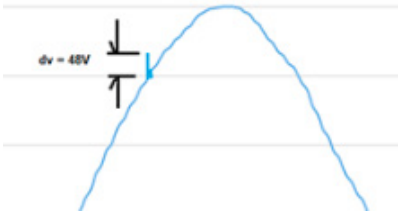
6.2.4 Sub-cycle Disturbance

A **Sub-Cycle Disturbance**, often referred to as a transient or surge, is a very short-duration (less than half a cycle) event of varying amplitude in voltage or current, potentially causing damage to electrical equipment.

PXQ detects Sub-Cycle Disturbance from instantaneous samples of Vag, Vbg, Vcg at 15.36kHz.

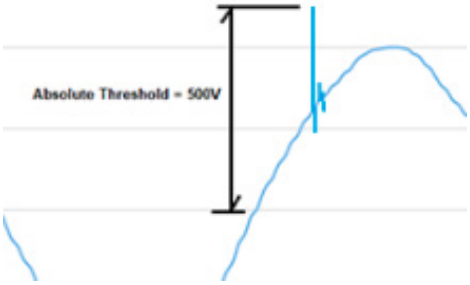
PXQ Settings > Triggers > Sub-Cycle Disturbance is disabled by default. If enabled, there are only two (2) settings (includingsuggested values) and option to enable Capture Waveform and/or Send E-Mail should a Sub-Cycle Disturbance event occur.

dv/dt Threshold = Nominal L-L Voltage / 20.0



dv/dt threshold defines the voltage change between samples required to activate this trigger.

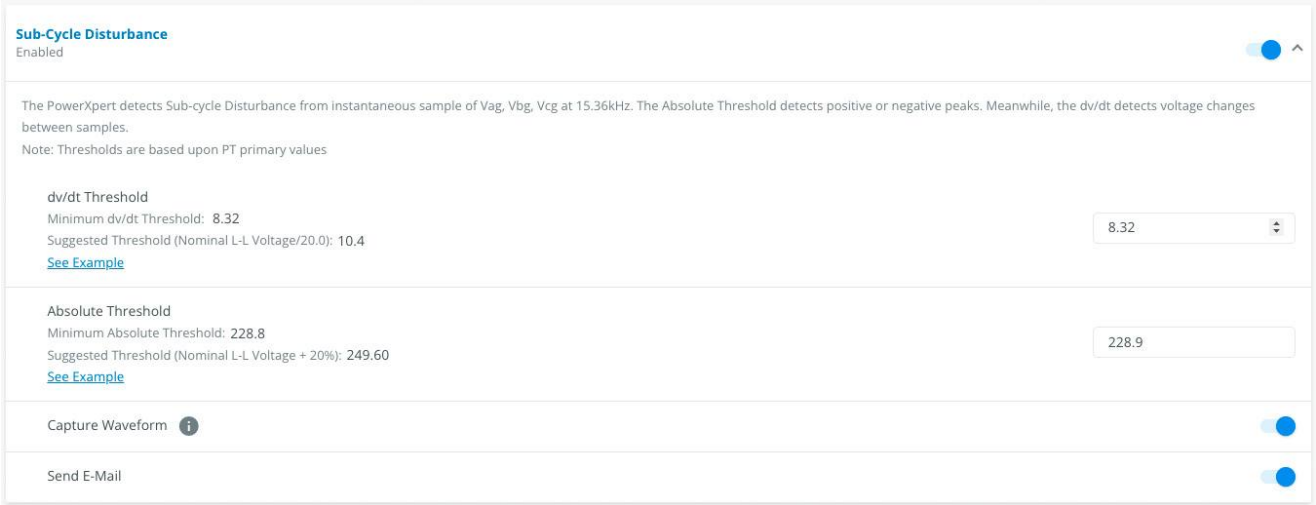
Absolute Threshold = Nominal L-L Voltage + 20.0%

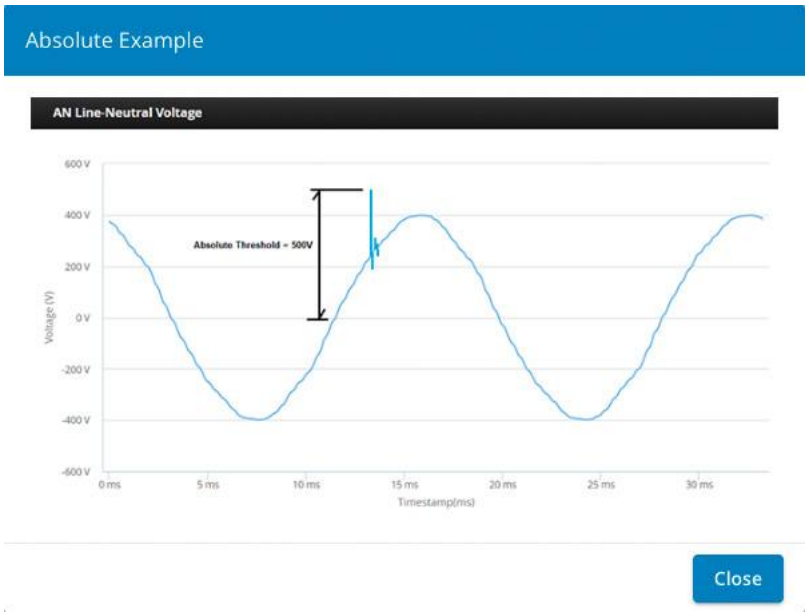


Absolute Threshold provides the positive or negative peaks required to activate this trigger.

Thresholds are based upon PT primary values.

Note: Thresholds are based on PT primary values. Nominal L-L Voltage is found on PXQ Settings > General > Quick Setup > Nominal L-L Voltage.

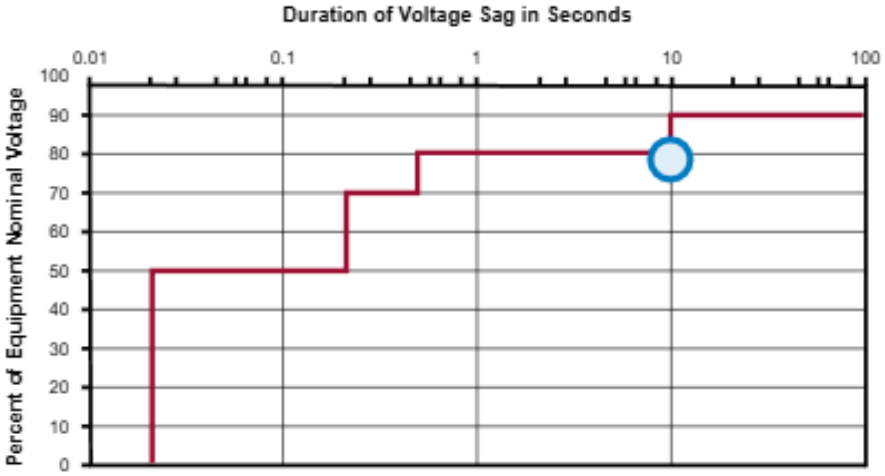




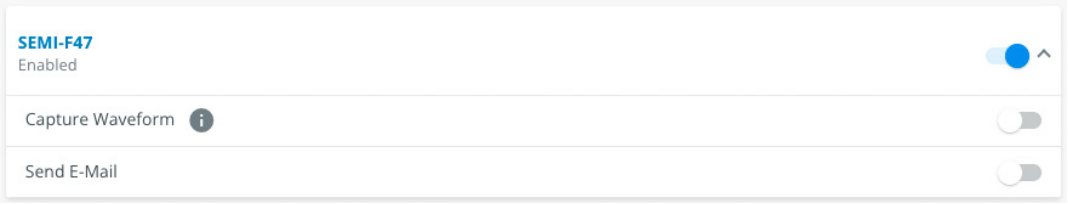
Sub-Cycle Disturbance events can be viewed on PXQ Timeline.

6.2.5 SEMI F47

Semiconductor wafer production is very sensitive to voltage sag (voltage dip) events and one voltage sag event could be costly. SEMI F47 is the specification for Semiconductor Processing Equipment voltage sag immunity and sets voltage sag limits that Semiconductor equipment can tolerate without creating any process upsets or shutdowns.



PXQ Settings > Triggers > SEMI-F47 is disabled by default. If enabled, there are no settings but there are options to enable Capture Waveform and/or Send E-Mail should a SEMI-F47 event occur.



SEMI-47 events can be viewed on PXQ Timeline.

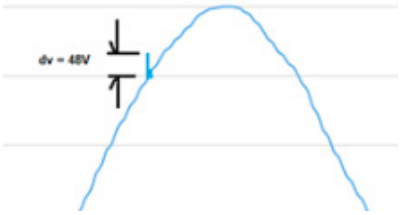
6.2.6 Fast Transient

Fast Transient detection requires the PXQ-ENH module.

PXQ Settings > Triggers > Fast Transient is disabled by default. If enabled, there are only three (3) settings (including suggestive settings) and the option to enable Capture Waveform and/or Send E-Mail should a Fast Transient event occur.

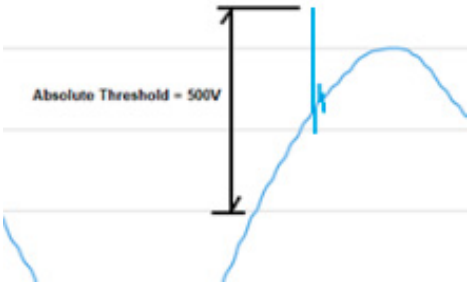
PXQ-ENH has a selectable sample rate of either 1MHz or 10MHz of PXQ-ENH inputs VAG, VBG, VCG.

dv/dt Threshold = Nominal L-L Voltage / 20.0



dv/dt threshold defines the voltage change between samples required to activate this trigger.

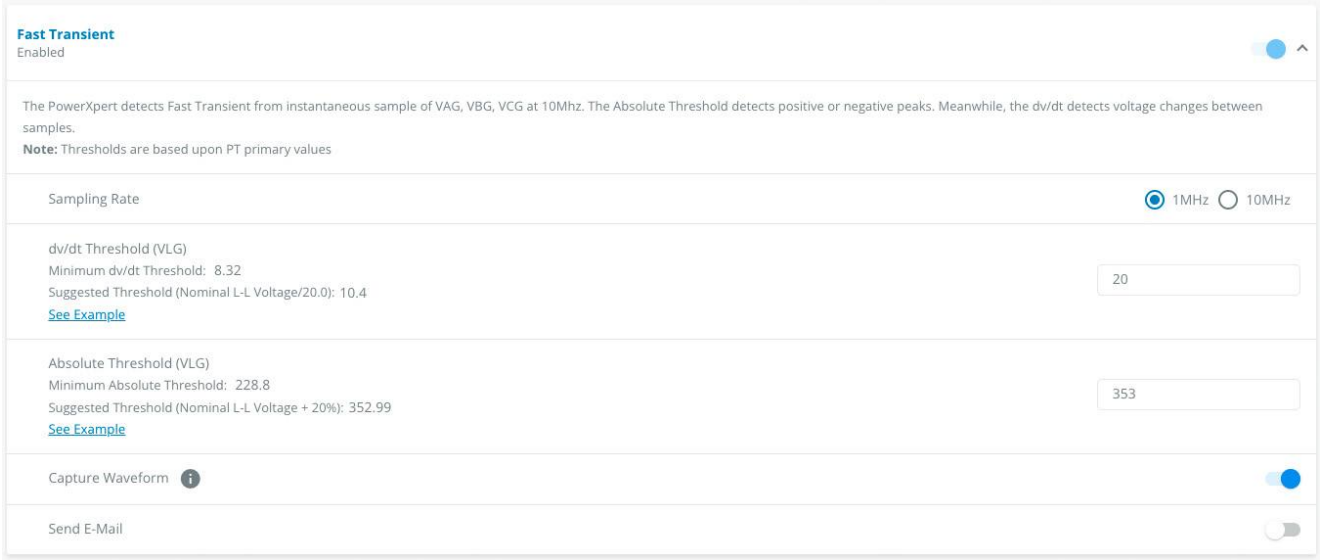
Absolute Threshold = Nominal L-L Voltage + 20.0%

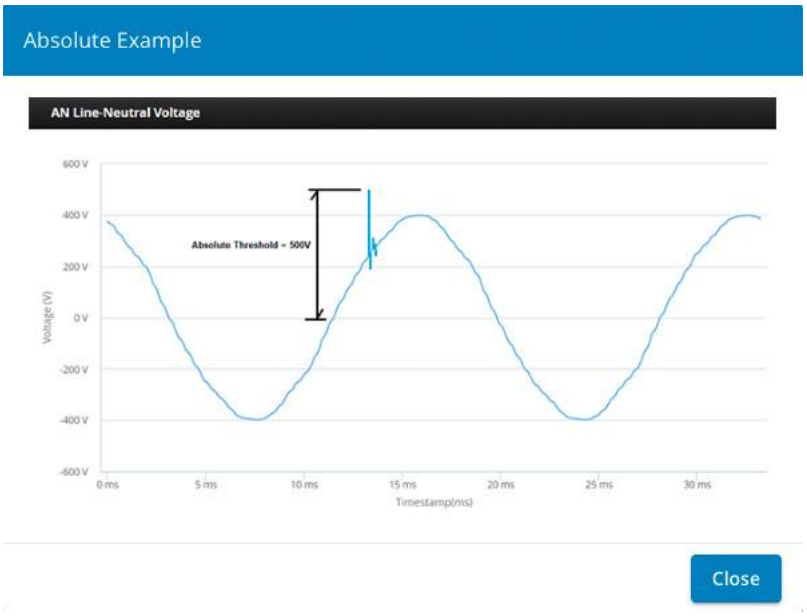


Absolute Threshold provides the positive or negative peaks required to activate this trigger.

Thresholds are based upon PT primary values.

Note: Thresholds are based on PT primary values. Nominal L-L Voltage is found on PXQ Settings > General > Quick Setup > Nominal L-L Voltage.

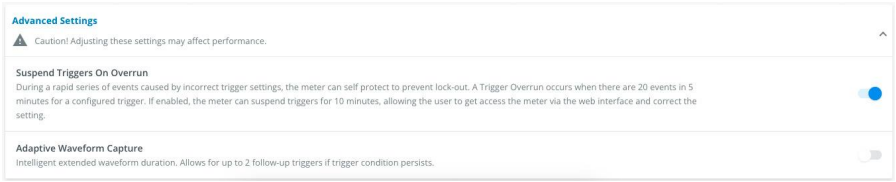




Fast Transient events can be viewed on PXQ Timeline.

6.2.7 Advanced Settings

PXQ Settings > Triggers > Advanced Settings provides two features that are disabled by default: Suspend Triggers On Overrun and Adaptive Waveform Capture.

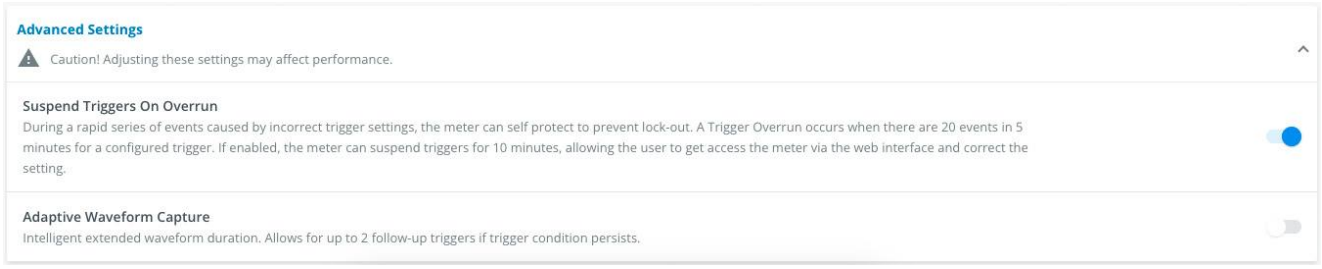


6.2.7.1 Suspend Triggers on Overrun

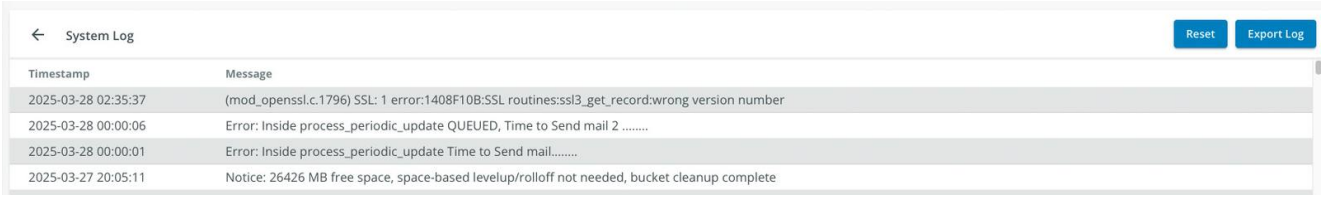
There can be times when a rapid series of events (Trigger Overrun) caused by incorrect trigger settings can overwhelm the PXQ causing response issues and possible lockup.

The PXQ defines a Trigger Overrun when a triggered event occurs 20 times in 5 minutes.

When Suspend Triggers On Overrun is enabled and a Trigger Overrun occurs, the PXQ will suspend that trigger for 10 minutes, allowing the user to get access the meter via the web interface and correct the setting.



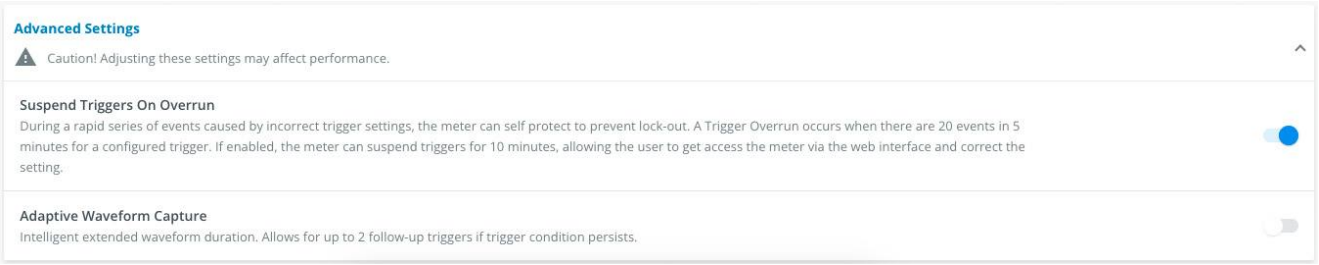
PXQ Settings > Diagnostics > Log > System Log will display a “Suspend Trigger” and corresponding “Trigger unsuspended” message.



6.2.7.2 Adaptive Waveform Capture

Adaptive Waveform Capture is disabled by default and associated with Out of Limits triggers with Waveform Capture enabled.

When Adaptive Waveform Capture is enabled, an Out of Limits trigger that is active for more than 1 second, the associated Waveform Capture COMTRADE file will remain open instead of creating additional COMTRADE files.



6.3 Network

6.3.1 Access Control

Access Control is used to enable desired protocols. Protocol configuration can be performed under Ethernet/LAN Setup.

6.3.1.1 BACnet/IP

Selecting BACnet/IP navigates the user to a separate BACnet Details pane.

BACNet/IP must first be enabled with the toggle. It is disabled by default.

A port number must be specified and Trusted designation may be made.

Optionally, a device name, location and description may be assigned from this window.

6.3.1.2 HTTP, HTTPs, MODBUS TCP, SNMP

The following communication protocols can be enabled under Access Control. They are disabled by default:

- HTTP (Default =80)
- HTTPs (Default = 443)
- MODBUS TCP – Cached Data server, (Default = 502) when enabled, additional settings under Ethernet/LAN Setup
- SNMP (Default = 161) when enabled, additional settings appear under Ethernet/LAN Setup.

Each protocol can be enabled with the toggle on the right hand side.

Each must be assigned a port and can be designated as trusted. Trusted Only is disabled by default. When Trusted Only is enabled, hosts/IP addresses must be added to the trusted Hosts field and only those hosts/IP address will have access to the PXQ.

6.3.1.3 SFTP

Designed by the Internet Engineering Task Force as an extension of Secure Shell (SSH), Secure File Transfer Protocol (SFTP) is a network protocol for secure access, transfer and management over a network.

SFTP is disabled by default and must be enabled before using.

If enabled, a User Name “ftp” must be created, go to PXQ Settings > Users > User Configuration > Add user.

SFTP service is accessible via port 22 using a SFTP client, e.g. WinSCP.

Once connected to the PXQ via SFTP, the following folders will be available.

Filename	Filesize	Filetype	Last modified	Permissions	Owner/Group
..					
trend		File folder	3/25/2025 8:02:01 PM	drwxrwxr-x	95 95
energy		File folder	3/25/2025 8:02:01 PM	drwxrwxr-x	95 95
ieee519		File folder	3/25/2025 8:00:00 PM	drwxrwxr-x	95 95
waveforms		File folder	3/25/2025 9:00:58 AM	drwxrwxr-x	2004 66
rms		File folder	3/25/2025 9:00:50 AM	drwxrwxr-x	2004 66
hsvm_waveforms		File folder	3/19/2025 1:22:24 PM	drwxrwxr-x	2004 66

Trend folder contains daily files in csv format, each containing 5-minute timestamped Minimum, Maximum, and Average VALUES for 201 monitored values.

Filename	Filesize	Filetype
..		
Trend_Log_2025_3_26.csv	318,984	Microsoft Excel Comma Separated Values File
Trend_Log_2025_3_25.csv	966,716	Microsoft Excel Comma Separated Values File
Trend_Log_2025_3_24.csv	972,054	Microsoft Excel Comma Separated Values File
Trend_Log_2025_3_23.csv	970,618	Microsoft Excel Comma Separated Values File
Trend_Log_2025_3_22.csv	971,602	Microsoft Excel Comma Separated Values File
Trend_Log_2025_3_21.csv	971,303	Microsoft Excel Comma Separated Values File
Trend_Log_2025_3_20.csv	964,993	Microsoft Excel Comma Separated Values File
Trend_Log_2025_3_19.csv	468,080	Microsoft Excel Comma Separated Values File

Energy folder contains daily files in csv format, each containing 5-minute time stamped, of Demand and Energy values (Forward, Reverse, Net, and Sum).

Energy folder also contains monthly files in csv format, each containing 5-minute time stamped data, of Pulse Counter (WAGES) values – snapshot of Total Accumulation and number of pulses counted.

Filename	Filesize	Filetype
..		
DI_scaled_counters_3_2025.csv	217,923	Microsoft Excel Comma Separated Values File
Energy_LP_2025_3_26.csv	11,669	Microsoft Excel Comma Separated Values File
Energy_LP_2025_3_25.csv	36,374	Microsoft Excel Comma Separated Values File
Energy_LP_2025_3_24.csv	36,627	Microsoft Excel Comma Separated Values File
Energy_LP_2025_3_23.csv	36,627	Microsoft Excel Comma Separated Values File
Energy_LP_2025_3_22.csv	36,627	Microsoft Excel Comma Separated Values File
Energy_LP_2025_3_21.csv	36,627	Microsoft Excel Comma Separated Values File
Energy_LP_2025_3_20.csv	36,382	Microsoft Excel Comma Separated Values File
Energy_LP_2025_3_19.csv	17,650	Microsoft Excel Comma Separated Values File

Waveforms, hsvm_waveforms, and rms folders contain COMTRADE files for captured waveforms. COMTRADE (Common Format for Transient Data Exchange) files are a standardized format and each waveform consists of three (3) files: data file (.dat), header file (*.hdr), and configuration file (.cfg). A COMTRADE Viewer, e.g. Wavewin, is required to view the COMTRADE Files for a specific waveform capture.

Waveforms are identified by the Sequence Number.

In the following example, PXQ Timeline shows Sequence 23 containing 7 events. The sftp folder Waveform identifies seven (7) waveforms: "23_0" to "23_7". The sftp folder rms identifies one (1) waveform "rms_seq_23".



The hsvm_waveforms folder contains COMTRADE files for captured waveforms pertaining to Fast Transients available through the PXQ-ENH module.

The ieee519 folder contains files associated with harmonic limitations.

IEEE 519 describes recommended practices and requirements for Harmonic Control in Electrical Power Systems providing recommended maximum voltage and current distortion limits.

6.3.1.4 Trusted Hosts

Trusted host IP Addresses can be entered in the field and added.

6.3.2 Ethernet/LAN Setup

6.3.2.1 Eth1 & Eth2

Selecting either Eth1 or Eth2 navigates to a separate Net Details setup screen.

An optional Account Name, Contact person and Asset ID can be recorded on the Net Details screen.

- Eth1 IPV4
  DHCP Enabled (default)
  Default IP = 192.168.199.11 / 255.255.255.0
- Eth2 IPV4
  DHCP Disabled (default)
  Default IP = 172.22.17.254 / 255.255.255.0

6.3.2.1.1 IPv4 Settings

Selecting IPv4 navigates to a separate IPv4 Settings setup screen.

IPv4 is disabled by default and must be enabled here before using.

IPv4 Settings also shows the port MAC ID and Link Status

DHCP for this port can be enabled here with the associated toggle. If this option is used, DHCP needs to be configured by navigating back to Net Details and selecting DHCP Settings.

Alternately, a static IP address can be configured by providing the following:

- IP Address
- Subnet Mask
- Default Gateway
- Preferred DNS Server
- Alternate DNS Server (optional)
- Domain Name

6.3.2.1.2 IPv6 Settings

Selecting IPv6 navigates to a separate IPv6 Settings setup screen.

IPv6 is disabled by default and must be enabled here before using.

The user must specify a gateway.

6.3.2.1.3 DHCP Settings

Selecting DHCP navigates to a separate DHCP Settings setup screen.

From this screen, the user has the option to enable use of NTP, DNS or the domain from DHCP.

The status of the DHCP Lease is given as True (connected, lease obtained) or False (lease not obtained).

6.3.2.2 SNMP Setup

PXQ supports both SNMP v1 and SNMPv3.

6.3.2.2.1 SNMPv1 Configuration

SNMP version 1 settings can be enabled from the SNMPv1 Details interface.

SNMP Community strings for both Read-only (for SNMP get operations) and Read/Write (for SNMP set operations) can be edited from this interface.

To restrict the hostname or IP address of the devices that can execute get and set requests, enter the desired address and select Add Host/IP Address.

Added addresses are displayed in a list and can be removed by selecting the checkbox next to each respective address and choosing Remove Selected.

6.3.2.2.2 SNMPv3 Configuration

SNMP version 3 settings can be enabled from the SNMPv3 Details interface.

For get operations, the Read-only username and passphrase can be entered, along with the option to Use Privacy to encrypt any communicated data.

For set operations, similar Read/Write credentials are available.

A Trap recipients community string can be edited.

To restrict the hostname or IP address of the devices that can execute get and set requests, enter the desired address, choose the V3 user, and select the desired security level.

The following security levels are supported:

- noAuthNoPriv – authorization based on username. No authentication password exchanged; communications not encrypted
- authNoPriv - password authentication is used; no encryption in communications.
- authPriv – password authentication is used; communications are encrypted.

6.3.2.2.3 MIBS

Available MIBS can be accessed by selecting the View MIBS button.

The following Management Information Bases (MIBs) provide information about the PXQ. Selecting the download button next to each downloads the MIBS file for use.

6.3.2.3.3 Modbus TCP - Cached Data Server

If enabled in Access Control, Modbus TCP settings can be edited.

A timeout can be defined in miliseconds.

Unsupported Read Requests – can be set to return an error or zeroes (Zero-Fill)

NaN (not a number) values can be expressed as zero.

The Device Mapping can be downloaded from this section by selecting view map.

6.3.2.4 BACnet/IP

If enabled in Access Control, BACnet/IP settings can be edited here.

For BBMD applications, an IPV4 address can be specified along with the BBMD's time to live, expressed in seconds).

The EPICS file is available to download by selecting Download File. This file is also available at www.eaton.com/PXQ.

6.3.3 COM1 & COM2 Setup

Modbus RTU is disabled by default and must be enabled here before using.

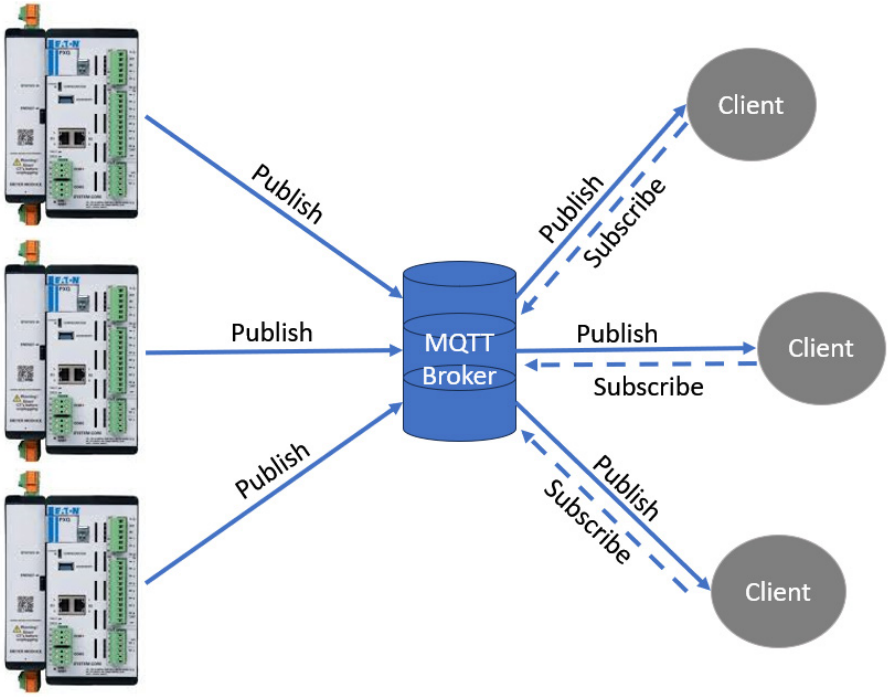
The following properties can be configured:

- Address
- Baud Rate (default: 9600)
- Stop Bits (default: 1)
- Parity (default: None)

6.3.4 Sub Synchronous Spectral Analysis Data Access (MQTTS) Setup

MQTT is a simple messaging protocol that works as a publish and subscribe system. PXQ publishes messages on a specific topic to a MQTT Broker and all clients that subscribe to that topic receive the message.

The MQTT broker is responsible for receiving all messages, filtering the messages, deciding who is interested in them, and then publishing the message to all subscribed clients.



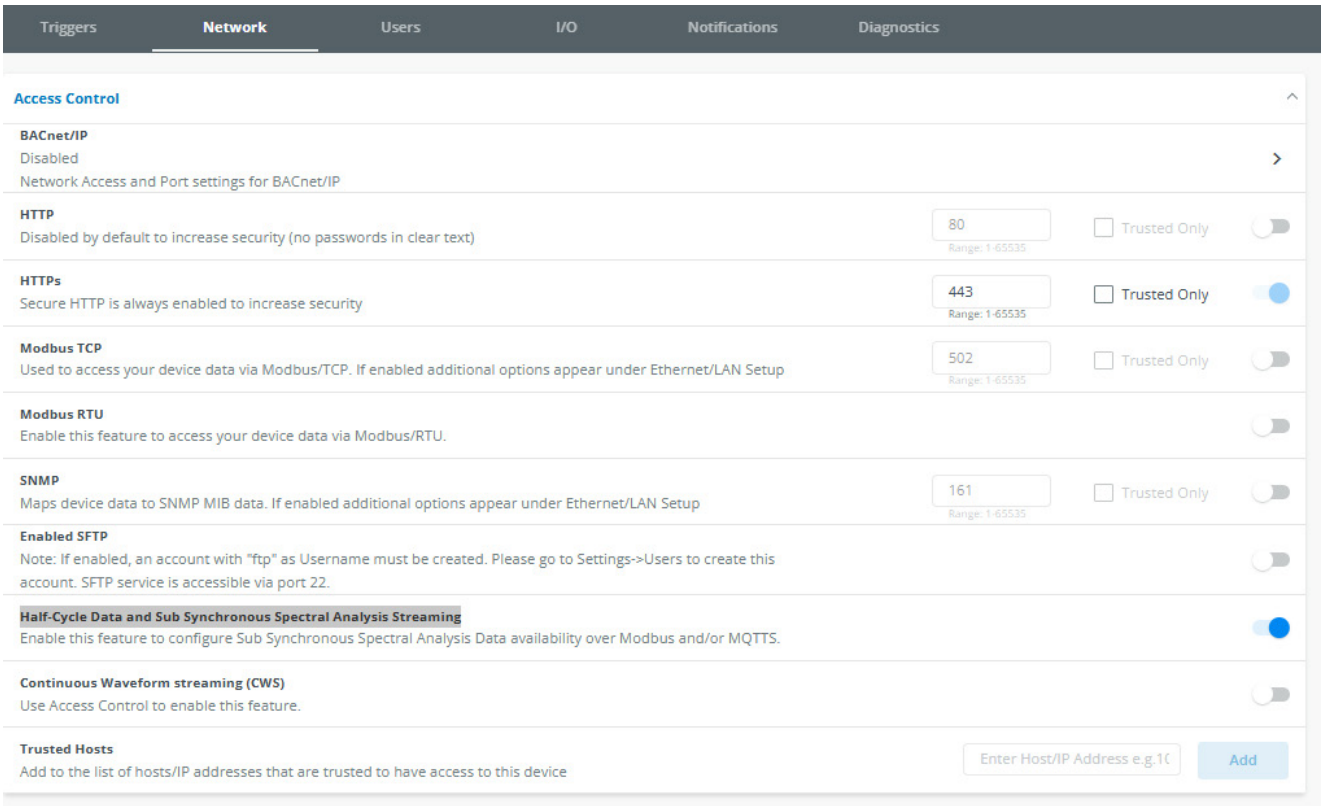
Real-time data pipeline to Sub Synchronous Spectral Analysis Data offers significant benefits in identifing and mitigating system instabilities that would cause equipment damage and network failures.

PXQ Half-Cycle Data and Sub Synchronous Spectral Analysis Streaming is disabled by default and enabled via PXQ WebUI > Settings > Network > Access Control.

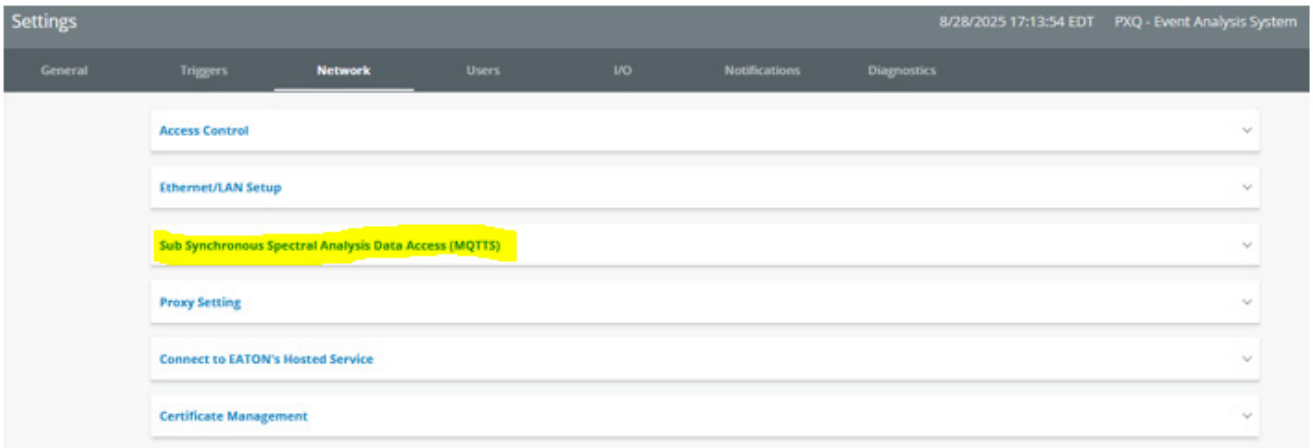
PXQ Half-Cycle Data and Sub Synchronous Spectral Analysis Streaming data source is the Sub-Synchronous Oscillation (SSO) Detection Setup, PXQ WebUI > Settings > Triggers > SSO Band Configuration.

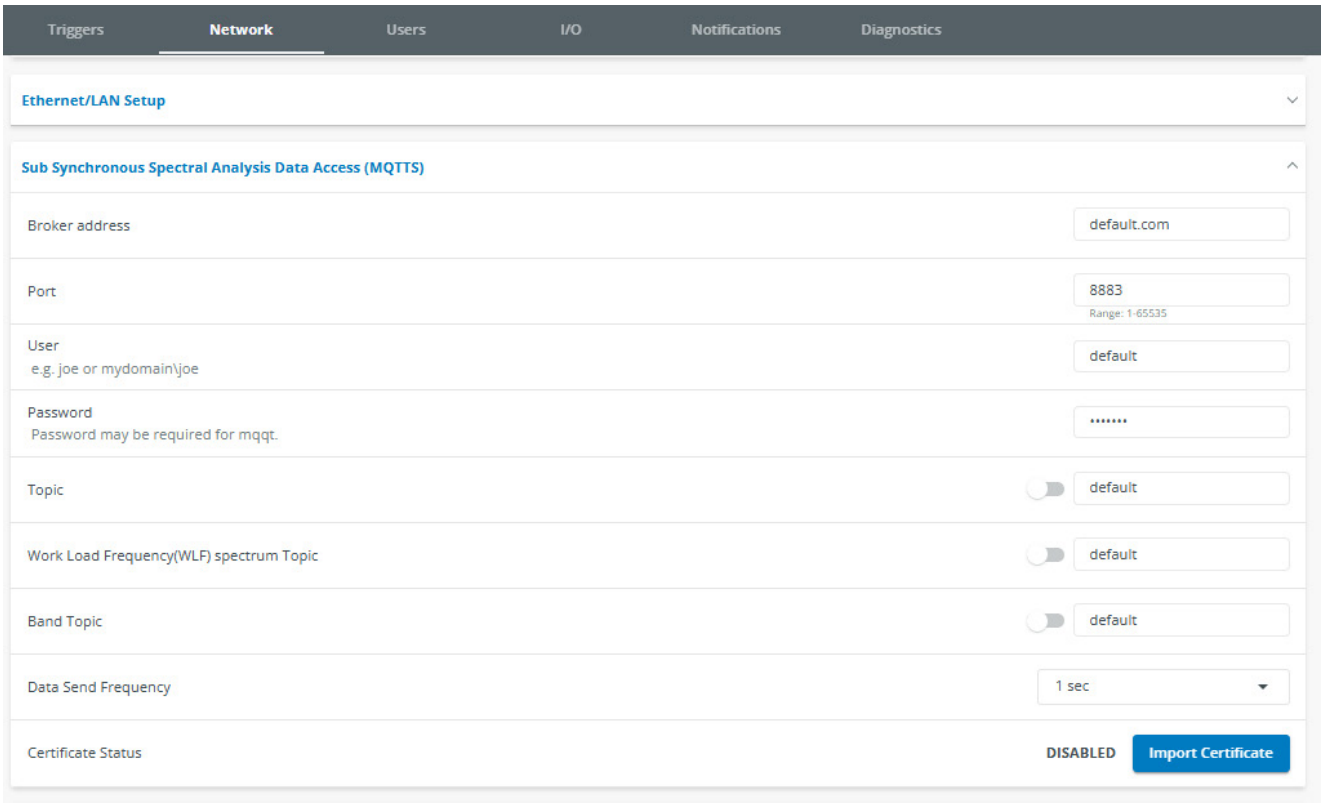
PXQ Publishes, a.k.a. Pushes, MQTT packet (JSON) containing Half Cycle Data to MQTT Broker:

- Timestamp with millisecond resolution
- 3-Phase Average RMS Voltage Line-Line
- 3-Phase Average RMS current
- 3-Phase Total/System Real Power (Watt)
- 3-Phase Total/System Reactive Power (VAR)
- 3-Phase Total/System Apparent Power (VA)
- 3-Phase Total Power Factor
- Frequency (Hz)



Once enabled “Sub Synchronous Spectral Analysis Data Access (MQTTS)” will appear for setup, PXQ WebUI > Settings > Network.





- Broker address: <Local MQTT Broker IP address>
- Port: 8883 (default setting, Port is the assigned port MQTT Broker is listening on)
- User: (MQTT Broker login supplied by local IT administrator)
- Password: (MQTT Broker login supplied by local IT administrator)

Note: MQTT Broker User & Password is used by PXQ to connect to Broker.

- Broker Address, Port, User & Password, are required to Import Broker Certificate to PXQ.
- Topic: <Max Length 64 alpha numeric characters, special chars include "/", "-", "_", ":" >

Note: An MQTT topic is a hierarchical string, similar to a file path, using forward slashes (/) to separate levels, guiding message routing between PXQ and MQTT Broker. Topics are case-sensitive.

- Work Load Frequency (WLF) spectrum Topic: <Max Length 64 alpha numeric characters, special chars include "/", "-", "_", ":" >

e.g. "PXQ1/Ab-1/Cd_3:64"

  - WLF Publish/Push FFT (Fourier Fast Transient) data to MQTT Broker:
  - FFT magnitudes: Amps, kW,
  - FFT Peaks: Current & Current-Frequency, kW & kW-Frequency

- Band Topic: <Max Length 64 alpha numeric characters, special chars include "/", "-", "_", ":" >
- e.g. "PXQ1/Ab-1/Cd_3:64"

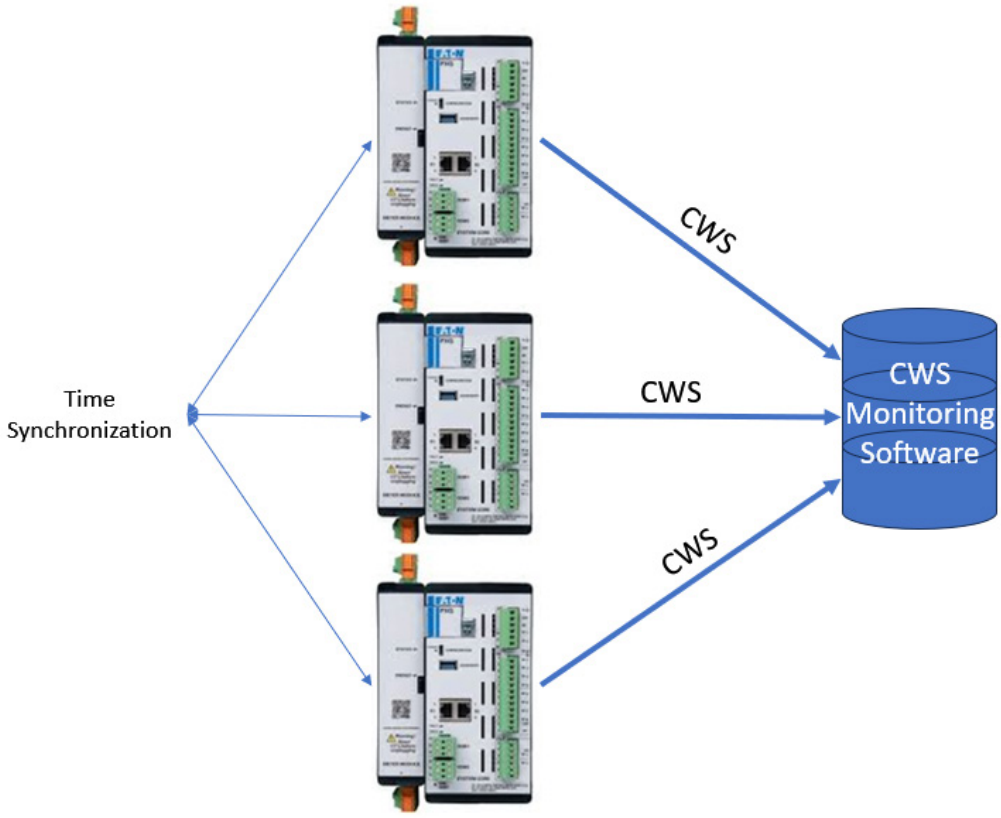
- Publish/Push Band data to MQTT Broker (only configured Bands) includes:
- Band # - Number from 1 to 50
- Band Limits - Separate fields for Band Start frequency & Band End frequency.
- Band Magnitudes - Separate fields for Amps & kW
- Band Status - Separate fields for Amps & kW (Can be “Normal”, “Cautionary”, “Critical”)

Note: Topic, WLF Topic, and Band Topic function independent of each other.

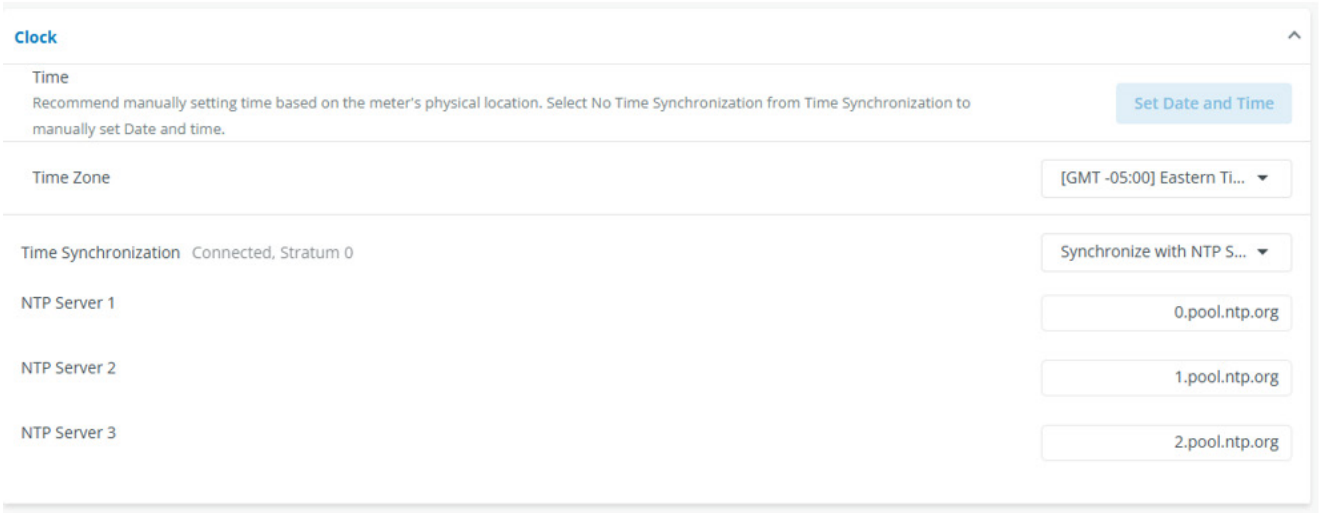
- Select frequency: 200 msec / 1sec (default is 1 sec)
  - Data Send Frequency (0.2 sec or 1 sec) data pushes for 1/2 cycle data.
- Import Certificate
  - Import Broker Certificate (e.g. ca.crt) to the PXQ, used to establish connection between PXQ and Broker over MQTT with TLS.

6.3.4.1 Continuous Waveform streaming (CWS) Setup - Data Streaming (3000 samples/second)

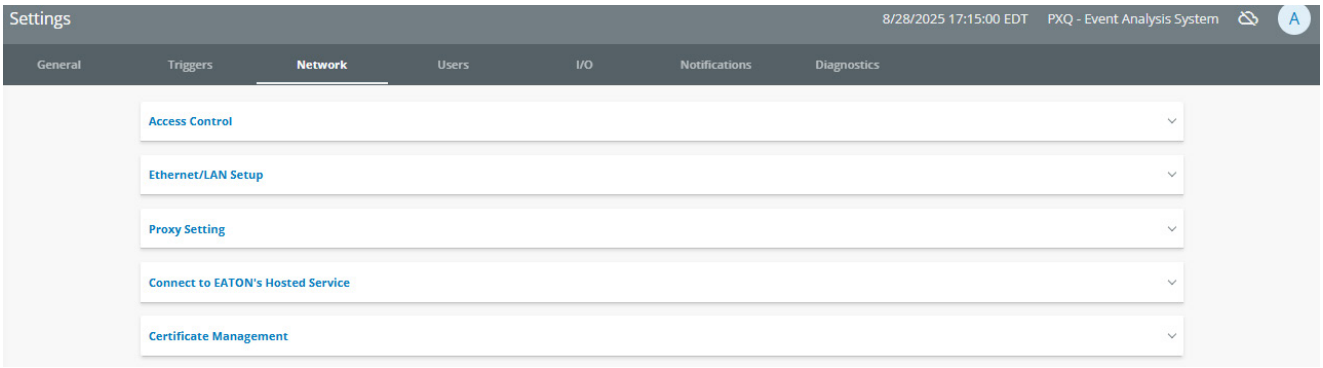
PXQ provides precise time-synchronized voltage and current waveform measurements in time-domain established via connections to high-accuracy time protocol. Operating at 3000 samples per cycle, PXQ CWS protocol provides high resolution voltage and current wave-shapes. This feature is based on IEEE C37.118 that was created out of the need for a standardized approach to synchrophasor measurements & data communications in power systems.

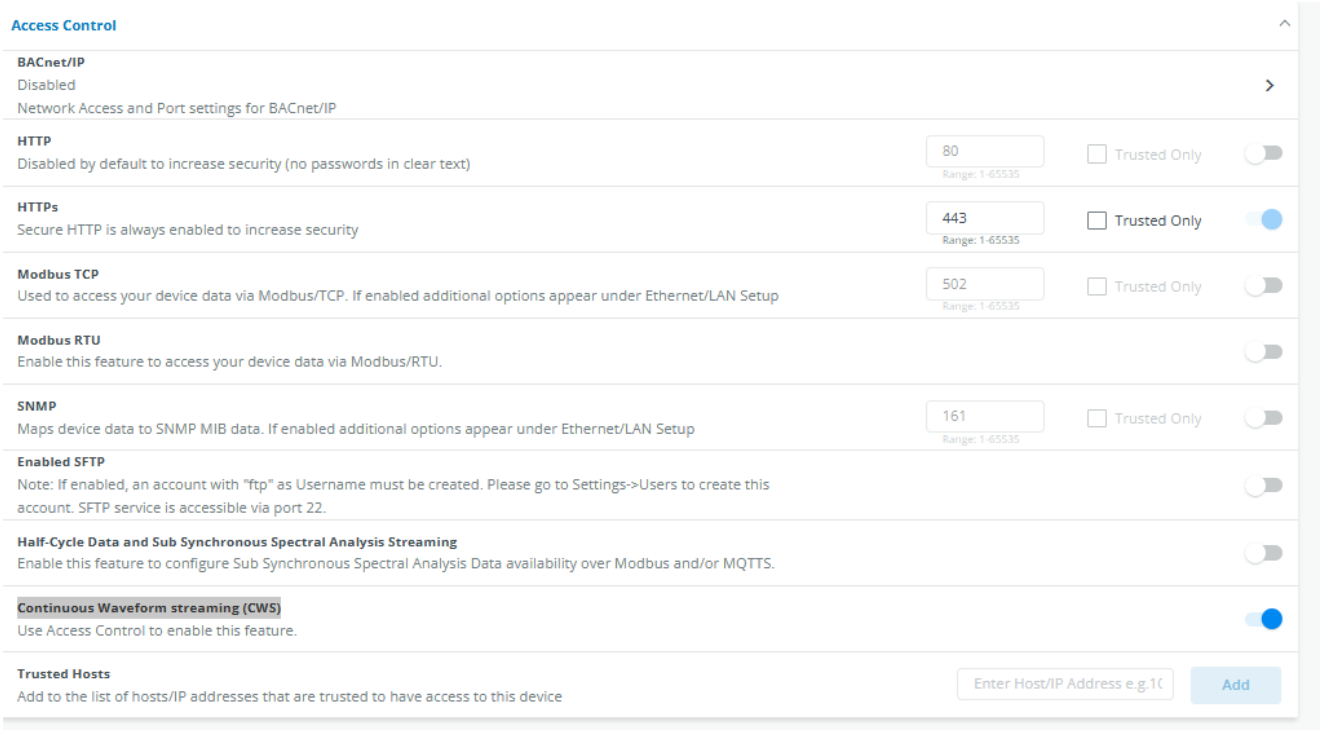


CWS Time Synchronization is accomplished via configuration of PXQ Clock, PXQ WebUI > Settings > General > Clock.

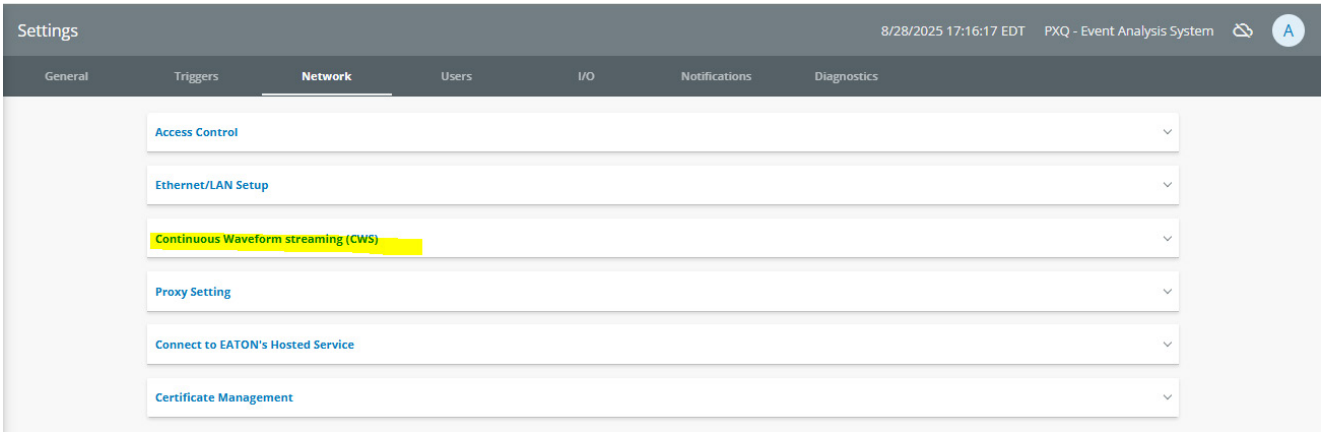


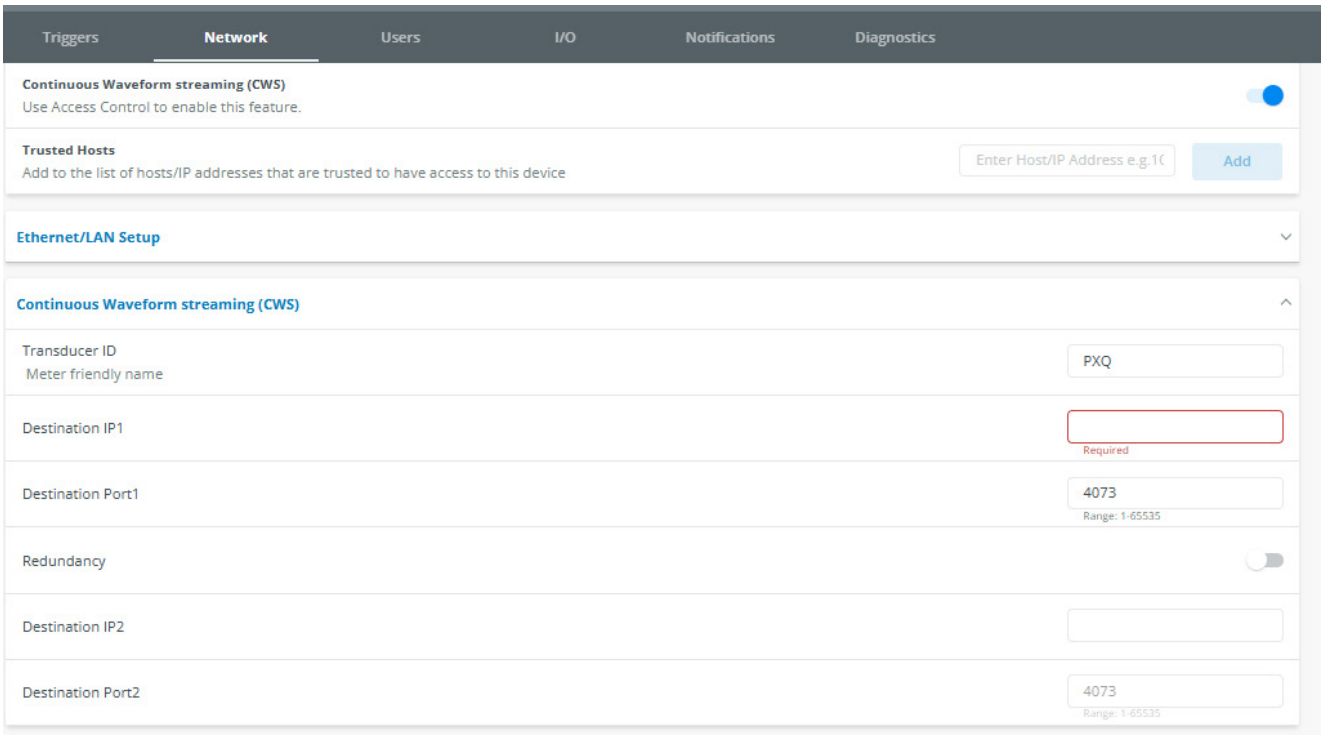
PXQ Continuous Waveform streaming (CWS) is disabled by default and enabled via PXQ WebUI > Settings > Network > Access Control > Continuous Waveform streaming (CWS).





After Continuous Waveform streaming (CWS) is enabled, Continuous Waveform streaming (CWS) will appear as a menu item under PXQ WebUI > Settings > Network where further configuration is required.





- Transducer ID: <Max Length 20 Alpha-Numeric Characters>
- Destination IP1: <UDP IP Address Required>
- Destination Port1: <UDP Port Range:1-65535, default is 4073>

Note: Destination IP is the IP address of CWS Monitoring Software.

- Redundancy: <disabled by default, supports a second Destination IP and Port for redundancy>
- Destination IP2: <UDP IP Address, only required if Redundancy is enabled>
- Destination Port1: <UDP Port Range:1-65535, default is 4073, only required if Redundancy is enabled >

Note: PXQ CWS protocol supports Destination IP1 equal to Destination IP2 with different port assignments to Destination Port 1 and Destination Port 2.

Depending upon PXQ Wiring Configuration setting (PXQ > Settings > General > Quick Setup), PXQ will stream either Line-Neutral Voltages or Line-Line Voltages along with the per Phase Currents.

IA	Phase A, Current, Instantaneous
IB	Phase B, Current, Instantaneous
IC	Phase C, Current, Instantaneous
VA or VAB	Phase A or AB, Voltage, Instantaneous
VB or VBC	Phase B or BC, Voltage, Instantaneous
VC or VCA	Phase C or CA, Voltage, Instantaneous

PXQ CWS is a low-overhead protocol based on IEEE C37.118 and COMTRADE standards. It supports unidirectional response-less streaming of data through separate Configuration and Data Frames. The "Destination IP" monitors for configuration frames, then will parse all following data messages that share a Channel ID with a known configuration frame.

Configuration Frame

Field	Size (Bytes)	Type	Description
Frame ID	2	Uint16	Frame type Designation and protocol version. Frame Type Designation 0x01 = Configuration Frame Protocol Version = 0x01
Size	4	Uint32	Total number of bytes that follow, not including FrameID and Size fields. (63 bytes + number of bytes in ChannelName)
ChannelID	8	Uint64	Device Serial Number and stream index.
PktCount	2	Uint16	Used to identify changes to the configuration. Starts at 0, increments when changed.
NumAnalogs	2	Uint16	Number of analog values per SampleSet. 4-byte int32 values. Analogs are as defined in Table J.2
NumDigitals	2	Uint16	Number of digitals per SampleSet. Grouped in blocks of 16 into a Uint16. This value is always zero.
SampleRate	4	Int32	A value of 3000 to indicate 3 ksps fixed sample rate.
Scalars	4 * NumAnalogs	Float32 array	One scalar per analog signal. The multiplier includes PTR or CTR * Analog to calculate scaled value.
ChannelName	2-21	String	Set to the value of the TID setting: String names are null terminated.
SignalNames	21	String	SignalName is the string containing names for each of the voltage and current signals, in order of transmission. Each name is delimited with a null character.

Data Frame

Field	Size (Bytes)	Type	Description
Frame ID	2	Uint16	Frame type Designation and protocol version. Frame Type Designation 0x00 = Configuration Frame Protocol Version = 0x01
Size	4	Uint32	Total packet size, not including FrameID and Size fields. (1218 bytes)
ChannelID	8	Uint64	Device Serial Number and stream index.
PktCount	2	Uint16	Continuously increments once per packet.
NumAnalogs	8	Int64	Nanoseconds since Epoch in UTC.
NumDigitals	1200	Int32 (Big Endian)	Data points need to be multiplied by scalars for each channel. The sample sets are grouped together, not serialized per channel. [Int32 x 6 for VA, VB, VC, IA, IB, IC] x 50

6.3.5 Proxy Setting

If use of a proxy is desired, settings can be enabled via Require Proxy.

The Proxy URI, user and password can be specified here.

6.3.6 Connect to Eaton's Cloud Hosted Service

In order to connect to one of Eaton's hosted cloud services (e.g. Brightlayer), configuration in both PXQ and the cloud service is required.

The connection must be enabled on the PXQ device by selecting Allow Connection to Eaton Hosted Server.

The Device GUID is the unique ID string for a particular device. This is to be copied and entered in the applicable field when prompted by the cloud service during setup.

A corresponding Connection String from the hosted service must be entered in the device under Connection String to Access the IoT Hub Server.

6.3.7 Certificate Management

Certificates play a crucial role in keeping communication secure. These certificates are sent out when communication starts. The other end will receive this certificate and will validate it. If the other end finds the certificate valid and trusted, that will allow for further communication. Otherwise, the device will reject the communication.

A certificate can be generated for one service (i.e. HTTPS service) by signing it from another service (i.e. PKI service).

User can generate self-signed signed certificate for a service (e.g. HTTPS).

User can Import, Export and Revoke certificates. They can also generate Certificate Signing Request (CSR) files, which can be CA signed and then re-imported into the Certificate Manager.

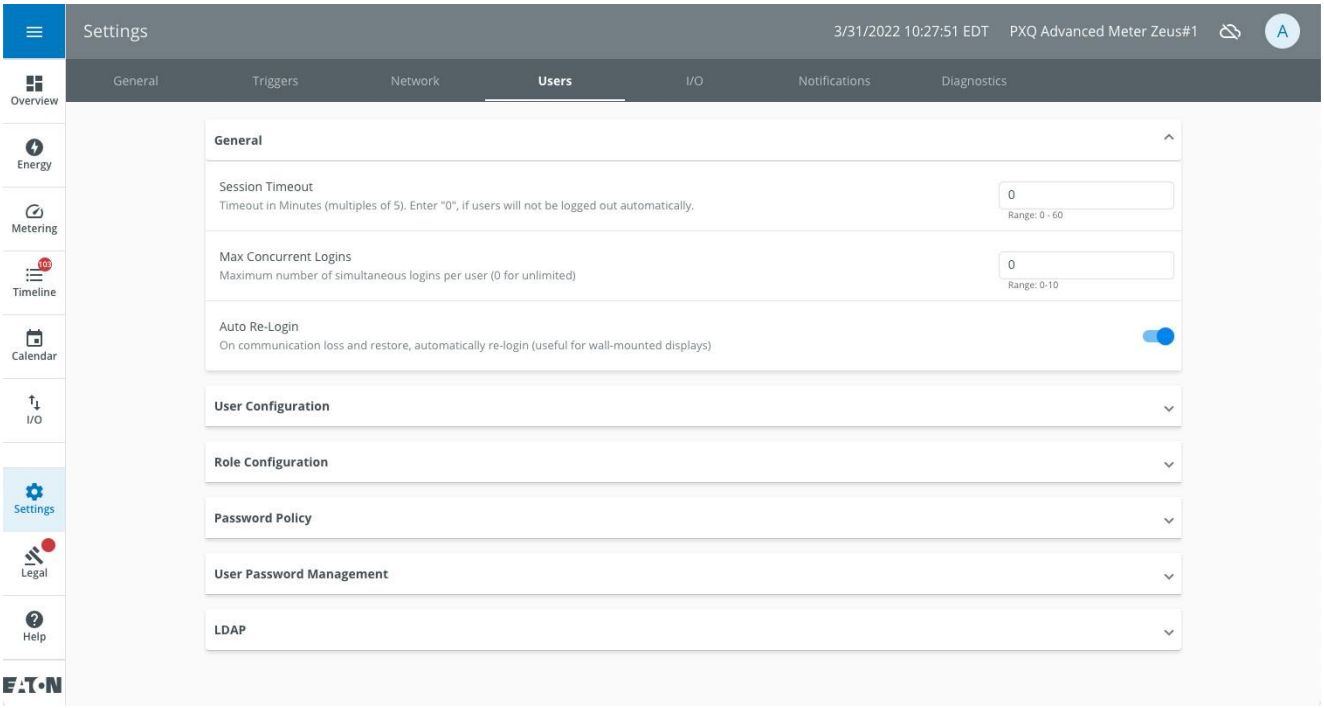
6.3.7.1 Certificate Upload

Users can get a CA-signed certificate from a third party and upload that for any service (e.g. HTTPS). These CA certificates can be used by the service to validate the issuer, and allow communications to begin.

Options are Server, CA, and Trusted Client.

6.4 Users

6.4.1 General

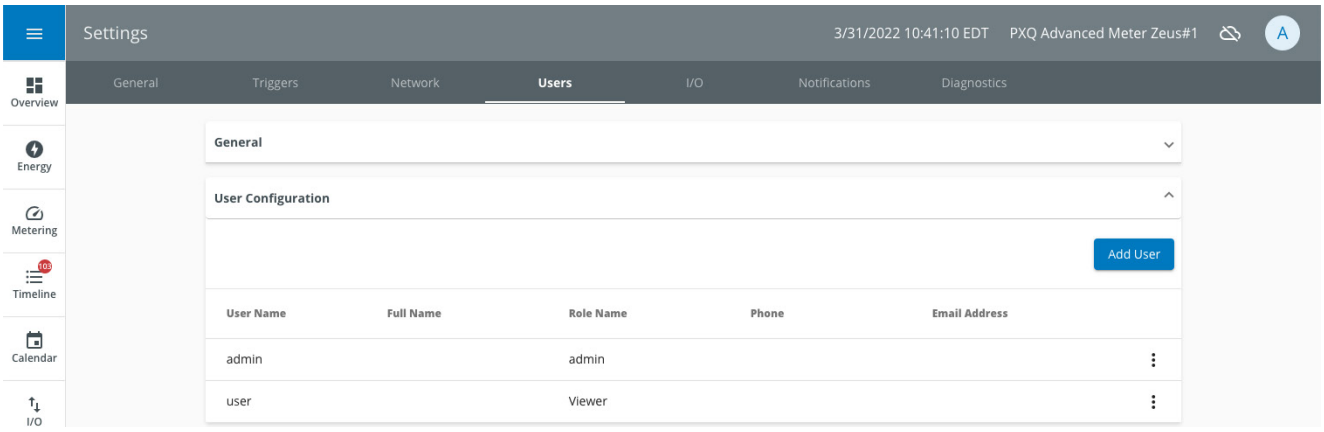


Session Timeout defines the number of minutes after which a user will automatically be logged out of the PXQ interface. This should be defined in multiples of 5. Setting this parameter to 0 provides no timeout.

Max Concurrent Logins defines the number of instances a given user can be logged into a particular PXQ.

Auto Re-Login enables a terminal to log back in if a connection is lost. This setting is intended for use on a local display and should not be enabled otherwise.

6.4.2 User Configuration



Users can be created with the Add User button.

For each new user, a case-sensitive user name, role assignment, email address and password must be provided. Name and contact phone number are optional.

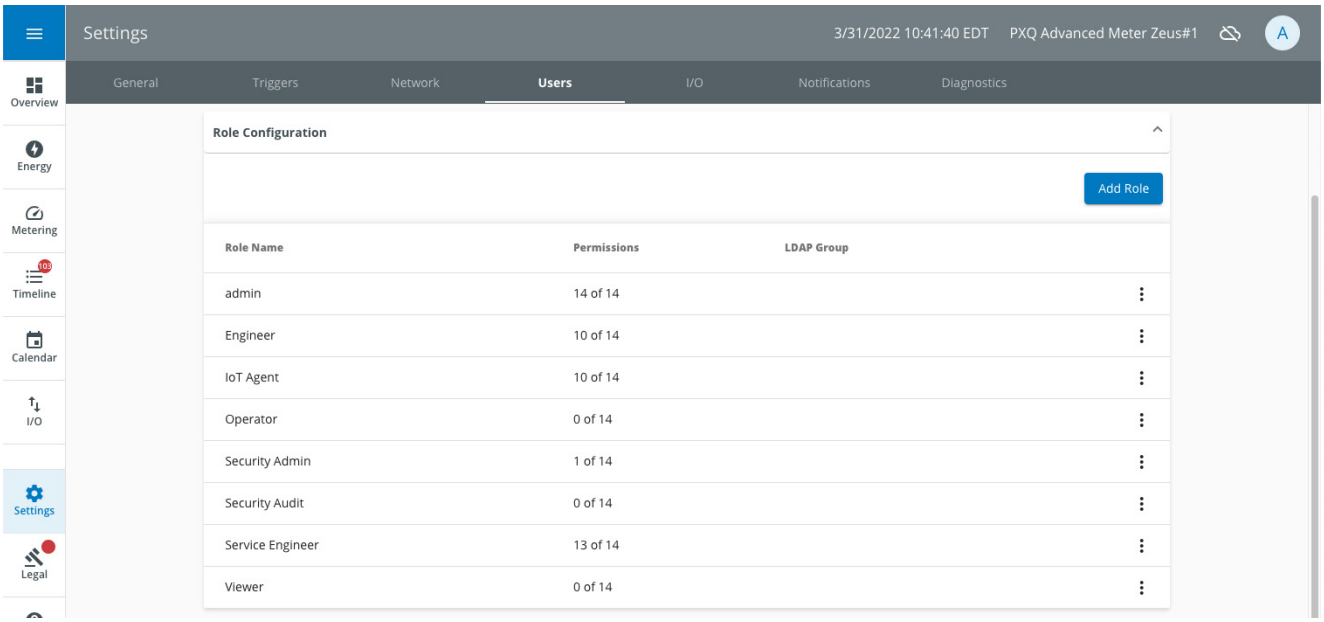
By default, two accounts have been created. Default credentials have been listed below.

- User name: admin
  Password: admin
- User name: user
  Password: user

To manage existing users and update any of the parameters, the three dots next to each user allows the user to view, edit or delete each profile.

A password is required to edit a profile.

6.4.3 Role Configuration



PXQ ships with 8 pre-configured roles. Each role has a subset of 14 permissions enabled.

By clicking on the three dots to the right of each role, each can be customized (renamed, or change assigned permissions) or deleted. Additional roles can be created with the Add Role button.

The following list of permissions is available to assign to a role.

- Security – making updates to users, roles and password policy (Settings > Users)
- Alarm – ability to acknowledge, delete or tag alarms, Capture Waveform, Capture High-speed waveform
- Metering Config – All Meter Configurations, Save/Restore Configuration (Settings > General & Triggers)
- I/O Config – allows changes to I/O settings (Settings > I/O)
- Demand Reset
- Energy Reset
- Clear Events & Waveforms - Erase Event Log
- Clear min/max
- Reset All Trends Data - Reset Latched Outputs, Reset Auto-Trigger Thresholds, Reset Input Counters, Reset Circuit Gauges, Reset PQ Health Index, Erase Event Log, Erase All Trend Data, Erase Profile Data
- Reboot – allows soft reboot of the PXQ
- Firmware Upgrade
- Set Date & Time
- Network – allows updates to settings on the Network tab
- Factory Reset – allows PXQ to be reset. Erases all data, configurations, triggers, users, network settings, etc.

6.4.4 Access Points

The access point feature allows you to pair a trusted machine (a PC, tablet, display, or HMI) with a PXQ for view-only access without requiring repeated logins.

This is most useful for pairing an HMI with a specific PXQ to use as a local display.

You can create up to 10 different access point profiles.

Creating an access point:

1. Select Add Access Point
2. In the dialog box, name the access point and select the desired viewer credential.

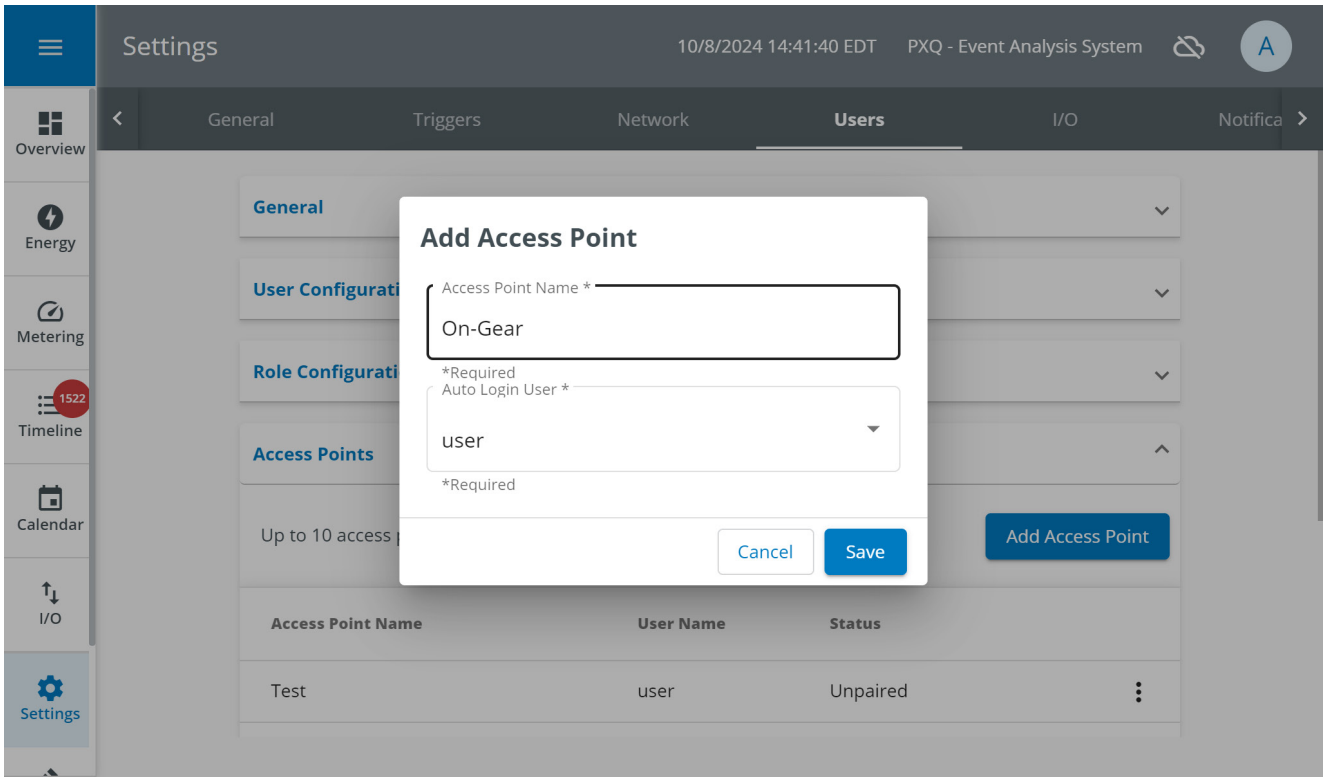
Note: Only Viewer role users are available to select.

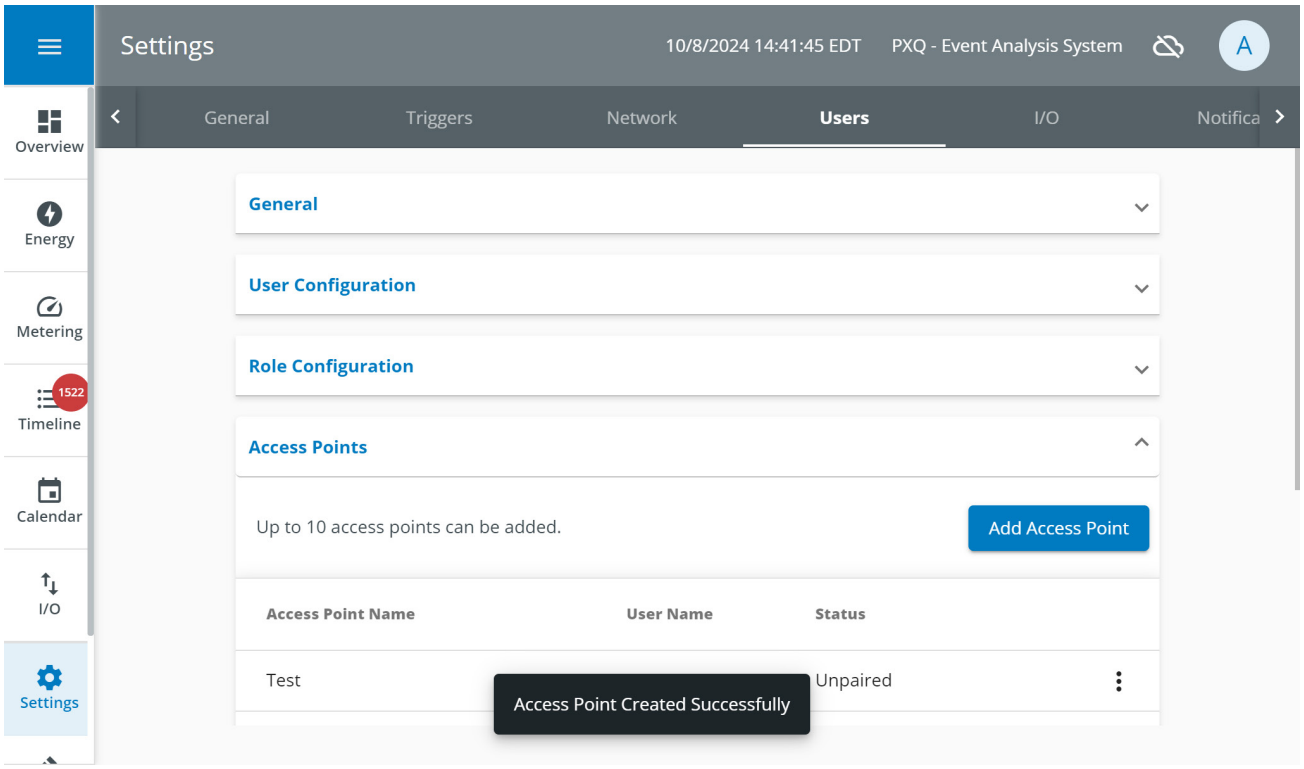
3. Click Save.
4. A status message will appear at the bottom of the screen stating: "Access Point Successfully Created". Your access point should now be visible in the table of access points.
5. Click on 3 dots next to the newly created access point and select Pair.
6. To complete pairing, follow the prompts that will guide you to Logout and Pair
7. Enter the access point account credentials. The PXQ will process, log out, and log in as the designated Viewer account. (You can verify this in the upper right corner of the display when the process is complete.)

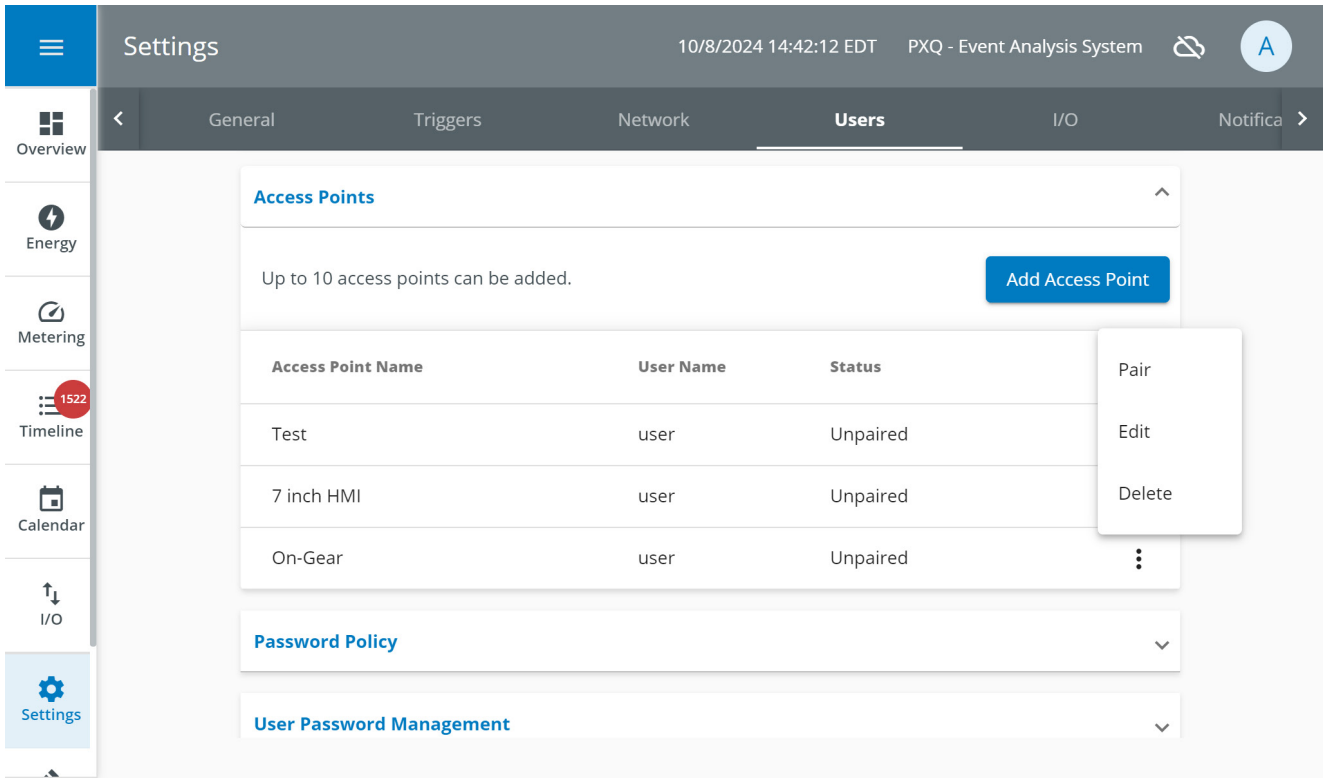
To optimize this feature for a smooth experience, your machine (particularly for HMI mounted in switchgear and displays intended for use as kiosks) should be configured with the following:

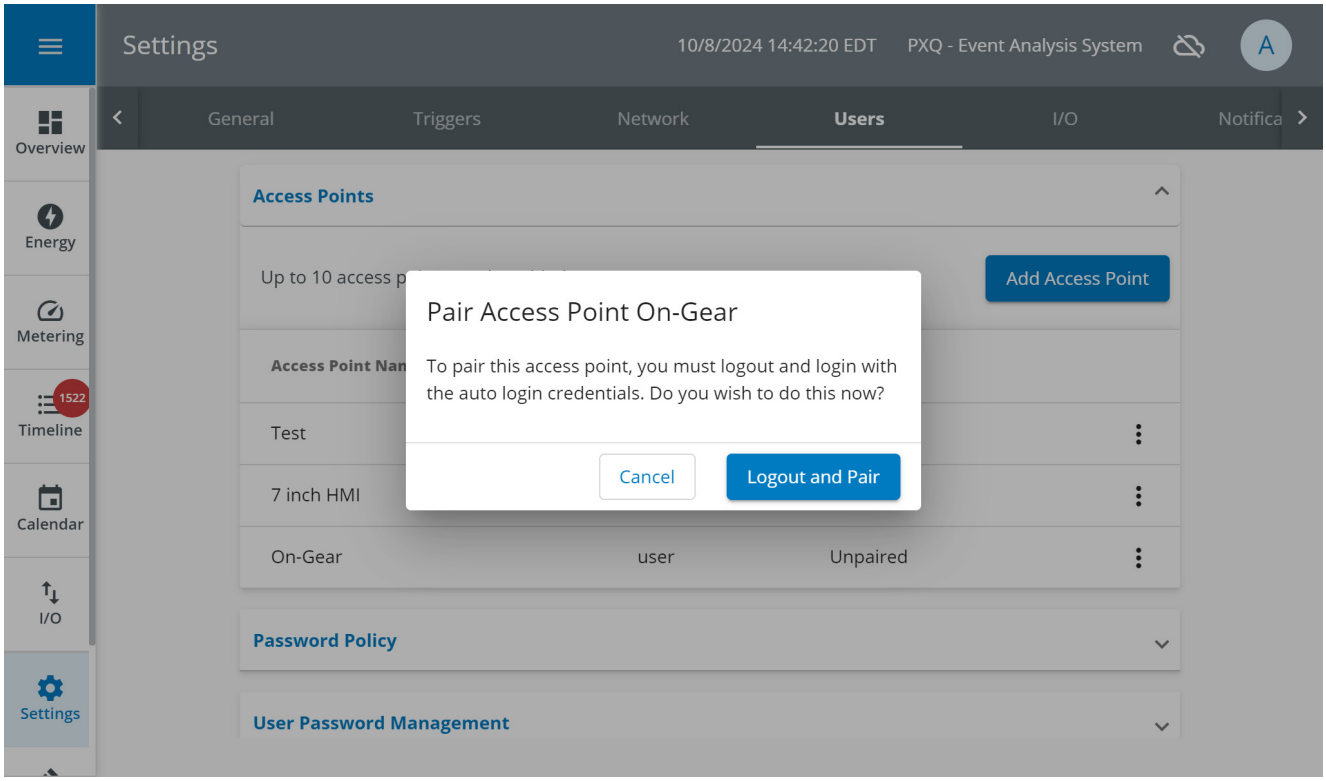
- Set your display to default to the IP address of the PXQ.
- SSL Certificate: Install the PXQ's self-signed SSL Certificate, add a third-party Certification Authority (CA)'s certificate to the PXQ, or configure your display to allow/ignore invalid SSL Certificates
- Set display sleep and timeout settings as desired.

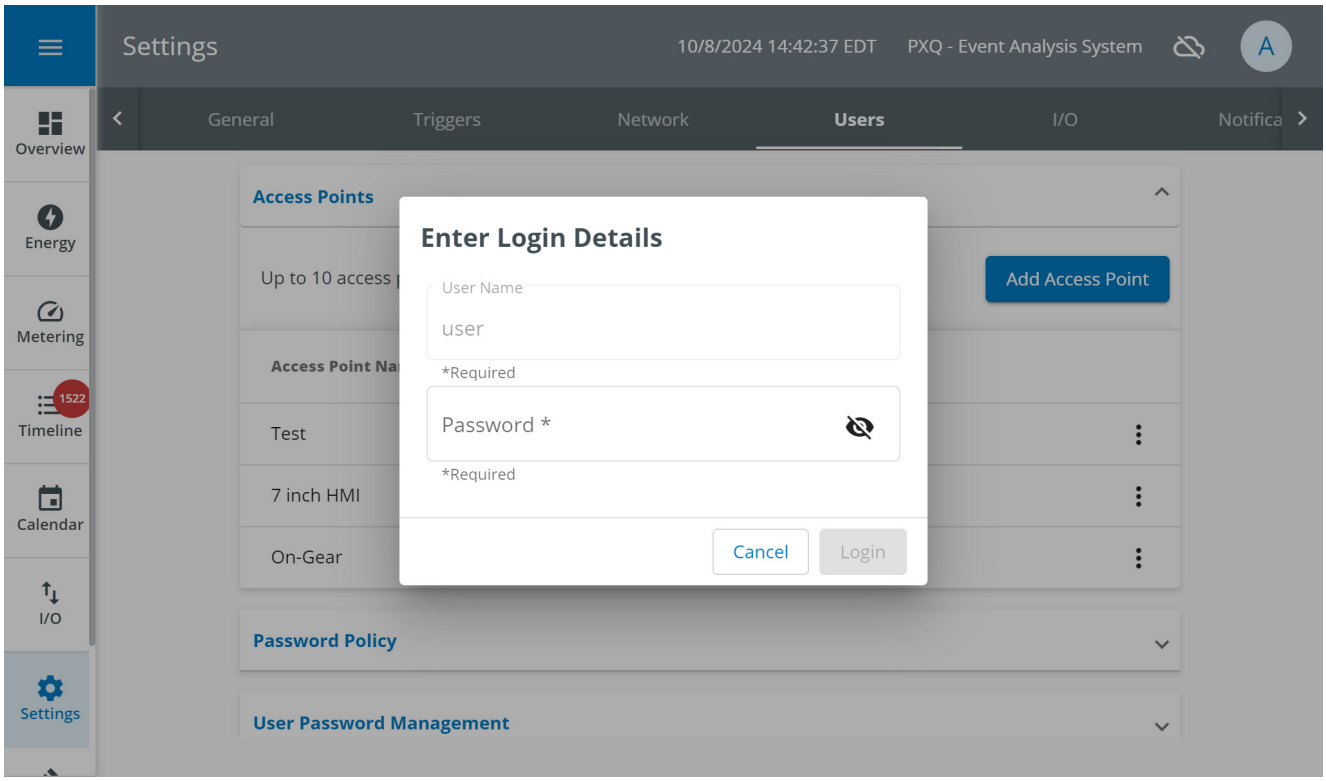
For more specific information, consult the user manual for your specific display or HMI.


Settings
10/8/2024 14:41:40 EDT
PXQ - Event Analysis System
A
General
Triggers
Network
Users
I/O
Notifica
Overview
Energy
Metering
Timeline
1522
Calendar
I/O
Settings
General
User Configurati
Role Configurati
Access Points
Up to 10 access
Add Access Point
Add Access Point
Access Point Name *
On-Gear
*Required
Auto Login User *
user
*Required
Cancel
Save
Access Point Name
User Name
Status
Test
user
Unpaired

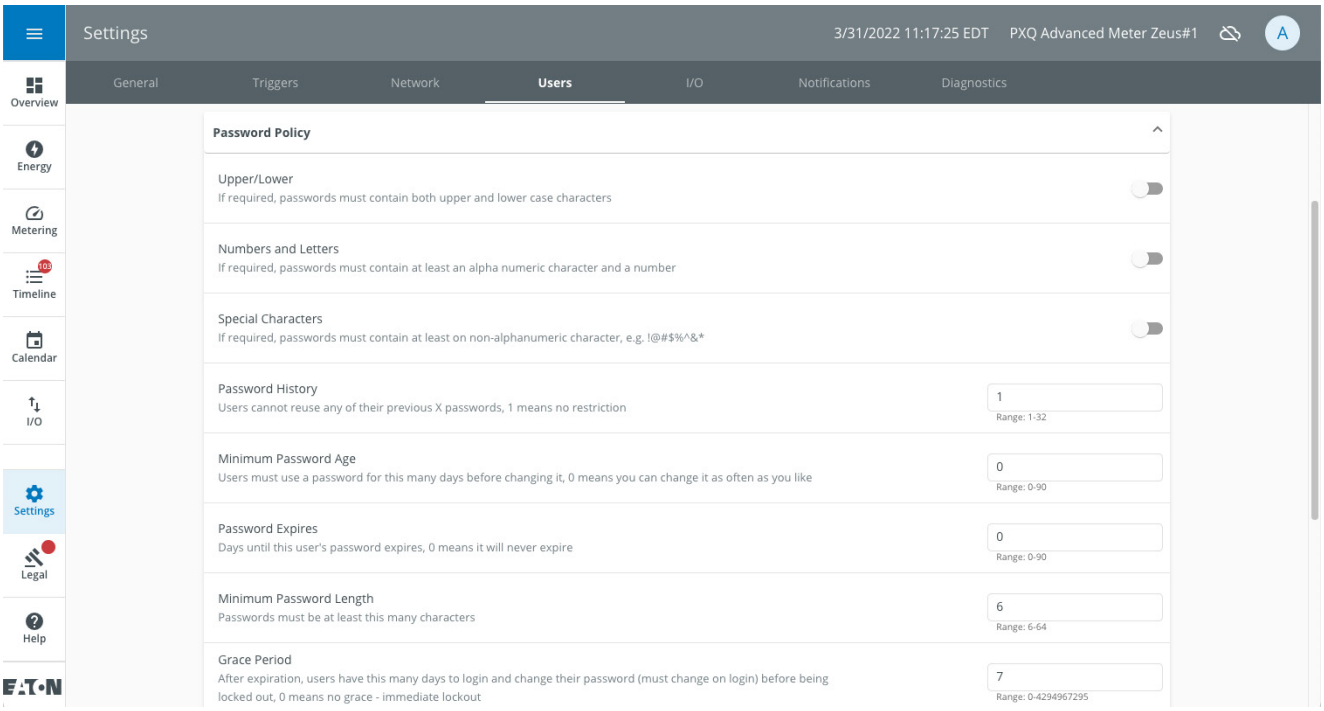

Settings
10/8/2024 14:41:45 EDT
PXQ - Event Analysis System
A
General
Triggers
Network
Users
I/O
Notifica
Overview
Energy
Metering
Timeline
1522
Calendar
I/O
Settings
General
User Configuration
Role Configuration
Access Points
Up to 10 access points can be added.
Add Access Point
Access Point Name
User Name
Status
Test
Unpaired
Access Point Created Successfully


Settings
10/8/2024 14:42:12 EDT
PXQ - Event Analysis System
A
Overview
Energy
Metering
Timeline
1522
Calendar
I/O
Settings
General
Triggers
Network
Users
I/O
Notifica
Access Points
Up to 10 access points can be added.
Add Access Point
Access Point Name
User Name
Status
Test
user
Unpaired
7 inch HMI
user
Unpaired
On-Gear
user
Unpaired
Pair
Edit
Delete
Password Policy
User Password Management


Settings
10/8/2024 14:42:20 EDT
PXQ - Event Analysis System
A
Overview
Energy
Metering
Timeline
1522
Calendar
I/O
Settings
General
Triggers
Network
Users
I/O
Notifica
Access Points
Add Access Point
Test
7 inch HMI
On-Gear
user
Unpaired
Password Policy
User Password Management
Pair Access Point On-Gear
To pair this access point, you must logout and login with the auto login credentials. Do you wish to do this now?
Cancel
Logout and Pair


Settings
10/8/2024 14:42:37 EDT
PXQ - Event Analysis System
A
Overview
Energy
Metering
Timeline
1522
Calendar
I/O
Settings
General
Triggers
Network
Users
I/O
Notifica
Access Points
Up to 10 access
Add Access Point
Access Point Na
Test
7 inch HMI
On-Gear
Password Policy
User Password Management
Enter Login Details
User Name
user
*Required
Password *
*Required
Cancel
Login

6.4.4 Password Policy



Password complexity and restrictions can be managed under Password Policy settings.

Password Complexity - A mix of upper/lowercase letters, letters and numbers, and/or special characters may be enforced by selecting the corresponding toggle.

Password History - The user cannot reuse any of their previous X passwords. A 1 means no restriction.

Minimum Password Age - Users must use a password for this many days before changing it, 0 means you can change it as often as desired

Password Expires - Days until this user's password expires, 0 means it will never expire

Minimum Password Length - Passwords must be at least this many characters

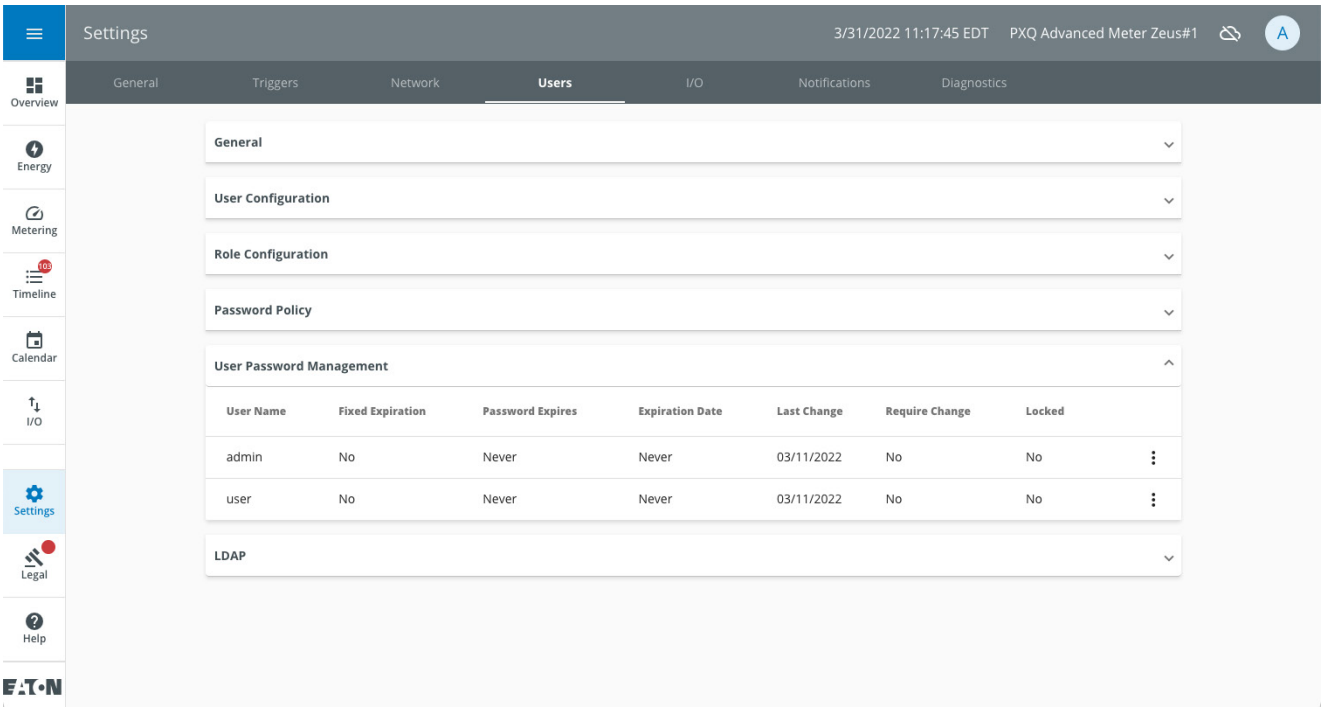
Grace Period - After expiration, users have this many days to login and change their password (must change on login) before being locked out, 0 means no grace - immediate lockout

Max Failed Attempts - Number of failed attempts to trigger a lockout, 0 means never lockout

Failed Login Attempt Window - Number of seconds after which the current count of failed attempts is reset, 0 means never lockout

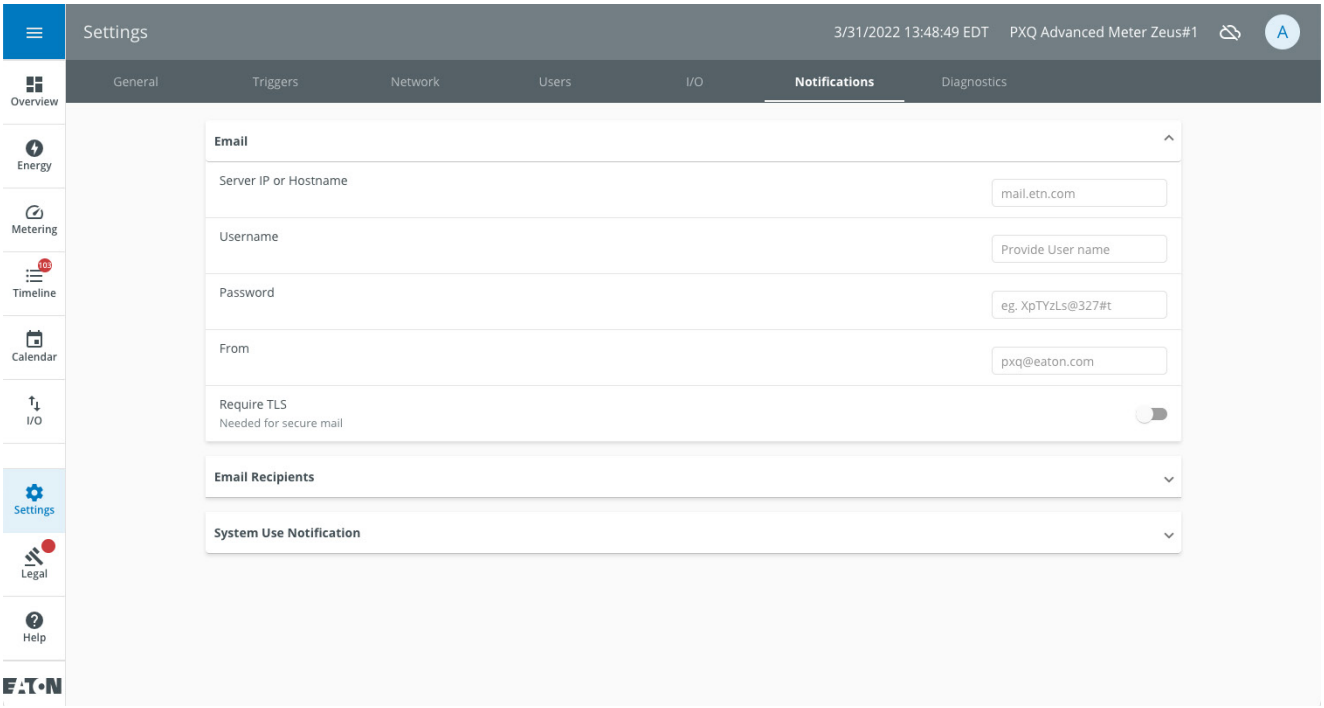
Failed Login Wait - After a lockout due to getting Max Failed Attempts within the Failed Login Attempt Window, this is the number of seconds until the lock will automatically be removed, 0 means never auto-unlock

6.4.5 User Password Management



To assist a user who has lost their credentials or been locked out of their account, select the three dots to the user. Here the user's account can be unlocked with the slider. A new password can be set. (This can be a temporary password, that does not meet complexity requirements and expires on next logon.)

6.5 Notifications



PXQ can be configured to send notifications to users periodically or on a triggered alarm. General Email server settings and periodic emails can be configured under **Settings > Notifications**. Emails on triggered alarm can be configured under **Settings > Triggers**. General server settings must be configured before either type of notification email can be sent.

Periodic notifications are sent at the start of each day at 12:01am, of the PXQ's configured time.

6.5.1 Email Settings

General email server settings are configured via Email Settings section. Your organization's IT department can help provide the correct values for these settings.

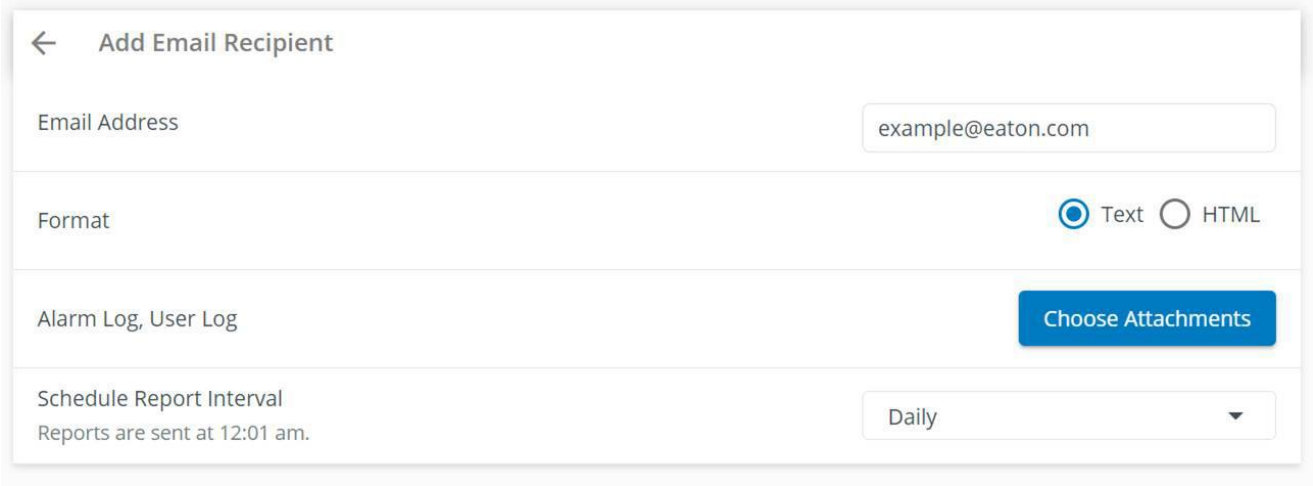
The following settings are required:

- Server IP or Hostname
- Username
- Password
- From (the email address corresponding to the username and password)
- TLS requirement – yes/no for this account

6.5.2 Email Recipients

Periodic emails and their recipients can be configured under Email Recipients.

Select Add Email to create a new email subscription. Only one subscription can be configured per email address.



The following settings can be used to configure the periodic notification subscription:

- **Email Address:** This field accepts only one email address. To deliver to a second recipient, create another notification via Add Email Recipient.
- **Format:** HTML or Text, describes the format of the email body.
- **List of selected attachments:** Any selected attachment is listed. Desired attachments may be configured by selecting Choose Attachments. By default, Alarm Log and User Log are selected. Available attachments are described below.
- **Schedule Report Interval:** Daily or Monthly. If Monthly is selected, user is prompted to choose which day of the month (1st, 2nd, 3rd, etc.) Notifications are sent at 12:01am on the selected day.
  - For a report that encompasses the calendar month, select for the notification to be sent on the 1st of each month. Since notifications are sent at 12:01am, all data up through midnight on the last day of the month will be captured.

The following attachments are available:

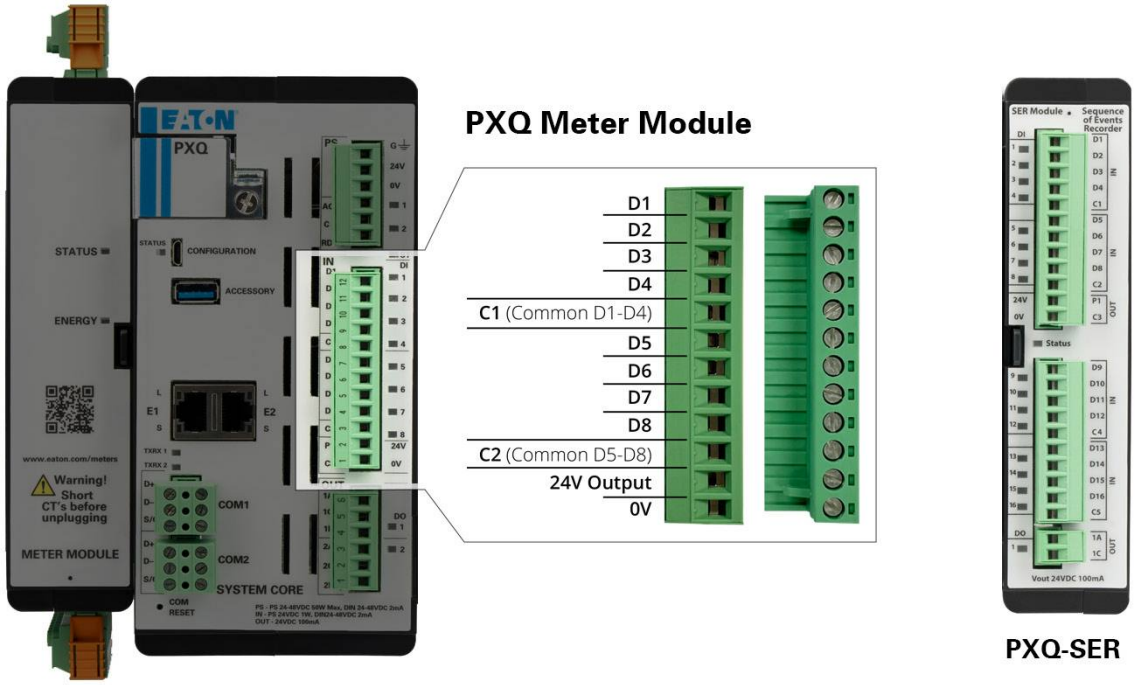
- Config Log
- Session Log
- Update Log
- User Log

6.6 I/O

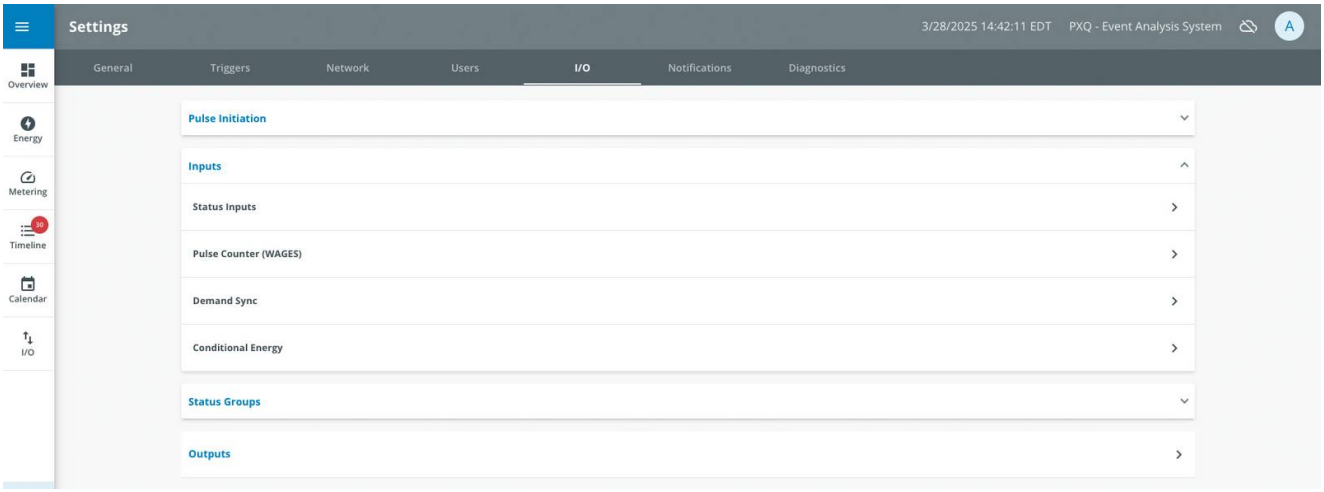
The PXQ Meter Module has 8 onboard Digital Inputs labeled as D1 through D8.

The optional PXQ-SER provides an additional 16 Digital Inputs labeled D1 through D16 (PXQ System Core supports maximum 2-PXQ-SER).

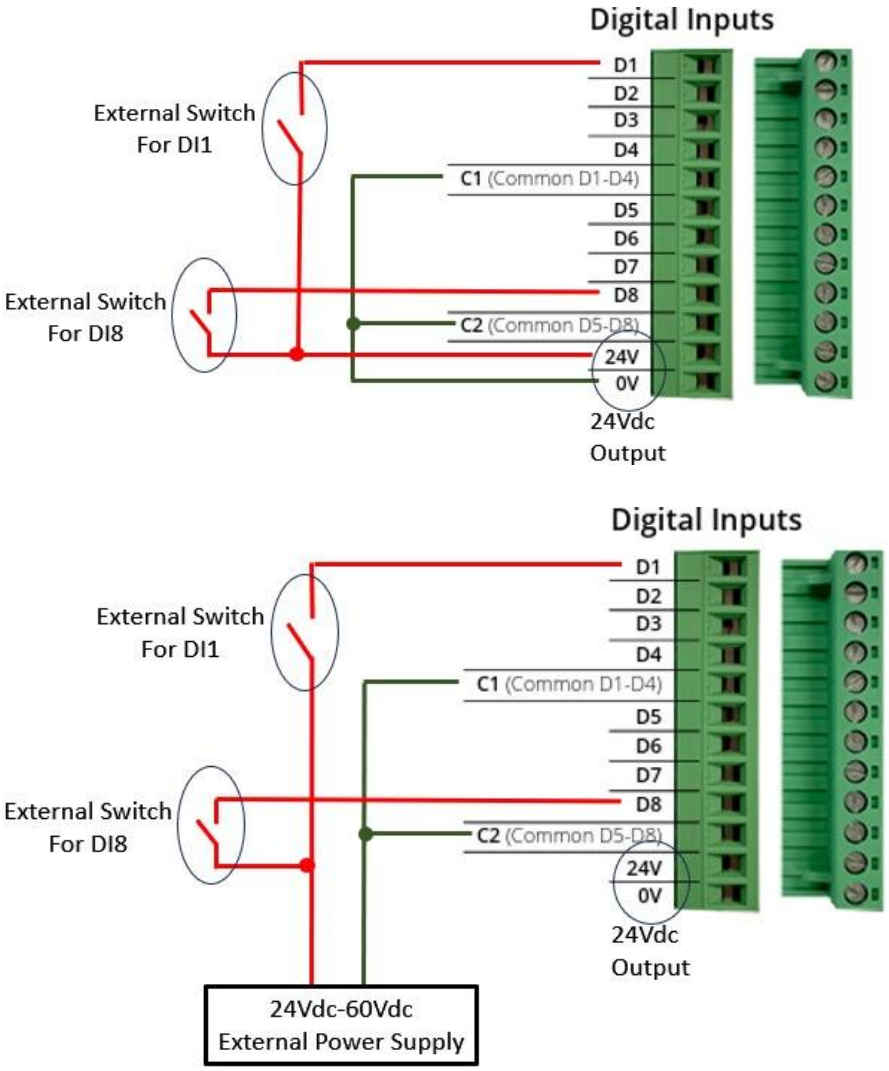
All Digital Inputs are disabled by default.



Digital Inputs are user configurable to support status outputs from Breakers, Control Power, UPS, SPD, Transformer Temperature Limit Switch, Alarm, Relay, Transfer Switch, etc.



Example Wiring Diagram



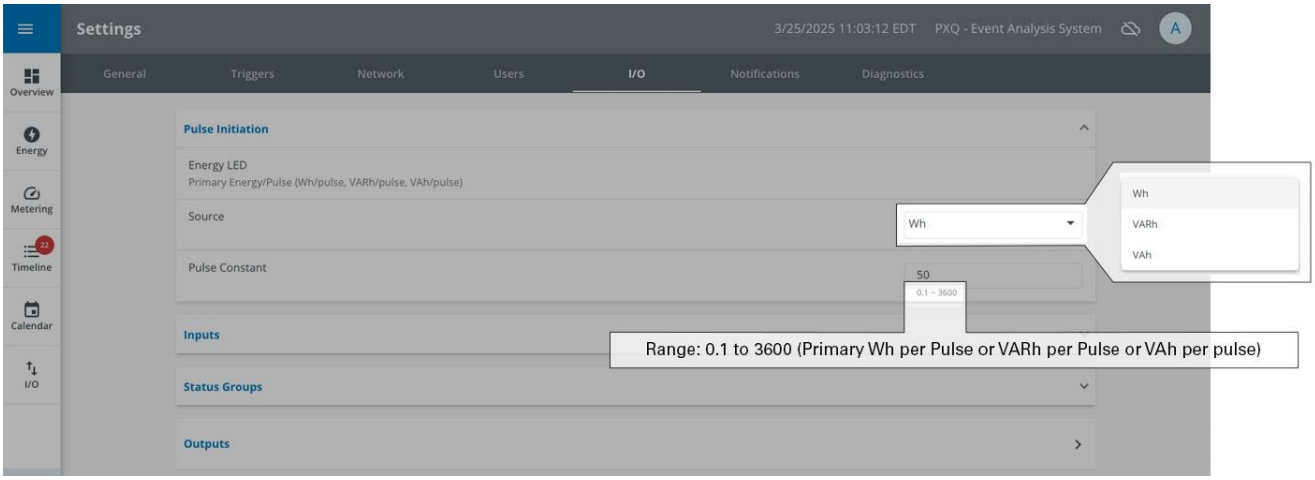
6.6.1 Pulse Initiation

The PXQ Energy LED is an optical pulse output LED indicating a certain amount of energy being metered. The amount of energy for each pulse depends on the PXQ configuration, i.e. CT ratio, PT ratio, Pulse Constant, Source (Wh, VARh, VAh).

The faster the PXQ Energy LED flashes the more energy being consumed by the load. By counting the number of flashes for a specific time period and then multiplying by the PXQ Pulse Constant, the amount of energy used during that time period can be determined. An optical (light) pulse sensor is typically used to count the pulses.

In Settings > I/O > Pulse Initiation, the PXQ Meter Module Energy LED can be configured.

There are two (2) settings (default settings shown):



Example Given using steady state Watt value:

Steady State Watts	Watt/Second (Steady State Watts / 3600 per Watt-Hour)	Pulse Constant	PXQ System Core Energy LED Pulse Rate (Seconds)	Description
90,000.00	25.00	25.0	1	pulse every 1 second at 25.0 watts
		50.0	2	pulse every 2 second at 50.0 watts
		75.0	3	pulse every 3 second at 75.0 watts
		100.0	4	pulse every 4 second at 100.0 watts
		125.0	5	pulse every 5 second at 125.0 watts
		150.0	6	pulse every 6 second at 150.0 watts
		175.0	7	pulse every 7 second at 175.0 watts
		200.0	8	pulse every 8 second at 200.0 watts
		225.0	9	pulse every 9 second at 225.0 watts
		250.0	10	pulse every 10 second at 250.0 watts

6.6.2 Inputs

The PXQ Meter Module has 8 onboard Digital Inputs labeled as D1 to D8. These are identified as Input 1 to Input 8 on drop downs associated with Settings > I/O > Inputs.

The optional PXQ-SER provides an additional 16 Digital Inputs labeled D1 through D16 (PXQ System Core supports maximum 2-PXQ-SER). These are identified as SER1 DI1 to SER1 DI16 and SER2 DI1 to SER2 DI16 on drop downs associated with Settings > I/O > Inputs.

All Settings > I/O > Inputs are disabled.

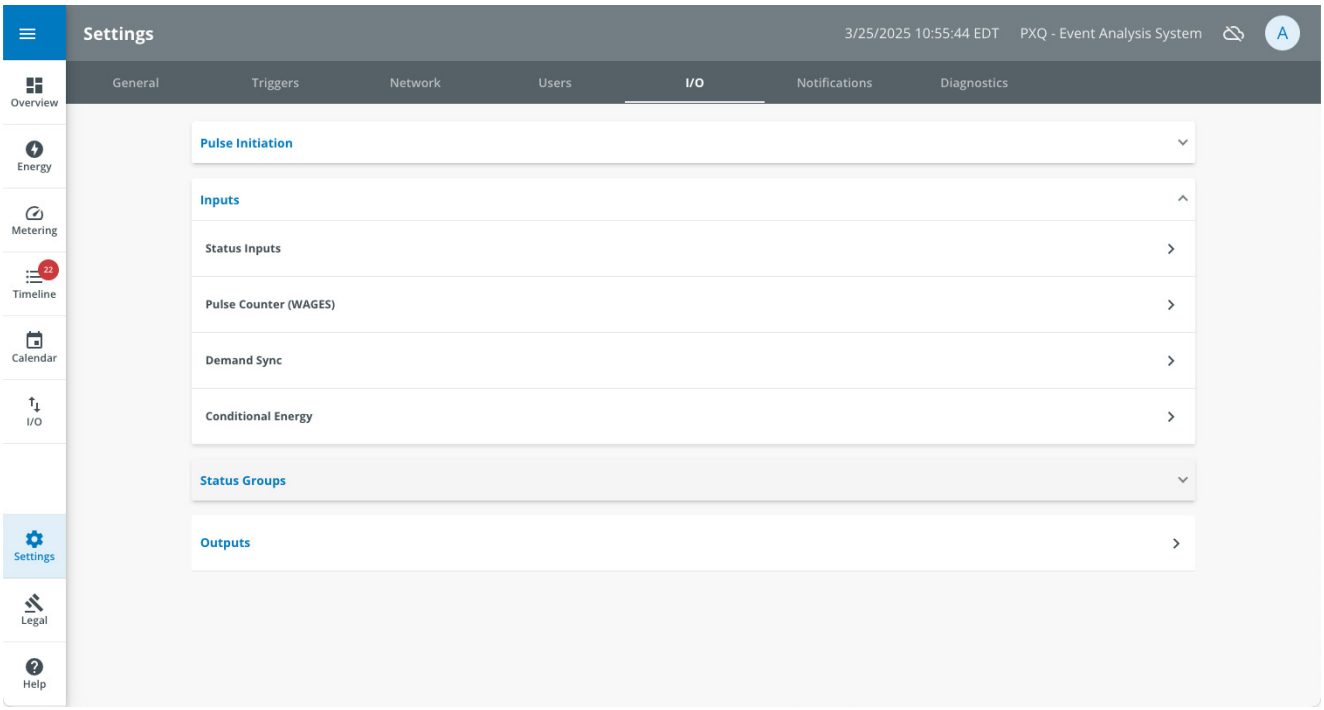
Digital Inputs can be configured as Status Input, Pulse Counter (WAGES), Demand Sync, or Conditional Energy.

Status Input Selection Types	Pulse Counter Selection types
Breakers (Open/Closed, Trip Status)	Water
Control Power	Air
UPS	Gas
SPD	Electric
Transformer Temperature Limit Switch	Steam
Alarm	Custom
Relay	
Transfer Switch	
Custom	

Demand Sync refers to an external signal, typically utility generated signal, connected to one (1) of the PXQ Digital Inputs and used by the PXQ for demand calculations. Any one (1) PXQ Digital Input can be designated as the Demand Sync.

Demand is the power consumption over a period of time (Fixed or Sliding) or in response to a designated Digital Input (Demand Sync). PXQ Demand Type supports Fixed (Total Interval = 1-60 minutes), Sliding (Total Interval = 1-60 minutes with Sub Interval = 1-30 minutes), and Demand Sync (dependent on designated Digital Input).

Conditional Energy or conditional peak demand refers to demand calculated based on a utility generated signal based on utility grid usage which is influenced by weather, season, time of day, and events. This utility signal is to be wired to the PXQ System Core Digital Input 8 (DI8) and when energized the PXQ will calculate the peak demand. Conditional Energy is displayed on the WebUI under Energy > Demand Comparison and Energy Comparison



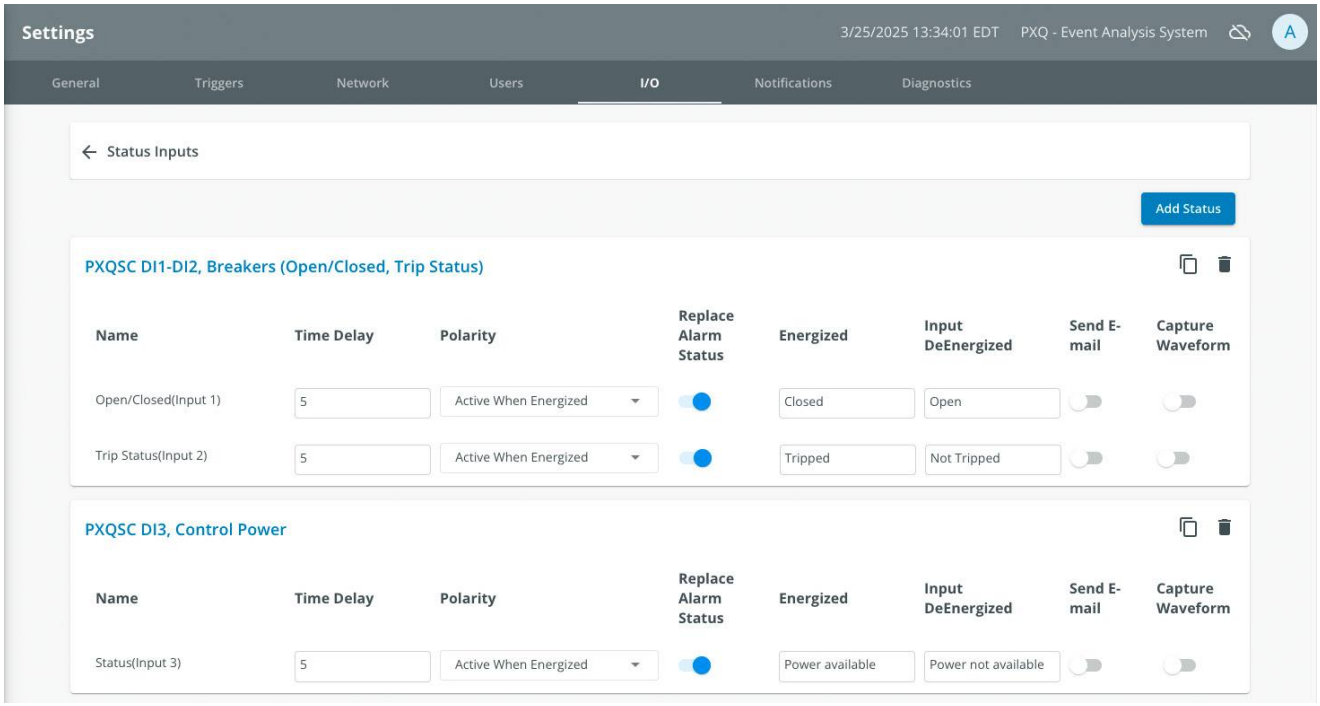
6.6.2.1 Status Inputs

In Settings > I/O > Inputs > Status Inputs and using Add Status, any of the PXQ System Core & PXQ-SER Digital Inputs can be configured for one of the following:

					Replace Alarm Status (default Enabled)		
Select	**Enter Custom Name[1]**	**Designate DI**	**Time Delay**	**Polarity**	**Energized[1]**	**Input DeEnergized[1]**	**Enable/Disable (default Disabled)**
Breakers (Open/ Closed, Trip Status	Custom Name[1]	Open/Closed DI	5-20 sec	**Active When Energized** Active When De-energized	**Closed**	**Open**	Send E-mail Capture Waveform
		Trip Status DI	5-20 sec	**Active When Energized** Active When De-energized	**Tripped**	**Not Tripped**	
Control Power	Custom Name[1]	Designate DI	5-20 sec	**Active When Energized** Active When De-energized	**Power Available**	**Power not Available**	Send E-mail Capture Waveform
UPS	Custom Name[1]	Designate DI	5-20 sec	**Active When Energized** Active When De-energized	**Normal**	**Bypass**	Send E-mail Capture Waveform
SPD	Custom Name[1]	Designate DI	5-20 sec	**Active When Energized** Active When De-energized	**Normal**	**Fault**	Send E-mail Capture Waveform
Transformer Temperature Limit Switch	Custom Name[1]	Warning Status DI	5-20 sec	**Active When Energized** Active When De-energized	**Normal**	**Warning**	Send E-mail Capture Waveform
		Critical Alarm DI	5-20 sec	**Active When Energized** Active When De-energized	**Normal**	**Critical**	
Alarm	Custom Name[1]	Designate DI	5-20 sec	**Active When Energized** Active When De-energized	**Normal**	**Alarm**	Send E-mail Capture Waveform
Relay	Custom Name[1]	Designate DI	5-20 sec	**Active When Energized** Active When De-energized	**Normal**	**Trip**	Send E-mail Capture Waveform

Transfer Switch	Custom Name[1]	Designate DI	5-20 sec	**Active When Energized** Active When De-energized	**Source 1**	**Source 2**	Send E-mail Capture Waveform
Custom	Custom Name[1]	Designate DI	5-20 sec	**Active When Energized** Active When De-energized	**Energized**	**DeEnergized**	Send E-mail Capture Waveform
	Custom Name[1]	Designate DI	5-20 sec	**Active When Energized** Active When De-energized	**Energized**	**DeEnergized**	Send E-mail Capture Waveform

[1]Custom Name, Energized, Input DeEnergized are 32-character maximum length
[2]Items in **Bold** represent default values



Replace Alarm Status is shown in Enabled position permitting user to change default verbiage for Energized and Input DeEnergized fields.

Once a Status Input has been created, the Name Field cannot be changed however all other fields can be changed.

The copy icon provides option to Copy Status Input.

The delete icon is used to delete a Status Input.

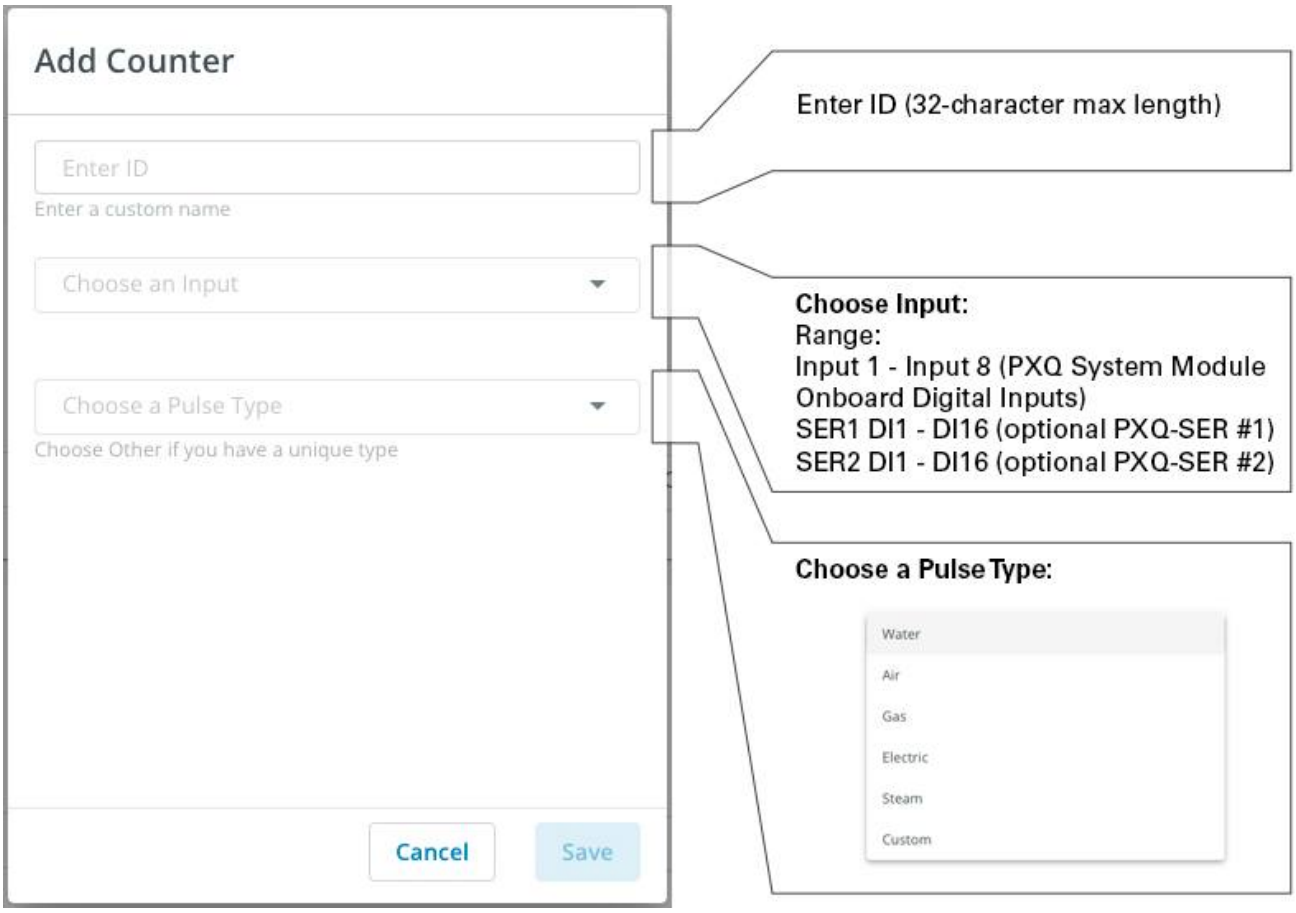
PXQ WebUI I/O > Status Inputs displays the status of all assigned Digital Inputs.

PXQ WebUI > I/O > Sequence of Events displays time stamp of Digital Input operations.

6.6.2.2 Pulse Counter (WAGES)

In Settings > I/O > Inputs > Pulse Counter (WAGES) and using Add Counter, any of PXQ System Core & PXQ-SER Digital Inputs can be designated as a Counter to monitor pulse meters, like those used in Water, Air, Gas, Electric, and Steam applications. This allows PXQ to be a central monitoring point for multiple meters.

A Pulse is counted when the Digital Input voltage is greater than 17.5Vdc (see Table 2.7 I/O Specification for details).



To define a Pulse Type, a Pulse Multiplier and Units must be defined.

The Pulse Multiplier determines the value of each pulse, e.g. if each pulse represents 10 gallons of water, the pulse multiplier would be 10, and the units would be Gallons.



This table displays the range and default settings for each chosen Pulse Type.

Choose Pulse Type	Pulse Multiplier Range: 1 - 1000	Unit 32 char max length	Cost Per Unit Range: 0.01 to 999,999,999,999.99	Delay Range: 5 – 20 milliseconds

Water	10	Gallons	$0.01	5
Air	1	cft	$0.01	5
Gas	1	cft	$0.001	5
Electric	1000	kWh	$0.1	5
Steam	1	lbs	$0.01	5
Custom	1		$0.01	5

Once a Pulse Counter is created, the Digital Input assignment cannot be changed however all other fields can be changed.

The delete icon is used to delete the Pulse Counter.

PXQ WebUI > I/O displays Pulse Counter (WAGES) data.

PXQ WebUI > I/O > Sequence of Events displays time stamp of Digital Input operations.

6.6.2.3 Demand Sync

In Settings > I/O > Inputs > Demand Sync, any one (1) PXQ Meter Module or PXQ-SER Digital Input can be designated as the Demand Sync.

Demand Sync refers to an external signal, typically utility generated signal, connected to one (1) of the PXQ Digital Inputs and used by the PXQ for demand calculations.

Demand is the power consumption over a period (Fixed or Sliding) or in response to a signal at a designated Status Input (Demand Sync). PXQ Demand Type supports Fixed (Total Interval = 1-60 minutes), Sliding (Total Interval = 1-60minutes with Sub Interval=1-30 minutes), and Demand Sync (dependent on designated Status Input).

There are two (2) settings for the Demand Sync:



The “Release Input” is used to remove/delete Demand Sync.

PXQ WebUI > Energy displays demand data.

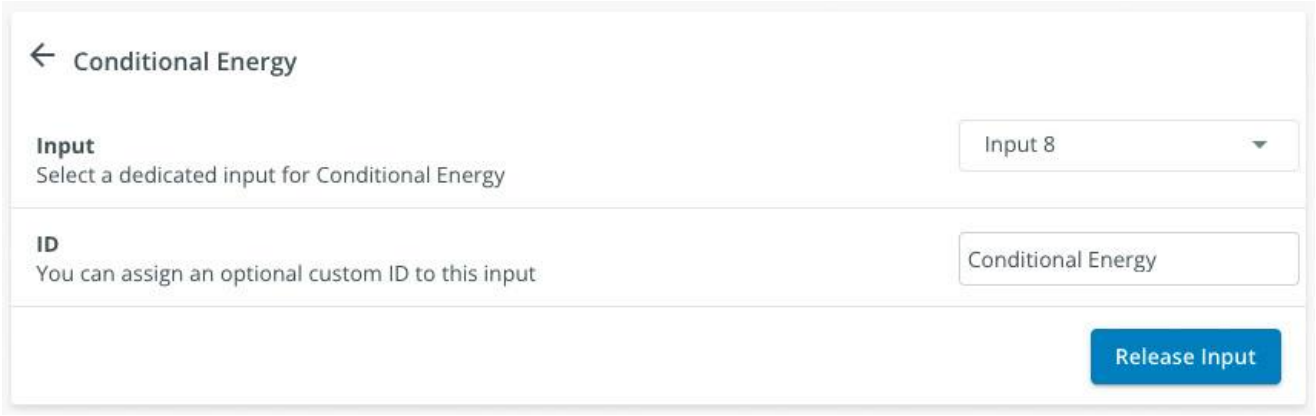
PXQ WebUI > I/O > Sequence of Events displays time stamp of Digital Input operations.

6.6.2.4 Conditional Energy

In Settings > I/O > Inputs > Conditional Energy, PXQ Meter Module Digital Input 8 can be configured for Conditional Energy.

Conditional Energy or conditional peak demand refers to demand calculated based on a utility generated signal based on utility grid usage which is influenced by weather, season, time of day, and events. This utility signal is to be wired to the PXQ System Core Digital Input 8 (DI8) and when energized the PXQ will calculate the peak demand.

There are two (2) settings for Conditional Energy:

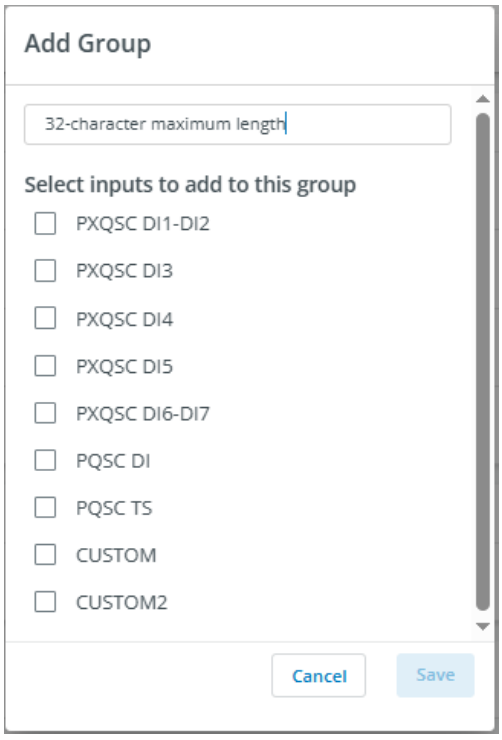


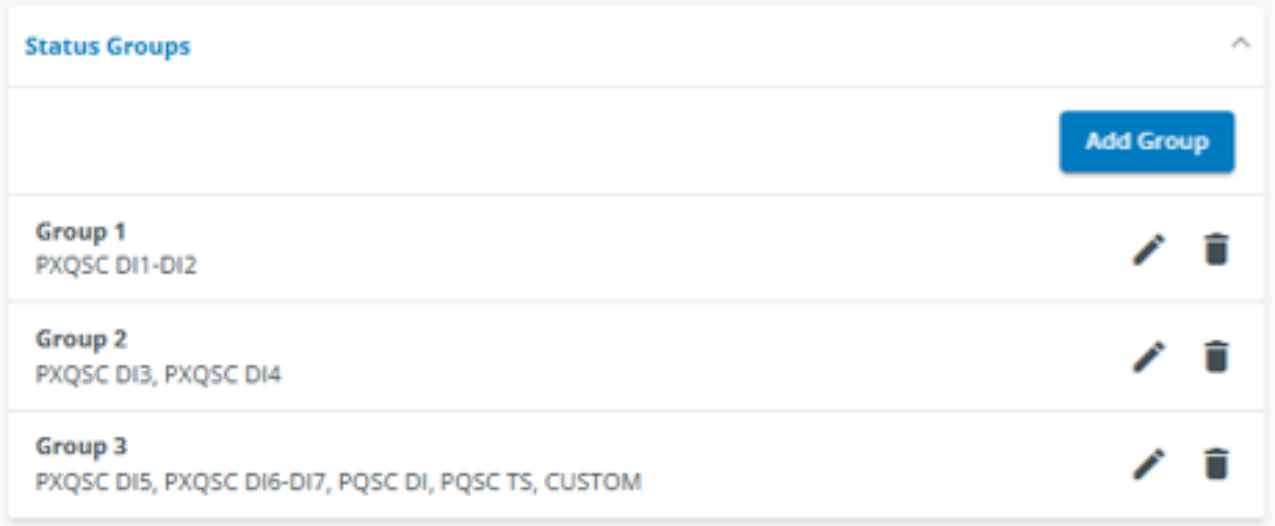
PXQ WebUI > Energy > Demand Comparison and Energy Comparison displays Conditional Energy data.

PXQ WebUI > I/O > Sequence of Events displays time stamp of Digital Input operations.

6.6.3 Status Groups

In Settings > I/O > Status Groups and using Add Group, Status Inputs can be grouped together for display purposes. Status Input can only be assigned to one (1) User Defined Group.

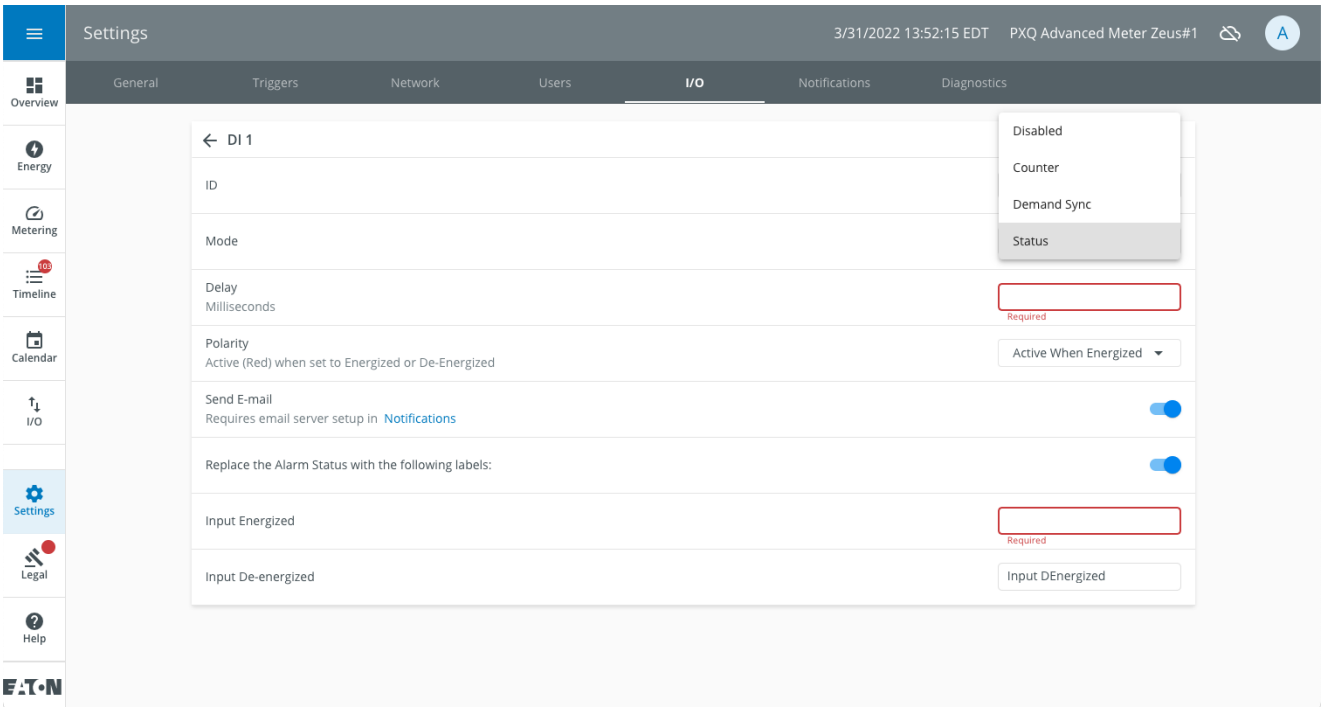




Once Groups are created, the edit button provides the option of removing Status Inputs from one Group and adding to another Group.

PXQ WebUI > I/O > Status Inputs > Sort by > User Defined Group displays User Defined Groups data.

6.6.1.3 Status

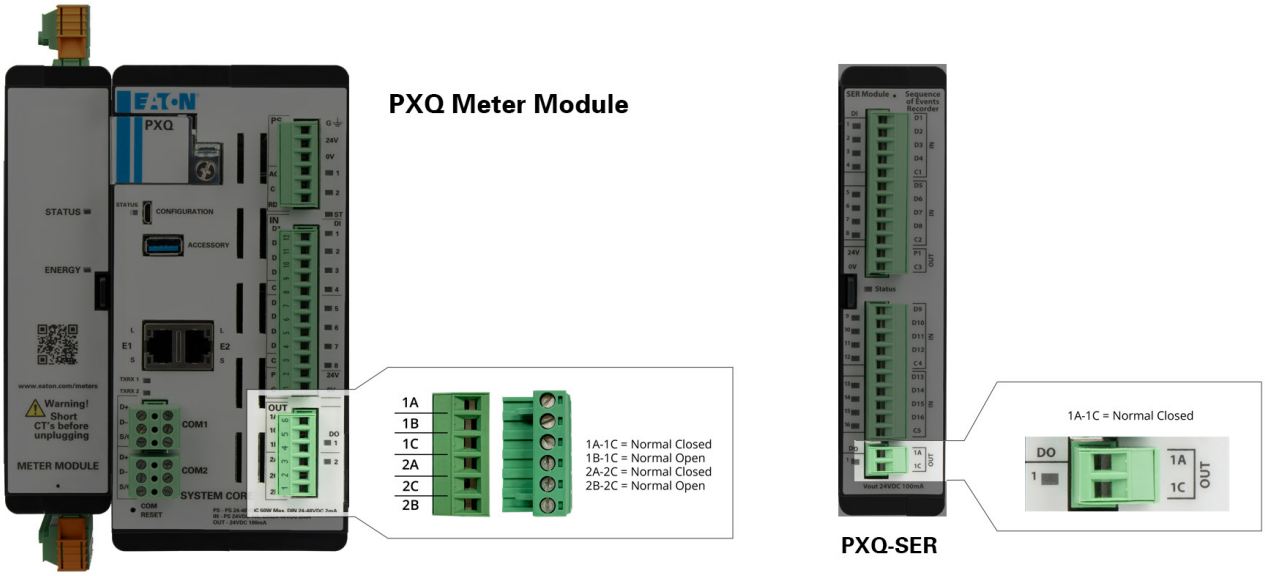


Any of the inputs can be configured as a status.

For a status input, the following parameters can be customized:

- Polarity:
  - Active when Energized
  - Active when De-Energized
- Send Email (Y/N)
- Replace the Alarm Status with custom Labels
- Customizable string for Input Energized
- Customizable string for Input De-Energized
- Preconfigured Input Types:
  - Breakers (Open/Closed, Trip Status)
  - Control Power
  - UPS
  - SPD
  - Transformer Temperature Limit Switch
  - Alarm
  - Relay
  - Transfer Switch
  - Custom

6.6.2 Outputs



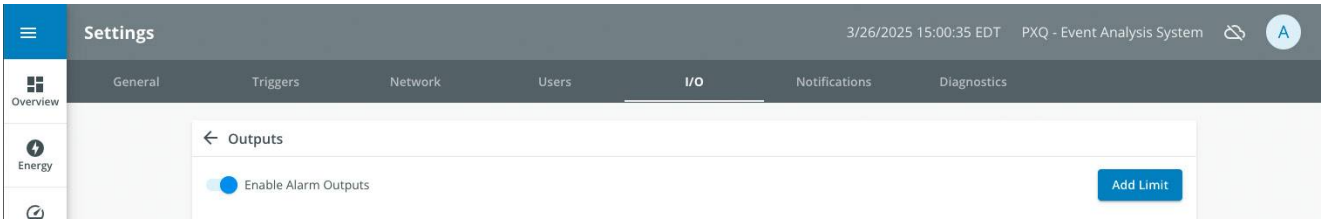
Relay Outputs are user configurable for Alarms (Out of Limit, Demand Overload, Sag, Swell, Subcycle Disturbance, SEMI-F47 and Fast Transient) or manual control (via Modbus commands).

Two Limits can be combined with the AND function (requiring both limits to be TRUE) to initiate the output. Each limit has Hold Time. This ensures the output is energized for a minimum duration. This is useful to signal a control action.

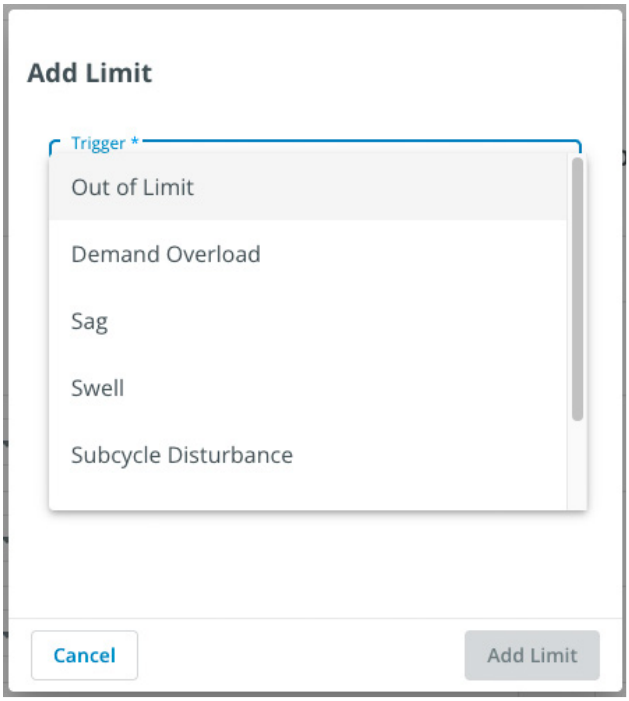
By default, the outputs are set for alarm enunciation. Once triggered by the assigned alarm condition, the form C contact will remain energized while the alarm condition is present (or if PXQ reboots or loses control power). Following a power-up or re-boot, trigger conditions will be re-evaluated, and the output will reenergize if the alarm condition is still present. This can be used as a signal input to another system that is tracking the status of an alarm condition.

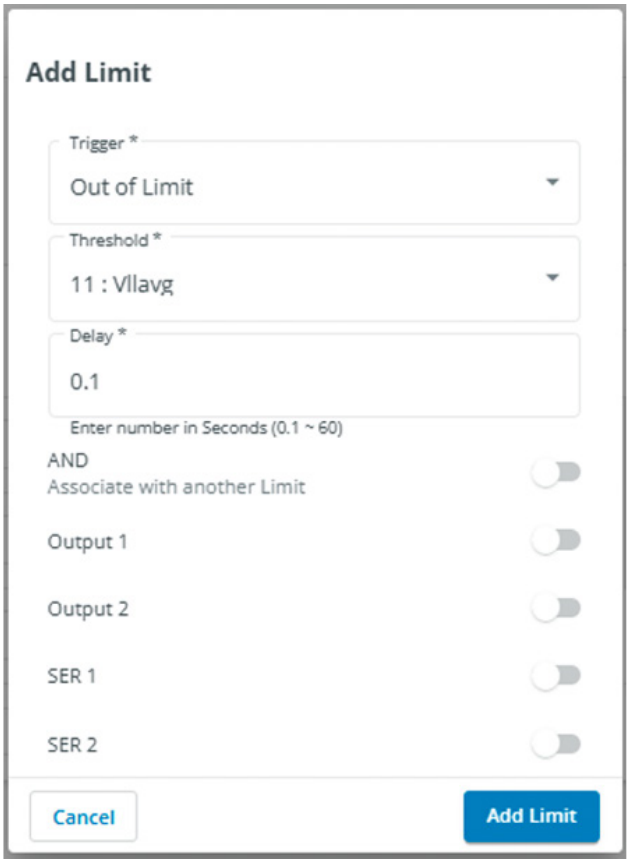
Outputs can be configured for Latched operation. When an output is set for Latched operation, it behaves as an electrically latched contact once an alarm trigger is detected. The output will remain energized until it is manually reset, or the PXQ is re-booted or loses control power. Latched mode operation is useful to maintain the annunciation of a temporary alarm condi¬tion that needs to be acknowledged by an operator.

Outputs may be configured for manual control. When Manual Control is enabled, the toggle turns blue.



In Settings > I/O > Outputs using Add Limit, Triggers (Out of Limit, Demand Overload, Sag, Swell, Subcycle Disturbance,SEMI-F47, Fast Transient) can be assigned to PXQ DO1, DO2, SER DO1.

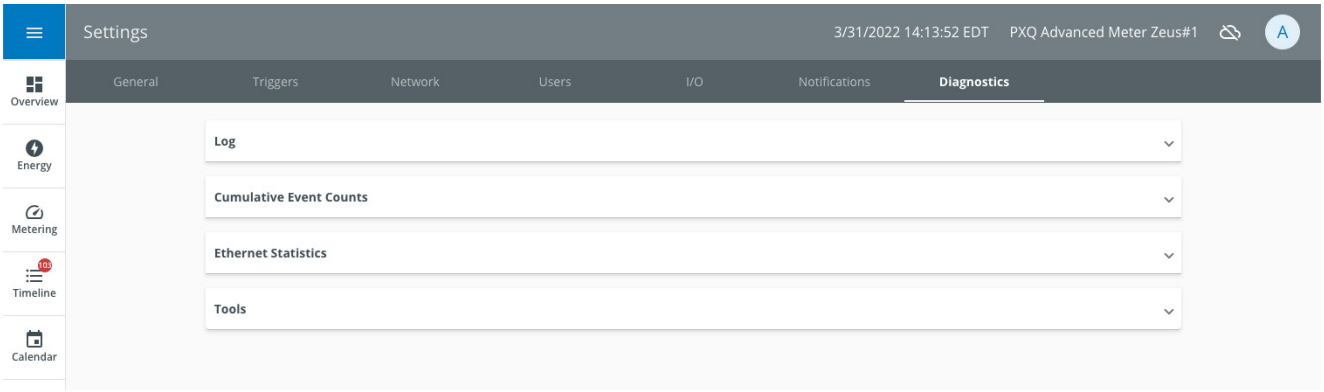




Fast Transient is only available when the PXQ-ENH is installed.

Before assigning Triggers to any of the outputs, trigger settings must be set using Settings > Triggers, see Section 6.2.

6.7 Diagnostics



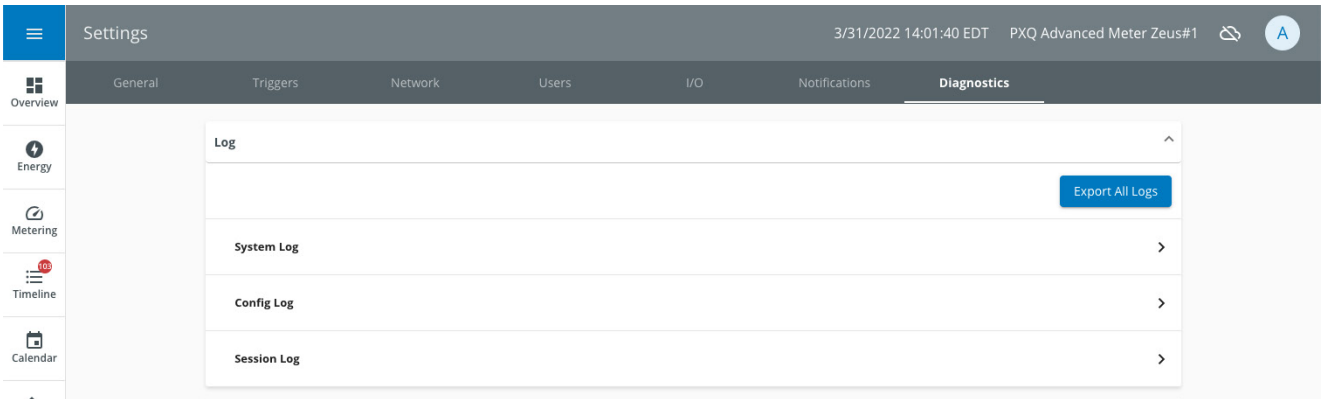
6.7.1 Logs

Logs are provided for product troubleshooting purposes. Each log may be exported as CSV (individually or all at once). Each Log may be reset.

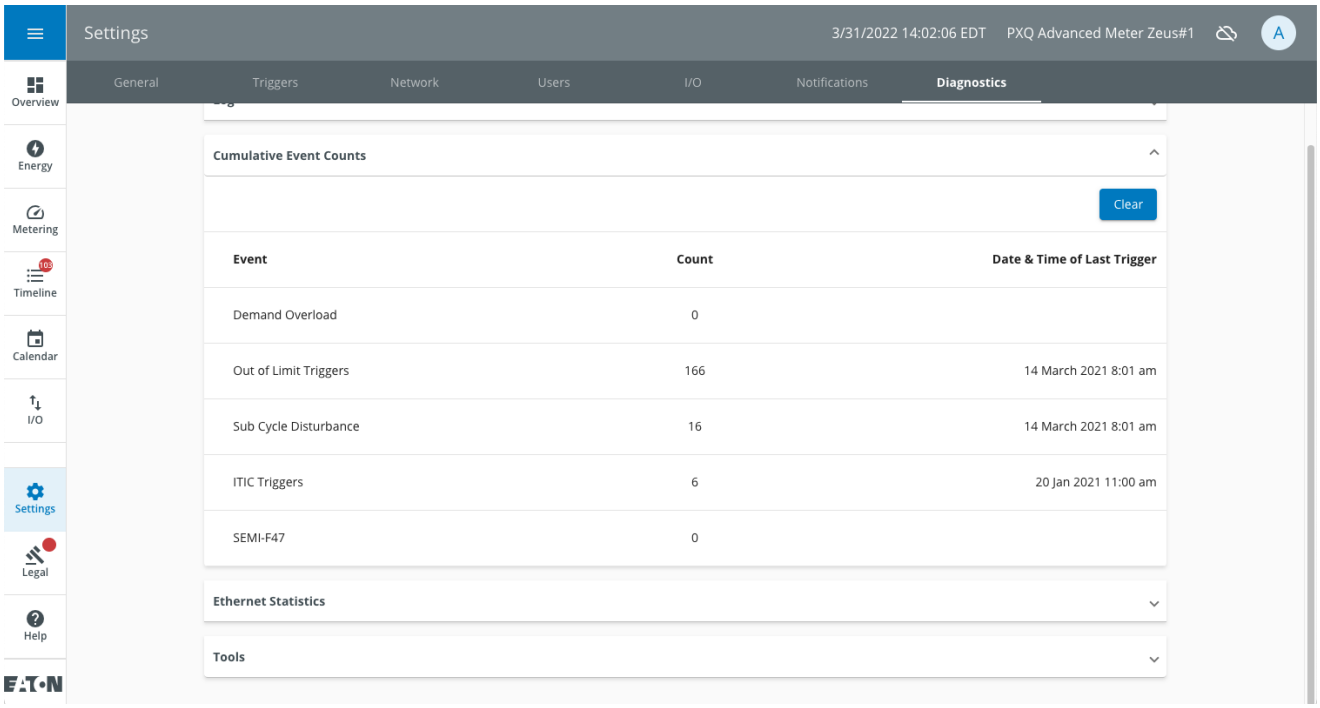
System Log
The System Log is used primarily for troubleshooting with technical support. It lists a history of various processes internal to PXQ including device access records.

Config Log
The Config Log is used primarily for troubleshooting with technical support. It lists a history of changes made to the PXQ by process.

Session Log
The Session Log provides a history of access requests by user and IP address.

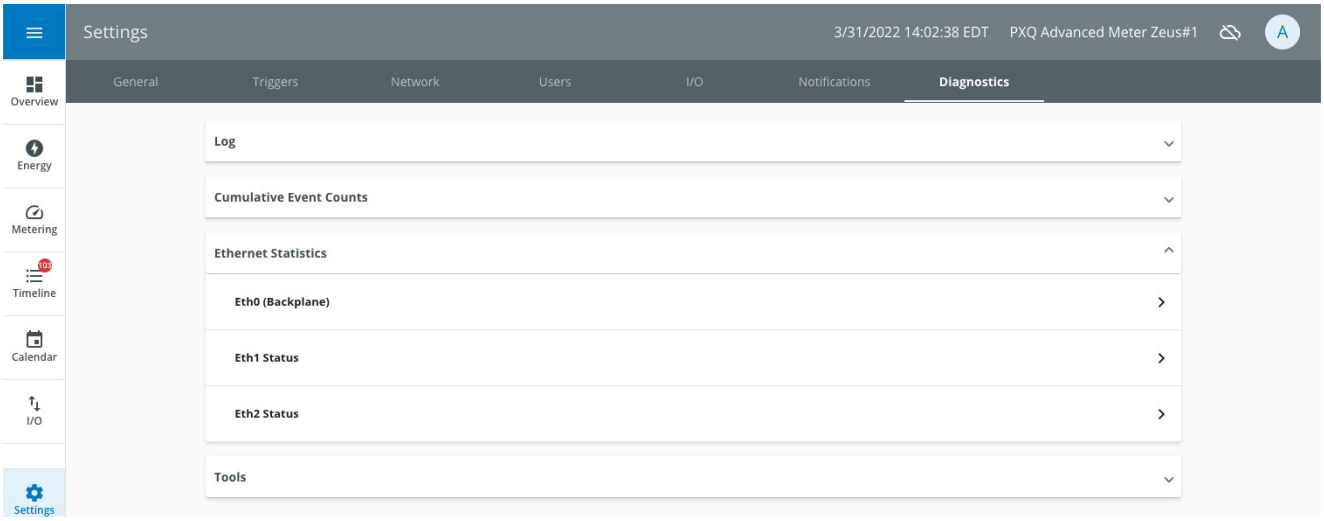


6.7.2 Cumulative Event Counts



The total number of events (since last reset) and date of most recent event is summarized by event type. These counts may be cleared, resetting each total to 0.

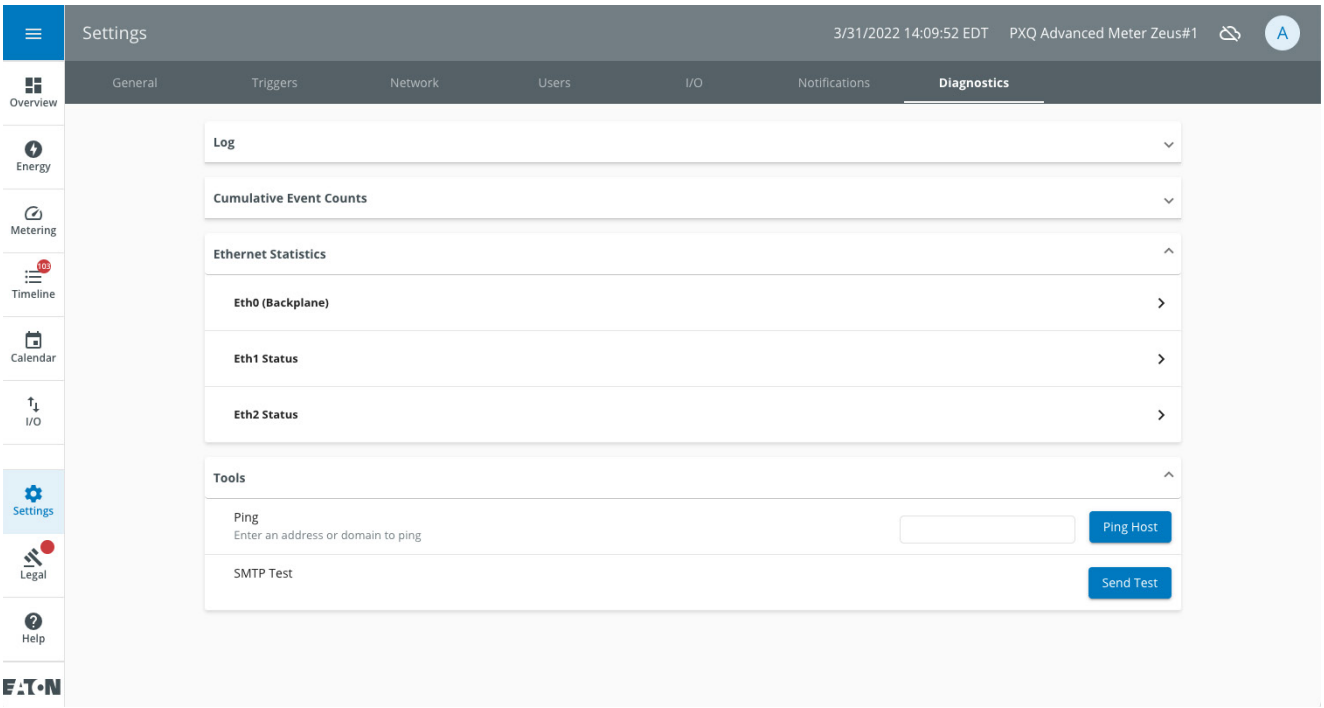
6.7.3 Ethernet Statistics



Eth0 Backplane
This section displays details about the PXQ's internal ethernet backplane that may be useful when troubleshooting the device.

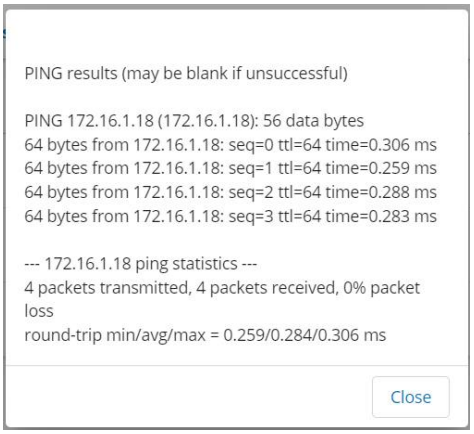
Eth1 & Eth2 Status
These sections provide communication statistics on communications transmissions and receipts. This information may be used when troubleshooting device communications during integration into a third-party software.

6.7.4 Tools



Ping
Located under Settings > Diagnostics > Tools.

This functionality can be used when troubleshooting integration to an upstream host or server. Enter the target IP address and select Ping host. Results may be blank if the ping is unsuccessful.

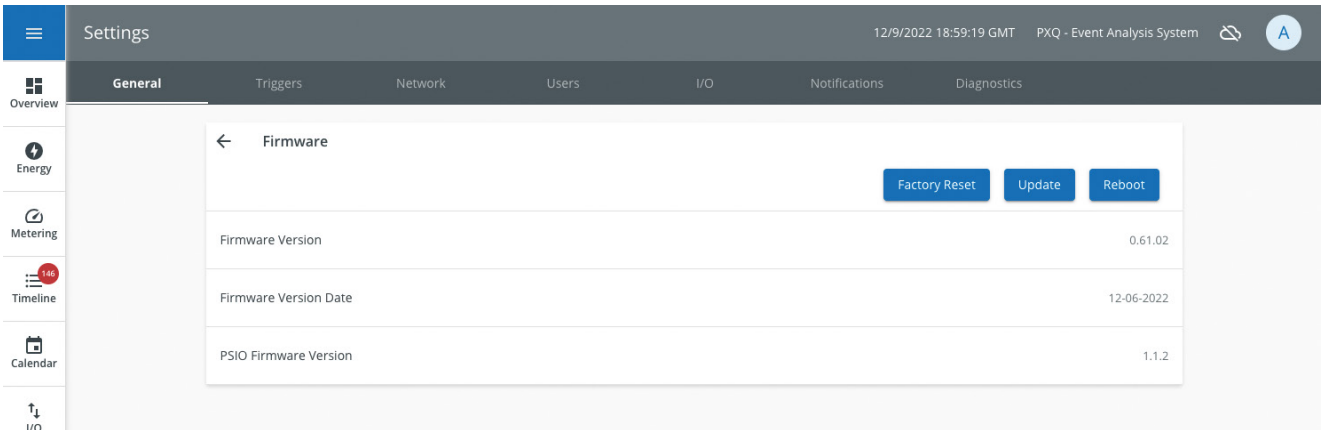


SMTP Test
The SMTP test can be used to confirm correct setup of the SMTP server. After configuring the SMTP server, click Send Test to send a test email.

7. Troubleshooting & Maintenance

Firmware

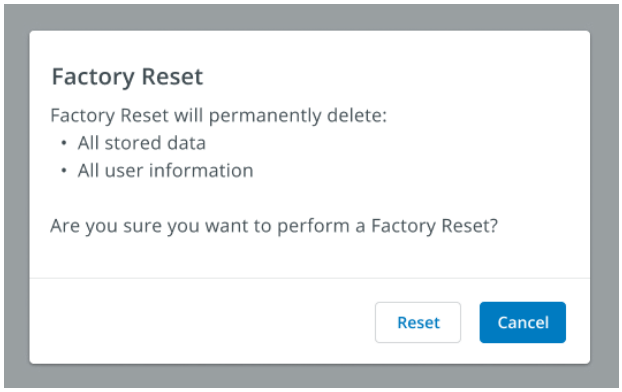
Navigate to **Settings > General > System > Firmware** to view details regarding your current version, as well as options for Factory Reset, Update and Reboot which are detailed in the following sections.



7.1 Factory Reset

A user with **admin** permissions may perform a factory reset from the web-user interface.

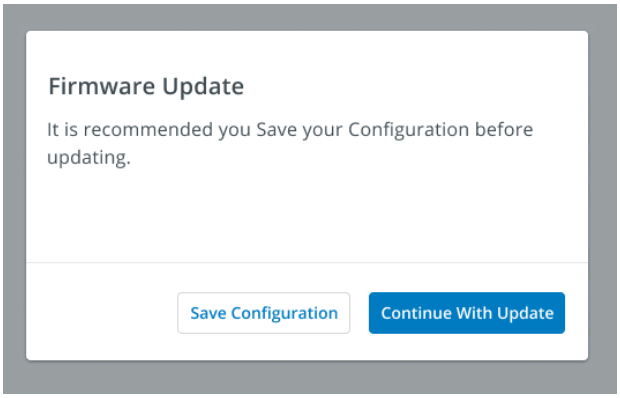
To reset the PXQ to factory defaults, select **Factory Reset.** You will see a prompt explaining that this action deletes all stored data and user information. To confirm and continue, select **Reset** from the prompt.



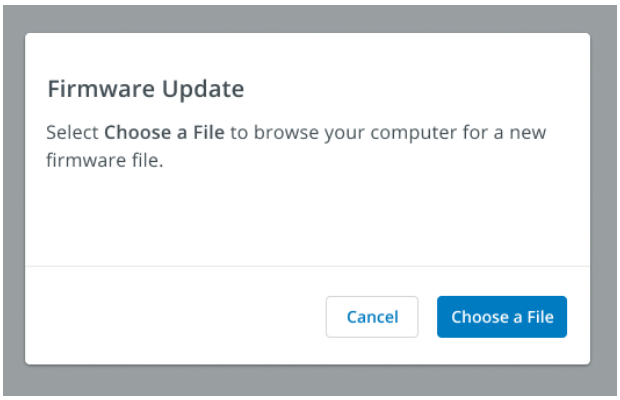
7.2 Update

Locate and download the firmware on the www.eaton.com/pxq website. Save the firmware update zip file to your PC.

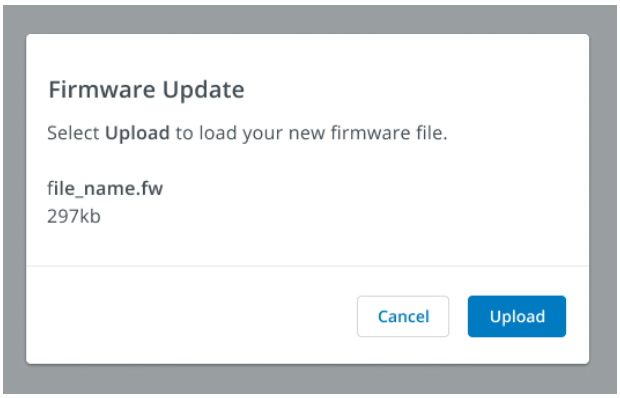
Next, select the **Update** button which will display a prompt suggesting you first save your current configuration from the Quick Setup section. If you have already done this, you can select **Continue wIth Update**.



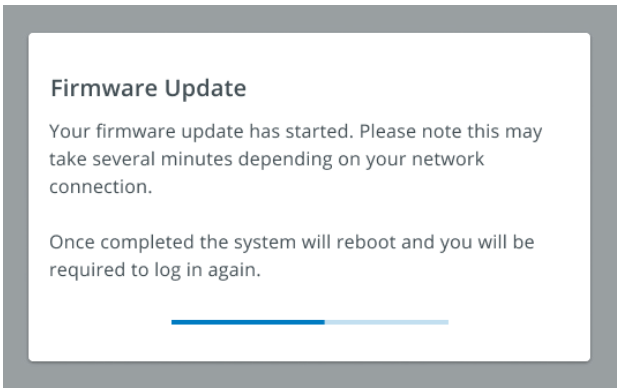
Next, select **Choose a file** to use your PC browser to find and select the file you downloaded.



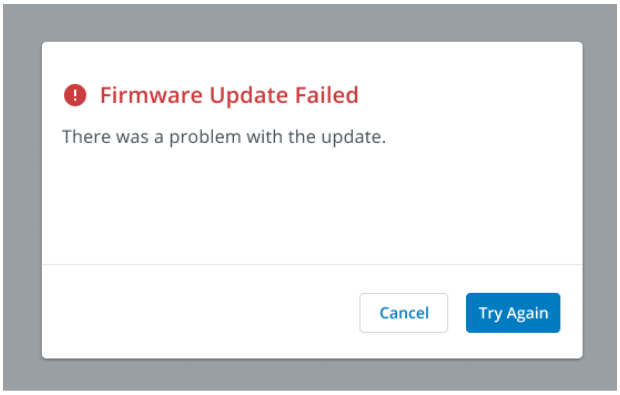
When you have selected the file, the prompt should display the name and file size. Click **Upload** to continue.

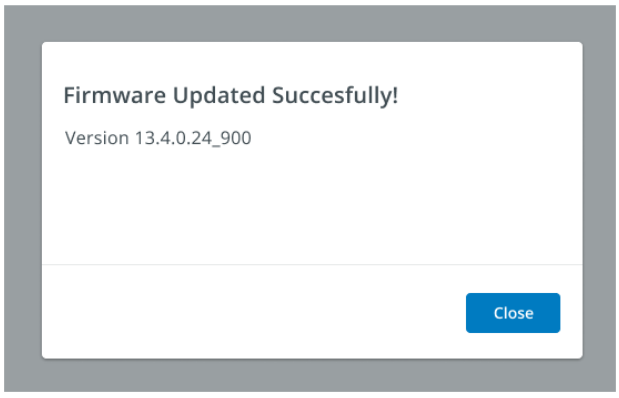


Once the upload has started the prompt will display a message to all users logged in. This alert will allow them the opportunity to finish work before the system will reboot.



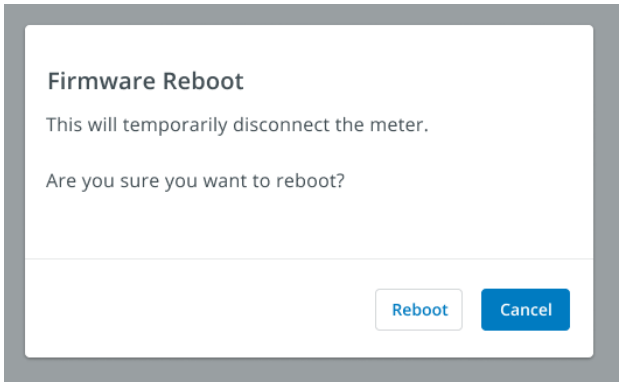
After the system has rebooted, you will need to login again. Upon login you will get notification if your update failed or was successful.





7.3 Reboot

To reboot your PXQ, select **Reboot**. This will display a prompt with instructions that the PXQ will need to temporarily disconnect. To confirm select **Reboot** from the prompt.



7.4 Com Reset

A recessed button labeled “COM RESET” provides a means of local access recovery during troubleshooting.

	Less than 5 sec	5-15 sec	15+ sec
	None - prevents unintentional reset by ignoring button engagement that is less than 5 seconds.	**Access recovery** - restores user accounts to as-shipped defaults, but retains data.	**Factory reset** - restores meter to as-shipped defaults. Data, settings, users, etc. are reset.
“Admin” account password reset to “admin”		•	•
Other users deleted		•	•
All Settings reset (including Network settings for Eth 1 & 2)			•
Data deleted			•

7.5 Maintenance and Care

PXQ is permanently calibrated at the factory. Further calibration in the field is not necessary. Calibration can be verified via the Energy LED test pulse.

The PXQ is designed to be a self-contained and maintenance-free unit. It is intended for service by factory trained personnel only. The PXQ should be stored in an environment that does not exceed the storage temperature range of -40° to 85° C. The environment should also be free of excess humidity. The device should be stored in its original packing material and container if not installed.

After installation, Remove the PXQ from line power when cleaning. Clean the unit with a clean dry cloth only. Do not use water or any liquid cleaning product.

7.6 Battery

The PXQ uses a CR1632 coin cell battery as a backup to maintain the time when the unit is not powered.

While battery replacement is not expected or required as part of regular maintenance and care, battery replacement may be desirable after leaving PXQ powered down for extended periods of time.

To replace the battery.

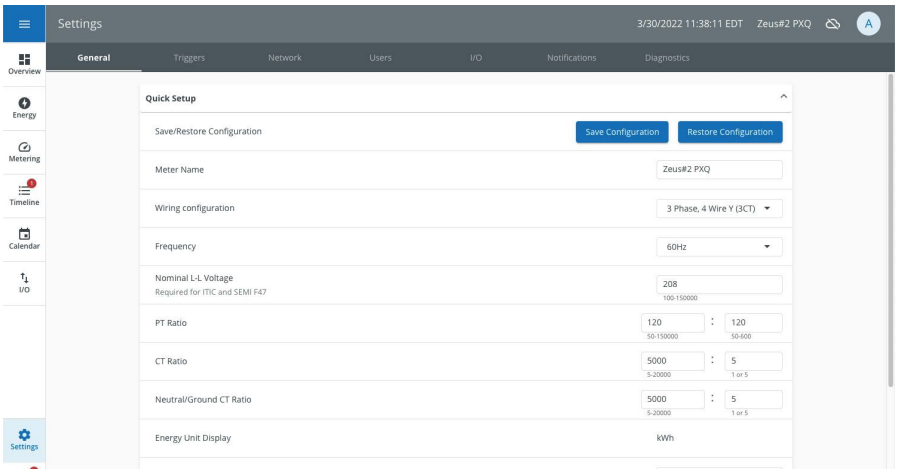
1. Unscrew the battery panel. Slide the panel to the left.
2. Do not turn. This is not a hinge.
3. Pull battery tray out to replace.

7.7 LED Status

The following statuses are presented for reference:

Module	LED	Behavior	Color	Status
System Core	Status	Blinking (1 sec On/ 1 sec Off)	Green	Normal
		Blinking	Red	Requires reboot
	ST	Blinking (1 sec On/ 1 sec Off)	Green	Normal
		Blinking (1 sec On/ 1 sec Off)	Red	Internal clock sync
Meter Module	Status	Solid	Green	Normal
Enhanced Capture Module	Status	Solid	Green	Initial power up
		Blinking (1 sec On/ 1 sec Off)	Green	Normal
		Blinking	Red	Unit booting
		Solid	Red	Error detected
System Core or Sequence of Events Recorder Module	One LED for each DI# or DO#	Solid	Green	Power initially applied
		Blinking (1 sec On/ 1 sec Off)	Green	Normal
		Fast blinking (200ms on / 200ms off)	Red	Excess temperatures warning or chatter detected
		Solid	Red	Power signal not received (Unit control power input may be low).
		Blinking (1 sec on / 1 sec off)	Red	SNTP not synchronized. Consider disabling SNTP in UI
		Slow blinking (3 sec on / 3 sec off)	Red	Not configured

7.8 Save and Restore Configurations



When the Quick Setup dropdown is expanded, configuration files can be downloaded to your local file system and then uploaded from your local file system to a PXQ. This is a handy convenience feature, especially if you're setting up multiple devices with either the same or similar settings. You can set up one PXQ, download the configuration file to your local computer, and then upload that same configuration file to other PXQs.

Note: The configuration file is actually a ZIP file that contains multiple files, each with configuration settings for a different aspect of the setup.

To download the current configuration file:

1. Expand the Quick Setup section.
2. Select Save Configuration.
3. Give the file a name (the dialog will fill in the .zip) extension.
4. Select Save.

To upload a configuration file to the PXQ:

1. Expand the Quick Setup section.
2. Select Restore Configuration.
3. Select a .zip file.
4. Click OK.

8. Communications

8.1 Modbus TCP

The following information pertains to all registers in this mapping:

- All registers are 1-based.
- Registers are presented with the most significant bit first (big-endian).
- Unless otherwise specified, all data is based on 200ms data intervals

DATE Format

All timestamp data is presented in a 3-register DATE format. The data is interpreted as hexadecimal, with 1 byte representing each parameter (2 per register):

- The first register is year and month
- The second register is day and hour
- The last register is for minutes and seconds

YYMM	DDHH	mmss

Here is an example of how to interpret the data:

The following registers represent: July 24, 2023, 3:50pm

YYMM	DDHH	mmss
1707h	180Fh	3200h

Step 1: Parse the hexadecimal registers and split into their respective bytes.

YY (year)	MM (month)	DD (day)	hh (hour)	mm (minutes)	ss (seconds)
17	07	18	0F	32	00

Step 2: Convert each byte to decimal, treating each parameter individually.

Year	Month	Day	Hour	Minutes	Seconds
17	07	18	0F	32	00

Step 3: Interpret as a timestamp

YY-MM-DD, hh:mm:ss > 23-07-24, 15:50:00 > July 24, 2023, 3:50pm

Device Information

These registers describe the device details of a specific PXQ. Details on the items marked with astrisk available at the end.

Description	Modbus TCP Register (1-based)	Register Count	Type ID
family	30000	32	STRING
model_series	30032	32	STRING
model	30064	32	STRING
name	30096	32	STRING
canonical_model_name	30128	32	STRING
network_address	30160	32	STRING
serial	30192	32	STRING
vendor	30224	32	STRING
asset_id	30256	32	STRING
location	30288	32	STRING
contact	30320	32	STRING
hardware_version	30352	32	STRING
software_version	30384	32	STRING
symbol_canonical_name	30416	32	STRING
Diagnosticuint32t	14	2	UINT32
Diagnosticfloat32t	26	2	FLOAT
ControlRegister	2901*	2	UINT32
DeviceName	4497	32	STRING
BASESubPartNumber1	4529	32	STRING
FWVersionStandard1	4567	32	STRING
MBProductID	4605*	2	UINT32
MBProductStatus	4607*	2	UINT32
MBCauseOfStatus	4609*	2	UINT32

PXM4000/6000/8000 compatibility registers

These registers are mapped to be the same as a PXM468K and supports retrofit applications.

Parameter	Units	Register Count	Type ID	Data Available in UI	Modbus TCP Register (1-based) (200ms data)
Ia	A	2	FLOAT	Trended	4611
Ib	A	2	FLOAT	Trended	4613
Ic	A	2	FLOAT	Trended	4615
Ig	A	2	FLOAT	Trended	4617
In	A	2	FLOAT	Trended	4619
Iavg	A	2	FLOAT	Trended	4621
Vab	V	2	FLOAT	Trended	4623
Vbc	V	2	FLOAT	Trended	4625
Vca	V	2	FLOAT	Trended	4627
VLL average	V	2	FLOAT	Trended	4629

Parameter	Units	Register Count	Type ID	Data Available in UI	Modbus TCP Register (1-based) (200ms data)
Van	V	2	FLOAT	Trended	4631
Vbn	V	2	FLOAT	Trended	4633
Vcn	V	2	FLOAT	Trended	4635
Vln average	V	2	FLOAT	Trended	4637
Vng	V	2	FLOAT	Trended	4639
Voltage unbalance	N/A	2	FLOAT	No	4641
Current unbalance	N/A	2	FLOAT	No	4643
Total real power	W	2	FLOAT	Trended	4651
Total reactive power	VAR	2	FLOAT	Trended	4653
Total apparent power	VA	2	FLOAT	Trended	4655
Frequency	Hz	2	FLOAT	Trended	4659
ITIC factor (system)	N/A	2	FLOAT	No	4661
K-factor (system)	N/A	2	FLOAT	Trended	4663
THD (system)	N/A	2	FLOAT	Trended	4665
Real Power, phase a	W	2	FLOAT	Trended	4667
Real Power, phase b	W	2	FLOAT	Trended	4669
Real Power, phase c	W	2	FLOAT	Trended	4671
Reactive power, phase a	VAR	2	FLOAT	Trended	4673
Reactive power, phase b	VAR	2	FLOAT	Trended	4675
Reactive power, phase c	VAR	2	FLOAT	Trended	4677
Apparent power, phase a	VA	2	FLOAT	Trended	4679
Apparent power, phase b	VA	2	FLOAT	Trended	4681
Apparent power, phase c	VA	2	FLOAT	Trended	4683
Displacement power factor, phase a	N/A	2	FLOAT	Trended	4685
Displacement power factor, phase b	N/A	2	FLOAT	Trended	4687
Displacement power factor, phase c	N/A	2	FLOAT	Trended	4689
Power factor, phase a	N/A	2	FLOAT	Trended	4691
Power factor, phase b	N/A	2	FLOAT	Trended	4693
Power factor, phase c	N/A	2	FLOAT	Trended	4695
Forward energy in kWh	kWh	2	UINT32	Trend w/ Comparison	6305
Reverse energy in kWh	kWh	2	UINT32	Trend w/ Comparison	6309
Total energy in kWh	kWh	2	UINT32	Trend w/ Comparison	6313
Forward apparent energy in kVAh	kVAh	2	UINT32	Trend w/ Comparison	6317
Reverse apparent energy in kVAh	kVAh	2	UINT32	Trend w/ Comparison	6321
Net apparent energy in kVAh	kVAh	2	INT32	Trend w/ Comparison	6325
Total reactive energy in kVARh	kVARh	2	UINT32	Trend w/ Comparison	6329

Basic electrical parameters and power quality

Trends for Voltages, Currents, Frequency, power factor available in Web UI under Metering > Meter. Comparisons for Energy available in Web UI under Energy > Energy Comparison. Trends for Power Qualiti parameters available in Web UI under Metering > Quality.

Parameter	Units	Register Count	Type ID	Data Available in UI	Modbus TCP Register (1-based) (200ms data)
System frequency	Hz	2	FLOAT	Trended	11004
Ig	A	2	FLOAT	Trended	11006
In	A	2	FLOAT	Trended	11008
Ia	A	2	FLOAT	Trended	11010
Ib	A	2	FLOAT	Trended	11012
Ic	A	2	FLOAT	Trended	11014
I average	A	2	FLOAT	Trended	11016
Vln average	V	2	FLOAT	Trended	11018
Vng	V	2	FLOAT	Trended	11020
Van	V	2	FLOAT	Trended	11022
Vab	V	2	FLOAT	Trended	11024
Vbn	V	2	FLOAT	Trended	11026
Vbc	V	2	FLOAT	Trended	11028
Vcn	V	2	FLOAT	Trended	11030
Vca	V	2	FLOAT	Trended	11032
VLL average	V	2	FLOAT	Trended	11034
Power factor, phase a	N/A	2	FLOAT	Trended	11036
Power factor, phase b	N/A	2	FLOAT	Trended	11038
Power factor, phase c	N/A	2	FLOAT	Trended	11040
Power factor, system	N/A	2	FLOAT	Trended	11042
Forward energy	kWh	2	UINT32	Trend w/ Comparison	11068
Reverse energy	kWh	2	UINT32	Trend w/ Comparison	11070
Net energy	kWh	2	INT32	Trend w/ Comparison	11072
Total energy	kWh	2	UINT32	Trend w/ Comparison	11074
Received reactive energy	kVARh	2	UINT32	Trend w/ Comparison	11076
Delivered reactive energy	kVARh	2	UINT32	Trend w/ Comparison	11078
Net reactive energy	kVARh	2	INT32	Trend w/ Comparison	11080
Total reactive energy	kVARh	2	UINT32	Trend w/ Comparison	11082
Forward apparent energy	kVAh	2	UINT32	Trend w/ Comparison	11084
Reverse apparent energy	kVAh	2	UINT32	Trend w/ Comparison	11086
Net apparent energy	kVAh	2	INT32	Trend w/ Comparison	11088
Total apparent energy	kVAh	2	UINT32	Trend w/ Comparison	11090
Crest factor, phase a	N/A	2	FLOAT	Trended	11092
Crest factor, phase b	N/A	2	FLOAT	Trended	11094
Crest factor, phase c	N/A	2	FLOAT	Trended	11096
Crest factor, system	N/A	2	FLOAT	Trended	11098
ITIC factor, phase a	N/A	2	FLOAT	No	11100

Parameter	Units	Register Count	Type ID	Data Available in UI	Modbus TCP Register (1-based) (200ms data)
ITIC factor, phase b	N/A	2	FLOAT	No	11102
ITIC factor, phase c	N/A	2	FLOAT	No	11104
ITIC factor, system	N/A	2	FLOAT	No	11106
Instantaneous flicker, Van	N/A	2	FLOAT	Trended	11108
Instantaneous flicker, Vbn	N/A	2	FLOAT	Trended	11110
Instantaneous flicker, Vcn	N/A	2	FLOAT	Trended	11112
Instantaneous flicker, Vab	N/A	2	FLOAT	Trended	11114
Instantaneous flicker, Vbc	N/A	2	FLOAT	Trended	11116
Instantaneous flicker, Vca	N/A	2	FLOAT	Trended	11118
Short term perceptibility / short term flicker (10 min), Van	N/A	2	FLOAT	Trended	11120
Short term perceptibility / short term flicker (10 min), Vbn	N/A	2	FLOAT	Trended	11122
Short term perceptibility / short term flicker (10 min), Vcn	N/A	2	FLOAT	Trended	11124
Short term perceptibility / short term flicker (10 min), Vab	N/A	2	FLOAT	Trended	11126
Short term perceptibility / short term flicker (10 min), Vbc	N/A	2	FLOAT	Trended	11128
Short term perceptibility / short term flicker (10 min), Vca	N/A	2	FLOAT	Trended	11130
K-factor, phase a	N/A	2	FLOAT	Trended	11132
K-factor, phase b	N/A	2	FLOAT	Trended	11134
K-factor, phase c	N/A	2	FLOAT	Trended	11136
K-factor, system	N/A	2	FLOAT	Trended	11138

Power and power factor

Three phase and system power (real, reactive, apparent), power factor. Data available in Web UI under Metering > Meter.

Parameter	Units	Register Count	Type ID	Data Available in UI	Modbus TCP Register (1-based) (200ms data)
Displacement power factor, phase a	N/A	2	FLOAT	Trended	11332
Displacement power factor, phase b	N/A	2	FLOAT	Trended	11334
Displacement power factor, phase c	N/A	2	FLOAT	Trended	11336
Displacement power factor, system	N/A	2	FLOAT	Trended	11338
Apparent power, phase a	VA	2	FLOAT	Trended	11340
Apparent power, phase b	VA	2	FLOAT	Trended	11342
Apparent power, phase c	VA	2	FLOAT	Trended	11344
Apparent power, system	VA	2	FLOAT	Trended	11346
Reactive power, phase a	VAR	2	FLOAT	Trended	11348
Reactive power, phase b	VAR	2	FLOAT	Trended	11350
Reactive power, phase c	VAR	2	FLOAT	Trended	11352
Reactive power, system	VAR	2	FLOAT	Trended	11354
Real power, phase a	W	2	FLOAT	Trended	11356

Parameter	Units	Register Count	Type ID	Data Available in UI	Modbus TCP Register (1-based) (200ms data)
Real power, phase b	W	2	FLOAT	Trended	11358
Real power, phase c	W	2	FLOAT	Trended	11360
Real power, system	W	2	FLOAT	Trended	11362

Min/Max/Average for power and power factor

Three phase and system power (real, reactive, apparent), power factor. Data Available in Web UI under Metering > Power.

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min since last reset	Max since last reset	Register Count	Type ID	Time Stamp Format	Timestamp of Min since last reset	Timestamp of Max since last reset
Real power, system	W	2	FLOAT	Displayed	11356	15030	17030	3	DATE	yymm|ddhh|mmss	15590	17590
Real power, phase a	W	2	FLOAT	Displayed	11358	15032	17032	3	DATE	yymm|ddhh|mmss	15596	17596
Real power, phase b	W	2	FLOAT	Displayed	11360	15034	17034	3	DATE	yymm|ddhh|mmss	15602	17602
Real power, phase c	W	2	FLOAT	Displayed	11362	15036	17036	3	DATE	yymm|ddhh|mmss	15608	17608
Reactive power, system	VAR	2	FLOAT	Displayed	11348	15038	17038	3	DATE	yymm|ddhh|mmss	15614	17614
Reactive power, phase a	VAR	2	FLOAT	Displayed	11350	15040	17040	3	DATE	yymm|ddhh|mmss	15620	17620
Reactive power, phase b	VAR	2	FLOAT	Displayed	11352	15042	17042	3	DATE	yymm|ddhh|mmss	15626	17626
Reactive power, phase c	VAR	2	FLOAT	Displayed	11354	15044	17044	3	DATE	yymm|ddhh|mmss	15632	17632
Apparent power, system	VA	2	FLOAT	Displayed	11340	15046	17046	3	DATE	yymm|ddhh|mmss	15638	17638
Apparent power, phase a	VA	2	FLOAT	Displayed	11342	15048	17048	3	DATE	yymm|ddhh|mmss	15644	17644
Apparent power, phase b	VA	2	FLOAT	Displayed	11344	15050	17050	3	DATE	yymm|ddhh|mmss	15650	17650
Apparent power, phase c	VA	2	FLOAT	Displayed	11346	15052	17052	3	DATE	yymm|ddhh|mmss	15656	17656
Apparent power factor, phase a	N/A	2	FLOAT	Displayed	11036	15054	17054	3	DATE	yymm|ddhh|mmss	15662	17662
Apparent power factor, phase b	N/A	2	FLOAT	Displayed	11038	15056	17056	3	DATE	yymm|ddhh|mmss	15668	17668
Apparent power factor, phase c	N/A	2	FLOAT	Displayed	11040	15058	17058	3	DATE	yymm|ddhh|mmss	15674	17674
Apparent power factor, system	N/A	2	FLOAT	Displayed	11042	15060	17060	3	DATE	yymm|ddhh|mmss	15680	17680
Displacement power factor, phase a	N/A	2	FLOAT	Displayed	11332	15062	17062	3	DATE	yymm|ddhh|mmss	15686	17686
Displacement power factor, phase b	N/A	2	FLOAT	Displayed	11334	15064	17064	3	DATE	yymm|ddhh|mmss	15692	17692
Displacement power factor, phase c	N/A	2	FLOAT	Displayed	11336	15066	17066	3	DATE	yymm|ddhh|mmss	15698	17698
Displacement power factor, system	N/A	2	FLOAT	Displayed	11338	15068	17068	3	DATE	yymm|ddhh|mmss	15704	17704

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min (last 5 minute interval)	Max (last 5 minute interval)	Avg (last 5 minute interval)	Min (last 15 minute interval)	Max (last 15 minute interval)	Avg (last 15 minute interval)
Van	V	2	FLOAT	Trended	11022	19006	19008	19010	21006	21008	21010
Vbn	V	2	FLOAT	Trended	11026	19012	19014	19016	21012	21014	21016

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min (last 5 minute interval)	Max (last 5 minute interval)	Avg (last 5 minute interval)	Min (last 15 minute interval)	Max (last 15 minute interval)	Avg (last 15 minute interval)
Vcn	V	2	FLOAT	Trended	11030	19018	19020	19022	21018	21020	21022
Vln average	V	2	FLOAT	Trended	11018	19024	19026	19028	21024	21026	21028
Vng	V	2	FLOAT	Trended	11020	19030	19032	19034	21030	21032	21034
Vab	V	2	FLOAT	Trended	11024	19036	19038	19040	21036	21038	21040
Vbc	V	2	FLOAT	Trended	11028	19042	19044	19046	21042	21044	21046
Vca	V	2	FLOAT	Trended	11032	19048	19050	19052	21048	21050	21052
Vll average	V	2	FLOAT	Trended	11034	19054	19056	19058	21054	21056	21058
Ia	A	2	FLOAT	Trended	11010	19060	19062	19064	21060	21062	21064
Ib	A	2	FLOAT	Trended	11012	19066	19068	19070	21066	21068	21070
Ic	A	2	FLOAT	Trended	11014	19072	19074	19076	21072	21074	21076
In	A	2	FLOAT	Trended	11008	19078	19080	19082	21078	21080	21082
Ig	A	2	FLOAT	Trended	11006	19084	19086	19088	21084	21086	21088
Frequency	Hz	2	FLOAT	Trended	11004	19090	19092	19094	21090	21092	21094

Min/Max/Average for power quality parameters

Flicker, instantaneous and short-term perceptibility, crest factor and K-factor. Data available in Web UI under Metering > Quality > Factor or Metering > Quality > Flicker.

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min since last reset	Max since last reset	Register Count	Type ID	Time Stamp Format	Timestamp of Min since last reset	Timestamp of Max since last reset
Flicker, instantaneous perceptibility, Van	N/A	2	FLOAT	Displayed	11108	15070	17070	3	DATE	yymm|ddhh|mmss	15710	17710
Flicker, instantaneous perceptibility, Vbn	N/A	2	FLOAT	Displayed	11110	15072	17072	3	DATE	yymm|ddhh|mmss	15716	17716
Flicker, instantaneous perceptibility, Vcn	N/A	2	FLOAT	Displayed	11112	15074	17074	3	DATE	yymm|ddhh|mmss	15722	17722
Flicker, instantaneous perceptibility, Vab	N/A	2	FLOAT	Displayed	11114	15076	17076	3	DATE	yymm|ddhh|mmss	15728	17728
Flicker, instantaneous perceptibility, Vbc	N/A	2	FLOAT	Displayed	11116	15078	17078	3	DATE	yymm|ddhh|mmss	15734	17734
Flicker, instantaneous perceptibility, Vca	N/A	2	FLOAT	Displayed	11118	15080	17080	3	DATE	yymm|ddhh|mmss	15740	17740
Flicker, short term (10 min) perceptibility, Van	N/A	2	FLOAT	Displayed	11120	15082	17082	3	DATE	yymm|ddhh|mmss	15746	17746
Flicker, short term (10 min) perceptibility, Vbn	N/A	2	FLOAT	Displayed	11122	15084	17084	3	DATE	yymm|ddhh|mmss	15752	17752
Flicker, short term (10 min) perceptibility, Vcn	N/A	2	FLOAT	Displayed	11124	15086	17086	3	DATE	yymm|ddhh|mmss	15758	17758
Flicker, short term (10 min) perceptibility, Vab	N/A	2	FLOAT	Displayed	11126	15088	17088	3	DATE	yymm|ddhh|mmss	15764	17764
Flicker, short term (10 min) perceptibility, Vbc	N/A	2	FLOAT	Displayed	11128	15090	17090	3	DATE	yymm|ddhh|mmss	15770	17770

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min since last reset	Max since last reset	Register Count	Type ID	Time Stamp Format	Timestamp of Min since last reset	Timestamp of Max since last reset
Flicker, short term (10 min) perceptibility, Vca	N/A	2	FLOAT	Displayed	11130	15092	17092	3	DATE	yymm|ddhh|mmss	15776	17776
Crest factor, phase a	N/A	2	FLOAT	Displayed	11092	15094	17094	3	DATE	yymm|ddhh|mmss	15782	17782
Crest factor, phase b	N/A	2	FLOAT	Displayed	11094	15096	17096	3	DATE	yymm|ddhh|mmss	15788	17788
Crest factor, phase c	N/A	2	FLOAT	Displayed	11096	15098	17098	3	DATE	yymm|ddhh|mmss	15794	17794
Crest factor, system	N/A	2	FLOAT	Displayed	11098	15100	17100	3	DATE	yymm|ddhh|mmss	15800	17800
K-factor, phase a	N/A	2	FLOAT	Displayed	11132	15102	17102	3	DATE	yymm|ddhh|mmss	15806	17806
K-factor, phase b	N/A	2	FLOAT	Displayed	11134	15104	17104	3	DATE	yymm|ddhh|mmss	15812	17812
K-factor, phase c	N/A	2	FLOAT	Displayed	11136	15106	17106	3	DATE	yymm|ddhh|mmss	15818	17818
K-factor, system	N/A	2	FLOAT	Displayed	11138	15108	17108	3	DATE	yymm|ddhh|mmss	15824	17824

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min (last 5 minute interval)	Max (last 5 minute interval)	Avg (last 5 minute interval)	Min (last 15 minute interval)	Max (last 15 minute interval)	Avg (last 15 minute interval)
Flicker, instantaneous perceptibility, Van	N/A	2	FLOAT	Trended	11108	19216	19218	19220	21216	21218	21220
Flicker, instantaneous perceptibility, Vbn	N/A	2	FLOAT	Trended	11110	19222	19224	19226	21222	21224	21226
Flicker, instantaneous perceptibility, Vcn	N/A	2	FLOAT	Trended	11112	19228	19230	19232	21228	21230	21232
Flicker, instantaneous perceptibility, Vab	N/A	2	FLOAT	Trended	11114	19234	19236	19238	21234	21236	21238
Flicker, instantaneous perceptibility, Vbc	N/A	2	FLOAT	Trended	11116	19240	19242	19244	21240	21242	21244
Flicker, instantaneous perceptibility, Vca	N/A	2	FLOAT	Trended	11118	19246	19248	19250	21246	21248	21250
Flicker, short term (10 min) perceptibility, Van	N/A	2	FLOAT	Trended	11120	19252	19254	19256	21252	21254	21256
Flicker, short term (10 min) perceptibility, Vbn	N/A	2	FLOAT	Trended	11122	19258	19260	19262	21258	21260	21262
Flicker, short term (10 min) perceptibility, Vcn	N/A	2	FLOAT	Trended	11124	19264	19266	19268	21264	21266	21268
Flicker, short term (10 min) perceptibility, Vab	N/A	2	FLOAT	Trended	11126	19270	19272	19274	21270	21272	21274
Flicker, short term (10 min) perceptibility, Vbc	N/A	2	FLOAT	Trended	11128	19276	19278	19280	21276	21278	21280
Flicker, short term (10 min) perceptibility, Vca	N/A	2	FLOAT	Trended	11130	19282	19284	19286	21282	21284	21286
Crest factor, phase a	N/A	2	FLOAT	Trended	11092	19288	19290	19292	21288	21290	21292

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min (last 5 minute interval)	Max (last 5 minute interval)	Avg (last 5 minute interval)	Min (last 15 minute interval)	Max (last 15 minute interval)	Avg (last 15 minute interval)
Crest factor, phase b	N/A	2	FLOAT	Trended	11094	19294	19296	19298	21294	21296	21298
Crest factor, phase c	N/A	2	FLOAT	Trended	11096	19300	19302	19304	21300	21302	21304
Crest factor, system	N/A	2	FLOAT	Trended	11098	19306	19308	19310	21306	21308	21310
K-factor, phase a	N/A	2	FLOAT	Trended	11132	19312	19314	19316	21312	21314	21316
K-factor, phase b	N/A	2	FLOAT	Trended	11134	19318	19320	19322	21318	21320	21322
K-factor, phase c	N/A	2	FLOAT	Trended	11136	19324	19326	19328	21324	21326	21328
K-factor, system	N/A	2	FLOAT	Trended	11138	19330	19332	19334	21330	21332	21334

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min (last 5 minute interval)	Max (last 5 minute interval)	Avg (last 5 minute interval)	Min (last 15 minute interval)	Max (last 15 minute interval)	Avg (last 15 minute interval)
Real power, system	W	2	FLOAT	Trended	11356	19096	19098	19100	21096	21098	21100
Real power, phase a	W	2	FLOAT	Trended	11358	19102	19104	19106	21102	21104	21106
Real power, phase b	W	2	FLOAT	Trended	11360	19108	19110	19112	21108	21110	21112
Real power, phase c	W	2	FLOAT	Trended	11362	19114	19116	19118	21114	21116	21118
Reactive power, system	VAR	2	FLOAT	Trended	11348	19120	19122	19124	21120	21122	21124
Reactive power, phase a	VAR	2	FLOAT	Trended	11350	19126	19128	19130	21126	21128	21130
Reactive power, phase b	VAR	2	FLOAT	Trended	11352	19132	19134	19136	21132	21134	21136
Reactive power, phase c	VAR	2	FLOAT	Trended	11354	19138	19140	19142	21138	21140	21142
Apparent power, system	VA	2	FLOAT	Trended	11340	19144	19146	19148	21144	21146	21148
Apparent power, phase a	VA	2	FLOAT	Trended	11342	19150	19152	19154	21150	21152	21154
Apparent power, phase b	VA	2	FLOAT	Trended	11344	19156	19158	19160	21156	21158	21160
Apparent power, phase c	VA	2	FLOAT	Trended	11346	19162	19164	19166	21162	21164	21166
Apparent power factor, phase a	N/A	2	FLOAT	Trended	11036	19168	19170	19172	21168	21170	21172
Apparent power factor, phase b	N/A	2	FLOAT	Trended	11038	19174	19176	19178	21174	21176	21178
Apparent power factor, phase c	N/A	2	FLOAT	Trended	11040	19180	19182	19184	21180	21182	21184
Apparent power factor, system	N/A	2	FLOAT	Trended	11042	19186	19188	19190	21186	21188	21190
Displacement power factor, phase a	N/A	2	FLOAT	Trended	11332	19192	19194	19196	21192	21194	21196
Displacement power factor, phase b	N/A	2	FLOAT	Trended	11334	19198	19200	19202	21198	21200	21202
Displacement power factor, phase c	N/A	2	FLOAT	Trended	11336	19204	19206	19208	21204	21206	21208
Displacement power factor, system	N/A	2	FLOAT	Trended	11338	19210	19212	19214	21210	21212	21214

Gapless Min/Max

These registers allow for min/max readings on a custom interval (determined by your poll-rate). Reading this register block (including 7000) resets the min/max values and starts a new interval. This result is a set of "gapless" min/max data, refreshed at your modbus poll-rate. These values are based on 200ms data.

Parameter	Units	Register Count	Type ID	Data Available in UI	Register (200ms data, updated at your poll rate)
Gapless Min/Max reset	N/A	2	UINT32	No	7000
Van, gapless min	V	2	FLOAT	No	7002
Vbn, gapless min	V	2	FLOAT	No	7004
Vcn, gapless min	V	2	FLOAT	No	7006
Vng, gapless min	V	2	FLOAT	No	7008
Vab, gapless min	V	2	FLOAT	No	7010
Vbc, gapless min	V	2	FLOAT	No	7012
Vca, gapless min	V	2	FLOAT	No	7014
Ia, gapless min	A	2	FLOAT	No	7016
Ib, gapless min	A	2	FLOAT	No	7018
Ic, gapless min	A	2	FLOAT	No	7020
In, gapless min	A	2	FLOAT	No	7022
Ig, gapless min	A	2	FLOAT	No	7024
Real power, system, gapless min	W	2	FLOAT	No	7026
Reactive power, system, gapless min	VAR	2	FLOAT	No	7028
Apparent power, system, gapless min	VA	2	FLOAT	No	7030
Van, gapless max	V	2	FLOAT	No	7032
Vbn, gapless max	V	2	FLOAT	No	7034
Vcn, gapless max	V	2	FLOAT	No	7036
Vng, gapless max	V	2	FLOAT	No	7038
Vab, gapless max	V	2	FLOAT	No	7040
Vbc, gapless max	V	2	FLOAT	No	7042
Vca, gapless max	V	2	FLOAT	No	7044
Ia, gapless max	A	2	FLOAT	No	7046
Ib, gapless max	A	2	FLOAT	No	7048
Ic, gapless max	A	2	FLOAT	No	7050
In, gapless max	A	2	FLOAT	No	7052
Ig, gapless max	A	2	FLOAT	No	7054
Real power, system, gapless min	W	2	FLOAT	No	7056
Reactive power, system, gapless min	VAR	2	FLOAT	No	7058

Parameter	Units	Register Count	Type ID	Data Available in UI	Register (200ms data, updated at your poll rate)
Apparent power, system, gapless min	VA	2	FLOAT	No	7060

Present Demands

These registers hold the present demand values for real/active power, reactive power and apparent power. There are no time stamps associated with this value, as these values reflect the current (present) time. Data displayed in Web UI under Energy > Demand Comparison.

Parameter	Units	Register Count	Type ID	Data Available in UI	Present Demand (200ms data)
Forward real power	W	2	FLOAT	Trend w/ Comparison	12000
Reverse real power	W	2	FLOAT	Trend w/ Comparison	12002
Net real power	W	2	FLOAT	Trend w/ Comparison	12004
Sum real power	W	2	FLOAT	Trend w/ Comparison	12006
Received reactive power	VAR	2	FLOAT	Trend w/ Comparison	12008
Delivered reactive power	VAR	2	FLOAT	Trend w/ Comparison	12010
Net reactive power	VAR	2	FLOAT	Trend w/ Comparison	12012
Sum reactive power	VAR	2	FLOAT	Trend w/ Comparison	12014
Forward apparent power	VA	2	FLOAT	Trend w/ Comparison	12016
Reverse apparent power	VA	2	FLOAT	Trend w/ Comparison	12018
Sum apparent power	VA	2	FLOAT	Trend w/ Comparison	12020
Average apparent power	VA	2	FLOAT	Trend w/ Comparison	12022

Peak Demands

*Conditional peak demand refers to demand calculated when Status Input 8 is configured for Conditional Energy and is energized. Data displayed in Web UI under Energy > Demand Comparison.

Parameter	Units	Register Count	Type ID	Data Available in UI	Peak Demand	Peak Demand so far this month	Peak Demand last month	Peak Demand since last reset	Peak Demand so far today	Conditional Peak Demand*
Forward real power	W	2	FLOAT	Displayed	12024	12120	12210	12306	12514	12402
Reverse real power	W	2	FLOAT	Displayed	12026	12122	12212	12308	12516	12404
Net real power	W	2	FLOAT	Displayed	12028	12124	12214	12310	12518	12406
Sum real power	W	2	FLOAT	Displayed	12030	12126	12216	12312	12520	12408
Received reactive power	VAR	2	FLOAT	Displayed	12032	12128	12218	12314	12522	12410
Delivered reactive power	VAR	2	FLOAT	Displayed	12034	12130	12220	12316	12524	12412
Net reactive power	VAR	2	FLOAT	Displayed	12036	12132	12222	12318	12526	12414
Sum reactive power	VAR	2	FLOAT	Displayed	12038	12134	12224	12320	12528	12416

Parameter	Units	Register Count	Type ID	Data Available in UI	Peak Demand	Peak Demand so far this month	Peak Demand last month	Peak Demand since last reset	Peak Demand so far today	Conditional Peak Demand*
Forward apparent power	VA	2	FLOAT	Displayed	12040	12136	12226	12322	12530	12418
Reverse apparent power	VA	2	FLOAT	Displayed	12042	12138	12228	12324	12532	12420
Sum apparent power	VA	2	FLOAT	Displayed	12044	12140	12230	12326	12534	12422
Average apparent power	VA	2	FLOAT	Displayed	12046	12142	12232	12328	12536	12424
In	A	2	FLOAT	Trended	11008	19078	19080	19082	21078	21080
Ig	A	2	FLOAT	Trended	11006	19084	19086	19088	21084	21086
Frequency	Hz	2	FLOAT	Trended	11004	19090	19092	19094	21090	21092

Parameter	Units	Register Count	Type ID	Data Available in UI	Peak Demand	Peak Demand so far this month	Peak Demand last month	Peak Demand since last reset	Peak Demand so far today	Conditional Peak Demand*
Forward real power	yymm|ddhh|mmss	3	DATE	Displayed	12048	12144	12234	12330	12562	12426
Reverse real power	yymm|ddhh|mmss	3	DATE	Displayed	12054	12150	12240	12336	12568	12432
Net real power	yymm|ddhh|mmss	3	DATE	Displayed	12060	12156	12246	12342	12574	12438
Sum real power	yymm|ddhh|mmss	3	DATE	Displayed	12066	12162	12252	12348	12580	12444
Received reactive power	yymm|ddhh|mmss	3	DATE	Displayed	12072	12168	12258	12354	12586	12450
Delivered reactive power	yymm|ddhh|mmss	3	DATE	Displayed	12078	N/A	12264	12360	12592	12456
Net reactive power	yymm|ddhh|mmss	3	DATE	Displayed	12084	12174	12270	12366	12598	12462
Sum reactive power	yymm|ddhh|mmss	3	DATE	Displayed	12090	12180	12276	12372	12604	12468
Forward apparent power	yymm|ddhh|mmss	3	DATE	Displayed	12096	12186	12282	12378	12610	12474
Reverse apparent power	yymm|ddhh|mmss	3	DATE	Displayed	12102	12192	12288	12384	12616	12480
Sum apparent power	yymm|ddhh|mmss	3	DATE	Displayed	12108	12198	12294	12390	12622	12486
Average apparent power	yymm|ddhh|mmss	3	DATE	Displayed	12114	12204	12300	12396	12628	12492
In	A	2	FLOAT	Trended	11008	19078	19080	19082	21078	21080
Ig	A	2	FLOAT	Trended	11006	19084	19086	19088	21084	21086
Frequency	Hz	2	FLOAT	Trended	11004	19090	19092	19094	21090	21092

Coincident Demands

*Conditional peak demand refers to demand calculated when Status Input 8 is configured for Conditional Energy and is energized. Data displayed in Web UI under Energy > Demand Comparison.

Parameter	Units	Register Count	Type ID	Data Available in UI	Coincident Forward Demand	Coincident Reverse Demand
Real power	W	2	FLOAT	No	12498	12506
Reverse real power	W	2	FLOAT	No	12500	12508
Reactive real power	VAR	2	FLOAT	No	12502	12510
Reverse reactive power	VAR	2	FLOAT	No	12504	12512
Coincident time stamp	yymm|ddhh|mmss	3	DATE	No	12048	12054

Digital inputs (basic info)

Applies to discrete digital inputs of all configurations. Reference sections below for additional data specific to each configuration. Data available in Web UI under I/O

Description	Register Count	Type ID	Data Available in UI	Modbus TCP Register (1-based)	Notes
Mode, DI1	2	UINT32	No	24000	
Mode, DI2	2	UINT32	No	24002	
Mode, DI3	2	UINT32	No	24004	Possible Values: 0:disabled, 1:counter, 2:status, 3:demand sync, 4:conditional energy; (Mode, DI8 ONLY)
Mode, DI4	2	UINT32	No	24006	
Mode, DI5	2	UINT32	No	24008	
Mode, DI6	2	UINT32	No	24010	
Mode, DI7	2	UINT32	No	24012	
Mode, DI8	2	UINT32	No	24014	
Label, DI1	32	STRING	Displayed	24104	
Label, DI2	32	STRING	Displayed	24136	
Label, DI3	32	STRING	Displayed	24168	
Label, DI4	32	STRING	Displayed	24200	User-defined name for DI
Label, DI5	32	STRING	Displayed	24232	
Label, DI6	32	STRING	Displayed	24264	
Label, DI7	32	STRING	Displayed	24296	
Label, DI8	32	STRING	Displayed	24328	
Status, DI1	2	UINT32	Displayed	24020	
Status, DI2	2	UINT32	Displayed	24022	
Status, DI3	2	UINT32	Displayed	24024	
Status, DI4	2	UINT32	Displayed	24026	Possible Values: 0:De-Energized, 1:Energized;
Status, DI5	2	UINT32	Displayed	24028	
Status, DI6	2	UINT32	Displayed	24030	
Status, DI7	2	UINT32	Displayed	24032	
Status, DI8	2	UINT32	Displayed	24034	

Digital inputs configured as pulse counters

Data pertaining to digital inputs configured as pulse counters. DI Mode: 1. Data available in Web UI under I/O

Description	Register Count	Type ID	Data Available in UI	Modbus TCP Register (1-based)	Notes
Label, DI1	32	STRING	Displayed	24104	
Label, DI2	32	STRING	Displayed	24136	
Label, DI3	32	STRING	Displayed	24168	
Label, DI4	32	STRING	Displayed	24200	User-defined name for DI
Label, DI5	32	STRING	Displayed	24232	
Label, DI6	32	STRING	Displayed	24264	
Label, DI7	32	STRING	Displayed	24296	
Label, DI8	32	STRING	Displayed	24328	
Raw count, DI1	2	UINT32	No	24040	
Raw count, DI2	2	UINT32	No	24042	
Raw count, DI3	2	UINT32	No	24044	
Raw count, DI4	2	UINT32	No	24046	Unscaled count since last reset
Raw count, DI5	2	UINT32	No	24048	
Raw count, DI6	2	UINT32	No	24050	
Raw count, DI7	2	UINT32	No	24052	
Raw count, DI8	2	UINT32	No	24054	
Multiplier, DI1	2	FLOAT	No	24056	
Multiplier, DI2	2	FLOAT	No	24058	
Multiplier, DI3	2	FLOAT	No	24060	
Multiplier, DI4	2	FLOAT	No	24062	User defined multiplier to convert unscaled count to WAGES
Multiplier, DI5	2	FLOAT	No	24064	
Multiplier, DI6	2	FLOAT	No	24066	
Multiplier, DI7	2	FLOAT	No	24068	
Multiplier, DI8	2	FLOAT	No	24070	
Scaled count, DI1	2	FLOAT	Trended	24072	
Scaled count, DI2	2	FLOAT	Trended	24074	
Scaled count, DI3	2	FLOAT	Trended	24076	
Scaled count, DI4	2	FLOAT	Trended	24078	Scaled count = Raw count x Multiplier
Scaled count, DI5	2	FLOAT	Trended	24080	
Scaled count, DI6	2	FLOAT	Trended	24082	
Scaled count, DI7	2	FLOAT	Trended	24084	
Scaled count, DI8	2	FLOAT	Trended	24086	

Description	Register Count	Type ID	Data Available in UI	Modbus TCP Register (1-based)	Notes
Units, DI1	32	STRING	Displayed	24872	
Units, DI2	32	STRING	Displayed	24904	
Units, DI3	32	STRING	Displayed	24936	
Units, DI4	32	STRING	Displayed	24968	User-defined units (show on Web Interface)
Units, DI5	32	STRING	Displayed	25000	
Units, DI6	32	STRING	Displayed	25032	
Units, DI7	32	STRING	Displayed	25064	
Units, DI8	32	STRING	Displayed	25096	

Digital inputs configured as status inputs

Data pertaining to digital inputs configured as pulse counters. DI Mode: 2. Data available in Web UI under I/O

Description	Register Count	Type ID	Data Available in UI	Modbus TCP Register (1-based)	Notes
Label, DI1	32	STRING	Displayed	24104	
Label, DI2	32	STRING	Displayed	24136	
Label, DI3	32	STRING	Displayed	24168	
Label, DI4	32	STRING	Displayed	24200	User-defined name for DI
Label, DI5	32	STRING	Displayed	24232	
Label, DI6	32	STRING	Displayed	24264	
Label, DI7	32	STRING	Displayed	24296	
Label, DI8	32	STRING	Displayed	24328	
Status, DI1	2	UINT32	Displayed	24020	
Status, DI2	2	UINT32	Displayed	24022	
Status, DI3	2	UINT32	Displayed	24024	
Status, DI4	2	UINT32	Displayed	24026	Possible Values: 0:De-Energized, 1:Energized;
Status, DI5	2	UINT32	Displayed	24028	
Status, DI6	2	UINT32	Displayed	24030	
Status, DI7	2	UINT32	Displayed	24032	
Status, DI8	2	UINT32	Displayed	24034	
Active label, DI1	32	STRING	Displayed	24360	
Active label, DI2	32	STRING	Displayed	24392	
Active label, DI3	32	STRING	Displayed	24424	
Active label, DI4	32	STRING	Displayed	24456	User defined label. Appears on onboard web-interface when input is Active
Active label, DI5	32	STRING	Displayed	24488	
Active label, DI6	32	STRING	Displayed	24520	
Active label, DI7	32	STRING	Displayed	24552	
Active label, DI8	32	STRING	Displayed	24584	

Description	Register Count	Type ID	Data Available in UI	Modbus TCP Register (1-based)	Notes
Inactive label, DI1	32	STRING	Displayed	24616	
Inactive label, DI2	32	STRING	Displayed	24648	
Inactive label, DI3	32	STRING	Displayed	24680	
Inactive label, DI4	32	STRING	Displayed	24712	User defined label. Appears on onboard web-interface when input is Inactive
Inactive label, DI5	32	STRING	Displayed	24744	
Inactive label, DI6	32	STRING	Displayed	24776	
Inactive label, DI7	32	STRING	Displayed	24808	
Inactive label, DI8	32	STRING	Displayed	24840	

Digital inputs configured for conditional energy

Shows energy accumulated in a given interval. Only Status Input 8 be configured as Conditional Energy input. Data available in Web UI under I/O.

Description	Register Count	Type ID	Data Available in UI	Modbus TCP Register (1-based)	Notes
Interval Accumulation, DI1	2	FLOAT	No	24088	
Interval Accumulation, DI2	2	FLOAT	No	24090	
Interval Accumulation, DI3	2	FLOAT	No	24092	
Interval Accumulation, DI4	2	FLOAT	No	24094	Accumulated total per minute. Visible only if configured as conditional energy
Interval Accumulation, DI5	2	FLOAT	No	24096	
Interval Accumulation, DI6	2	FLOAT	No	24098	
Interval Accumulation, DI7	2	FLOAT	No	24100	
Interval Accumulation, DI8	2	FLOAT	No	24102	

Digital outputs

Data available in Web UI under I/O.

Description	Register Count	Type ID	Data Available in UI	Modbus TCP Register (1-based)	Notes
Mode, DO1	2	UINT32	Displayed	24016	
Mode, DO2	2	UINT32	Displayed	24018	Possible values are: 0:normal, 1:latched;
Status, DO1	2	UINT32	Displayed	24036	
Status, DO2	2	UINT32	Displayed	24038	
Interval Accumulation, DI5	2	FLOAT	No	24096	
Interval Accumulation, DI6	2	FLOAT	No	24098	Possible values are: 0:de-energized (NC-closed/NO-open), 1:energized (NC-open/NO-closed);
Interval Accumulation, DI7	2	FLOAT	No	24100	
Interval Accumulation, DI8	2	FLOAT	No	24102	

Phasors

Phase angles for phasors. Note that Van is reference and designated as 0. Phasor data available in Web UI under Metering > Phasor

Description	Register Count	Type ID	Data Available in UI	Modbus TCP Register (1-based)
Phasor Angle for Ia	degrees	2	FLOAT	11386
Phasor Angle for Ib	degrees	2	FLOAT	11388
Phasor Angle for Ic	degrees	2	FLOAT	11390
Phasor Angle for Van	degrees	2	FLOAT	11392
Phasor Angle for Vbn	degrees	2	FLOAT	11394
Phasor Angle for Vcn	degrees	2	FLOAT	11396
Phasor Angle for Vab	degrees	2	FLOAT	11398
Phasor Angle for Vbc	degrees	2	FLOAT	11400
Phasor Angle for Vca	degrees	2	FLOAT	11402

Total Harmonics

Trends available in the Web UI under Metering > Quality.

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min since last reset	Max since last reset	Register Count	Type ID	Time Stamp Format	Timestamp of Min since last reset	Timestamp of Max since last reset
Ia, Total Harmonics	A	2	FLOAT	Displayed	11140	15112	17112	3	DATE	yymm|ddhh|mmss	15836	17836
Ib, Total Harmonics	A	2	FLOAT	Displayed	11142	15114	17114	3	DATE	yymm|ddhh|mmss	15842	17842
Ic, Total Harmonics	A	2	FLOAT	Displayed	11144	15116	17116	3	DATE	yymm|ddhh|mmss	15848	17848
Vab, Total Harmonics	V	2	FLOAT	Displayed	11146	15118	17118	3	DATE	yymm|ddhh|mmss	15854	17854
Vbc, Total Harmonics	V	2	FLOAT	Displayed	11148	15120	17120	3	DATE	yymm|ddhh|mmss	15860	17860
Vca, Total Harmonics	V	2	FLOAT	Displayed	11150	15122	17122	3	DATE	yymm|ddhh|mmss	15866	17866
Van, Total Harmonics	V	2	FLOAT	Displayed	11152	15124	17124	3	DATE	yymm|ddhh|mmss	15872	17872
Vbn, Total Harmonics	V	2	FLOAT	Displayed	11154	15126	17126	3	DATE	yymm|ddhh|mmss	15878	17878
Vcn, Total Harmonics	V	2	FLOAT	Displayed	11156	15128	17128	3	DATE	yymm|ddhh|mmss	15884	17884
In, Total Harmonics	A	2	FLOAT	Displayed	11158	15130	17130	3	DATE	yymm|ddhh|mmss	15890	17890
Ia, Total Harmonics, percentage	%	2	FLOAT	Displayed	11160	15132	17132	3	DATE	yymm|ddhh|mmss	15896	17896
Ib, Total Harmonics, percentage	%	2	FLOAT	Displayed	11162	15134	17134	3	DATE	yymm|ddhh|mmss	15902	17902
Ic, Total Harmonics, percentage	%	2	FLOAT	Displayed	11164	15136	17136	3	DATE	yymm|ddhh|mmss	15908	17908
Vab, Total Harmonics, percentage	%	2	FLOAT	Displayed	11166	15138	17138	3	DATE	yymm|ddhh|mmss	15914	17914
Vbc, Total Harmonics, percentage	%	2	FLOAT	Displayed	11168	15140	17140	3	DATE	yymm|ddhh|mmss	15920	17920
Vca, Total Harmonics, percentage	%	2	FLOAT	Displayed	11170	15142	17142	3	DATE	yymm|ddhh|mmss	15926	17926
Van, Total Harmonics, percentage	%	2	FLOAT	Displayed	11172	15144	17144	3	DATE	yymm|ddhh|mmss	15932	17932
Vbn, Total Harmonics, percentage	%	2	FLOAT	Displayed	11174	15146	17146	3	DATE	yymm|ddhh|mmss	15938	17938
Vcn, Total Harmonics, percentage	%	2	FLOAT	Displayed	11176	15148	17148	3	DATE	yymm|ddhh|mmss	15944	17944
In, Total Harmonics, percentage	%	2	FLOAT	Displayed	11178	15150	17150	3	DATE	yymm|ddhh|mmss	15950	17950
Ia, Total Demand Distortion (TDD), percentage	%	2	FLOAT	Displayed	11180	15152	17152	3	DATE	yymm|ddhh|mmss	15956	17956
Ib, Total Demand Distortion (TDD), percentage	%	2	FLOAT	Displayed	11182	15154	17154	3	DATE	yymm|ddhh|mmss	15962	17962
Ic, Total Demand Distortion (TDD), percentage	%	2	FLOAT	Displayed	11184	15156	17156	3	DATE	yymm|ddhh|mmss	15968	17968
In, Total Demand Distortion (TDD), percentage	%	2	FLOAT	Displayed	11186	15158	17158	3	DATE	yymm|ddhh|mmss	15974	17974

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min (last 5 minute interval)	Max (last 5 minute interval)	Avg (last 5 minute interval)	Min (last 15 minute interval)	Max (last 15 minute interval)	Avg (last 15 minute interval)
Ia, Total Harmonics	A	2	FLOAT	Trended	11140	19336	19338	19340	21336	21338	21340
Ib, Total Harmonics	A	2	FLOAT	Trended	11142	19342	19344	19346	21342	21344	21346
Ic, Total Harmonics	A	2	FLOAT	Trended	11144	19348	19350	19352	21348	21350	21352
Vab, Total Harmonics	V	2	FLOAT	Trended	11146	19354	19356	19358	21354	21356	21358
Vbc, Total Harmonics	V	2	FLOAT	Trended	11148	19360	19362	19364	21360	21362	21364
Vca, Total Harmonics	V	2	FLOAT	Trended	11150	19366	19368	19370	21366	21368	21370
Van, Total Harmonics	V	2	FLOAT	Trended	11152	19372	19374	19376	21372	21374	21376
Vbn, Total Harmonics	V	2	FLOAT	Trended	11154	19378	19380	19382	21378	21380	21382
Vcn, Total Harmonics	V	2	FLOAT	Trended	11156	19384	19386	19388	21384	21386	21388
In, Total Harmonics	A	2	FLOAT	Trended	11158	19390	19392	19394	21390	21392	21394
Ia, Total Harmonics, percentage	%	2	FLOAT	Trended	11160	19396	19398	19400	21396	21398	21400
Ib, Total Harmonics, percentage	%	2	FLOAT	Trended	11162	19402	19404	19406	21402	21404	21406
Ic, Total Harmonics, percentage	%	2	FLOAT	Trended	11164	19408	19410	19412	21408	21410	21412
Vab, Total Harmonics, percentage	%	2	FLOAT	Trended	11166	19414	19416	19418	21414	21416	21418
Vbc, Total Harmonics, percentage	%	2	FLOAT	Trended	11168	19420	19422	19424	21420	21422	21424
Vca, Total Harmonics, percentage	%	2	FLOAT	Trended	11170	19426	19428	19430	21426	21428	21430
Van, Total Harmonics, percentage	%	2	FLOAT	Trended	11172	19432	19434	19436	21432	21434	21436
Vbn, Total Harmonics, percentage	%	2	FLOAT	Trended	11174	19438	19440	19442	21438	21440	21442
Vcn, Total Harmonics, percentage	%	2	FLOAT	Trended	11176	19444	19446	19448	21444	21446	21448
In, Total Harmonics, percentage	%	2	FLOAT	Trended	11178	19450	19452	19454	21450	21452	21454
Ia, Total Demand Distortion (TDD), percentage	%	2	FLOAT	Trended	11180	19456	19458	19460	21456	21458	21460
Ib, Total Demand Distortion (TDD), percentage	%	2	FLOAT	Trended	11182	19462	19464	19466	21462	21464	21466
Ic, Total Demand Distortion (TDD), percentage	%	2	FLOAT	Trended	11184	19468	19470	19472	21468	21470	21472
In, Total Demand Distortion (TDD), percentage	%	2	FLOAT	Trended	11186	19474	19476	19478	21474	21476	21478

Even Harmonics

Trends available in the Web UI under Metering > Quality.

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min since last reset	Max since last reset	Register Count	Type ID	Time Stamp Format	Timestamp of Min since last reset	Timestamp of Max since last reset
Ia, Even Harmonics	A	2	FLOAT	Displayed	11188	15160	17160	3	DATE	yymm|ddhh|mmss	15980	17980
Ib, Even Harmonics	A	2	FLOAT	Displayed	11190	15162	17162	3	DATE	yymm|ddhh|mmss	15986	17986
Ic, Even Harmonics	A	2	FLOAT	Displayed	11192	15164	17164	3	DATE	yymm|ddhh|mmss	15992	17992
Vab, Even Harmonics	V	2	FLOAT	Displayed	11194	15166	17166	3	DATE	yymm|ddhh|mmss	15998	17998
Vbc, Even Harmonics	V	2	FLOAT	Displayed	11196	15168	17168	3	DATE	yymm|ddhh|mmss	16004	18004
Vca, Even Harmonics	V	2	FLOAT	Displayed	11198	15170	17170	3	DATE	yymm|ddhh|mmss	16010	18010
Van, Even Harmonics	V	2	FLOAT	Displayed	11200	15172	17172	3	DATE	yymm|ddhh|mmss	16016	18016
Vbn, Even Harmonics	V	2	FLOAT	Displayed	11202	15174	17174	3	DATE	yymm|ddhh|mmss	16022	18022
Vcn, Even Harmonics	V	2	FLOAT	Displayed	11204	15176	17176	3	DATE	yymm|ddhh|mmss	16028	18028
In, Even Harmonics	A	2	FLOAT	Displayed	11206	15178	17178	3	DATE	yymm|ddhh|mmss	16034	18034
Ia, Even Harmonics, percentage	%	2	FLOAT	Displayed	11208	15180	17180	3	DATE	yymm|ddhh|mmss	16040	18040
Ib, Even Harmonics, percentage	%	2	FLOAT	Displayed	11210	15182	17182	3	DATE	yymm|ddhh|mmss	16046	18046
Ic, Even Harmonics, percentage	%	2	FLOAT	Displayed	11212	15184	17184	3	DATE	yymm|ddhh|mmss	16052	18052
Vab, Even Harmonics, percentage	%	2	FLOAT	Displayed	11214	15186	17186	3	DATE	yymm|ddhh|mmss	16058	18058
Vbc, Even Harmonics, percentage	%	2	FLOAT	Displayed	11216	15188	17188	3	DATE	yymm|ddhh|mmss	16064	18064
Vca, Even Harmonics, percentage	%	2	FLOAT	Displayed	11218	15190	17190	3	DATE	yymm|ddhh|mmss	16070	18070
Van, Even Harmonics, percentage	%	2	FLOAT	Displayed	11220	15192	17192	3	DATE	yymm|ddhh|mmss	16076	18076
Vbn, Even Harmonics, percentage	%	2	FLOAT	Displayed	11222	15194	17194	3	DATE	yymm|ddhh|mmss	16082	18082
Vcn, Even Harmonics, percentage	%	2	FLOAT	Displayed	11224	15196	17196	3	DATE	yymm|ddhh|mmss	16088	18088
In, Even Harmonics, percentage	%	2	FLOAT	Displayed	11226	15198	17198	3	DATE	yymm|ddhh|mmss	16094	18094
Ia, Even TDD, percentage	%	2	FLOAT	Displayed	11228	15200	17200	3	DATE	yymm|ddhh|mmss	16100	18100
Ib, Even TDD, percentage	%	2	FLOAT	Displayed	11230	15202	17202	3	DATE	yymm|ddhh|mmss	16106	18106
Ic, Even TDD, percentage	%	2	FLOAT	Displayed	11232	15204	17204	3	DATE	yymm|ddhh|mmss	16112	18112
In, Even TDD, percentage	%	2	FLOAT	Displayed	11234	15206	17206	3	DATE	yymm|ddhh|mmss	16118	18118

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min (last 5 minute interval)	Max (last 5 minute interval)	Avg (last 5 minute interval)	Min (last 15 minute interval)	Max (last 15 minute interval)	Avg (last 15 minute interval)
Ia, Even Harmonics	A	2	FLOAT	Trended	11188	19480	19482	19484	21480	21482	21484
Ib, Even Harmonics	A	2	FLOAT	Trended	11190	19486	19488	19490	21486	21488	21490
Ic, Even Harmonics	A	2	FLOAT	Trended	11192	19492	19494	19496	21492	21494	21496
Vab, Even Harmonics	V	2	FLOAT	Trended	11194	19498	19500	19502	21498	21500	21502
Vbc, Even Harmonics	V	2	FLOAT	Trended	11196	19504	19506	19508	21504	21506	21508
Vca, Even Harmonics	V	2	FLOAT	Trended	11198	19510	19512	19514	21510	21512	21514
Van, Even Harmonics	V	2	FLOAT	Trended	11200	19516	19518	19520	21516	21518	21520
Vbn, Even Harmonics	V	2	FLOAT	Trended	11202	19522	19524	19526	21522	21524	21526
Vcn, Even Harmonics	V	2	FLOAT	Trended	11204	19528	19530	19532	21528	21530	21532
In, Even Harmonics	A	2	FLOAT	Trended	11206	19534	19536	19538	21534	21536	21538
Ia, Even Harmonics, percentage	%	2	FLOAT	Trended	11208	19540	19542	19544	21540	21542	21544
Ib, Even Harmonics, percentage	%	2	FLOAT	Trended	11210	19546	19548	19550	21546	21548	21550
Ic, Even Harmonics, percentage	%	2	FLOAT	Trended	11212	19552	19554	19556	21552	21554	21556
Vab, Even Harmonics, percentage	%	2	FLOAT	Trended	11214	19558	19560	19562	21558	21560	21562
Vbc, Even Harmonics, percentage	%	2	FLOAT	Trended	11216	19564	19566	19568	21564	21566	21568
Vca, Even Harmonics, percentage	%	2	FLOAT	Trended	11218	19570	19572	19574	21570	21572	21574
Van, Even Harmonics, percentage	%	2	FLOAT	Trended	11220	19576	19578	19580	21576	21578	21580
Vbn, Even Harmonics, percentage	%	2	FLOAT	Trended	11222	19582	19584	19586	21582	21584	21586
Vcn, Even Harmonics, percentage	%	2	FLOAT	Trended	11224	19588	19590	19592	21588	21590	21592
In, Even Harmonics, percentage	%	2	FLOAT	Trended	11226	19594	19596	19598	21594	21596	21598
Ia, Even TDD, percentage	%	2	FLOAT	Trended	11228	19600	19602	19604	21600	21602	21604
Ib, Even TDD, percentage	%	2	FLOAT	Trended	11230	19606	19608	19610	21606	21608	21610
Ic, Even TDD, percentage	%	2	FLOAT	Trended	11232	19612	19614	19616	21612	21614	21616
In, Even TDD, percentage	%	2	FLOAT	Trended	11234	19618	19620	19622	21618	21620	21622

Odd Harmonics

Trends available in the Web UI under Metering > Quality.

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min since last reset	Max since last reset	Register Count	Type ID	Time Stamp Format	Timestamp of Min since last reset	Timestamp of Max since last reset
Ia, Odd Harmonics	A	2	FLOAT	Displayed	11236	15208	17208	3	DATE	yymm|ddhh|mmss	16124	18124
Ib, Odd Harmonics	A	2	FLOAT	Displayed	11238	15210	17210	3	DATE	yymm|ddhh|mmss	16130	18130
Ic, Odd Harmonics	A	2	FLOAT	Displayed	11240	15212	17212	3	DATE	yymm|ddhh|mmss	16136	18136
Vab, Odd Harmonics	V	2	FLOAT	Displayed	11242	15214	17214	3	DATE	yymm|ddhh|mmss	16142	18142
Vbc, Odd Harmonics	V	2	FLOAT	Displayed	11244	15216	17216	3	DATE	yymm|ddhh|mmss	16148	18148
Vca, Odd Harmonics	V	2	FLOAT	Displayed	11246	15218	17218	3	DATE	yymm|ddhh|mmss	16154	18154
Van, Odd Harmonics	V	2	FLOAT	Displayed	11248	15220	17220	3	DATE	yymm|ddhh|mmss	16160	18160
Vbn, Odd Harmonics	V	2	FLOAT	Displayed	11250	15222	17222	3	DATE	yymm|ddhh|mmss	16166	18166
Vcn, Odd Harmonics	V	2	FLOAT	Displayed	11252	15224	17224	3	DATE	yymm|ddhh|mmss	16172	18172
In, Odd Harmonics	A	2	FLOAT	Displayed	11254	15226	17226	3	DATE	yymm|ddhh|mmss	16178	18178
Ia, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11256	15228	17228	3	DATE	yymm|ddhh|mmss	16184	18184
Ib, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11258	15230	17230	3	DATE	yymm|ddhh|mmss	16190	18190
Ic, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11260	15232	17232	3	DATE	yymm|ddhh|mmss	16196	18196
Vab, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11262	15234	17234	3	DATE	yymm|ddhh|mmss	16202	18202
Vbc, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11264	15236	17236	3	DATE	yymm|ddhh|mmss	16208	18208
Vca, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11266	15238	17238	3	DATE	yymm|ddhh|mmss	16214	18214
Van, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11268	15240	17240	3	DATE	yymm|ddhh|mmss	16220	18220
Vbn, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11270	15242	17242	3	DATE	yymm|ddhh|mmss	16226	18226
Vcn, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11272	15244	17244	3	DATE	yymm|ddhh|mmss	16232	18232
In, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11274	15246	17246	3	DATE	yymm|ddhh|mmss	16238	18238
Ia, Odd TDD, percentage	%	2	FLOAT	Displayed	11276	15248	17248	3	DATE	yymm|ddhh|mmss	16244	18244
Ib, Odd TDD, percentage	%	2	FLOAT	Displayed	11278	15250	17250	3	DATE	yymm|ddhh|mmss	16250	18250
Ic, Odd TDD, percentage	%	2	FLOAT	Displayed	11280	15252	17252	3	DATE	yymm|ddhh|mmss	16256	18256
In, Odd TDD, percentage	%	2	FLOAT	Displayed	11282	15254	17254	3	DATE	yymm|ddhh|mmss	16262	18262

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min (last 5 minute interval)	Max (last 5 minute interval)	Avg (last 5 minute interval)	Min (last 15 minute interval)	Max (last 15 minute interval)	Avg (last 15 minute interval)
Ia, Odd Harmonics	A	2	FLOAT	Trended	11236	19624	19626	19628	21624	21626	21628
Ib, Odd Harmonics	A	2	FLOAT	Trended	11238	19630	19632	19634	21630	21632	21634
Ic, Odd Harmonics	A	2	FLOAT	Trended	11240	19636	19638	19640	21636	21638	21640
Vab, Odd Harmonics	V	2	FLOAT	Trended	11242	19642	19644	19646	21642	21644	21646
Vbc, Odd Harmonics	V	2	FLOAT	Trended	11244	19648	19650	19652	21648	21650	21652
Vca, Odd Harmonics	V	2	FLOAT	Trended	11246	19654	19656	19658	21654	21656	21658
Van, Odd Harmonics	V	2	FLOAT	Trended	11248	19660	19662	19664	21660	21662	21664
Vbn, Odd Harmonics	V	2	FLOAT	Trended	11250	19666	19668	19670	21666	21668	21670
Vcn, Odd Harmonics	V	2	FLOAT	Trended	11252	19672	19674	19676	21672	21674	21676
In, Odd Harmonics	A	2	FLOAT	Trended	11254	19678	19680	19682	21678	21680	21682
Ia, Odd Harmonics, percentage	%	2	FLOAT	Trended	11256	19684	19686	19688	21684	21686	21688
Ib, Odd Harmonics, percentage	%	2	FLOAT	Trended	11258	19690	19692	19694	21690	21692	21694
Ic, Odd Harmonics, percentage	%	2	FLOAT	Trended	11260	19696	19698	19700	21696	21698	21700
Vab, Odd Harmonics, percentage	%	2	FLOAT	Trended	11262	19702	19704	19706	21702	21704	21706
Vbc, Odd Harmonics, percentage	%	2	FLOAT	Trended	11264	19708	19710	19712	21708	21710	21712
Vca, Odd Harmonics, percentage	%	2	FLOAT	Trended	11266	19714	19716	19718	21714	21716	21718
Van, Odd Harmonics, percentage	%	2	FLOAT	Trended	11268	19720	19722	19724	21720	21722	21724
Vbn, Odd Harmonics, percentage	%	2	FLOAT	Trended	11270	19726	19728	19730	21726	21728	21730
Vcn, Odd Harmonics, percentage	%	2	FLOAT	Trended	11272	19732	19734	19736	21732	21734	21736
In, Odd Harmonics, percentage	%	2	FLOAT	Trended	11274	19738	19740	19742	21738	21740	21742
Ia, Odd TDD, percentage	%	2	FLOAT	Trended	11276	19744	19746	19748	21744	21746	21748
Ib, Odd TDD, percentage	%	2	FLOAT	Trended	11278	19750	19752	19754	21750	21752	21754
Ic, Odd TDD, percentage	%	2	FLOAT	Trended	11280	19756	19758	19760	21756	21758	21760
In, Odd TDD, percentage	%	2	FLOAT	Trended	11282	19762	19764	19766	21762	21764	21766

Interharmonics

Trends available in the Web UI under Metering > Quality.

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min since last reset	Max since last reset	Register Count	Type ID	Time Stamp Format	Timestamp of Min since last reset	Timestamp of Max since last reset
Ia, Odd Harmonics	A	2	FLOAT	Displayed	11236	15208	17208	3	DATE	yymm|ddhh|mmss	16124	18124
Ib, Odd Harmonics	A	2	FLOAT	Displayed	11238	15210	17210	3	DATE	yymm|ddhh|mmss	16130	18130
Ic, Odd Harmonics	A	2	FLOAT	Displayed	11240	15212	17212	3	DATE	yymm|ddhh|mmss	16136	18136
Vab, Odd Harmonics	V	2	FLOAT	Displayed	11242	15214	17214	3	DATE	yymm|ddhh|mmss	16142	18142
Vbc, Odd Harmonics	V	2	FLOAT	Displayed	11244	15216	17216	3	DATE	yymm|ddhh|mmss	16148	18148
Vca, Odd Harmonics	V	2	FLOAT	Displayed	11246	15218	17218	3	DATE	yymm|ddhh|mmss	16154	18154
Van, Odd Harmonics	V	2	FLOAT	Displayed	11248	15220	17220	3	DATE	yymm|ddhh|mmss	16160	18160
Vbn, Odd Harmonics	V	2	FLOAT	Displayed	11250	15222	17222	3	DATE	yymm|ddhh|mmss	16166	18166
Vcn, Odd Harmonics	V	2	FLOAT	Displayed	11252	15224	17224	3	DATE	yymm|ddhh|mmss	16172	18172
In, Odd Harmonics	A	2	FLOAT	Displayed	11254	15226	17226	3	DATE	yymm|ddhh|mmss	16178	18178
Ia, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11256	15228	17228	3	DATE	yymm|ddhh|mmss	16184	18184
Ib, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11258	15230	17230	3	DATE	yymm|ddhh|mmss	16190	18190
Ic, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11260	15232	17232	3	DATE	yymm|ddhh|mmss	16196	18196
Vab, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11262	15234	17234	3	DATE	yymm|ddhh|mmss	16202	18202
Vbc, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11264	15236	17236	3	DATE	yymm|ddhh|mmss	16208	18208
Vca, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11266	15238	17238	3	DATE	yymm|ddhh|mmss	16214	18214
Van, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11268	15240	17240	3	DATE	yymm|ddhh|mmss	16220	18220
Vbn, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11270	15242	17242	3	DATE	yymm|ddhh|mmss	16226	18226
Vcn, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11272	15244	17244	3	DATE	yymm|ddhh|mmss	16232	18232
In, Odd Harmonics, percentage	%	2	FLOAT	Displayed	11274	15246	17246	3	DATE	yymm|ddhh|mmss	16238	18238
Ia, Odd TDD, percentage	%	2	FLOAT	Displayed	11276	15248	17248	3	DATE	yymm|ddhh|mmss	16244	18244
Ib, Odd TDD, percentage	%	2	FLOAT	Displayed	11278	15250	17250	3	DATE	yymm|ddhh|mmss	16250	18250
Ic, Odd TDD, percentage	%	2	FLOAT	Displayed	11280	15252	17252	3	DATE	yymm|ddhh|mmss	16256	18256
In, Odd TDD, percentage	%	2	FLOAT	Displayed	11282	15254	17254	3	DATE	yymm|ddhh|mmss	16262	18262

Individual harmonic magnitudes

Individual harmonic magnitudes for VLL and phase currents are available for odd harmonic magnitudes from 1st to 31st. The harmonic spectrum data is available in the Web UI under Metering > Quality > Harmonics (drop-down).

Harmonic Order	Register Count	Type ID	Data Available in UI	Vab	Vbc	Vca	Ia	Ib	Ic
1st	2	FLOAT	No	26000	26032	26064	26096	26128	26160
3rd	2	FLOAT	No	26002	26034	26066	26098	26130	26162
5th	2	FLOAT	No	26004	26036	26068	26100	26132	26164
7th	2	FLOAT	No	26006	26038	26070	26102	26134	26166
9th	2	FLOAT	No	26008	26040	26072	26104	26136	26168
11th	2	FLOAT	No	26010	26042	26074	26106	26138	26170
13th	2	FLOAT	No	26012	26044	26076	26108	26140	26172
15th	2	FLOAT	No	26014	26046	26078	26110	26142	26174
17th	2	FLOAT	No	26016	26048	26080	26112	26144	26176
19th	2	FLOAT	No	26018	26050	26082	26114	26146	26178
21st	2	FLOAT	No	26020	26052	26084	26116	26148	26180
23rd	2	FLOAT	No	26022	26054	26086	26118	26150	26182
25th	2	FLOAT	No	26024	26056	26088	26120	26152	26184
27th	2	FLOAT	No	26026	26058	26090	26122	26154	26186
29th	2	FLOAT	No	26028	26060	26092	26124	26156	26188
31st	2	FLOAT	No	26030	26062	26094	26126	26158	26190

Supra-Harmonic frequency spectrum

Supraharmonics for VLN, VLL, phase and neutral currents from 2khz to 9kHz with 200 Hz bins per IEC 61000-4-7.

Harmonic Frequency Bin	Register Count	Type ID	Data Available in UI	Van	Vbn	Vcn	Vab	Vbc	Vca	Ia	Ib	Ic	In
2100 Hz	2	FLOAT	No	26192	26262	26332	26402	26472	26542	26612	26682	26752	26822
2300 Hz	2	FLOAT	No	26194	26264	26334	26404	26474	26544	26614	26684	26754	26824
2500 Hz	2	FLOAT	No	26196	26266	26336	26406	26476	26546	26616	26686	26756	26826
2700 Hz	2	FLOAT	No	26198	26268	26338	26408	26478	26548	26618	26688	26758	26828
2900 Hz	2	FLOAT	No	26200	26270	26340	26410	26480	26550	26620	26690	26760	26830
3100 Hz	2	FLOAT	No	26202	26272	26342	26412	26482	26552	26622	26692	26762	26832
3300 Hz	2	FLOAT	No	26204	26274	26344	26414	26484	26554	26624	26694	26764	26834
3500 Hz	2	FLOAT	No	26206	26276	26346	26416	26486	26556	26626	26696	26766	26836
3700 Hz	2	FLOAT	No	26208	26278	26348	26418	26488	26558	26628	26698	26768	26838
3900 Hz	2	FLOAT	No	26210	26280	26350	26420	26490	26560	26630	26700	26770	26840
4100 Hz	2	FLOAT	No	26212	26282	26352	26422	26492	26562	26632	26702	26772	26842
4300 Hz	2	FLOAT	No	26214	26284	26354	26424	26494	26564	26634	26704	26774	26844
4500 Hz	2	FLOAT	No	26216	26286	26356	26426	26496	26566	26636	26706	26776	26846
4700 Hz	2	FLOAT	No	26218	26288	26358	26428	26498	26568	26638	26708	26778	26848
4900 Hz	2	FLOAT	No	26220	26290	26360	26430	26500	26570	26640	26710	26780	26850
5100 Hz	2	FLOAT	No	26222	26292	26362	26432	26502	26572	26642	26712	26782	26852
5300 Hz	2	FLOAT	No	26224	26294	26364	26434	26504	26574	26644	26714	26784	26854
5500 Hz	2	FLOAT	No	26226	26296	26366	26436	26506	26576	26646	26716	26786	26856
5700 Hz	2	FLOAT	No	26228	26298	26368	26438	26508	26578	26648	26718	26788	26858
5900 Hz	2	FLOAT	No	26230	26300	26370	26440	26510	26580	26650	26720	26790	26860
6100 Hz	2	FLOAT	No	26232	26302	26372	26442	26512	26582	26652	26722	26792	26862
6300 Hz	2	FLOAT	No	26234	26304	26374	26444	26514	26584	26654	26724	26794	26864
6500 Hz	2	FLOAT	No	26236	26306	26376	26446	26516	26586	26656	26726	26796	26866
6700 Hz	2	FLOAT	No	26238	26308	26378	26448	26518	26588	26658	26728	26798	26868
6900 Hz	2	FLOAT	No	26240	26310	26380	26450	26520	26590	26660	26730	26800	26870
7100 Hz	2	FLOAT	No	26242	26312	26382	26452	26522	26592	26662	26732	26802	26872
7300 Hz	2	FLOAT	No	26244	26314	26384	26454	26524	26594	26664	26734	26804	26874
7500 Hz	2	FLOAT	No	26246	26316	26386	26456	26526	26596	26666	26736	26806	26876
7700 Hz	2	FLOAT	No	26248	26318	26388	26458	26528	26598	26668	26738	26808	26878
7900 Hz	2	FLOAT	No	26250	26320	26390	26460	26530	26600	26670	26740	26810	26880

Harmonic Frequency Bin	Register Count	Type ID	Data Available in UI	Van	Vbn	Vcn	Vab	Vbc	Vca	Ia	Ib	Ic	In
8100 Hz	2	FLOAT	No	26252	26322	26392	26462	26532	26602	26672	26742	26812	26882
8300 Hz	2	FLOAT	No	26254	26324	26394	26464	26534	26604	26674	26744	26814	26884
8500 Hz	2	FLOAT	No	26256	26326	26396	26466	26536	26606	26676	26746	26816	26886
8700 Hz	2	FLOAT	No	26258	26328	26398	26468	26538	26608	26678	26748	26818	26888
8900 Hz	2	FLOAT	No	26260	26330	26400	26470	26540	26610	26680	26750	26820	26890

Trended Harmonic Magnitudes

Trend data for selected harmonic magnitudes. User can select which channel (VLN, VLL, I) and harmonic order (1st to 85th) to trend under Settings > General> Metering> Harmonic Channel Selection. Trends and data are available under Metering > Quality > Selected Harmonics.

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min since last reset	Max since last reset	Register Count	Type ID	Time Stamp Format	Timestamp of Min since last reset	Timestamp of Max since last reset
Magnitude, Configured Harmonic 1	depends on configured channel	2	FLOAT	Displayed if configured	27000	15304	17304	3	DATE	yymm|ddhh|mmss	16412	18412
Magnitude, Configured Harmonic 2	depends on configured channel	2	FLOAT	Displayed if configured	27002	15306	17306	3	DATE	yymm|ddhh|mmss	16418	18418
Magnitude, Configured Harmonic 3	depends on configured channel	2	FLOAT	Displayed if configured	27004	15308	17308	3	DATE	yymm|ddhh|mmss	16424	18424
Magnitude, Configured Harmonic 4	depends on configured channel	2	FLOAT	Displayed if configured	27006	15310	17310	3	DATE	yymm|ddhh|mmss	16430	18430
Magnitude, Configured Harmonic 5	depends on configured channel	2	FLOAT	Displayed if configured	27008	15312	17312	3	DATE	yymm|ddhh|mmss	16436	18436
Magnitude, Configured Harmonic 6	depends on configured channel	2	FLOAT	Displayed if configured	27010	15314	17314	3	DATE	yymm|ddhh|mmss	16442	18442
Magnitude, Configured Harmonic 7	depends on configured channel	2	FLOAT	Displayed if configured	27012	15316	17316	3	DATE	yymm|ddhh|mmss	16448	18448
Magnitude, Configured Harmonic 8	depends on configured channel	2	FLOAT	Displayed if configured	27014	15318	17318	3	DATE	yymm|ddhh|mmss	16454	18454
Magnitude, Configured Harmonic 9	depends on configured channel	2	FLOAT	Displayed if configured	27016	15320	17320	3	DATE	yymm|ddhh|mmss	16460	18460
Magnitude, Configured Harmonic 10	depends on configured channel	2	FLOAT	Displayed if configured	27018	15322	17322	3	DATE	yymm|ddhh|mmss	16466	18466
Magnitude, Configured Harmonic 11	depends on configured channel	2	FLOAT	Displayed if configured	27020	15324	17324	3	DATE	yymm|ddhh|mmss	16472	18472
Magnitude, Configured Harmonic 12	depends on configured channel	2	FLOAT	Displayed if configured	27022	15326	17326	3	DATE	yymm|ddhh|mmss	16478	18478
Magnitude, Configured Harmonic 13	depends on configured channel	2	FLOAT	Displayed if configured	27024	15328	17328	3	DATE	yymm|ddhh|mmss	16484	18484
Magnitude, Configured Harmonic 14	depends on configured channel	2	FLOAT	Displayed if configured	27026	15330	17330	3	DATE	yymm|ddhh|mmss	16490	18490
Magnitude, Configured Harmonic 15	depends on configured channel	2	FLOAT	Displayed if configured	27028	15332	17332	3	DATE	yymm|ddhh|mmss	16496	18496
Magnitude, Configured Harmonic 16	depends on configured channel	2	FLOAT	Displayed if configured	27030	15334	17334	3	DATE	yymm|ddhh|mmss	16502	18502
Magnitude, Configured Harmonic 17	depends on configured channel	2	FLOAT	Displayed if configured	27032	15336	17336	3	DATE	yymm|ddhh|mmss	16508	18508
Magnitude, Configured Harmonic 18	depends on configured channel	2	FLOAT	Displayed if configured	27034	15338	17338	3	DATE	yymm|ddhh|mmss	16514	18514
Magnitude, Configured Harmonic 19	depends on configured channel	2	FLOAT	Displayed if configured	27036	15340	17340	3	DATE	yymm|ddhh|mmss	16520	18520

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min since last reset	Max since last reset	Register Count	Type ID	Time Stamp Format	Timestamp of Min since last reset	Timestamp of Max since last reset
Magnitude, Configured Harmonic 20	depends on configured channel	2	FLOAT	Displayed if configured	27038	15342	17342	3	DATE	yymm|ddhh|mmss	16526	18526
Magnitude, Configured Harmonic 21	depends on configured channel	2	FLOAT	Displayed if configured	27040	15344	17344	3	DATE	yymm|ddhh|mmss	16532	18532
Magnitude, Configured Harmonic 22	depends on configured channel	2	FLOAT	Displayed if configured	27042	15346	17346	3	DATE	yymm|ddhh|mmss	16538	18538
Magnitude, Configured Harmonic 23	depends on configured channel	2	FLOAT	Displayed if configured	27044	15348	17348	3	DATE	yymm|ddhh|mmss	16544	18544
Magnitude, Configured Harmonic 24	depends on configured channel	2	FLOAT	Displayed if configured	27046	15350	17350	3	DATE	yymm|ddhh|mmss	16550	18550
Magnitude, Configured Harmonic 25	depends on configured channel	2	FLOAT	Displayed if configured	27048	15352	17352	3	DATE	yymm|ddhh|mmss	16556	18556
Magnitude, Configured Harmonic 26	depends on configured channel	2	FLOAT	Displayed if configured	27050	15354	17354	3	DATE	yymm|ddhh|mmss	16562	18562
Magnitude, Configured Harmonic 27	depends on configured channel	2	FLOAT	Displayed if configured	27052	15356	17356	3	DATE	yymm|ddhh|mmss	16568	18568
Magnitude, Configured Harmonic 28	depends on configured channel	2	FLOAT	Displayed if configured	27054	15358	17358	3	DATE	yymm|ddhh|mmss	16574	18574
Magnitude, Configured Harmonic 29	depends on configured channel	2	FLOAT	Displayed if configured	27056	15360	17360	3	DATE	yymm|ddhh|mmss	16580	18580
Magnitude, Configured Harmonic 30	depends on configured channel	2	FLOAT	Displayed if configured	27058	15362	17362	3	DATE	yymm|ddhh|mmss	16586	18586
Magnitude, Configured Harmonic 31	depends on configured channel	2	FLOAT	Displayed if configured	27060	15364	17364	3	DATE	yymm|ddhh|mmss	16592	18592
Magnitude, Configured Harmonic 32	depends on configured channel	2	FLOAT	Displayed if configured	27062	15366	17366	3	DATE	yymm|ddhh|mmss	16598	18598
Magnitude, Configured Harmonic 33	depends on configured channel	2	FLOAT	Displayed if configured	27064	15368	17368	3	DATE	yymm|ddhh|mmss	16604	18604
Magnitude, Configured Harmonic 34	depends on configured channel	2	FLOAT	Displayed if configured	27066	15370	17370	3	DATE	yymm|ddhh|mmss	16610	18610
Magnitude, Configured Harmonic 35	depends on configured channel	2	FLOAT	Displayed if configured	27068	15372	17372	3	DATE	yymm|ddhh|mmss	16616	18616
Magnitude, Configured Harmonic 36	depends on configured channel	2	FLOAT	Displayed if configured	27070	15374	17374	3	DATE	yymm|ddhh|mmss	16622	18622
Magnitude, Configured Harmonic 37	depends on configured channel	2	FLOAT	Displayed if configured	27072	15376	17376	3	DATE	yymm|ddhh|mmss	16628	18628
Magnitude, Configured Harmonic 38	depends on configured channel	2	FLOAT	Displayed if configured	27074	15378	17378	3	DATE	yymm|ddhh|mmss	16634	18634
Magnitude, Configured Harmonic 39	depends on configured channel	2	FLOAT	Displayed if configured	27076	15380	17380	3	DATE	yymm|ddhh|mmss	16640	18640
Magnitude, Configured Harmonic 40	depends on configured channel	2	FLOAT	Displayed if configured	27078	15382	17382	3	DATE	yymm|ddhh|mmss	16646	18646

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min since last reset	Max since last reset	Register Count	Type ID	Time Stamp Format	Timestamp of Min since last reset	Timestamp of Max since last reset
Magnitude, Configured Harmonic 41	depends on configured channel	2	FLOAT	Displayed if configured	27080	15384	17384	3	DATE	yymm|ddhh|mmss	16652	18652
Magnitude, Configured Harmonic 42	depends on configured channel	2	FLOAT	Displayed if configured	27082	15386	17386	3	DATE	yymm|ddhh|mmss	16658	18658
Magnitude, Configured Harmonic 43	depends on configured channel	2	FLOAT	Displayed if configured	27084	15388	17388	3	DATE	yymm|ddhh|mmss	16664	18664
Magnitude, Configured Harmonic 44	depends on configured channel	2	FLOAT	Displayed if configured	27086	15390	17390	3	DATE	yymm|ddhh|mmss	16670	18670
Magnitude, Configured Harmonic 45	depends on configured channel	2	FLOAT	Displayed if configured	27088	15392	17392	3	DATE	yymm|ddhh|mmss	16676	18676
Magnitude, Configured Harmonic 46	depends on configured channel	2	FLOAT	Displayed if configured	27090	15394	17394	3	DATE	yymm|ddhh|mmss	16682	18682
Magnitude, Configured Harmonic 47	depends on configured channel	2	FLOAT	Displayed if configured	27092	15396	17396	3	DATE	yymm|ddhh|mmss	16688	18688
Magnitude, Configured Harmonic 48	depends on configured channel	2	FLOAT	Displayed if configured	27094	15398	17398	3	DATE	yymm|ddhh|mmss	16694	18694
Magnitude, Configured Harmonic 49	depends on configured channel	2	FLOAT	Displayed if configured	27096	15400	17400	3	DATE	yymm|ddhh|mmss	16700	18700
Magnitude, Configured Harmonic 50	depends on configured channel	2	FLOAT	Displayed if configured	27098	15402	17402	3	DATE	yymm|ddhh|mmss	16706	18706

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min (last 5 minute interval)	Max (last 5 minute interval)	Avg (last 5 minute interval)
Magnitude, Configured Harmonic 1	depends on configured channel	2	FLOAT	Trended if configured	27000	19912	19914	19916
Magnitude, Configured Harmonic 2	depends on configured channel	2	FLOAT	Trended if configured	27002	19918	19920	19922
Magnitude, Configured Harmonic 3	depends on configured channel	2	FLOAT	Trended if configured	27004	19924	19926	19928
Magnitude, Configured Harmonic 4	depends on configured channel	2	FLOAT	Trended if configured	27006	19930	19932	19934
Magnitude, Configured Harmonic 5	depends on configured channel	2	FLOAT	Trended if configured	27008	19936	19938	19940
Magnitude, Configured Harmonic 6	depends on configured channel	2	FLOAT	Trended if configured	27010	19942	19944	19946
Magnitude, Configured Harmonic 7	depends on configured channel	2	FLOAT	Trended if configured	27012	19948	19950	19952
Magnitude, Configured Harmonic 8	depends on configured channel	2	FLOAT	Trended if configured	27014	19954	19956	19958
Magnitude, Configured Harmonic 9	depends on configured channel	2	FLOAT	Trended if configured	27016	19960	19962	19964

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min (last 5 minute interval)	Max (last 5 minute interval)	Avg (last 5 minute interval)
Magnitude, Configured Harmonic 10	depends on configured channel	2	FLOAT	Trended if configured	27018	19966	19968	19970
Magnitude, Configured Harmonic 11	depends on configured channel	2	FLOAT	Trended if configured	27020	19972	19974	19976
Magnitude, Configured Harmonic 12	depends on configured channel	2	FLOAT	Trended if configured	27022	19978	19980	19982
Magnitude, Configured Harmonic 13	depends on configured channel	2	FLOAT	Trended if configured	27024	19984	19986	19988
Magnitude, Configured Harmonic 14	depends on configured channel	2	FLOAT	Trended if configured	27026	19990	19992	19994
Magnitude, Configured Harmonic 15	depends on configured channel	2	FLOAT	Trended if configured	27028	19996	19998	20000
Magnitude, Configured Harmonic 16	depends on configured channel	2	FLOAT	Trended if configured	27030	20002	20004	20006
Magnitude, Configured Harmonic 17	depends on configured channel	2	FLOAT	Trended if configured	27032	20008	20010	20012
Magnitude, Configured Harmonic 18	depends on configured channel	2	FLOAT	Trended if configured	27034	20014	20016	20018
Magnitude, Configured Harmonic 19	depends on configured channel	2	FLOAT	Trended if configured	27036	20020	20022	20024
Magnitude, Configured Harmonic 20	depends on configured channel	2	FLOAT	Trended if configured	27038	20026	20028	20030
Magnitude, Configured Harmonic 21	depends on configured channel	2	FLOAT	Trended if configured	27040	20032	20034	20036
Magnitude, Configured Harmonic 22	depends on configured channel	2	FLOAT	Trended if configured	27042	20038	20040	20042
Magnitude, Configured Harmonic 23	depends on configured channel	2	FLOAT	Trended if configured	27044	20044	20046	20048
Magnitude, Configured Harmonic 24	depends on configured channel	2	FLOAT	Trended if configured	27046	20050	20052	20054
Magnitude, Configured Harmonic 25	depends on configured channel	2	FLOAT	Trended if configured	27048	20056	20058	20060
Magnitude, Configured Harmonic 26	depends on configured channel	2	FLOAT	Trended if configured	27050	20062	20064	20066
Magnitude, Configured Harmonic 27	depends on configured channel	2	FLOAT	Trended if configured	27052	20068	20070	20072
Magnitude, Configured Harmonic 28	depends on configured channel	2	FLOAT	Trended if configured	27054	20074	20076	20078
Magnitude, Configured Harmonic 29	depends on configured channel	2	FLOAT	Trended if configured	27056	20080	20082	20084
Magnitude, Configured Harmonic 30	depends on configured channel	2	FLOAT	Trended if configured	27058	20086	20088	20090

Parameter	Units	Register Count	Type ID	Data Available in UI	Value (200ms data)	Min (last 5 minute interval)	Max (last 5 minute interval)	Avg (last 5 minute interval)
Magnitude, Configured Harmonic 31	depends on configured channel	2	FLOAT	Trended if configured	27060	20092	20094	20096
Magnitude, Configured Harmonic 32	depends on configured channel	2	FLOAT	Trended if configured	27062	20098	20100	20102
Magnitude, Configured Harmonic 33	depends on configured channel	2	FLOAT	Trended if configured	27064	20104	20106	20108
Magnitude, Configured Harmonic 34	depends on configured channel	2	FLOAT	Trended if configured	27066	20110	20112	20114
Magnitude, Configured Harmonic 35	depends on configured channel	2	FLOAT	Trended if configured	27068	20116	20118	20120
Magnitude, Configured Harmonic 36	depends on configured channel	2	FLOAT	Trended if configured	27070	20122	20124	20126
Magnitude, Configured Harmonic 37	depends on configured channel	2	FLOAT	Trended if configured	27072	20128	20130	20132
Magnitude, Configured Harmonic 38	depends on configured channel	2	FLOAT	Trended if configured	27074	20134	20136	20138
Magnitude, Configured Harmonic 39	depends on configured channel	2	FLOAT	Trended if configured	27076	20140	20142	20144
Magnitude, Configured Harmonic 40	depends on configured channel	2	FLOAT	Trended if configured	27078	20146	20148	20150
Magnitude, Configured Harmonic 41	depends on configured channel	2	FLOAT	Trended if configured	27080	20152	20154	20156
Magnitude, Configured Harmonic 42	depends on configured channel	2	FLOAT	Trended if configured	27082	20158	20160	20162
Magnitude, Configured Harmonic 43	depends on configured channel	2	FLOAT	Trended if configured	27084	20164	20166	20168
Magnitude, Configured Harmonic 44	depends on configured channel	2	FLOAT	Trended if configured	27086	20170	20172	20174
Magnitude, Configured Harmonic 45	depends on configured channel	2	FLOAT	Trended if configured	27088	20176	20178	20180
Magnitude, Configured Harmonic 46	depends on configured channel	2	FLOAT	Trended if configured	27090	20182	20184	20186
Magnitude, Configured Harmonic 47	depends on configured channel	2	FLOAT	Trended if configured	27092	20188	20190	20192
Magnitude, Configured Harmonic 48	depends on configured channel	2	FLOAT	Trended if configured	27094	20194	20196	20198
Magnitude, Configured Harmonic 49	depends on configured channel	2	FLOAT	Trended if configured	27096	20200	20202	20204
Magnitude, Configured Harmonic 50	depends on configured channel	2	FLOAT	Trended if configured	27098	20206	20208	20210

Subsynchronous oscillation detection (SSO)

Sub-Synchronous Oscillations (SSO) are oscillatory phenomena that occur in power systems at frequencies below the system's synchronous frequency (50Hz or 60Hz). Electrical SSO are caused by dynamic, nonlinear, and poorly damped power loads. Rapid and significant changes in power demand causes oscillations of voltage and current that interact with the natural frequencies of the power system. SSOs can cause equipment damage, particularly to generator shafts and transformers, and can cause widespread brownouts and blackouts.

These registers provide an alternative to the MQTT topics for data integration.

Parameter	Register Count	Type ID	Data Available in UI	Band status	Current magnitude (A)	Power magnitude (kW)
SSO band 1	2	Status: UINT32 / Magnitude: FLOAT	No	27508	27608	27708
SSO band 2	2	Status: UINT32 / Magnitude: FLOAT	No	27510	27610	27710
SSO band 3	2	Status: UINT32 / Magnitude: FLOAT	No	27512	27612	27712
SSO band 4	2	Status: UINT32 / Magnitude: FLOAT	No	27514	27614	27714
SSO band 5	2	Status: UINT32 / Magnitude: FLOAT	No	27516	27616	27716
SSO band 6	2	Status: UINT32 / Magnitude: FLOAT	No	27518	27618	27718
SSO band 7	2	Status: UINT32 / Magnitude: FLOAT	No	27520	27620	27720
SSO band 8	2	Status: UINT32 / Magnitude: FLOAT	No	27522	27622	27722
SSO band 9	2	Status: UINT32 / Magnitude: FLOAT	No	27524	27624	27724
SSO band 10	2	Status: UINT32 / Magnitude: FLOAT	No	27526	27626	27726
SSO band 11	2	Status: UINT32 / Magnitude: FLOAT	No	27528	27628	27728
SSO band 12	2	Status: UINT32 / Magnitude: FLOAT	No	27530	27630	27730
SSO band 13	2	Status: UINT32 / Magnitude: FLOAT	No	27532	27632	27732
SSO band 14	2	Status: UINT32 / Magnitude: FLOAT	No	27534	27634	27734
SSO band 15	2	Status: UINT32 / Magnitude: FLOAT	No	27536	27636	27736
SSO band 16	2	Status: UINT32 / Magnitude: FLOAT	No	27538	27638	27738
SSO band 17	2	Status: UINT32 / Magnitude: FLOAT	No	27540	27640	27740
SSO band 18	2	Status: UINT32 / Magnitude: FLOAT	No	27542	27642	27742
SSO band 19	2	Status: UINT32 / Magnitude: FLOAT	No	27544	27644	27744
SSO band 20	2	Status: UINT32 / Magnitude: FLOAT	No	27546	27646	27746
SSO band 21	2	Status: UINT32 / Magnitude: FLOAT	No	27548	27648	27748
SSO band 22	2	Status: UINT32 / Magnitude: FLOAT	No	27550	27650	27750

Parameter	Register Count	Type ID	Data Available in UI	Band status	Current magnitude (A)	Power magnitude (kW)
SSO band 23	2	Status: UINT32 / Magnitude: FLOAT	No	27552	27652	27752
SSO band 24	2	Status: UINT32 / Magnitude: FLOAT	No	27554	27654	27754
SSO band 25	2	Status: UINT32 / Magnitude: FLOAT	No	27556	27656	27756
SSO band 26	2	Status: UINT32 / Magnitude: FLOAT	No	27558	27658	27758
SSO band 27	2	Status: UINT32 / Magnitude: FLOAT	No	27560	27660	27760
SSO band 28	2	Status: UINT32 / Magnitude: FLOAT	No	27562	27662	27762
SSO band 29	2	Status: UINT32 / Magnitude: FLOAT	No	27564	27664	27764
SSO band 30	2	Status: UINT32 / Magnitude: FLOAT	No	27566	27666	27766
SSO band 31	2	Status: UINT32 / Magnitude: FLOAT	No	27568	27668	27768
SSO band 32	2	Status: UINT32 / Magnitude: FLOAT	No	27570	27670	27770
SSO band 33	2	Status: UINT32 / Magnitude: FLOAT	No	27572	27672	27772
SSO band 34	2	Status: UINT32 / Magnitude: FLOAT	No	27574	27674	27774
SSO band 35	2	Status: UINT32 / Magnitude: FLOAT	No	27576	27676	27776
SSO band 36	2	Status: UINT32 / Magnitude: FLOAT	No	27578	27678	27778
SSO band 37	2	Status: UINT32 / Magnitude: FLOAT	No	27580	27680	27780
SSO band 38	2	Status: UINT32 / Magnitude: FLOAT	No	27582	27682	27782
SSO band 39	2	Status: UINT32 / Magnitude: FLOAT	No	27584	27684	27784
SSO band 40	2	Status: UINT32 / Magnitude: FLOAT	No	27586	27686	27786
SSO band 41	2	Status: UINT32 / Magnitude: FLOAT	No	27588	27688	27788
SSO band 42	2	Status: UINT32 / Magnitude: FLOAT	No	27590	27690	27790
SSO band 43	2	Status: UINT32 / Magnitude: FLOAT	No	27592	27692	27792
SSO band 44	2	Status: UINT32 / Magnitude: FLOAT	No	27594	27694	27794
SSO band 45	2	Status: UINT32 / Magnitude: FLOAT	No	27596	27696	27796
SSO band 46	2	Status: UINT32 / Magnitude: FLOAT	No	27598	27698	27798
SSO band 47	2	Status: UINT32 / Magnitude: FLOAT	No	27600	27700	27800
SSO band 48	2	Status: UINT32 / Magnitude: FLOAT	No	27602	27702	27802

Parameter	Register Count	Type ID	Data Available in UI	Band status	Current magnitude (A)	Power magnitude (kW)
SSO band 49	2	Status: UINT32 / Magnitude: FLOAT	No	27604	27704	27804
SSO band 50	2	Status: UINT32 / Magnitude: FLOAT	No	27606	27706	27806

Time protocol statuses

Only NTP or PTP can be used at a given time. Only one should show as "synchronized".

Parameter	Register Count	Type ID	Register	Notes
NTP Status	2	UINT32	28000	0:not-synchronized; 1:synchronized
PTP Status	2	UINT32	28002	0:not-synchronized; 1:synchronized

Troubleshooting statistics for configuration USB port

Available in web UI under Settings > Diagnostics.

Parameter	Register Count	Type ID	Register
Bytes received	2	UINT32	28020
Packets received	2	UINT32	28022
Errors received	2	UINT32	28024
Packets dropped (received)	2	UINT32	28026
Overruns received	2	UINT32	28028
Errors received	2	UINT32	28030
Bytes transmitted	2	UINT32	28032
Packets transmitted	2	UINT32	28034
Errors transmitted	2	UINT32	28036
Packets dropped (transmitted)	2	UINT32	28038
Overruns (transmitted)	2	UINT32	28040
Errors Transmitted	2	UINT32	28042

Troubleshooting statistics for Ethernet 1 and Ethernet 2

Available in web UI under Settings > Diagnostics

Parameter	Register Count	Type ID	Register
Bytes received, Eth 1	2	UINT32	28068
Packets received, Eth 1	2	UINT32	28070
Errors received, Eth 1	2	UINT32	28072
Packets dropped (received), Eth 1	2	UINT32	28074
Overruns received, Eth 1	2	UINT32	28076
Errors received, Eth 1	2	UINT32	28078
Bytes transmitted, Eth 1	2	UINT32	28080
Packets transmitted, Eth 1	2	UINT32	28082

Parameter	Register Count	Type ID	Register
Errors transmitted, Eth 1	2	UINT32	28084
Packets dropped (transmitted), Eth 1	2	UINT32	28086
Overruns (transmitted), Eth 1	2	UINT32	28088
Errors Transmitted, Eth 1	2	UINT32	28090
Bytes received, Eth 2	2	UINT32	28092
Packets received, Eth 2	2	UINT32	28094
Errors received, Eth 2	2	UINT32	28096
Packets dropped (received), Eth 2	2	UINT32	28098
Overruns received, Eth 2	2	UINT32	28100
Errors received, Eth 2	2	UINT32	28102
Bytes transmitted, Eth 2	2	UINT32	28104
Packets transmitted, Eth 2	2	UINT32	28106
Errors transmitted, Eth 2	2	UINT32	28108
Packets dropped (transmitted), Eth 2	2	UINT32	28110
Overruns (transmitted), Eth 2	2	UINT32	28112
Errors Transmitted, Eth 2	2	UINT32	28114

Miscellaneous

Parameter	Register Count	Type ID	Register
Firmware version	32	STRING	10000
Firmware version date	32	STRING	10032
PSIO (internal version info for troubleshooting only)	32	STRING	10064
PXQ System Core catalog number	32	STRING	10096
PXQ System Core style number	32	STRING	10128
PXQ System Core Hardware Revision	32	STRING	10160
PXQ System Core Serial Number	32	STRING	10192
PXQ System Core date code	32	STRING	10224
Dip-Switch Position (internal use only)	2	UINT32	10256
MAC address, Eth1	32	STRING	10264
MAC address, Eth2	32	STRING	10296
IP address, Eth1	32	STRING	10330
IP subnet mask, Eth1	32	STRING	10362
IP address, Eth2	32	STRING	10394
IP subnet mask, Eth2	32	STRING	10426
Default Gateway, Eth1	32	STRING	10458
IPv6 Address, Eth 1	32	STRING	10490
IPv6 Address, Eth 2	32	STRING	10554
IPv6 default gateway, Eth 1	32	STRING	10618
10 second frequency aggregation, per IEC 61000-4-30	2	FLOAT	29000

Register 2901 (Modbus RTU Controls)

Standard Control Definitions	BYTE3	BYTE2	BYTE1	BYTE0
Reset peak demand (kW, kvar, kVA, Amps)	0	0	0	4
Reset all min/max values	0	0	1	4
Reset discrete/digital input counters (WAGES)	0	0	1	6
Capture waveform	0	3	0	1
Acknowledge triggered events (clear unread events flag from status)	0	3	0	4
Activate relay output #X (0-1) for relays 1-2, respectively	0	4	1	X
De-Activate relay output #X (0-1) for relays 1-2, respectively	0	4	2	X

Register 4605 (Product ID – internal use)

High Byte								Lowe Byte							
P5	P4	P3	P2	P1	P0	V3	V2	V1	V0	D5	D4	D3	D2	D1	D0

Where:

- D5-D0 = Division Code
- V3-V0 = Comm Version
- P5-P0 = Product ID

Register 4607 (Product Status)

High Byte (Primary Status Code)								Low Byte (Secondary Status Code)							
P7	P6	P5	P4	P3	P2	P1	P0	S7	S6	S5	S4	S3	S2	S1	S0
4:Alarmed, 9:Operational;								Powered up							

Register 4609 (Product Cause of Status

High Byte (Primary Status Code)								Low Byte (Secondary Status Code)							
0	0	0	0	0	C10	C9	C8	C7	C6	C5	C4	C3	C2	C1	C0

Code	Reason for Status	Code	Reason for Status
0	Unknown	29	Current Demand
1	Normal Operating Mode	30	Total Harmonic Distortion
2	Discrete Input #1	42	Multiple causes
11	Overvoltage	65	Reverse power
12	Undervoltage	81	Discrete Input #2
15	Underfrequency	83	Discrete Input #3
16	Overfrequency	86	Discrete Input #4
19	Apparent power factor	87	Discrete Input #5
20	Displacement power factor	88	Discrete Input #6
23	Watt	89	Discrete Input #7
24	VA	90	Discrete Input #8
25	VAR	117	ITIC Voltage Sag / Semi F47

Code	Reason for Status	Code	Reason for Status
26	Power Demand	118	ITIC Voltage Swell
27	VA Demand	129	Manual Event / Modbus
28	VAR Demand	142	Voltage Transient

8.2 BACnet IP

Eaton's PXQ offers comprehensive world-class power and energy measurements and monitoring that reduce day-to-day operating cost. PXQ can be configured to communicate over IP (Annex J) with the option of registering as a foreign device for participation in wide-area BACnet inter-networks. PXQ conforms to the ASHRAE 135 Standard and are regularly tested in accordance with the ASHRAE 135-1 Testing Standard.

The following document is intended for building automation professionals (technicians, engineers, and end-users alike) that will integrate PXQ into their control system. The table(s) provides commonly needed information for data residing in the device including BACnet Object identifiers, helpful descriptions, and native names that are observed by way of "discovery".

Object ID	Object Type	Name
1200	0_AnalogInput	mControlRegister
1201	0_AnalogInput	mMBProductID
1202	0_AnalogInput	mMBProductStatus
1203	0_AnalogInput	mMBCauseOfStatus
1204	0_AnalogInput	mIa200
1205	0_AnalogInput	mIb200
1206	0_AnalogInput	mIc200
1207	0_AnalogInput	mIg200
1208	0_AnalogInput	mIn200
1209	0_AnalogInput	mIavg200
1210	0_AnalogInput	mVab200
1211	0_AnalogInput	mVbc200
1212	0_AnalogInput	mVca200
1213	0_AnalogInput	mVllavg200
1214	0_AnalogInput	mVan200
1215	0_AnalogInput	mVbn200
1216	0_AnalogInput	mVcn200
1217	0_AnalogInput	mVlnavg200
1218	0_AnalogInput	mVoltageUnbalance
1219	0_AnalogInput	mCurrentUnbalance
1220	0_AnalogInput	mPtotal
1221	0_AnalogInput	mQtotal
1222	0_AnalogInput	mStotal
1223	0_AnalogInput	mFreq200ms
1224	0_AnalogInput	mIFsys
1225	0_AnalogInput	mKFsys
1226	0_AnalogInput	mTHDF
1227	0_AnalogInput	mPa
1228	0_AnalogInput	mPb
1229	0_AnalogInput	mPc
1230	0_AnalogInput	mQa
1231	0_AnalogInput	mQb
1232	0_AnalogInput	mQc

Object ID	Object Type	Name
1233	0_AnalogInput	mSa
1234	0_AnalogInput	mSb
1235	0_AnalogInput	mSc
1236	0_AnalogInput	mDPFa
1237	0_AnalogInput	mDPFb
1238	0_AnalogInput	mDPFc
1239	0_AnalogInput	mg200MinMaxResetCnt
1240	0_AnalogInput	mgVan200Min
1241	0_AnalogInput	mgVbn200Min
1242	0_AnalogInput	mgVcn200Min
1243	0_AnalogInput	mgVng200Min
1244	0_AnalogInput	mgVab200Min
1245	0_AnalogInput	mgVbc200Min
1246	0_AnalogInput	mgVca200Min
1247	0_AnalogInput	mgIa200Min
1248	0_AnalogInput	mgIb200Min
1249	0_AnalogInput	mgIc200Min
1250	0_AnalogInput	mgIn200Min
1251	0_AnalogInput	mgIg200Min
1252	0_AnalogInput	mgPtotal200Min
1253	0_AnalogInput	mgQtotal200Min
1254	0_AnalogInput	mgStotal200Min
1255	0_AnalogInput	mgVan200Max
1256	0_AnalogInput	mgVbn200Max
1257	0_AnalogInput	mgVcn200Max
1258	0_AnalogInput	mgVng200Max
1259	0_AnalogInput	mgVab200Max
1260	0_AnalogInput	mgVbc200Max
1261	0_AnalogInput	mgVca200Max
1262	0_AnalogInput	mgIa200Max
1263	0_AnalogInput	mgIb200Max
1264	0_AnalogInput	mgIc200Max
1265	0_AnalogInput	mgIn200Max
1266	0_AnalogInput	mgIg200Max
1267	0_AnalogInput	mgPtotal200Max
1268	0_AnalogInput	mgQtotal200Max
1269	0_AnalogInput	mgStotal200Max
1270	0_AnalogInput	mDIPSW
1271	0_AnalogInput	mDataLink0Address
1272	0_AnalogInput	mDataLink1Address
1273	0_AnalogInput	mDataLink2Address

Object ID	Object Type	Name
1274	0_AnalogInput	mPfA200ms
1275	0_AnalogInput	mPfB200ms
1276	0_AnalogInput	mPfC200ms
1277	0_AnalogInput	mPf200ms
1278	0_AnalogInput	mForwardWsec
1279	0_AnalogInput	mReverseWsec
1280	0_AnalogInput	mNetWsec
1281	0_AnalogInput	mTotalWsec
1282	0_AnalogInput	mVARsecDelivered
1283	0_AnalogInput	mVARsecReceived
1284	0_AnalogInput	mNetVARsec
1285	0_AnalogInput	mTotalVARsec
1286	0_AnalogInput	mVAsecForward
1287	0_AnalogInput	mVAsecReverse
1288	0_AnalogInput	mVAsecNet
1289	0_AnalogInput	mVAsecTotal
1290	0_AnalogInput	mForwardEnergy
1291	0_AnalogInput	mReverseEnergy
1292	0_AnalogInput	mNetEnergy
1293	0_AnalogInput	mTotalEnergy
1294	0_AnalogInput	mVARhReceived
1295	0_AnalogInput	mVARhDelivered
1296	0_AnalogInput	mNetVARh
1297	0_AnalogInput	mTotalVARh
1298	0_AnalogInput	mkVAhForward
1299	0_AnalogInput	mkVAhReverse
1300	0_AnalogInput	mkVAhNet
1301	0_AnalogInput	mkVAhTotal
1302	0_AnalogInput	mCFa
1303	0_AnalogInput	mCFb
1304	0_AnalogInput	mCFc
1305	0_AnalogInput	mCFsys
1306	0_AnalogInput	mIFa
1307	0_AnalogInput	mIFb
1308	0_AnalogInput	mIFc
1309	0_AnalogInput	mIFsys
1310	0_AnalogInput	mPinstVan
1311	0_AnalogInput	mPinstVbn
1312	0_AnalogInput	mPinstVcn
1313	0_AnalogInput	mPinstVab
1314	0_AnalogInput	mPinstVbc

Object ID	Object Type	Name
1315	0_AnalogInput	mPinstVca
1316	0_AnalogInput	mPstVan
1317	0_AnalogInput	mPstVab
1318	0_AnalogInput	mPstVbn
1319	0_AnalogInput	mPstVbc
1320	0_AnalogInput	mPstVcn
1321	0_AnalogInput	mPstVca
1322	0_AnalogInput	mKFa
1323	0_AnalogInput	mKFb
1324	0_AnalogInput	mKFc
1325	0_AnalogInput	mKFsys
1326	0_AnalogInput	mTotalHarmonicIa
1327	0_AnalogInput	mTotalHarmonicIb
1328	0_AnalogInput	mTotalHarmonicIc
1329	0_AnalogInput	mTotalHarmonicVab
1330	0_AnalogInput	mTotalHarmonicVbc
1331	0_AnalogInput	mTotalHarmonicVca
1332	0_AnalogInput	mTotalHarmonicVan
1333	0_AnalogInput	mTotalHarmonicVbn
1334	0_AnalogInput	mTotalHarmonicVcn
1335	0_AnalogInput	mTotalHarmonicIn
1336	0_AnalogInput	mTotalHarmonicIaPct
1337	0_AnalogInput	mTotalHarmonicIbPct
1338	0_AnalogInput	mTotalHarmonicIcPct
1339	0_AnalogInput	mTotalHarmonicVabPct
1340	0_AnalogInput	mTotalHarmonicVbcPct
1341	0_AnalogInput	mTotalHarmonicVcaPct
1342	0_AnalogInput	mTotalHarmonicVanPct
1343	0_AnalogInput	mTotalHarmonicVbnPct
1344	0_AnalogInput	mTotalHarmonicVcnPct
1345	0_AnalogInput	mTotalHarmonicInPct
1346	0_AnalogInput	mTDDIaPct
1347	0_AnalogInput	mTDDIbPct
1348	0_AnalogInput	mTDDIcPct
1349	0_AnalogInput	mTDDInPct
1350	0_AnalogInput	mEvenHarmonicIa
1351	0_AnalogInput	mEvenHarmonicIb
1352	0_AnalogInput	mEvenHarmonicIc
1353	0_AnalogInput	mEvenHarmonicVab
1354	0_AnalogInput	mEvenHarmonicVbc
1355	0_AnalogInput	mEvenHarmonicVca

Object ID	Object Type	Name
1356	0_AnalogInput	mEvenHarmonicVan
1357	0_AnalogInput	mEvenHarmonicVbn
1358	0_AnalogInput	mEvenHarmonicVcn
1359	0_AnalogInput	mEvenHarmonicIn
1360	0_AnalogInput	mEvenHarmonicIaPct
1361	0_AnalogInput	mEvenHarmonicIbPct
1362	0_AnalogInput	mEvenHarmonicIcPct
1363	0_AnalogInput	mEvenHarmonicVabPct
1364	0_AnalogInput	mEvenHarmonicVbcPct
1365	0_AnalogInput	mEvenHarmonicVcaPct
1366	0_AnalogInput	mEvenHarmonicVanPct
1367	0_AnalogInput	mEvenHarmonicVbnPct
1368	0_AnalogInput	mEvenHarmonicVcnPct
1369	0_AnalogInput	mEvenHarmonicInPct
1370	0_AnalogInput	mEvenTDDIaPct
1371	0_AnalogInput	mEvenTDDIbPct
1372	0_AnalogInput	mEvenTDDIcPct
1373	0_AnalogInput	mEvenTDDInPct
1374	0_AnalogInput	mOddHarmonicIa
1375	0_AnalogInput	mOddHarmonicIb
1376	0_AnalogInput	mOddHarmonicIc
1377	0_AnalogInput	mOddHarmonicVab
1378	0_AnalogInput	mOddHarmonicVbc
1379	0_AnalogInput	mOddHarmonicVca
1380	0_AnalogInput	mOddHarmonicVan
1381	0_AnalogInput	mOddHarmonicVbn
1382	0_AnalogInput	mOddHarmonicVcn
1383	0_AnalogInput	mOddHarmonicIn
1384	0_AnalogInput	mOddHarmonicIaPct
1385	0_AnalogInput	mOddHarmonicIbPct
1386	0_AnalogInput	mOddHarmonicIcPct
1387	0_AnalogInput	mOddHarmonicVabPct
1388	0_AnalogInput	mOddHarmonicVbcPct
1389	0_AnalogInput	mOddHarmonicVcaPct
1390	0_AnalogInput	mOddHarmonicVanPct
1391	0_AnalogInput	mOddHarmonicVbnPct
1392	0_AnalogInput	mOddHarmonicVcnPct
1393	0_AnalogInput	mOddHarmonicInPct
1394	0_AnalogInput	mOddTDDIaPct
1395	0_AnalogInput	mOddTDDIbPct
1396	0_AnalogInput	mOddTDDIcPct

Object ID	Object Type	Name
1397	0_AnalogInput	mOddTDDInPct
1398	0_AnalogInput	mInterHarmonicIa
1399	0_AnalogInput	mInterHarmonicIb
1400	0_AnalogInput	mInterHarmonicIc
1401	0_AnalogInput	mInterHarmonicVab
1402	0_AnalogInput	mInterHarmonicVbc
1403	0_AnalogInput	mInterHarmonicVca
1404	0_AnalogInput	mInterHarmonicVan
1405	0_AnalogInput	mInterHarmonicVbn
1406	0_AnalogInput	mInterHarmonicVcn
1407	0_AnalogInput	mInterHarmonicIn
1408	0_AnalogInput	mInterHarmonicIaPct
1409	0_AnalogInput	mInterHarmonicIbPct
1410	0_AnalogInput	mInterHarmonicIcPct
1411	0_AnalogInput	mInterHarmonicVabPct
1412	0_AnalogInput	mInterHarmonicVbcPct
1413	0_AnalogInput	mInterHarmonicVcaPct
1414	0_AnalogInput	mInterHarmonicVanPct
1415	0_AnalogInput	mInterHarmonicVbnPct
1416	0_AnalogInput	mInterHarmonicVcnPct
1417	0_AnalogInput	mInterHarmonicInPct
1418	0_AnalogInput	mInterTDDIaPct
1419	0_AnalogInput	mInterTDDIbPct
1420	0_AnalogInput	mInterTDDIcPct
1421	0_AnalogInput	mInterTDDInPct
1422	0_AnalogInput	mDPFa
1423	0_AnalogInput	mDPFb
1424	0_AnalogInput	mDPFc
1425	0_AnalogInput	mdpf200ms
1426	0_AnalogInput	mSa
1427	0_AnalogInput	mSb
1428	0_AnalogInput	mSc
1429	0_AnalogInput	mStotal
1430	0_AnalogInput	mQa
1431	0_AnalogInput	mQb
1432	0_AnalogInput	mQc
1433	0_AnalogInput	mQtotal
1434	0_AnalogInput	mPa
1435	0_AnalogInput	mPb
1436	0_AnalogInput	mPc
1437	0_AnalogInput	mPtotal

Object ID	Object Type	Name
1438	0_AnalogInput	mAngleIa
1439	0_AnalogInput	mAngleIb
1440	0_AnalogInput	mAngleIc
1441	0_AnalogInput	mAngleVbn
1442	0_AnalogInput	mAngleVcn
1443	0_AnalogInput	mAngleVab
1444	0_AnalogInput	mAngleVbc
1445	0_AnalogInput	mAngleVca
1446	0_AnalogInput	mPosSeqI200ms
1447	0_AnalogInput	mNegSeqI200ms
1448	0_AnalogInput	mZeroSeqI200ms
1449	0_AnalogInput	mPosSeqV200ms
1450	0_AnalogInput	mNegSeqV200ms
1451	0_AnalogInput	mZeroSeqV200ms
1452	0_AnalogInput	mUnderDevVan
1453	0_AnalogInput	mUnderDevVbn
1454	0_AnalogInput	mUnderDevVcn
1455	0_AnalogInput	mUnderDevVab
1456	0_AnalogInput	mUnderDevVbc
1457	0_AnalogInput	mUnderDevVca
1458	0_AnalogInput	mOverDevVan
1459	0_AnalogInput	mOverDevVbn
1460	0_AnalogInput	mOverDevVcn
1461	0_AnalogInput	mOverDevVab
1462	0_AnalogInput	mOverDevVbc
1463	0_AnalogInput	mVan200MinReg
1464	0_AnalogInput	mVbn200MinReg
1465	0_AnalogInput	mVcn200MinReg
1466	0_AnalogInput	mVlnavg200MinReg
1467	0_AnalogInput	mVng200MinReg
1468	0_AnalogInput	mVab200MinReg
1469	0_AnalogInput	mVbc200MinReg
1470	0_AnalogInput	mVca200MinReg
1471	0_AnalogInput	mVllavg200MinReg
1472	0_AnalogInput	mIa200MinReg
1473	0_AnalogInput	mIb200MinReg
1474	0_AnalogInput	mIc200MinReg
1475	0_AnalogInput	mIn200MinReg
1476	0_AnalogInput	mIg200MinReg
1477	0_AnalogInput	mFsysMinReg
1478	0_AnalogInput	mPsysMinReg

Object ID	Object Type	Name
1479	0_AnalogInput	mPaMinReg
1480	0_AnalogInput	mPbMinReg
1481	0_AnalogInput	mPcMinReg
1482	0_AnalogInput	mQsysMinReg
1483	0_AnalogInput	mQaMinReg
1484	0_AnalogInput	mQbMinReg
1485	0_AnalogInput	mQcMinReg
1486	0_AnalogInput	mSsysMinReg
1487	0_AnalogInput	mSaMinReg
1488	0_AnalogInput	mSbMinReg
1489	0_AnalogInput	mScMinReg
1490	0_AnalogInput	mAPFaMinReg
1491	0_AnalogInput	mAPFbMinReg
1492	0_AnalogInput	mAPFcMinReg
1493	0_AnalogInput	mAPFsysMinReg
1494	0_AnalogInput	mDPFaMinReg
1495	0_AnalogInput	mDPFbMinReg
1496	0_AnalogInput	mDPFcMinReg
1497	0_AnalogInput	mDPFsysMinReg
1498	0_AnalogInput	mFlickerPerceptVanMinReg
1499	0_AnalogInput	mFlickerPerceptVbnMinReg
1500	0_AnalogInput	mFlickerPerceptVcnMinReg
1501	0_AnalogInput	mFlickerPerceptVabMinReg
1502	0_AnalogInput	mFlickerPerceptVbcMinReg
1503	0_AnalogInput	mFlickerPerceptVcaMinReg
1504	0_AnalogInput	mPstVanMinReg
1505	0_AnalogInput	mPstVabMinReg
1506	0_AnalogInput	mPstVbnMinReg
1507	0_AnalogInput	mPstVbcMinReg
1508	0_AnalogInput	mPstVcnMinReg
1509	0_AnalogInput	mPstVcaMinReg
1510	0_AnalogInput	mCFaMinReg
1511	0_AnalogInput	mCFbMinReg
1512	0_AnalogInput	mCFcMinReg
1513	0_AnalogInput	mCFsysMinReg
1514	0_AnalogInput	mKFaMinReg
1515	0_AnalogInput	mKFbMinReg
1516	0_AnalogInput	mKFcMinReg
1517	0_AnalogInput	mKFsysMinReg
1518	0_AnalogInput	mTotalHarmonicIaMinReg
1519	0_AnalogInput	mTotalHarmonicIbMinReg

Object ID	Object Type	Name
1520	0_AnalogInput	mTotalHarmonicIcMinReg
1521	0_AnalogInput	mTotalHarmonicVabMinReg
1522	0_AnalogInput	mTotalHarmonicVbcMinReg
1523	0_AnalogInput	mTotalHarmonicVcaMinReg
1524	0_AnalogInput	mTotalHarmonicVanMinReg
1525	0_AnalogInput	mTotalHarmonicVbnMinReg
1526	0_AnalogInput	mTotalHarmonicVcnMinReg
1527	0_AnalogInput	mTotalHarmonicInMinReg
1528	0_AnalogInput	mTotalHarmonicIaPctMinReg
1529	0_AnalogInput	mTotalHarmonicIbPctMinReg
1530	0_AnalogInput	mTotalHarmonicIcPctMinReg
1531	0_AnalogInput	mTotalHarmonicVabPctMinReg
1532	0_AnalogInput	mTotalHarmonicVbcPctMinReg
1533	0_AnalogInput	mTotalHarmonicVcaPctMinReg
1534	0_AnalogInput	mTotalHarmonicVanPctMinReg
1535	0_AnalogInput	mTotalHarmonicVbnPctMinReg
1536	0_AnalogInput	mTotalHarmonicVcnPctMinReg
1537	0_AnalogInput	mTotalHarmonicInPctMinReg
1538	0_AnalogInput	mTDDIaPctMinReg
1539	0_AnalogInput	mTDDIbPctMinReg
1540	0_AnalogInput	mTDDIcPctMinReg
1541	0_AnalogInput	mTDDInPctMinReg
1542	0_AnalogInput	mEvenHarmonicIaMinReg
1543	0_AnalogInput	mEvenHarmonicIbMinReg
1544	0_AnalogInput	mEvenHarmonicIcMinReg
1545	0_AnalogInput	mEvenHarmonicVabMinReg
1546	0_AnalogInput	mEvenHarmonicVbcMinReg
1547	0_AnalogInput	mEvenHarmonicVcaMinReg
1548	0_AnalogInput	mEvenHarmonicVanMinReg
1549	0_AnalogInput	mEvenHarmonicVbnMinReg
1550	0_AnalogInput	mEvenHarmonicVcnMinReg
1551	0_AnalogInput	mEvenHarmonicInMinReg
1552	0_AnalogInput	mEvenHarmonicIaPctMinReg
1553	0_AnalogInput	mEvenHarmonicIbPctMinReg
1554	0_AnalogInput	mEvenHarmonicIcPctMinReg
1555	0_AnalogInput	mEvenHarmonicVabPctMinReg
1556	0_AnalogInput	mEvenHarmonicVbcPctMinReg
1557	0_AnalogInput	mEvenHarmonicVcaPctMinReg
1558	0_AnalogInput	mEvenHarmonicVanPctMinReg
1559	0_AnalogInput	mEvenHarmonicVbnPctMinReg
1560	0_AnalogInput	mEvenHarmonicVcnPctMinReg

Object ID	Object Type	Name
1561	0_AnalogInput	mEvenHarmonicInPctMinReg
1562	0_AnalogInput	mEvenTDDIaPctMinReg
1563	0_AnalogInput	mEvenTDDIbPctMinReg
1564	0_AnalogInput	mEvenTDDIcPctMinReg
1565	0_AnalogInput	mEvenTDDInPctMinReg
1566	0_AnalogInput	mOddHarmonicIaMinReg
1567	0_AnalogInput	mOddHarmonicIbMinReg
1568	0_AnalogInput	mOddHarmonicIcMinReg
1569	0_AnalogInput	mOddHarmonicVabMinReg
1570	0_AnalogInput	mOddHarmonicVbcMinReg
1571	0_AnalogInput	mOddHarmonicVcaMinReg
1572	0_AnalogInput	mOddHarmonicVanMinReg
1573	0_AnalogInput	mOddHarmonicVbnMinReg
1574	0_AnalogInput	mOddHarmonicVcnMinReg
1575	0_AnalogInput	mOddHarmonicInMinReg
1576	0_AnalogInput	mOddHarmonicIaPctMinReg
1577	0_AnalogInput	mOddHarmonicIbPctMinReg
1578	0_AnalogInput	mOddHarmonicIcPctMinReg
1579	0_AnalogInput	mOddHarmonicVabPctMinReg
1580	0_AnalogInput	mOddHarmonicVbcPctMinReg
1581	0_AnalogInput	mOddHarmonicVcaPctMinReg
1582	0_AnalogInput	mOddHarmonicVanPctMinReg
1583	0_AnalogInput	mOddHarmonicVbnPctMinReg
1584	0_AnalogInput	mOddHarmonicVcnPctMinReg
1585	0_AnalogInput	mOddHarmonicInPctMinReg
1586	0_AnalogInput	mOddTDDIaPctMinReg
1587	0_AnalogInput	mOddTDDIbPctMinReg
1588	0_AnalogInput	mOddTDDIcPctMinReg
1589	0_AnalogInput	mOddTDDInPctMinReg
1590	0_AnalogInput	mInterHarmonicIaMinReg
1591	0_AnalogInput	mInterHarmonicIbMinReg
1592	0_AnalogInput	mInterHarmonicIcMinReg
1593	0_AnalogInput	mInterHarmonicVabMinReg
1594	0_AnalogInput	mInterHarmonicVbcMinReg
1595	0_AnalogInput	mInterHarmonicVcaMinReg
1596	0_AnalogInput	mInterHarmonicVanMinReg
1597	0_AnalogInput	mInterHarmonicVbnMinReg
1598	0_AnalogInput	mInterHarmonicVcnMinReg
1599	0_AnalogInput	mInterHarmonicInMinReg
1600	0_AnalogInput	mInterHarmonicIaPctMinReg
1601	0_AnalogInput	mInterHarmonicIbPctMinReg

Object ID	Object Type	Name
1602	0_AnalogInput	mInterHarmonicIcPctMinReg
1603	0_AnalogInput	mInterHarmonicVabPctMinReg
1604	0_AnalogInput	mInterHarmonicVbcPctMinReg
1605	0_AnalogInput	mInterHarmonicVcaPctMinReg
1606	0_AnalogInput	mInterHarmonicVanPctMinReg
1607	0_AnalogInput	mInterHarmonicVbnPctMinReg
1608	0_AnalogInput	mInterHarmonicVcnPctMinReg
1609	0_AnalogInput	mInterHarmonicInPctMinReg
1610	0_AnalogInput	mInterTDDIaPctMinReg
1611	0_AnalogInput	mInterTDDIbPctMinReg
1612	0_AnalogInput	mInterTDDIcPctMinReg
1613	0_AnalogInput	mInterTDDInPctMinReg
1614	0_AnalogInput	mSelectHarmonicMag1MinReg
1615	0_AnalogInput	mSelectHarmonicMag2MinReg
1616	0_AnalogInput	mSelectHarmonicMag3MinReg
1617	0_AnalogInput	mSelectHarmonicMag4MinReg
1618	0_AnalogInput	mSelectHarmonicMag5MinReg
1619	0_AnalogInput	mSelectHarmonicMag6MinReg
1620	0_AnalogInput	mSelectHarmonicMag7MinReg
1621	0_AnalogInput	mSelectHarmonicMag8MinReg
1622	0_AnalogInput	mSelectHarmonicMag9MinReg
1623	0_AnalogInput	mSelectHarmonicMag10MinReg
1624	0_AnalogInput	mSelectHarmonicMag11MinReg
1625	0_AnalogInput	mSelectHarmonicMag12MinReg
1626	0_AnalogInput	mSelectHarmonicMag13MinReg
1627	0_AnalogInput	mSelectHarmonicMag14MinReg
1628	0_AnalogInput	mSelectHarmonicMag15MinReg
1629	0_AnalogInput	mSelectHarmonicMag16MinReg
1630	0_AnalogInput	mSelectHarmonicMag17MinReg
1631	0_AnalogInput	mSelectHarmonicMag18MinReg
1632	0_AnalogInput	mSelectHarmonicMag19MinReg
1633	0_AnalogInput	mSelectHarmonicMag20MinReg
1634	0_AnalogInput	mSelectHarmonicMag21MinReg
1635	0_AnalogInput	mSelectHarmonicMag22MinReg
1636	0_AnalogInput	mSelectHarmonicMag23MinReg
1637	0_AnalogInput	mSelectHarmonicMag24MinReg
1638	0_AnalogInput	mSelectHarmonicMag25MinReg
1639	0_AnalogInput	mSelectHarmonicMag26MinReg
1640	0_AnalogInput	mSelectHarmonicMag27MinReg
1641	0_AnalogInput	mSelectHarmonicMag28MinReg
1642	0_AnalogInput	mSelectHarmonicMag29MinReg

Object ID	Object Type	Name
1643	0_AnalogInput	mSelectHarmonicMag30MinReg
1644	0_AnalogInput	mSelectHarmonicMag31MinReg
1645	0_AnalogInput	mSelectHarmonicMag32MinReg
1646	0_AnalogInput	mSelectHarmonicMag33MinReg
1647	0_AnalogInput	mSelectHarmonicMag34MinReg
1648	0_AnalogInput	mSelectHarmonicMag35MinReg
1649	0_AnalogInput	mSelectHarmonicMag36MinReg
1650	0_AnalogInput	mSelectHarmonicMag37MinReg
1651	0_AnalogInput	mSelectHarmonicMag38MinReg
1652	0_AnalogInput	mSelectHarmonicMag39MinReg
1653	0_AnalogInput	mSelectHarmonicMag40MinReg
1654	0_AnalogInput	mSelectHarmonicMag41MinReg
1655	0_AnalogInput	mSelectHarmonicMag42MinReg
1656	0_AnalogInput	mSelectHarmonicMag43MinReg
1657	0_AnalogInput	mSelectHarmonicMag44MinReg
1658	0_AnalogInput	mSelectHarmonicMag45MinReg
1659	0_AnalogInput	mSelectHarmonicMag46MinReg
1660	0_AnalogInput	mSelectHarmonicMag47MinReg
1661	0_AnalogInput	mSelectHarmonicMag48MinReg
1662	0_AnalogInput	mSelectHarmonicMag49MinReg
1663	0_AnalogInput	mSelectHarmonicMag50MinReg
1664	0_AnalogInput	mVan200MinRegTime
1665	0_AnalogInput	mVbn200MinRegTime
1666	0_AnalogInput	mVcn200MinRegTime
1667	0_AnalogInput	mVlnavg200MinRegTime
1668	0_AnalogInput	mVng200MinRegTime
1669	0_AnalogInput	mVab200MinRegTime
1670	0_AnalogInput	mVbc200MinRegTime
1671	0_AnalogInput	mVca200MinRegTime
1672	0_AnalogInput	mVllavg200MinRegTime
1673	0_AnalogInput	mIa200MinRegTime
1674	0_AnalogInput	mIb200MinRegTime
1675	0_AnalogInput	mIc200MinRegTime
1676	0_AnalogInput	mIn200MinRegTime
1677	0_AnalogInput	mIg200MinRegTime
1678	0_AnalogInput	mFsysMinRegTime
1679	0_AnalogInput	mPsysMinRegTime
1680	0_AnalogInput	mPaMinRegTime
1681	0_AnalogInput	mPbMinRegTime
1682	0_AnalogInput	mPcMinRegTime
1683	0_AnalogInput	mQsysMinRegTime

Object ID	Object Type	Name
1684	0_AnalogInput	mQaMinRegTime
1685	0_AnalogInput	mQbMinRegTime
1686	0_AnalogInput	mQcMinRegTime
1687	0_AnalogInput	mSsysMinRegTime
1688	0_AnalogInput	mSaMinRegTime
1689	0_AnalogInput	mSbMinRegTime
1690	0_AnalogInput	mScMinRegTime
1691	0_AnalogInput	mAPFaMinRegTime
1692	0_AnalogInput	mAPFbMinRegTime
1693	0_AnalogInput	mAPFcMinRegTime
1694	0_AnalogInput	mAPFsysMinRegTime
1695	0_AnalogInput	mDPFaMinRegTime
1696	0_AnalogInput	mDPFbMinRegTime
1697	0_AnalogInput	mDPFcMinRegTime
1698	0_AnalogInput	mDPFsysMinRegTime
1699	0_AnalogInput	mFlickerPerceptVanMinRegTime
1700	0_AnalogInput	mFlickerPerceptVbnMinRegTime
1701	0_AnalogInput	mFlickerPerceptVcnMinRegTime
1702	0_AnalogInput	mFlickerPerceptVabMinRegTime
1703	0_AnalogInput	mFlickerPerceptVbcMinRegTime
1704	0_AnalogInput	mFlickerPerceptVcaMinRegTime
1705	0_AnalogInput	mPstVanMinRegTime
1706	0_AnalogInput	mPstVabMinRegTime
1707	0_AnalogInput	mPstVbnMinRegTime
1708	0_AnalogInput	mPstVbcMinRegTime
1709	0_AnalogInput	mPstVcnMinRegTime
1710	0_AnalogInput	mPstVcaMinRegTime
1711	0_AnalogInput	mCFaMinRegTime
1712	0_AnalogInput	mCFbMinRegTime
1713	0_AnalogInput	mCFcMinRegTime
1714	0_AnalogInput	mCFsysMinRegTime
1715	0_AnalogInput	mKFaMinRegTime
1716	0_AnalogInput	mKFbMinRegTime
1717	0_AnalogInput	mKFcMinRegTime
1718	0_AnalogInput	mKFsysMinRegTime
1719	0_AnalogInput	mTotalHarmonicIaMinRegTime
1720	0_AnalogInput	mTotalHarmonicIbMinRegTime
1721	0_AnalogInput	mTotalHarmonicIcMinRegTime
1722	0_AnalogInput	mTotalHarmonicVabMinRegTime
1723	0_AnalogInput	mTotalHarmonicVbcMinRegTime
1724	0_AnalogInput	mTotalHarmonicVcaMinRegTime

Object ID	Object Type	Name
1725	0_AnalogInput	mTotalHarmonicVanMinRegTime
1726	0_AnalogInput	mTotalHarmonicVbnMinRegTime
1727	0_AnalogInput	mTotalHarmonicVcnMinRegTime
1728	0_AnalogInput	mTotalHarmonicInMinRegTime
1729	0_AnalogInput	mTotalHarmonicIaPctMinRegTime
1730	0_AnalogInput	mTotalHarmonicIbPctMinRegTime
1731	0_AnalogInput	mTotalHarmonicIcPctMinRegTime
1732	0_AnalogInput	mTotalHarmonicVabPctMinRegTime
1733	0_AnalogInput	mTotalHarmonicVbcPctMinRegTime
1734	0_AnalogInput	mTotalHarmonicVcaPctMinRegTime
1735	0_AnalogInput	mTotalHarmonicVanPctMinRegTime
1736	0_AnalogInput	mTotalHarmonicVbnPctMinRegTime
1737	0_AnalogInput	mTotalHarmonicVcnPctMinRegTime
1738	0_AnalogInput	mTotalHarmonicInPctMinRegTime
1739	0_AnalogInput	mTDDIaPctMinRegTime
1740	0_AnalogInput	mTDDIbPctMinRegTime
1741	0_AnalogInput	mTDDIcPctMinRegTime
1742	0_AnalogInput	mTDDInPctMinRegTime
1743	0_AnalogInput	mEvenHarmonicIaMinRegTime
1744	0_AnalogInput	mEvenHarmonicIbMinRegTime
1745	0_AnalogInput	mEvenHarmonicIcMinRegTime
1746	0_AnalogInput	mEvenHarmonicVabMinRegTime
1747	0_AnalogInput	mEvenHarmonicVbcMinRegTime
1748	0_AnalogInput	mEvenHarmonicVcaMinRegTime
1749	0_AnalogInput	mEvenHarmonicVanMinRegTime
1750	0_AnalogInput	mEvenHarmonicVbnMinRegTime
1751	0_AnalogInput	mEvenHarmonicVcnMinRegTime
1752	0_AnalogInput	mEvenHarmonicInMinRegTime
1753	0_AnalogInput	mEvenHarmonicIaPctMinRegTime
1754	0_AnalogInput	mEvenHarmonicIbPctMinRegTime
1755	0_AnalogInput	mEvenHarmonicIcPctMinRegTime
1756	0_AnalogInput	mEvenHarmonicVabPctMinRegTime
1757	0_AnalogInput	mEvenHarmonicVbcPctMinRegTime
1758	0_AnalogInput	mEvenHarmonicVcaPctMinRegTime
1759	0_AnalogInput	mEvenHarmonicVanPctMinRegTime
1760	0_AnalogInput	mEvenHarmonicVbnPctMinRegTime
1761	0_AnalogInput	mEvenHarmonicVcnPctMinRegTime
1762	0_AnalogInput	mEvenHarmonicInPctMinRegTime
1763	0_AnalogInput	mEvenTDDIaPctMinRegTime
1764	0_AnalogInput	mEvenTDDIbPctMinRegTime
1765	0_AnalogInput	mEvenTDDIcPctMinRegTime

Object ID	Object Type	Name
1766	0_AnalogInput	mEvenTDDInPctMinRegTime
1767	0_AnalogInput	mOddHarmonicIaMinRegTime
1768	0_AnalogInput	mOddHarmonicIbMinRegTime
1769	0_AnalogInput	mOddHarmonicIcMinRegTime
1770	0_AnalogInput	mOddHarmonicVabMinRegTime
1771	0_AnalogInput	mOddHarmonicVbcMinRegTime
1772	0_AnalogInput	mOddHarmonicVcaMinRegTime
1773	0_AnalogInput	mOddHarmonicVanMinRegTime
1774	0_AnalogInput	mOddHarmonicVbnMinRegTime
1775	0_AnalogInput	mOddHarmonicVcnMinRegTime
1776	0_AnalogInput	mOddHarmonicInMinRegTime
1777	0_AnalogInput	mOddHarmonicIaPctMinRegTime
1778	0_AnalogInput	mOddHarmonicIbPctMinRegTime
1779	0_AnalogInput	mOddHarmonicIcPctMinRegTime
1780	0_AnalogInput	mOddHarmonicVabPctMinRegTime
1781	0_AnalogInput	mOddHarmonicVbcPctMinRegTime
1782	0_AnalogInput	mOddHarmonicVcaPctMinRegTime
1783	0_AnalogInput	mOddHarmonicVanPctMinRegTime
1784	0_AnalogInput	mOddHarmonicVbnPctMinRegTime
1785	0_AnalogInput	mOddHarmonicVcnPctMinRegTime
1786	0_AnalogInput	mOddHarmonicInPctMinRegTime
1787	0_AnalogInput	mOddTDDIaPctMinRegTime
1788	0_AnalogInput	mOddTDDIbPctMinRegTime
1789	0_AnalogInput	mOddTDDIcPctMinRegTime
1790	0_AnalogInput	mOddTDDInPctMinRegTime
1791	0_AnalogInput	mInterHarmonicIaMinRegTime
1792	0_AnalogInput	mInterHarmonicIbMinRegTime
1793	0_AnalogInput	mInterHarmonicIcMinRegTime
1794	0_AnalogInput	mInterHarmonicVabMinRegTime
1795	0_AnalogInput	mInterHarmonicVbcMinRegTime
1796	0_AnalogInput	mInterHarmonicVcaMinRegTime
1797	0_AnalogInput	mInterHarmonicVanMinRegTime
1798	0_AnalogInput	mInterHarmonicVbnMinRegTime
1799	0_AnalogInput	mInterHarmonicVcnMinRegTime
1800	0_AnalogInput	mInterHarmonicInMinRegTime
1801	0_AnalogInput	mInterHarmonicIaPctMinRegTime
1802	0_AnalogInput	mInterHarmonicIbPctMinRegTime
1803	0_AnalogInput	mInterHarmonicIcPctMinRegTime
1804	0_AnalogInput	mInterHarmonicVabPctMinRegTime
1805	0_AnalogInput	mInterHarmonicVbcPctMinRegTime
1806	0_AnalogInput	mInterHarmonicVcaPctMinRegTime

Object ID	Object Type	Name
1807	0_AnalogInput	mInterHarmonicVanPctMinRegTime
1808	0_AnalogInput	mInterHarmonicVbnPctMinRegTime
1809	0_AnalogInput	mInterHarmonicVcnPctMinRegTime
1810	0_AnalogInput	mInterHarmonicInPctMinRegTime
1811	0_AnalogInput	mInterTDDIaPctMinRegTime
1812	0_AnalogInput	mInterTDDIbPctMinRegTime
1813	0_AnalogInput	mInterTDDIcPctMinRegTime
1814	0_AnalogInput	mInterTDDInPctMinRegTime
1815	0_AnalogInput	mSelectHarmonicMag1MinRegTime
1816	0_AnalogInput	mSelectHarmonicMag2MinRegTime
1817	0_AnalogInput	mSelectHarmonicMag3MinRegTime
1818	0_AnalogInput	mSelectHarmonicMag4MinRegTime
1819	0_AnalogInput	mSelectHarmonicMag5MinRegTime
1820	0_AnalogInput	mSelectHarmonicMag6MinRegTime
1821	0_AnalogInput	mSelectHarmonicMag7MinRegTime
1822	0_AnalogInput	mSelectHarmonicMag8MinRegTime
1823	0_AnalogInput	mSelectHarmonicMag9MinRegTime
1824	0_AnalogInput	mSelectHarmonicMag10MinRegTime
1825	0_AnalogInput	mSelectHarmonicMag11MinRegTime
1826	0_AnalogInput	mSelectHarmonicMag12MinRegTime
1827	0_AnalogInput	mSelectHarmonicMag13MinRegTime
1828	0_AnalogInput	mSelectHarmonicMag14MinRegTime
1829	0_AnalogInput	mSelectHarmonicMag15MinRegTime
1830	0_AnalogInput	mSelectHarmonicMag16MinRegTime
1831	0_AnalogInput	mSelectHarmonicMag17MinRegTime
1832	0_AnalogInput	mSelectHarmonicMag18MinRegTime
1833	0_AnalogInput	mSelectHarmonicMag19MinRegTime
1834	0_AnalogInput	mSelectHarmonicMag20MinRegTime
1835	0_AnalogInput	mSelectHarmonicMag21MinRegTime
1836	0_AnalogInput	mSelectHarmonicMag22MinRegTime
1837	0_AnalogInput	mSelectHarmonicMag23MinRegTime
1838	0_AnalogInput	mSelectHarmonicMag24MinRegTime
1839	0_AnalogInput	mSelectHarmonicMag25MinRegTime
1840	0_AnalogInput	mSelectHarmonicMag26MinRegTime
1841	0_AnalogInput	mSelectHarmonicMag27MinRegTime
1842	0_AnalogInput	mSelectHarmonicMag28MinRegTime
1843	0_AnalogInput	mSelectHarmonicMag29MinRegTime
1844	0_AnalogInput	mSelectHarmonicMag30MinRegTime
1845	0_AnalogInput	mSelectHarmonicMag31MinRegTime
1846	0_AnalogInput	mSelectHarmonicMag32MinRegTime
1847	0_AnalogInput	mSelectHarmonicMag33MinRegTime

Object ID	Object Type	Name
1848	0_AnalogInput	mSelectHarmonicMag34MinRegTime
1849	0_AnalogInput	mSelectHarmonicMag35MinRegTime
1850	0_AnalogInput	mSelectHarmonicMag36MinRegTime
1851	0_AnalogInput	mSelectHarmonicMag37MinRegTime
1852	0_AnalogInput	mSelectHarmonicMag38MinRegTime
1853	0_AnalogInput	mSelectHarmonicMag39MinRegTime
1854	0_AnalogInput	mSelectHarmonicMag40MinRegTime
1855	0_AnalogInput	mSelectHarmonicMag41MinRegTime
1856	0_AnalogInput	mSelectHarmonicMag42MinRegTime
1857	0_AnalogInput	mSelectHarmonicMag43MinRegTime
1858	0_AnalogInput	mSelectHarmonicMag44MinRegTime
1859	0_AnalogInput	mSelectHarmonicMag45MinRegTime
1860	0_AnalogInput	mSelectHarmonicMag46MinRegTime
1861	0_AnalogInput	mSelectHarmonicMag47MinRegTime
1862	0_AnalogInput	mSelectHarmonicMag48MinRegTime
1863	0_AnalogInput	mSelectHarmonicMag49MinRegTime
1864	0_AnalogInput	mSelectHarmonicMag50MinRegTime
1865	0_AnalogInput	mVan200MaxReg
1866	0_AnalogInput	mVbn200MaxReg
1867	0_AnalogInput	mVcn200MaxReg
1868	0_AnalogInput	mVlnavg200MaxReg
1869	0_AnalogInput	mVng200MaxReg
1870	0_AnalogInput	mVab200MaxReg
1871	0_AnalogInput	mVbc200MaxReg
1872	0_AnalogInput	mVca200MaxReg
1873	0_AnalogInput	mVllavg200MaxReg
1874	0_AnalogInput	mIa200MaxReg
1875	0_AnalogInput	mIb200MaxReg
1876	0_AnalogInput	mIc200MaxReg
1877	0_AnalogInput	mIn200MaxReg
1878	0_AnalogInput	mIg200MaxReg
1879	0_AnalogInput	mFsysMaxReg
1880	0_AnalogInput	mPsysMaxReg
1881	0_AnalogInput	mPaMaxReg
1882	0_AnalogInput	mPbMaxReg
1883	0_AnalogInput	mPcMaxReg
1884	0_AnalogInput	mQsysMaxReg
1885	0_AnalogInput	mQaMaxReg
1886	0_AnalogInput	mQbMaxReg
1887	0_AnalogInput	mQcMaxReg
1888	0_AnalogInput	mSsysMaxReg

Object ID	Object Type	Name
1889	0_AnalogInput	mSaMaxReg
1890	0_AnalogInput	mSbMaxReg
1891	0_AnalogInput	mScMaxReg
1892	0_AnalogInput	mAPFaMaxReg
1893	0_AnalogInput	mAPFbMaxReg
1894	0_AnalogInput	mAPFcMaxReg
1895	0_AnalogInput	mAPFsysMaxReg
1896	0_AnalogInput	mDPFaMaxReg
1897	0_AnalogInput	mDPFbMaxReg
1898	0_AnalogInput	mDPFcMaxReg
1899	0_AnalogInput	mDPFsysMaxReg
1900	0_AnalogInput	mFlickerPerceptVanMaxReg
1901	0_AnalogInput	mFlickerPerceptVbnMaxReg
1902	0_AnalogInput	mFlickerPerceptVcnMaxReg
1903	0_AnalogInput	mFlickerPerceptVabMaxReg
1904	0_AnalogInput	mFlickerPerceptVbcMaxReg
1905	0_AnalogInput	mFlickerPerceptVcaMaxReg
1906	0_AnalogInput	mPstVanMaxReg
1907	0_AnalogInput	mPstVabMaxReg
1908	0_AnalogInput	mPstVbnMaxReg
1909	0_AnalogInput	mPstVbcMaxReg
1910	0_AnalogInput	mPstVcnMaxReg
1911	0_AnalogInput	mPstVcaMaxReg
1912	0_AnalogInput	mCFaMaxReg
1913	0_AnalogInput	mCFbMaxReg
1914	0_AnalogInput	mCFcMaxReg
1915	0_AnalogInput	mCFsysMaxReg
1916	0_AnalogInput	mKFaMaxReg
1917	0_AnalogInput	mKFbMaxReg
1918	0_AnalogInput	mKFcMaxReg
1919	0_AnalogInput	mKFsysMaxReg
1920	0_AnalogInput	mTotalHarmonicIaMaxReg
1921	0_AnalogInput	mTotalHarmonicIbMaxReg
1922	0_AnalogInput	mTotalHarmonicIcMaxReg
1923	0_AnalogInput	mTotalHarmonicVabMaxReg
1924	0_AnalogInput	mTotalHarmonicVbcMaxReg
1925	0_AnalogInput	mTotalHarmonicVcaMaxReg
1926	0_AnalogInput	mTotalHarmonicVanMaxReg
1927	0_AnalogInput	mTotalHarmonicVbnMaxReg
1928	0_AnalogInput	mTotalHarmonicVcnMaxReg
1929	0_AnalogInput	mTotalHarmonicInMaxReg

Object ID	Object Type	Name
1930	0_AnalogInput	mTotalHarmonicIaPctMaxReg
1931	0_AnalogInput	mTotalHarmonicIbPctMaxReg
1932	0_AnalogInput	mTotalHarmonicIcPctMaxReg
1933	0_AnalogInput	mTotalHarmonicVabPctMaxReg
1934	0_AnalogInput	mTotalHarmonicVbcPctMaxReg
1935	0_AnalogInput	mTotalHarmonicVcaPctMaxReg
1936	0_AnalogInput	mTotalHarmonicVanPctMaxReg
1937	0_AnalogInput	mTotalHarmonicVbnPctMaxReg
1938	0_AnalogInput	mTotalHarmonicVcnPctMaxReg
1939	0_AnalogInput	mTotalHarmonicInPctMaxReg
1940	0_AnalogInput	mTDDIaPctMaxReg
1941	0_AnalogInput	mTDDIbPctMaxReg
1942	0_AnalogInput	mTDDIcPctMaxReg
1943	0_AnalogInput	mTDDInPctMaxReg
1944	0_AnalogInput	mEvenHarmonicIaMaxReg
1945	0_AnalogInput	mEvenHarmonicIbMaxReg
1946	0_AnalogInput	mEvenHarmonicIcMaxReg
1947	0_AnalogInput	mEvenHarmonicVabMaxReg
1948	0_AnalogInput	mEvenHarmonicVbcMaxReg
1949	0_AnalogInput	mEvenHarmonicVcaMaxReg
1950	0_AnalogInput	mEvenHarmonicVanMaxReg
1951	0_AnalogInput	mEvenHarmonicVbnMaxReg
1952	0_AnalogInput	mEvenHarmonicVcnMaxReg
1953	0_AnalogInput	mEvenHarmonicInMaxReg
1954	0_AnalogInput	mEvenHarmonicIaPctMaxReg
1955	0_AnalogInput	mEvenHarmonicIbPctMaxReg
1956	0_AnalogInput	mEvenHarmonicIcPctMaxReg
1957	0_AnalogInput	mEvenHarmonicVabPctMaxReg
1958	0_AnalogInput	mEvenHarmonicVbcPctMaxReg
1959	0_AnalogInput	mEvenHarmonicVcaPctMaxReg
1960	0_AnalogInput	mEvenHarmonicVanPctMaxReg
1961	0_AnalogInput	mEvenHarmonicVbnPctMaxReg
1962	0_AnalogInput	mEvenHarmonicVcnPctMaxReg
1963	0_AnalogInput	mEvenHarmonicInPctMaxReg
1964	0_AnalogInput	mEvenTDDIaPctMaxReg
1965	0_AnalogInput	mEvenTDDIbPctMaxReg
1966	0_AnalogInput	mEvenTDDIcPctMaxReg
1967	0_AnalogInput	mEvenTDDInPctMaxReg
1968	0_AnalogInput	mOddHarmonicIaMaxReg
1969	0_AnalogInput	mOddHarmonicIbMaxReg
1970	0_AnalogInput	mOddHarmonicIcMaxReg

Object ID	Object Type	Name
1971	0_AnalogInput	mOddHarmonicVabMaxReg
1972	0_AnalogInput	mOddHarmonicVbcMaxReg
1973	0_AnalogInput	mOddHarmonicVcaMaxReg
1974	0_AnalogInput	mOddHarmonicVanMaxReg
1975	0_AnalogInput	mOddHarmonicVbnMaxReg
1976	0_AnalogInput	mOddHarmonicVcnMaxReg
1977	0_AnalogInput	mOddHarmonicInMaxReg
1978	0_AnalogInput	mOddHarmonicIaPctMaxReg
1979	0_AnalogInput	mOddHarmonicIbPctMaxReg
1980	0_AnalogInput	mOddHarmonicIcPctMaxReg
1981	0_AnalogInput	mOddHarmonicVabPctMaxReg
1982	0_AnalogInput	mOddHarmonicVbcPctMaxReg
1983	0_AnalogInput	mOddHarmonicVcaPctMaxReg
1984	0_AnalogInput	mOddHarmonicVanPctMaxReg
1985	0_AnalogInput	mOddHarmonicVbnPctMaxReg
1986	0_AnalogInput	mOddHarmonicVcnPctMaxReg
1987	0_AnalogInput	mOddHarmonicInPctMaxReg
1988	0_AnalogInput	mOddTDDIaPctMaxReg
1989	0_AnalogInput	mOddTDDIbPctMaxReg
1990	0_AnalogInput	mOddTDDIcPctMaxReg
1991	0_AnalogInput	mOddTDDInPctMaxReg
1992	0_AnalogInput	mInterHarmonicIaMaxReg
1993	0_AnalogInput	mInterHarmonicIbMaxReg
1994	0_AnalogInput	mInterHarmonicIcMaxReg
1995	0_AnalogInput	mInterHarmonicVabMaxReg
1996	0_AnalogInput	mInterHarmonicVbcMaxReg
1997	0_AnalogInput	mInterHarmonicVcaMaxReg
1998	0_AnalogInput	mInterHarmonicVanMaxReg
1999	0_AnalogInput	mInterHarmonicVbnMaxReg
2000	0_AnalogInput	mInterHarmonicVcnMaxReg
2001	0_AnalogInput	mInterHarmonicInMaxReg
2002	0_AnalogInput	mInterHarmonicIaPctMaxReg
2003	0_AnalogInput	mInterHarmonicIbPctMaxReg
2004	0_AnalogInput	mInterHarmonicIcPctMaxReg
2005	0_AnalogInput	mInterHarmonicVabPctMaxReg
2006	0_AnalogInput	mInterHarmonicVbcPctMaxReg
2007	0_AnalogInput	mInterHarmonicVcaPctMaxReg
2008	0_AnalogInput	mInterHarmonicVanPctMaxReg
2009	0_AnalogInput	mInterHarmonicVbnPctMaxReg
2010	0_AnalogInput	mInterHarmonicVcnPctMaxReg
2011	0_AnalogInput	mInterHarmonicInPctMaxReg

Object ID	Object Type	Name
2012	0_AnalogInput	mInterTDDIaPctMaxReg
2013	0_AnalogInput	mInterTDDIbPctMaxReg
2014	0_AnalogInput	mInterTDDIcPctMaxReg
2015	0_AnalogInput	mInterTDDInPctMaxReg
2016	0_AnalogInput	mSelectHarmonicMag1MaxReg
2017	0_AnalogInput	mSelectHarmonicMag2MaxReg
2018	0_AnalogInput	mSelectHarmonicMag3MaxReg
2019	0_AnalogInput	mSelectHarmonicMag4MaxReg
2020	0_AnalogInput	mSelectHarmonicMag5MaxReg
2021	0_AnalogInput	mSelectHarmonicMag6MaxReg
2022	0_AnalogInput	mSelectHarmonicMag7MaxReg
2023	0_AnalogInput	mSelectHarmonicMag8MaxReg
2024	0_AnalogInput	mSelectHarmonicMag9MaxReg
2025	0_AnalogInput	mSelectHarmonicMag10MaxReg
2026	0_AnalogInput	mSelectHarmonicMag11MaxReg
2027	0_AnalogInput	mSelectHarmonicMag12MaxReg
2028	0_AnalogInput	mSelectHarmonicMag13MaxReg
2029	0_AnalogInput	mSelectHarmonicMag14MaxReg
2030	0_AnalogInput	mSelectHarmonicMag15MaxReg
2031	0_AnalogInput	mSelectHarmonicMag16MaxReg
2032	0_AnalogInput	mSelectHarmonicMag17MaxReg
2033	0_AnalogInput	mSelectHarmonicMag18MaxReg
2034	0_AnalogInput	mSelectHarmonicMag19MaxReg
2035	0_AnalogInput	mSelectHarmonicMag20MaxReg
2036	0_AnalogInput	mSelectHarmonicMag21MaxReg
2037	0_AnalogInput	mSelectHarmonicMag22MaxReg
2038	0_AnalogInput	mSelectHarmonicMag23MaxReg
2039	0_AnalogInput	mSelectHarmonicMag24MaxReg
2040	0_AnalogInput	mSelectHarmonicMag25MaxReg
2041	0_AnalogInput	mSelectHarmonicMag26MaxReg
2042	0_AnalogInput	mSelectHarmonicMag27MaxReg
2043	0_AnalogInput	mSelectHarmonicMag28MaxReg
2044	0_AnalogInput	mSelectHarmonicMag29MaxReg
2045	0_AnalogInput	mSelectHarmonicMag30MaxReg
2046	0_AnalogInput	mSelectHarmonicMag31MaxReg
2047	0_AnalogInput	mSelectHarmonicMag32MaxReg
2048	0_AnalogInput	mSelectHarmonicMag33MaxReg
2049	0_AnalogInput	mSelectHarmonicMag34MaxReg
2050	0_AnalogInput	mSelectHarmonicMag35MaxReg
2051	0_AnalogInput	mSelectHarmonicMag36MaxReg
2052	0_AnalogInput	mSelectHarmonicMag37MaxReg

Object ID	Object Type	Name
2053	0_AnalogInput	mSelectHarmonicMag38MaxReg
2054	0_AnalogInput	mSelectHarmonicMag39MaxReg
2055	0_AnalogInput	mSelectHarmonicMag40MaxReg
2056	0_AnalogInput	mSelectHarmonicMag41MaxReg
2057	0_AnalogInput	mSelectHarmonicMag42MaxReg
2058	0_AnalogInput	mSelectHarmonicMag43MaxReg
2059	0_AnalogInput	mSelectHarmonicMag44MaxReg
2060	0_AnalogInput	mSelectHarmonicMag45MaxReg
2061	0_AnalogInput	mSelectHarmonicMag46MaxReg
2062	0_AnalogInput	mSelectHarmonicMag47MaxReg
2063	0_AnalogInput	mSelectHarmonicMag48MaxReg
2064	0_AnalogInput	mSelectHarmonicMag49MaxReg
2065	0_AnalogInput	mSelectHarmonicMag50MaxReg
2066	0_AnalogInput	mVan200MaxRegTime
2067	0_AnalogInput	mVbn200MaxRegTime
2068	0_AnalogInput	mVcn200MaxRegTime
2069	0_AnalogInput	mVlnavg200MaxRegTime
2070	0_AnalogInput	mVng200MaxRegTime
2071	0_AnalogInput	mVab200MaxRegTime
2072	0_AnalogInput	mVbc200MaxRegTime
2073	0_AnalogInput	mVca200MaxRegTime
2074	0_AnalogInput	mVllavg200MaxRegTime
2075	0_AnalogInput	mIa200MaxRegTime
2076	0_AnalogInput	mIb200MaxRegTime
2077	0_AnalogInput	mIc200MaxRegTime
2078	0_AnalogInput	mIn200MaxRegTime
2079	0_AnalogInput	mIg200MaxRegTime
2080	0_AnalogInput	mFsysMaxRegTime
2081	0_AnalogInput	mPsysMaxRegTime
2082	0_AnalogInput	mPaMaxRegTime
2083	0_AnalogInput	mPbMaxRegTime
2084	0_AnalogInput	mPcMaxRegTime
2085	0_AnalogInput	mQsysMaxRegTime
2086	0_AnalogInput	mQaMaxRegTime
2087	0_AnalogInput	mQbMaxRegTime
2088	0_AnalogInput	mQcMaxRegTime
2089	0_AnalogInput	mSsysMaxRegTime
2090	0_AnalogInput	mSaMaxRegTime
2091	0_AnalogInput	mSbMaxRegTime
2092	0_AnalogInput	mScMaxRegTime
2093	0_AnalogInput	mAPFaMaxRegTime

Object ID	Object Type	Name
2094	0_AnalogInput	mAPFbMaxRegTime
2095	0_AnalogInput	mAPFcMaxRegTime
2096	0_AnalogInput	mAPFsysMaxRegTime
2097	0_AnalogInput	mDPFaMaxRegTime
2098	0_AnalogInput	mDPFbMaxRegTime
2099	0_AnalogInput	mDPFcMaxRegTime
2100	0_AnalogInput	mDPFsysMaxRegTime
2101	0_AnalogInput	mFlickerPerceptVanMaxRegTime
2102	0_AnalogInput	mFlickerPerceptVbnMaxRegTime
2103	0_AnalogInput	mFlickerPerceptVcnMaxRegTime
2104	0_AnalogInput	mFlickerPerceptVabMaxRegTime
2105	0_AnalogInput	mFlickerPerceptVbcMaxRegTime
2106	0_AnalogInput	mFlickerPerceptVcaMaxRegTime
2107	0_AnalogInput	mPstVanMaxRegTime
2108	0_AnalogInput	mPstVabMaxRegTime
2109	0_AnalogInput	mPstVbnMaxRegTime
2110	0_AnalogInput	mPstVbcMaxRegTime
2111	0_AnalogInput	mPstVcnMaxRegTime
2112	0_AnalogInput	mPstVcaMaxRegTime
2113	0_AnalogInput	mCFaMaxRegTime
2114	0_AnalogInput	mCFbMaxRegTime
2115	0_AnalogInput	mCFcMaxRegTime
2116	0_AnalogInput	mCFsysMaxRegTime
2117	0_AnalogInput	mKFaMaxRegTime
2118	0_AnalogInput	mKFbMaxRegTime
2119	0_AnalogInput	mKFcMaxRegTime
2120	0_AnalogInput	mKFsysMaxRegTime
2121	0_AnalogInput	mTotalHarmonicIaMaxRegTime
2122	0_AnalogInput	mTotalHarmonicIbMaxRegTime
2123	0_AnalogInput	mTotalHarmonicIcMaxRegTime
2124	0_AnalogInput	mTotalHarmonicVabMaxRegTime
2125	0_AnalogInput	mTotalHarmonicVbcMaxRegTime
2126	0_AnalogInput	mTotalHarmonicVcaMaxRegTime
2127	0_AnalogInput	mTotalHarmonicVanMaxRegTime
2128	0_AnalogInput	mTotalHarmonicVbnMaxRegTime
2129	0_AnalogInput	mTotalHarmonicVcnMaxRegTime
2130	0_AnalogInput	mTotalHarmonicInMaxRegTime
2131	0_AnalogInput	mTotalHarmonicIaPctMaxRegTime
2132	0_AnalogInput	mTotalHarmonicIbPctMaxRegTime
2133	0_AnalogInput	mTotalHarmonicIcPctMaxRegTime
2134	0_AnalogInput	mTotalHarmonicVabPctMaxRegTime

Object ID	Object Type	Name
2135	0_AnalogInput	mTotalHarmonicVbcPctMaxRegTime
2136	0_AnalogInput	mTotalHarmonicVcaPctMaxRegTime
2137	0_AnalogInput	mTotalHarmonicVanPctMaxRegTime
2138	0_AnalogInput	mTotalHarmonicVbnPctMaxRegTime
2139	0_AnalogInput	mTotalHarmonicVcnPctMaxRegTime
2140	0_AnalogInput	mTotalHarmonicInPctMaxRegTime
2141	0_AnalogInput	mTDDIaPctMaxRegTime
2142	0_AnalogInput	mTDDIbPctMaxRegTime
2143	0_AnalogInput	mTDDIcPctMaxRegTime
2144	0_AnalogInput	mTDDInPctMaxRegTime
2145	0_AnalogInput	mEvenHarmonicIaMaxRegTime
2146	0_AnalogInput	mEvenHarmonicIbMaxRegTime
2147	0_AnalogInput	mEvenHarmonicIcMaxRegTime
2148	0_AnalogInput	mEvenHarmonicVabMaxRegTime
2149	0_AnalogInput	mEvenHarmonicVbcMaxRegTime
2150	0_AnalogInput	mEvenHarmonicVcaMaxRegTime
2151	0_AnalogInput	mEvenHarmonicVanMaxRegTime
2152	0_AnalogInput	mEvenHarmonicVbnMaxRegTime
2153	0_AnalogInput	mEvenHarmonicVcnMaxRegTime
2154	0_AnalogInput	mEvenHarmonicInMaxRegTime
2155	0_AnalogInput	mEvenHarmonicIaPctMaxRegTime
2156	0_AnalogInput	mEvenHarmonicIbPctMaxRegTime
2157	0_AnalogInput	mEvenHarmonicIcPctMaxRegTime
2158	0_AnalogInput	mEvenHarmonicVabPctMaxRegTime
2159	0_AnalogInput	mEvenHarmonicVbcPctMaxRegTime
2160	0_AnalogInput	mEvenHarmonicVcaPctMaxRegTime
2161	0_AnalogInput	mEvenHarmonicVanPctMaxRegTime
2162	0_AnalogInput	mEvenHarmonicVbnPctMaxRegTime
2163	0_AnalogInput	mEvenHarmonicVcnPctMaxRegTime
2164	0_AnalogInput	mEvenHarmonicInPctMaxRegTime
2165	0_AnalogInput	mEvenTDDIaPctMaxRegTime
2166	0_AnalogInput	mEvenTDDIbPctMaxRegTime
2167	0_AnalogInput	mEvenTDDIcPctMaxRegTime
2168	0_AnalogInput	mEvenTDDInPctMaxRegTime
2169	0_AnalogInput	mOddHarmonicIaMaxRegTime
2170	0_AnalogInput	mOddHarmonicIbMaxRegTime
2171	0_AnalogInput	mOddHarmonicIcMaxRegTime
2172	0_AnalogInput	mOddHarmonicVabMaxRegTime
2173	0_AnalogInput	mOddHarmonicVbcMaxRegTime
2174	0_AnalogInput	mOddHarmonicVcaMaxRegTime
2175	0_AnalogInput	mOddHarmonicVanMaxRegTime

Object ID	Object Type	Name
2176	0_AnalogInput	mOddHarmonicVbnMaxRegTime
2177	0_AnalogInput	mOddHarmonicVcnMaxRegTime
2178	0_AnalogInput	mOddHarmonicInMaxRegTime
2179	0_AnalogInput	mOddHarmonicIaPctMaxRegTime
2180	0_AnalogInput	mOddHarmonicIbPctMaxRegTime
2181	0_AnalogInput	mOddHarmonicIcPctMaxRegTime
2182	0_AnalogInput	mOddHarmonicVabPctMaxRegTime
2183	0_AnalogInput	mOddHarmonicVbcPctMaxRegTime
2184	0_AnalogInput	mOddHarmonicVcaPctMaxRegTime
2185	0_AnalogInput	mOddHarmonicVanPctMaxRegTime
2186	0_AnalogInput	mOddHarmonicVbnPctMaxRegTime
2187	0_AnalogInput	mOddHarmonicVcnPctMaxRegTime
2188	0_AnalogInput	mOddHarmonicInPctMaxRegTime
2189	0_AnalogInput	mOddTDDIaPctMaxRegTime
2190	0_AnalogInput	mOddTDDIbPctMaxRegTime
2191	0_AnalogInput	mOddTDDIcPctMaxRegTime
2192	0_AnalogInput	mOddTDDInPctMaxRegTime
2193	0_AnalogInput	mInterHarmonicIaMaxRegTime
2194	0_AnalogInput	mInterHarmonicIbMaxRegTime
2195	0_AnalogInput	mInterHarmonicIcMaxRegTime
2196	0_AnalogInput	mInterHarmonicVabMaxRegTime
2197	0_AnalogInput	mInterHarmonicVbcMaxRegTime
2198	0_AnalogInput	mInterHarmonicVcaMaxRegTime
2199	0_AnalogInput	mInterHarmonicVanMaxRegTime
2200	0_AnalogInput	mInterHarmonicVbnMaxRegTime
2201	0_AnalogInput	mInterHarmonicVcnMaxRegTime
2202	0_AnalogInput	mInterHarmonicInMaxRegTime
2203	0_AnalogInput	mInterHarmonicIaPctMaxRegTime
2204	0_AnalogInput	mInterHarmonicIbPctMaxRegTime
2205	0_AnalogInput	mInterHarmonicIcPctMaxRegTime
2206	0_AnalogInput	mInterHarmonicVabPctMaxRegTime
2207	0_AnalogInput	mInterHarmonicVbcPctMaxRegTime
2208	0_AnalogInput	mInterHarmonicVcaPctMaxRegTime
2209	0_AnalogInput	mInterHarmonicVanPctMaxRegTime
2210	0_AnalogInput	mInterHarmonicVbnPctMaxRegTime
2211	0_AnalogInput	mInterHarmonicVcnPctMaxRegTime
2212	0_AnalogInput	mInterHarmonicInPctMaxRegTime
2213	0_AnalogInput	mInterTDDIaPctMaxRegTime
2214	0_AnalogInput	mInterTDDIbPctMaxRegTime
2215	0_AnalogInput	mInterTDDIcPctMaxRegTime
2216	0_AnalogInput	mInterTDDInPctMaxRegTime

Object ID	Object Type	Name
2217	0_AnalogInput	mSelectHarmonicMag1MaxRegTime
2218	0_AnalogInput	mSelectHarmonicMag2MaxRegTime
2219	0_AnalogInput	mSelectHarmonicMag3MaxRegTime
2220	0_AnalogInput	mSelectHarmonicMag4MaxRegTime
2221	0_AnalogInput	mSelectHarmonicMag5MaxRegTime
2222	0_AnalogInput	mSelectHarmonicMag6MaxRegTime
2223	0_AnalogInput	mSelectHarmonicMag7MaxRegTime
2224	0_AnalogInput	mSelectHarmonicMag8MaxRegTime
2225	0_AnalogInput	mSelectHarmonicMag9MaxRegTime
2226	0_AnalogInput	mSelectHarmonicMag10MaxRegTime
2227	0_AnalogInput	mSelectHarmonicMag11MaxRegTime
2228	0_AnalogInput	mSelectHarmonicMag12MaxRegTime
2229	0_AnalogInput	mSelectHarmonicMag13MaxRegTime
2230	0_AnalogInput	mSelectHarmonicMag14MaxRegTime
2231	0_AnalogInput	mSelectHarmonicMag15MaxRegTime
2232	0_AnalogInput	mSelectHarmonicMag16MaxRegTime
2233	0_AnalogInput	mSelectHarmonicMag17MaxRegTime
2234	0_AnalogInput	mSelectHarmonicMag18MaxRegTime
2235	0_AnalogInput	mSelectHarmonicMag19MaxRegTime
2236	0_AnalogInput	mSelectHarmonicMag20MaxRegTime
2237	0_AnalogInput	mSelectHarmonicMag21MaxRegTime
2238	0_AnalogInput	mSelectHarmonicMag22MaxRegTime
2239	0_AnalogInput	mSelectHarmonicMag23MaxRegTime
2240	0_AnalogInput	mSelectHarmonicMag24MaxRegTime
2241	0_AnalogInput	mSelectHarmonicMag25MaxRegTime
2242	0_AnalogInput	mSelectHarmonicMag26MaxRegTime
2243	0_AnalogInput	mSelectHarmonicMag27MaxRegTime
2244	0_AnalogInput	mSelectHarmonicMag28MaxRegTime
2245	0_AnalogInput	mSelectHarmonicMag29MaxRegTime
2246	0_AnalogInput	mSelectHarmonicMag30MaxRegTime
2247	0_AnalogInput	mSelectHarmonicMag31MaxRegTime
2248	0_AnalogInput	mSelectHarmonicMag32MaxRegTime
2249	0_AnalogInput	mSelectHarmonicMag33MaxRegTime
2250	0_AnalogInput	mSelectHarmonicMag34MaxRegTime
2251	0_AnalogInput	mSelectHarmonicMag35MaxRegTime
2252	0_AnalogInput	mSelectHarmonicMag36MaxRegTime
2253	0_AnalogInput	mSelectHarmonicMag37MaxRegTime
2254	0_AnalogInput	mSelectHarmonicMag38MaxRegTime
2255	0_AnalogInput	mSelectHarmonicMag39MaxRegTime
2256	0_AnalogInput	mSelectHarmonicMag40MaxRegTime
2257	0_AnalogInput	mSelectHarmonicMag41MaxRegTime

Object ID	Object Type	Name
2258	0_AnalogInput	mSelectHarmonicMag42MaxRegTime
2259	0_AnalogInput	mSelectHarmonicMag43MaxRegTime
2260	0_AnalogInput	mSelectHarmonicMag44MaxRegTime
2261	0_AnalogInput	mSelectHarmonicMag45MaxRegTime
2262	0_AnalogInput	mSelectHarmonicMag46MaxRegTime
2263	0_AnalogInput	mSelectHarmonicMag47MaxRegTime
2264	0_AnalogInput	mSelectHarmonicMag48MaxRegTime
2265	0_AnalogInput	mSelectHarmonicMag49MaxRegTime
2266	0_AnalogInput	mSelectHarmonicMag50MaxRegTime
2267	0_AnalogInput	mFwdPwrDmdPres
2268	0_AnalogInput	mRevPwrDmdPres
2269	0_AnalogInput	mNetPwrDmdPres
2270	0_AnalogInput	mSumPwrDmdPres
2271	0_AnalogInput	mRecReactivePwrDmdPres
2272	0_AnalogInput	mDelReactivePwrDmdPres
2273	0_AnalogInput	mNetReactivePwrDmdPres
2274	0_AnalogInput	mSumReactivePwrDmdPres
2275	0_AnalogInput	mFwdApparentPwrDmdPres
2276	0_AnalogInput	mRevApparentPwrDmdPres
2277	0_AnalogInput	mSumApparentPwrDmdPres
2278	0_AnalogInput	mAvgCurrentDmdPres
2279	0_AnalogInput	mFwdPwrDmdPeak
2280	0_AnalogInput	mRevPwrDmdPeak
2281	0_AnalogInput	mNetPwrDmdPeak
2282	0_AnalogInput	mSumPwrDmdPeak
2283	0_AnalogInput	mRecReactivePwrDmdPeak
2284	0_AnalogInput	mDelReactivePwrDmdPeak
2285	0_AnalogInput	mNetReactivePwrDmdPeak
2286	0_AnalogInput	mSumReactivePwrDmdPeak
2287	0_AnalogInput	mFwdApparentPwrDmdPeak
2288	0_AnalogInput	mRevApparentPwrDmdPeak
2289	0_AnalogInput	mSumApparentPwrDmdPeak
2290	0_AnalogInput	mAvgCurrentDmdPeak
2291	0_AnalogInput	mFwdPwrDmdPeakTime
2292	0_AnalogInput	mRevPwrDmdPeakTime
2293	0_AnalogInput	mNetPwrDmdPeakTime
2294	0_AnalogInput	mSumPwrDmdPeakTime
2295	0_AnalogInput	mRecReactivePwrDmdPeakTime
2296	0_AnalogInput	mDelReactivePwrDmdPeakTime
2297	0_AnalogInput	mNetReactivePwrDmdPeakTime
2298	0_AnalogInput	mSumReactivePwrDmdPeakTime

Object ID	Object Type	Name
2299	0_AnalogInput	mFwdApparentPwrDmdPeakTime
2300	0_AnalogInput	mRevApparentPwrDmdPeakTime
2301	0_AnalogInput	mSumApparentPwrDmdPeakTime
2302	0_AnalogInput	mAvgCurrentDmdPeakTime
2303	0_AnalogInput	mFwdPwrDmdPresMoPeak
2304	0_AnalogInput	mRevPwrDmdPresMoPeak
2305	0_AnalogInput	mNetPwrDmdPresMoPeak
2306	0_AnalogInput	mSumPwrDmdPresMoPeak
2307	0_AnalogInput	mRecReactivePwrDmdPresMoPeak
2308	0_AnalogInput	mDelReactivePwrDmdPresMoPeak
2309	0_AnalogInput	mNetReactivePwrDmdPresMoPeak
2310	0_AnalogInput	mSumReactivePwrDmdPresMoPeak
2311	0_AnalogInput	mFwdApparentPwrDmdPresMoPeak
2312	0_AnalogInput	mRevApparentPwrDmdPresMoPeak
2313	0_AnalogInput	mSumApparentPwrDmdPresMoPeak
2314	0_AnalogInput	mAvgCurrentDmdPresMoPeak
2315	0_AnalogInput	mFwdPwrDmdPresMoPeakTime
2316	0_AnalogInput	mRevPwrDmdPresMoPeakTime
2317	0_AnalogInput	mNetPwrDmdPresMoPeakTime
2318	0_AnalogInput	mSumPwrDmdPresMoPeakTime
2319	0_AnalogInput	mRecReactivePwrDmdPresMoPeakTime
2320	0_AnalogInput	mNetReactivePwrDmdPresMoPeakTime
2321	0_AnalogInput	mSumReactivePwrDmdPresMoPeakTime
2322	0_AnalogInput	mFwdApparentPwrDmdPresMoPeakTime
2323	0_AnalogInput	mRevApparentPwrDmdPresMoPeakTime
2324	0_AnalogInput	mSumApparentPwrDmdPresMoPeakTime
2325	0_AnalogInput	mAvgCurrentDmdPresMoPeakTime
2326	0_AnalogInput	mFwdPwrDmdLastMoPeak
2327	0_AnalogInput	mRevPwrDmdLastMoPeak
2328	0_AnalogInput	mNetPwrDmdLastMoPeak
2329	0_AnalogInput	mSumPwrDmdLastMoPeak
2330	0_AnalogInput	mRecReactivePwrDmdLastMoPeak
2331	0_AnalogInput	mDelReactivePwrDmdLastMoPeak
2332	0_AnalogInput	mNetReactivePwrDmdLastMoPeak
2333	0_AnalogInput	mSumReactivePwrDmdLastMoPeak
2334	0_AnalogInput	mFwdApparentPwrDmdLastMoPeak
2335	0_AnalogInput	mRevApparentPwrDmdLastMoPeak
2336	0_AnalogInput	mSumApparentPwrDmdLastMoPeak
2337	0_AnalogInput	mAvgCurrentDmdLastMoPeak
2338	0_AnalogInput	mFwdPwrDmdLastMoPeakTime
2339	0_AnalogInput	mRevPwrDmdLastMoPeakTime

Object ID	Object Type	Name
2340	0_AnalogInput	mNetPwrDmdLastMoPeakTime
2341	0_AnalogInput	mSumPwrDmdLastMoPeakTime
2342	0_AnalogInput	mRecReactivePwrDmdLastMoPeakTime
2343	0_AnalogInput	mDelReactivePwrDmdLastMoPeakTime
2344	0_AnalogInput	mNetReactivePwrDmdLastMoPeakTime
2345	0_AnalogInput	mSumReactivePwrDmdLastMoPeakTime
2346	0_AnalogInput	mFwdApparentPwrDmdLastMoPeakTime
2347	0_AnalogInput	mRevApparentPwrDmdLastMoPeakTime
2348	0_AnalogInput	mSumApparentPwrDmdLastMoPeakTime
2349	0_AnalogInput	mAvgCurrentDmdLastMoPeakTime
2350	0_AnalogInput	mFwdPwrDmdLastResetPeak
2351	0_AnalogInput	mRevPwrDmdLastResetPeak
2352	0_AnalogInput	mNetPwrDmdLastResetPeak
2353	0_AnalogInput	mSumPwrDmdLastResetPeak
2354	0_AnalogInput	mRecReactivePwrDmdLastResetPeak
2355	0_AnalogInput	mDelReactivePwrDmdLastResetPeak
2356	0_AnalogInput	mNetReactivePwrDmdLastResetPeak
2357	0_AnalogInput	mSumReactivePwrDmdLastResetPeak
2358	0_AnalogInput	mFwdApparentPwrDmdLastResetPeak
2359	0_AnalogInput	mRevApparentPwrDmdLastResetPeak
2360	0_AnalogInput	mSumApparentPwrDmdLastResetPeak
2361	0_AnalogInput	mAvgCurrentDmdLastResetPeak
2362	0_AnalogInput	mFwdPwrDmdLastResetPeakTime
2363	0_AnalogInput	mRevPwrDmdLastResetPeakTime
2364	0_AnalogInput	mNetPwrDmdLastResetPeakTime
2365	0_AnalogInput	mSumPwrDmdLastResetPeakTime
2366	0_AnalogInput	mRecReactivePwrDmdLastResetPeakTime
2367	0_AnalogInput	mDelReactivePwrDmdLastResetPeakTime
2368	0_AnalogInput	mNetReactivePwrDmdLastResetPeakTime
2369	0_AnalogInput	mSumReactivePwrDmdLastResetPeakTime
2370	0_AnalogInput	mFwdApparentPwrDmdLastResetPeakTime
2371	0_AnalogInput	mRevApparentPwrDmdLastResetPeakTime
2372	0_AnalogInput	mSumApparentPwrDmdLastResetPeakTime
2373	0_AnalogInput	mAvgCurrentDmdLastResetPeakTime
2374	0_AnalogInput	mFwdPwrDmdCondPeak
2375	0_AnalogInput	mRevPwrDmdCondPeak
2376	0_AnalogInput	mNetPwrDmdCondPeak
2377	0_AnalogInput	mSumPwrDmdCondPeak
2378	0_AnalogInput	mRecReactivePwrDmdCondPeak
2379	0_AnalogInput	mDelReactivePwrDmdCondPeak
2380	0_AnalogInput	mNetReactivePwrDmdCondPeak

Object ID	Object Type	Name
2381	0_AnalogInput	mSumReactivePwrDmdCondPeak
2382	0_AnalogInput	mFwdApparentPwrDmdCondPeak
2383	0_AnalogInput	mRevApparentPwrDmdCondPeak
2384	0_AnalogInput	mSumApparentPwrDmdCondPeak
2385	0_AnalogInput	mAvgCurrentDmdCondPeak
2386	0_AnalogInput	mFwdPwrDmdCondPeakTime
2387	0_AnalogInput	mRevPwrDmdCondPeakTime
2388	0_AnalogInput	mNetPwrDmdCondPeakTime
2389	0_AnalogInput	mSumPwrDmdCondPeakTime
2390	0_AnalogInput	mRecReactivePwrDmdCondPeakTime
2391	0_AnalogInput	mDelReactivePwrDmdCondPeakTime
2392	0_AnalogInput	mNetReactivePwrDmdCondPeakTime
2393	0_AnalogInput	mSumReactivePwrDmdCondPeakTime
2394	0_AnalogInput	mFwdApparentPwrDmdCondPeakTime
2395	0_AnalogInput	mRevApparentPwrDmdCondPeakTime
2396	0_AnalogInput	mSumApparentPwrDmdCondPeakTime
2397	0_AnalogInput	mAvgCurrentDmdCondPeakTime
2398	0_AnalogInput	mDIModeDI1
2399	0_AnalogInput	mDIModeDI2
2400	0_AnalogInput	mDIModeDI3
2401	0_AnalogInput	mDIModeDI4
2402	0_AnalogInput	mDIModeDI5
2403	0_AnalogInput	mDIModeDI6
2404	0_AnalogInput	mDIModeDI7
2405	0_AnalogInput	mDIModeDI8
2406	0_AnalogInput	mDigitalInputDI1
2407	0_AnalogInput	mDigitalInputDI2
2408	0_AnalogInput	mDigitalInputDI3
2409	0_AnalogInput	mDigitalInputDI4
2410	0_AnalogInput	mDigitalInputDI5
2411	0_AnalogInput	mDigitalInputDI6
2412	0_AnalogInput	mDigitalInputDI7
2413	0_AnalogInput	mDigitalInputDI8
2414	0_AnalogInput	mDORawStateDO1
2415	0_AnalogInput	mDORawStateDO2
2416	0_AnalogInput	mDICounterRawCountDI1
2417	0_AnalogInput	mDICounterRawCountDI2
2418	0_AnalogInput	mDICounterRawCountDI3
2419	0_AnalogInput	mDICounterRawCountDI4
2420	0_AnalogInput	mDICounterRawCountDI5
2421	0_AnalogInput	mDICounterRawCountDI6

Object ID	Object Type	Name
2422	0_AnalogInput	mDICounterRawCountDI7
2423	0_AnalogInput	mDICounterRawCountDI8
2424	0_AnalogInput	mDIMultiplierDI1
2425	0_AnalogInput	mDIMultiplierDI2
2426	0_AnalogInput	mDIMultiplierDI3
2427	0_AnalogInput	mDIMultiplierDI4
2428	0_AnalogInput	mDIMultiplierDI5
2429	0_AnalogInput	mDIMultiplierDI6
2430	0_AnalogInput	mDIMultiplierDI7
2431	0_AnalogInput	mDIMultiplierDI8
2432	0_AnalogInput	mDIScaledDI1
2433	0_AnalogInput	mDIScaledDI2
2434	0_AnalogInput	mDIScaledDI3
2435	0_AnalogInput	mDIScaledDI4
2436	0_AnalogInput	mDIScaledDI5
2437	0_AnalogInput	mDIScaledDI6
2438	0_AnalogInput	mDIScaledDI7
2439	0_AnalogInput	mDIScaledDI8
2440	0_AnalogInput	mDIIntervalAccumDI1
2441	0_AnalogInput	mDIIntervalAccumDI2
2442	0_AnalogInput	mDIIntervalAccumDI3
2443	0_AnalogInput	mDIIntervalAccumDI4
2444	0_AnalogInput	mDIIntervalAccumDI5
2445	0_AnalogInput	mDIIntervalAccumDI6
2446	0_AnalogInput	mDIIntervalAccumDI7
2447	0_AnalogInput	mDIIntervalAccumDI8
2448	0_AnalogInput	mDILabelDI1
2449	0_AnalogInput	mDILabelDI2
2450	0_AnalogInput	mDILabelDI3
2451	0_AnalogInput	mDILabelDI4
2452	0_AnalogInput	mDILabelDI5
2453	0_AnalogInput	mDILabelDI6
2454	0_AnalogInput	mDILabelDI7
2455	0_AnalogInput	mDILabelDI8
2456	0_AnalogInput	mDIActiveDI1
2457	0_AnalogInput	mDIActiveDI2
2458	0_AnalogInput	mDIActiveDI3
2459	0_AnalogInput	mDIActiveDI4
2460	0_AnalogInput	mDIActiveDI5
2461	0_AnalogInput	mDIActiveDI6
2462	0_AnalogInput	mDIActiveDI7

Object ID	Object Type	Name
2463	0_AnalogInput	mDIActiveDI8
2464	0_AnalogInput	mDIInactiveDI1
2465	0_AnalogInput	mDIInactiveDI2
2466	0_AnalogInput	mDIInactiveDI3
2467	0_AnalogInput	mDIInactiveDI4
2468	0_AnalogInput	mDIInactiveDI5
2469	0_AnalogInput	mDIInactiveDI6
2470	0_AnalogInput	mDIInactiveDI7
2471	0_AnalogInput	mDIInactiveDI8
2472	0_AnalogInput	mDIUnitsDI1
2473	0_AnalogInput	mDIUnitsDI2
2474	0_AnalogInput	mDIUnitsDI3
2475	0_AnalogInput	mDIUnitsDI4
2476	0_AnalogInput	mDIUnitsDI5
2477	0_AnalogInput	mDIUnitsDI6
2478	0_AnalogInput	mDIUnitsDI7
2479	0_AnalogInput	mDIUnitsDI8
2480	0_AnalogInput	mMin5minVan200
2481	0_AnalogInput	mMax5minVan200
2482	0_AnalogInput	mAvg5minVan200
2483	0_AnalogInput	mMin5minVbn200
2484	0_AnalogInput	mMax5minVbn200
2485	0_AnalogInput	mAvg5minVbn200
2486	0_AnalogInput	mMin5minVcn200
2487	0_AnalogInput	mMax5minVcn200
2488	0_AnalogInput	mAvg5minVcn200
2489	0_AnalogInput	mMin5minVlnavg200
2490	0_AnalogInput	mMax5minVlnavg200
2491	0_AnalogInput	mAvg5minVlnavg200
2492	0_AnalogInput	mMin5minVng200
2493	0_AnalogInput	mMax5minVng200
2494	0_AnalogInput	mAvg5minVng200
2495	0_AnalogInput	mMin5minVab200
2496	0_AnalogInput	mMax5minVab200
2497	0_AnalogInput	mAvg5minVab200
2498	0_AnalogInput	mMin5minVbc200
2499	0_AnalogInput	mMax5minVbc200
2500	0_AnalogInput	mAvg5minVbc200
2501	0_AnalogInput	mMin5minVca200
2502	0_AnalogInput	mMax5minVca200
2503	0_AnalogInput	mAvg5minVca200

Object ID	Object Type	Name
2504	0_AnalogInput	mMin5minVllavg200
2505	0_AnalogInput	mMax5minVllavg200
2506	0_AnalogInput	mAvg5minVllavg200
2507	0_AnalogInput	mMin5minIa200
2508	0_AnalogInput	mMax5minIa200
2509	0_AnalogInput	mAvg5minIa200
2510	0_AnalogInput	mMin5minIb200
2511	0_AnalogInput	mMax5minIb200
2512	0_AnalogInput	mAvg5minIb200
2513	0_AnalogInput	mMin5minIc200
2514	0_AnalogInput	mMax5minIc200
2515	0_AnalogInput	mAvg5minIc200
2516	0_AnalogInput	mMin5minIn200
2517	0_AnalogInput	mMax5minIn200
2518	0_AnalogInput	mAvg5minIn200
2519	0_AnalogInput	mMin5minIg200
2520	0_AnalogInput	mMax5minIg200
2521	0_AnalogInput	mAvg5minIg200
2522	0_AnalogInput	mMin5minFreq200ms
2523	0_AnalogInput	mMax5minFreq200ms
2524	0_AnalogInput	mAvg5minFreq200ms
2525	0_AnalogInput	mMin5minPsys
2526	0_AnalogInput	mMax5minPsys
2527	0_AnalogInput	mAvg5minPsys
2528	0_AnalogInput	mMin5minPa
2529	0_AnalogInput	mMax5minPa
2530	0_AnalogInput	mAvg5minPa
2531	0_AnalogInput	mMin5minPb
2532	0_AnalogInput	mMax5minPb
2533	0_AnalogInput	mAvg5minPb
2534	0_AnalogInput	mMin5minPc
2535	0_AnalogInput	mMax5minPc
2536	0_AnalogInput	mAvg5minPc
2537	0_AnalogInput	mMin5minQsys
2538	0_AnalogInput	mMax5minQsys
2539	0_AnalogInput	mAvg5minQsys
2540	0_AnalogInput	mMin5minQa
2541	0_AnalogInput	mMax5minQa
2542	0_AnalogInput	mAvg5minQa
2543	0_AnalogInput	mMin5minQb
2544	0_AnalogInput	mMax5minQb

Object ID	Object Type	Name
2545	0_AnalogInput	mAvg5minQb
2546	0_AnalogInput	mMin5minQc
2547	0_AnalogInput	mMax5minQc
2548	0_AnalogInput	mAvg5minQc
2549	0_AnalogInput	mMin5minSsys
2550	0_AnalogInput	mMax5minSsys
2551	0_AnalogInput	mAvg5minSsys
2552	0_AnalogInput	mMin5minSa
2553	0_AnalogInput	mMax5minSa
2554	0_AnalogInput	mAvg5minSa
2555	0_AnalogInput	mMin5minSb
2556	0_AnalogInput	mMax5minSb
2557	0_AnalogInput	mAvg5minSb
2558	0_AnalogInput	mMin5minSc
2559	0_AnalogInput	mMax5minSc
2560	0_AnalogInput	mAvg5minSc
2561	0_AnalogInput	mMin5minPfA200ms
2562	0_AnalogInput	mMax5minPfA200ms
2563	0_AnalogInput	mAvg5minPfA200ms
2564	0_AnalogInput	mMin5minPfB200ms
2565	0_AnalogInput	mMax5minPfB200ms
2566	0_AnalogInput	mAvg5minPfB200ms
2567	0_AnalogInput	mMin5minPfC200ms
2568	0_AnalogInput	mMax5minPfC200ms
2569	0_AnalogInput	mAvg5minPfC200ms
2570	0_AnalogInput	mMin5minPf200ms
2571	0_AnalogInput	mMax5minPf200ms
2572	0_AnalogInput	mAvg5minPf200ms
2573	0_AnalogInput	mMin5mindpfA200ms
2574	0_AnalogInput	mMax5mindpfA200ms
2575	0_AnalogInput	mAvg5mindpfA200ms
2576	0_AnalogInput	mMin5mindpfB200ms
2577	0_AnalogInput	mMax5mindpfB200ms
2578	0_AnalogInput	mAvg5mindpfB200ms
2579	0_AnalogInput	mMin5mindpfC200ms
2580	0_AnalogInput	mMax5mindpfC200ms
2581	0_AnalogInput	mAvg5mindpfC200ms
2582	0_AnalogInput	mMin5mindpf200ms
2583	0_AnalogInput	mMax5mindpf200ms
2584	0_AnalogInput	mAvg5mindpf200ms
2585	0_AnalogInput	mMin5minPinstVan

Object ID	Object Type	Name
2586	0_AnalogInput	mMax5minPinstVan
2587	0_AnalogInput	mAvg5minPinstVan
2588	0_AnalogInput	mMin5minPinstVbn
2589	0_AnalogInput	mMax5minPinstVbn
2590	0_AnalogInput	mAvg5minPinstVbn
2591	0_AnalogInput	mMin5minPinstVcn
2592	0_AnalogInput	mMax5minPinstVcn
2593	0_AnalogInput	mAvg5minPinstVcn
2594	0_AnalogInput	mMin5minPinstVab
2595	0_AnalogInput	mMax5minPinstVab
2596	0_AnalogInput	mAvg5minPinstVab
2597	0_AnalogInput	mMin5minPinstVbc
2598	0_AnalogInput	mMax5minPinstVbc
2599	0_AnalogInput	mAvg5minPinstVbc
2600	0_AnalogInput	mMin5minPinstVca
2601	0_AnalogInput	mMax5minPinstVca
2602	0_AnalogInput	mAvg5minPinstVca
2603	0_AnalogInput	mMin5minPstVan
2604	0_AnalogInput	mMax5minPstVan
2605	0_AnalogInput	mAvg5minPstVan
2606	0_AnalogInput	mMin5minPstVab
2607	0_AnalogInput	mMax5minPstVab
2608	0_AnalogInput	mAvg5minPstVab
2609	0_AnalogInput	mMin5minPstVbn
2610	0_AnalogInput	mMax5minPstVbn
2611	0_AnalogInput	mAvg5minPstVbn
2612	0_AnalogInput	mMin5minPstVbc
2613	0_AnalogInput	mMax5minPstVbc
2614	0_AnalogInput	mAvg5minPstVbc
2615	0_AnalogInput	mMin5minPstVcn
2616	0_AnalogInput	mMax5minPstVcn
2617	0_AnalogInput	mAvg5minPstVcn
2618	0_AnalogInput	mMin5minPstVca
2619	0_AnalogInput	mMax5minPstVca
2620	0_AnalogInput	mAvg5minPstVca
2621	0_AnalogInput	mMin5minCFa
2622	0_AnalogInput	mMax5minCFa
2623	0_AnalogInput	mAvg5minCFa
2624	0_AnalogInput	mMin5minCFb
2625	0_AnalogInput	mMax5minCFb
2626	0_AnalogInput	mAvg5minCFb

Object ID	Object Type	Name
2627	0_AnalogInput	mMin5minCFc
2628	0_AnalogInput	mMax5minCFc
2629	0_AnalogInput	mAvg5minCFc
2630	0_AnalogInput	mMin5minCFsys
2631	0_AnalogInput	mMax5minCFsys
2632	0_AnalogInput	mAvg5minCFsys
2633	0_AnalogInput	mMin5minKFa
2634	0_AnalogInput	mMax5minKFa
2635	0_AnalogInput	mAvg5minKFa
2636	0_AnalogInput	mMin5minKFb
2637	0_AnalogInput	mMax5minKFb
2638	0_AnalogInput	mAvg5minKFb
2639	0_AnalogInput	mMin5minKFc
2640	0_AnalogInput	mMax5minKFc
2641	0_AnalogInput	mAvg5minKFc
2642	0_AnalogInput	mMin5minKFsys
2643	0_AnalogInput	mMax5minKFsys
2644	0_AnalogInput	mAvg5minKFsys
2645	0_AnalogInput	mMin5minTotalHarmonicIa
2646	0_AnalogInput	mMax5minTotalHarmonicIa
2647	0_AnalogInput	mAvg5minTotalHarmonicIa
2648	0_AnalogInput	mMin5minTotalHarmonicIb
2649	0_AnalogInput	mMax5minTotalHarmonicIb
2650	0_AnalogInput	mAvg5minTotalHarmonicIb
2651	0_AnalogInput	mMin5minTotalHarmonicIc
2652	0_AnalogInput	mMax5minTotalHarmonicIc
2653	0_AnalogInput	mAvg5minTotalHarmonicIc
2654	0_AnalogInput	mMin5minTotalHarmonicVab
2655	0_AnalogInput	mMax5minTotalHarmonicVab
2656	0_AnalogInput	mAvg5minTotalHarmonicVab
2657	0_AnalogInput	mMin5minTotalHarmonicVbc
2658	0_AnalogInput	mMax5minTotalHarmonicVbc
2659	0_AnalogInput	mAvg5minTotalHarmonicVbc
2660	0_AnalogInput	mMin5minTotalHarmonicVca
2661	0_AnalogInput	mMax5minTotalHarmonicVca
2662	0_AnalogInput	mAvg5minTotalHarmonicVca
2663	0_AnalogInput	mMin5minTotalHarmonicVan
2664	0_AnalogInput	mMax5minTotalHarmonicVan
2665	0_AnalogInput	mAvg5minTotalHarmonicVan
2666	0_AnalogInput	mMin5minTotalHarmonicVbn
2667	0_AnalogInput	mMax5minTotalHarmonicVbn

Object ID	Object Type	Name
2668	0_AnalogInput	mAvg5minTotalHarmonicVbn
2669	0_AnalogInput	mMin5minTotalHarmonicVcn
2670	0_AnalogInput	mMax5minTotalHarmonicVcn
2671	0_AnalogInput	mAvg5minTotalHarmonicVcn
2672	0_AnalogInput	mMin5minTotalHarmonicIn
2673	0_AnalogInput	mMax5minTotalHarmonicIn
2674	0_AnalogInput	mAvg5minTotalHarmonicIn
2675	0_AnalogInput	mMin5minTotalHarmonicIaPct
2676	0_AnalogInput	mMax5minTotalHarmonicIaPct
2677	0_AnalogInput	mAvg5minTotalHarmonicIaPct
2678	0_AnalogInput	mMin5minTotalHarmonicIbPct
2679	0_AnalogInput	mMax5minTotalHarmonicIbPct
2680	0_AnalogInput	mAvg5minTotalHarmonicIbPct
2681	0_AnalogInput	mMin5minTotalHarmonicIcPct
2682	0_AnalogInput	mMax5minTotalHarmonicIcPct
2683	0_AnalogInput	mAvg5minTotalHarmonicIcPct
2684	0_AnalogInput	mMin5minTotalHarmonicVabPct
2685	0_AnalogInput	mMax5minTotalHarmonicVabPct
2686	0_AnalogInput	mAvg5minTotalHarmonicVabPct
2687	0_AnalogInput	mMin5minTotalHarmonicVbcPct
2688	0_AnalogInput	mMax5minTotalHarmonicVbcPct
2689	0_AnalogInput	mAvg5minTotalHarmonicVbcPct
2690	0_AnalogInput	mMin5minTotalHarmonicVcaPct
2691	0_AnalogInput	mMax5minTotalHarmonicVcaPct
2692	0_AnalogInput	mAvg5minTotalHarmonicVcaPct
2693	0_AnalogInput	mMin5minTotalHarmonicVanPct
2694	0_AnalogInput	mMax5minTotalHarmonicVanPct
2695	0_AnalogInput	mAvg5minTotalHarmonicVanPct
2696	0_AnalogInput	mMin5minTotalHarmonicVbnPct
2697	0_AnalogInput	mMax5minTotalHarmonicVbnPct
2698	0_AnalogInput	mAvg5minTotalHarmonicVbnPct
2699	0_AnalogInput	mMin5minTotalHarmonicVcnPct
2700	0_AnalogInput	mMax5minTotalHarmonicVcnPct
2701	0_AnalogInput	mAvg5minTotalHarmonicVcnPct
2702	0_AnalogInput	mMin5minTotalHarmonicInPct
2703	0_AnalogInput	mMax5minTotalHarmonicInPct
2704	0_AnalogInput	mAvg5minTotalHarmonicInPct
2705	0_AnalogInput	mMin5minTDDIaPct
2706	0_AnalogInput	mMax5minTDDIaPct
2707	0_AnalogInput	mAvg5minTDDIaPct
2708	0_AnalogInput	mMin5minTDDIbPct

Object ID	Object Type	Name
2709	0_AnalogInput	mMax5minTDDIbPct
2710	0_AnalogInput	mAvg5minTDDIbPct
2711	0_AnalogInput	mMin5minTDDIcPct
2712	0_AnalogInput	mMax5minTDDIcPct
2713	0_AnalogInput	mAvg5minTDDIcPct
2714	0_AnalogInput	mMin5minTDDInPct
2715	0_AnalogInput	mMax5minTDDInPct
2716	0_AnalogInput	mAvg5minTDDInPct
2717	0_AnalogInput	mMin5minEvenHarmonicIa
2718	0_AnalogInput	mMax5minEvenHarmonicIa
2719	0_AnalogInput	mAvg5minEvenHarmonicIa
2720	0_AnalogInput	mMin5minEvenHarmonicIb
2721	0_AnalogInput	mMax5minEvenHarmonicIb
2722	0_AnalogInput	mAvg5minEvenHarmonicIb
2723	0_AnalogInput	mMin5minEvenHarmonicIc
2724	0_AnalogInput	mMax5minEvenHarmonicIc
2725	0_AnalogInput	mAvg5minEvenHarmonicIc
2726	0_AnalogInput	mMin5minEvenHarmonicVab
2727	0_AnalogInput	mMax5minEvenHarmonicVab
2728	0_AnalogInput	mAvg5minEvenHarmonicVab
2729	0_AnalogInput	mMin5minEvenHarmonicVbc
2730	0_AnalogInput	mMax5minEvenHarmonicVbc
2731	0_AnalogInput	mAvg5minEvenHarmonicVbc
2732	0_AnalogInput	mMin5minEvenHarmonicVca
2733	0_AnalogInput	mMax5minEvenHarmonicVca
2734	0_AnalogInput	mAvg5minEvenHarmonicVca
2735	0_AnalogInput	mMin5minEvenHarmonicVan
2736	0_AnalogInput	mMax5minEvenHarmonicVan
2737	0_AnalogInput	mAvg5minEvenHarmonicVan
2738	0_AnalogInput	mMin5minEvenHarmonicVbn
2739	0_AnalogInput	mMax5minEvenHarmonicVbn
2740	0_AnalogInput	mAvg5minEvenHarmonicVbn
2741	0_AnalogInput	mMin5minEvenHarmonicVcn
2742	0_AnalogInput	mMax5minEvenHarmonicVcn
2743	0_AnalogInput	mAvg5minEvenHarmonicVcn
2744	0_AnalogInput	mMin5minEvenHarmonicIn
2745	0_AnalogInput	mMax5minEvenHarmonicIn
2746	0_AnalogInput	mAvg5minEvenHarmonicIn
2747	0_AnalogInput	mMin5minEvenHarmonicIaPct
2748	0_AnalogInput	mMax5minEvenHarmonicIaPct
2749	0_AnalogInput	mAvg5minEvenHarmonicIaPct

Object ID	Object Type	Name
2750	0_AnalogInput	mMin5minEvenHarmonicIbPct
2751	0_AnalogInput	mMax5minEvenHarmonicIbPct
2752	0_AnalogInput	mAvg5minEvenHarmonicIbPct
2753	0_AnalogInput	mMin5minEvenHarmonicIcPct
2754	0_AnalogInput	mMax5minEvenHarmonicIcPct
2755	0_AnalogInput	mAvg5minEvenHarmonicIcPct
2756	0_AnalogInput	mMin5minEvenHarmonicVabPct
2757	0_AnalogInput	mMax5minEvenHarmonicVabPct
2758	0_AnalogInput	mAvg5minEvenHarmonicVabPct
2759	0_AnalogInput	mMin5minEvenHarmonicVbcPct
2760	0_AnalogInput	mMax5minEvenHarmonicVbcPct
2761	0_AnalogInput	mAvg5minEvenHarmonicVbcPct
2762	0_AnalogInput	mMin5minEvenHarmonicVcaPct
2763	0_AnalogInput	mMax5minEvenHarmonicVcaPct
2764	0_AnalogInput	mAvg5minEvenHarmonicVcaPct
2765	0_AnalogInput	mMin5minEvenHarmonicVanPct
2766	0_AnalogInput	mMax5minEvenHarmonicVanPct
2767	0_AnalogInput	mAvg5minEvenHarmonicVanPct
2768	0_AnalogInput	mMin5minEvenHarmonicVbnPct
2769	0_AnalogInput	mMax5minEvenHarmonicVbnPct
2770	0_AnalogInput	mAvg5minEvenHarmonicVbnPct
2771	0_AnalogInput	mMin5minEvenHarmonicVcnPct
2772	0_AnalogInput	mMax5minEvenHarmonicVcnPct
2773	0_AnalogInput	mAvg5minEvenHarmonicVcnPct
2774	0_AnalogInput	mMin5minEvenHarmonicInPct
2775	0_AnalogInput	mMax5minEvenHarmonicInPct
2776	0_AnalogInput	mAvg5minEvenHarmonicInPct
2777	0_AnalogInput	mMin5minEvenTDDIaPct
2778	0_AnalogInput	mMax5minEvenTDDIaPct
2779	0_AnalogInput	mAvg5minEvenTDDIaPct
2780	0_AnalogInput	mMin5minEvenTDDIbPct
2781	0_AnalogInput	mMax5minEvenTDDIbPct
2782	0_AnalogInput	mAvg5minEvenTDDIbPct
2783	0_AnalogInput	mMin5minEvenTDDIcPct
2784	0_AnalogInput	mMax5minEvenTDDIcPct
2785	0_AnalogInput	mAvg5minEvenTDDIcPct
2786	0_AnalogInput	mMin5minEvenTDDInPct
2787	0_AnalogInput	mMax5minEvenTDDInPct
2788	0_AnalogInput	mAvg5minEvenTDDInPct
2789	0_AnalogInput	mMin5minOddHarmonicIa
2790	0_AnalogInput	mMax5minOddHarmonicIa

Object ID	Object Type	Name
2791	0_AnalogInput	mAvg5minOddHarmonicIa
2792	0_AnalogInput	mMin5minOddHarmonicIb
2793	0_AnalogInput	mMax5minOddHarmonicIb
2794	0_AnalogInput	mAvg5minOddHarmonicIb
2795	0_AnalogInput	mMin5minOddHarmonicIc
2796	0_AnalogInput	mMax5minOddHarmonicIc
2797	0_AnalogInput	mAvg5minOddHarmonicIc
2798	0_AnalogInput	mMin5minOddHarmonicVab
2799	0_AnalogInput	mMax5minOddHarmonicVab
2800	0_AnalogInput	mAvg5minOddHarmonicVab
2801	0_AnalogInput	mMin5minOddHarmonicVbc
2802	0_AnalogInput	mMax5minOddHarmonicVbc
2803	0_AnalogInput	mAvg5minOddHarmonicVbc
2804	0_AnalogInput	mMin5minOddHarmonicVca
2805	0_AnalogInput	mMax5minOddHarmonicVca
2806	0_AnalogInput	mAvg5minOddHarmonicVca
2807	0_AnalogInput	mMin5minOddHarmonicVan
2808	0_AnalogInput	mMax5minOddHarmonicVan
2809	0_AnalogInput	mAvg5minOddHarmonicVan
2810	0_AnalogInput	mMin5minOddHarmonicVbn
2811	0_AnalogInput	mMax5minOddHarmonicVbn
2812	0_AnalogInput	mAvg5minOddHarmonicVbn
2813	0_AnalogInput	mMin5minOddHarmonicVcn
2814	0_AnalogInput	mMax5minOddHarmonicVcn
2815	0_AnalogInput	mAvg5minOddHarmonicVcn
2816	0_AnalogInput	mMin5minOddHarmonicIn
2817	0_AnalogInput	mMax5minOddHarmonicIn
2818	0_AnalogInput	mAvg5minOddHarmonicIn
2819	0_AnalogInput	mMin5minOddHarmonicIaPct
2820	0_AnalogInput	mMax5minOddHarmonicIaPct
2821	0_AnalogInput	mAvg5minOddHarmonicIaPct
2822	0_AnalogInput	mMin5minOddHarmonicIbPct
2823	0_AnalogInput	mMax5minOddHarmonicIbPct
2824	0_AnalogInput	mAvg5minOddHarmonicIbPct
2825	0_AnalogInput	mMin5minOddHarmonicIcPct
2826	0_AnalogInput	mMax5minOddHarmonicIcPct
2827	0_AnalogInput	mAvg5minOddHarmonicIcPct
2828	0_AnalogInput	mMin5minOddHarmonicVabPct
2829	0_AnalogInput	mMax5minOddHarmonicVabPct
2830	0_AnalogInput	mAvg5minOddHarmonicVabPct
2831	0_AnalogInput	mMin5minOddHarmonicVbcPct

Object ID	Object Type	Name
2832	0_AnalogInput	mMax5minOddHarmonicVbcPct
2833	0_AnalogInput	mAvg5minOddHarmonicVbcPct
2834	0_AnalogInput	mMin5minOddHarmonicVcaPct
2835	0_AnalogInput	mMax5minOddHarmonicVcaPct
2836	0_AnalogInput	mAvg5minOddHarmonicVcaPct
2837	0_AnalogInput	mMin5minOddHarmonicVanPct
2838	0_AnalogInput	mMax5minOddHarmonicVanPct
2839	0_AnalogInput	mAvg5minOddHarmonicVanPct
2840	0_AnalogInput	mMin5minOddHarmonicVbnPct
2841	0_AnalogInput	mMax5minOddHarmonicVbnPct
2842	0_AnalogInput	mAvg5minOddHarmonicVbnPct
2843	0_AnalogInput	mMin5minOddHarmonicVcnPct
2844	0_AnalogInput	mMax5minOddHarmonicVcnPct
2845	0_AnalogInput	mAvg5minOddHarmonicVcnPct
2846	0_AnalogInput	mMin5minOddHarmonicInPct
2847	0_AnalogInput	mMax5minOddHarmonicInPct
2848	0_AnalogInput	mAvg5minOddHarmonicInPct
2849	0_AnalogInput	mMin5minOddTDDIaPct
2850	0_AnalogInput	mMax5minOddTDDIaPct
2851	0_AnalogInput	mAvg5minOddTDDIaPct
2852	0_AnalogInput	mMin5minOddTDDIbPct
2853	0_AnalogInput	mMax5minOddTDDIbPct
2854	0_AnalogInput	mAvg5minOddTDDIbPct
2855	0_AnalogInput	mMin5minOddTDDIcPct
2856	0_AnalogInput	mMax5minOddTDDIcPct
2857	0_AnalogInput	mAvg5minOddTDDIcPct
2858	0_AnalogInput	mMin5minOddTDDInPct
2859	0_AnalogInput	mMax5minOddTDDInPct
2860	0_AnalogInput	mAvg5minOddTDDInPct
2861	0_AnalogInput	mMin5minInterHarmonicIa
2862	0_AnalogInput	mMax5minInterHarmonicIa
2863	0_AnalogInput	mAvg5minInterHarmonicIa
2864	0_AnalogInput	mMin5minInterHarmonicIb
2865	0_AnalogInput	mMax5minInterHarmonicIb
2866	0_AnalogInput	mAvg5minInterHarmonicIb
2867	0_AnalogInput	mMin5minInterHarmonicIc
2868	0_AnalogInput	mMax5minInterHarmonicIc
2869	0_AnalogInput	mAvg5minInterHarmonicIc
2870	0_AnalogInput	mMin5minInterHarmonicVab
2871	0_AnalogInput	mMax5minInterHarmonicVab
2872	0_AnalogInput	mAvg5minInterHarmonicVab

Object ID	Object Type	Name
2873	0_AnalogInput	mMin5minInterHarmonicVbc
2874	0_AnalogInput	mMax5minInterHarmonicVbc
2875	0_AnalogInput	mAvg5minInterHarmonicVbc
2876	0_AnalogInput	mMin5minInterHarmonicVca
2877	0_AnalogInput	mMax5minInterHarmonicVca
2878	0_AnalogInput	mAvg5minInterHarmonicVca
2879	0_AnalogInput	mMin5minInterHarmonicVan
2880	0_AnalogInput	mMax5minInterHarmonicVan
2881	0_AnalogInput	mAvg5minInterHarmonicVan
2882	0_AnalogInput	mMin5minInterHarmonicVbn
2883	0_AnalogInput	mMax5minInterHarmonicVbn
2884	0_AnalogInput	mAvg5minInterHarmonicVbn
2885	0_AnalogInput	mMin5minInterHarmonicVcn
2886	0_AnalogInput	mMax5minInterHarmonicVcn
2887	0_AnalogInput	mAvg5minInterHarmonicVcn
2888	0_AnalogInput	mMin5minInterHarmonicIn
2889	0_AnalogInput	mMax5minInterHarmonicIn
2890	0_AnalogInput	mAvg5minInterHarmonicIn
2891	0_AnalogInput	mMin5minInterHarmonicIaPct
2892	0_AnalogInput	mMax5minInterHarmonicIaPct
2893	0_AnalogInput	mAvg5minInterHarmonicIaPct
2894	0_AnalogInput	mMin5minInterHarmonicIbPct
2895	0_AnalogInput	mMax5minInterHarmonicIbPct
2896	0_AnalogInput	mAvg5minInterHarmonicIbPct
2897	0_AnalogInput	mMin5minInterHarmonicIcPct
2898	0_AnalogInput	mMax5minInterHarmonicIcPct
2899	0_AnalogInput	mAvg5minInterHarmonicIcPct
2900	0_AnalogInput	mMin5minInterHarmonicVabPct
2901	0_AnalogInput	mMax5minInterHarmonicVabPct
2902	0_AnalogInput	mAvg5minInterHarmonicVabPct
2903	0_AnalogInput	mMin5minInterHarmonicVbcPct
2904	0_AnalogInput	mMax5minInterHarmonicVbcPct
2905	0_AnalogInput	mAvg5minInterHarmonicVbcPct
2906	0_AnalogInput	mMin5minInterHarmonicVcaPct
2907	0_AnalogInput	mMax5minInterHarmonicVcaPct
2908	0_AnalogInput	mAvg5minInterHarmonicVcaPct
2909	0_AnalogInput	mMin5minInterHarmonicVanPct
2910	0_AnalogInput	mMax5minInterHarmonicVanPct
2911	0_AnalogInput	mAvg5minInterHarmonicVanPct
2912	0_AnalogInput	mMin5minInterHarmonicVbnPct
2913	0_AnalogInput	mMax5minInterHarmonicVbnPct

Object ID	Object Type	Name
2914	0_AnalogInput	mAvg5minInterHarmonicVbnPct
2915	0_AnalogInput	mMin5minInterHarmonicVcnPct
2916	0_AnalogInput	mMax5minInterHarmonicVcnPct
2917	0_AnalogInput	mAvg5minInterHarmonicVcnPct
2918	0_AnalogInput	mMin5minInterHarmonicInPct
2919	0_AnalogInput	mMax5minInterHarmonicInPct
2920	0_AnalogInput	mAvg5minInterHarmonicInPct
2921	0_AnalogInput	mMin5minInterTDDIaPct
2922	0_AnalogInput	mMax5minInterTDDIaPct
2923	0_AnalogInput	mAvg5minInterTDDIaPct
2924	0_AnalogInput	mMin5minInterTDDIbPct
2925	0_AnalogInput	mMax5minInterTDDIbPct
2926	0_AnalogInput	mAvg5minInterTDDIbPct
2927	0_AnalogInput	mMin5minInterTDDIcPct
2928	0_AnalogInput	mMax5minInterTDDIcPct
2929	0_AnalogInput	mAvg5minInterTDDIcPct
2930	0_AnalogInput	mMin5minInterTDDInPct
2931	0_AnalogInput	mMax5minInterTDDInPct
2932	0_AnalogInput	mAvg5minInterTDDInPct
2933	0_AnalogInput	mMin5minSelectHarmonicMag1
2934	0_AnalogInput	mMax5minSelectHarmonicMag1
2935	0_AnalogInput	mAvg5minSelectHarmonicMag1
2936	0_AnalogInput	mMin5minSelectHarmonicMag2
2937	0_AnalogInput	mMax5minSelectHarmonicMag2
2938	0_AnalogInput	mAvg5minSelectHarmonicMag2
2939	0_AnalogInput	mMin5minSelectHarmonicMag3
2940	0_AnalogInput	mMax5minSelectHarmonicMag3
2941	0_AnalogInput	mAvg5minSelectHarmonicMag3
2942	0_AnalogInput	mMin5minSelectHarmonicMag4
2943	0_AnalogInput	mMax5minSelectHarmonicMag4
2944	0_AnalogInput	mAvg5minSelectHarmonicMag4
2945	0_AnalogInput	mMin5minSelectHarmonicMag5
2946	0_AnalogInput	mMax5minSelectHarmonicMag5
2947	0_AnalogInput	mAvg5minSelectHarmonicMag5
2948	0_AnalogInput	mMin5minSelectHarmonicMag6
2949	0_AnalogInput	mMax5minSelectHarmonicMag6
2950	0_AnalogInput	mAvg5minSelectHarmonicMag6
2951	0_AnalogInput	mMin5minSelectHarmonicMag7
2952	0_AnalogInput	mMax5minSelectHarmonicMag7
2953	0_AnalogInput	mAvg5minSelectHarmonicMag7
2954	0_AnalogInput	mMin5minSelectHarmonicMag8

Object ID	Object Type	Name
2955	0_AnalogInput	mMax5minSelectHarmonicMag8
2956	0_AnalogInput	mAvg5minSelectHarmonicMag8
2957	0_AnalogInput	mMin5minSelectHarmonicMag9
2958	0_AnalogInput	mMax5minSelectHarmonicMag9
2959	0_AnalogInput	mAvg5minSelectHarmonicMag9
2960	0_AnalogInput	mMin5minSelectHarmonicMag10
2961	0_AnalogInput	mMax5minSelectHarmonicMag10
2962	0_AnalogInput	mAvg5minSelectHarmonicMag10
2963	0_AnalogInput	mMin5minSelectHarmonicMag11
2964	0_AnalogInput	mMax5minSelectHarmonicMag11
2965	0_AnalogInput	mAvg5minSelectHarmonicMag11
2966	0_AnalogInput	mMin5minSelectHarmonicMag12
2967	0_AnalogInput	mMax5minSelectHarmonicMag12
2968	0_AnalogInput	mAvg5minSelectHarmonicMag12
2969	0_AnalogInput	mMin5minSelectHarmonicMag13
2970	0_AnalogInput	mMax5minSelectHarmonicMag13
2971	0_AnalogInput	mAvg5minSelectHarmonicMag13
2972	0_AnalogInput	mMin5minSelectHarmonicMag14
2973	0_AnalogInput	mMax5minSelectHarmonicMag14
2974	0_AnalogInput	mAvg5minSelectHarmonicMag14
2975	0_AnalogInput	mMin5minSelectHarmonicMag15
2976	0_AnalogInput	mMax5minSelectHarmonicMag15
2977	0_AnalogInput	mAvg5minSelectHarmonicMag15
2978	0_AnalogInput	mMin5minSelectHarmonicMag16
2979	0_AnalogInput	mMax5minSelectHarmonicMag16
2980	0_AnalogInput	mAvg5minSelectHarmonicMag16
2981	0_AnalogInput	mMin5minSelectHarmonicMag17
2982	0_AnalogInput	mMax5minSelectHarmonicMag17
2983	0_AnalogInput	mAvg5minSelectHarmonicMag17
2984	0_AnalogInput	mMin5minSelectHarmonicMag18
2985	0_AnalogInput	mMax5minSelectHarmonicMag18
2986	0_AnalogInput	mAvg5minSelectHarmonicMag18
2987	0_AnalogInput	mMin5minSelectHarmonicMag19
2988	0_AnalogInput	mMax5minSelectHarmonicMag19
2989	0_AnalogInput	mAvg5minSelectHarmonicMag19
2990	0_AnalogInput	mMin5minSelectHarmonicMag20
2991	0_AnalogInput	mMax5minSelectHarmonicMag20
2992	0_AnalogInput	mAvg5minSelectHarmonicMag20
2993	0_AnalogInput	mMin5minSelectHarmonicMag21
2994	0_AnalogInput	mMax5minSelectHarmonicMag21
2995	0_AnalogInput	mAvg5minSelectHarmonicMag21

Object ID	Object Type	Name
2996	0_AnalogInput	mMin5minSelectHarmonicMag22
2997	0_AnalogInput	mMax5minSelectHarmonicMag22
2998	0_AnalogInput	mAvg5minSelectHarmonicMag22
2999	0_AnalogInput	mMin5minSelectHarmonicMag23
3000	0_AnalogInput	mMax5minSelectHarmonicMag23
3001	0_AnalogInput	mAvg5minSelectHarmonicMag23
3002	0_AnalogInput	mMin5minSelectHarmonicMag24
3003	0_AnalogInput	mMax5minSelectHarmonicMag24
3004	0_AnalogInput	mAvg5minSelectHarmonicMag24
3005	0_AnalogInput	mMin5minSelectHarmonicMag25
3006	0_AnalogInput	mMax5minSelectHarmonicMag25
3007	0_AnalogInput	mAvg5minSelectHarmonicMag25
3008	0_AnalogInput	mMin5minSelectHarmonicMag26
3009	0_AnalogInput	mMax5minSelectHarmonicMag26
3010	0_AnalogInput	mAvg5minSelectHarmonicMag26
3011	0_AnalogInput	mMin5minSelectHarmonicMag27
3012	0_AnalogInput	mMax5minSelectHarmonicMag27
3013	0_AnalogInput	mAvg5minSelectHarmonicMag27
3014	0_AnalogInput	mMin5minSelectHarmonicMag28
3015	0_AnalogInput	mMax5minSelectHarmonicMag28
3016	0_AnalogInput	mAvg5minSelectHarmonicMag28
3017	0_AnalogInput	mMin5minSelectHarmonicMag29
3018	0_AnalogInput	mMax5minSelectHarmonicMag29
3019	0_AnalogInput	mAvg5minSelectHarmonicMag29
3020	0_AnalogInput	mMin5minSelectHarmonicMag30
3021	0_AnalogInput	mMax5minSelectHarmonicMag30
3022	0_AnalogInput	mAvg5minSelectHarmonicMag30
3023	0_AnalogInput	mMin5minSelectHarmonicMag31
3024	0_AnalogInput	mMax5minSelectHarmonicMag31
3025	0_AnalogInput	mAvg5minSelectHarmonicMag31
3026	0_AnalogInput	mMin5minSelectHarmonicMag32
3027	0_AnalogInput	mMax5minSelectHarmonicMag32
3028	0_AnalogInput	mAvg5minSelectHarmonicMag32
3029	0_AnalogInput	mMin5minSelectHarmonicMag33
3030	0_AnalogInput	mMax5minSelectHarmonicMag33
3031	0_AnalogInput	mAvg5minSelectHarmonicMag33
3032	0_AnalogInput	mMin5minSelectHarmonicMag34
3033	0_AnalogInput	mMax5minSelectHarmonicMag34
3034	0_AnalogInput	mAvg5minSelectHarmonicMag34
3035	0_AnalogInput	mMin5minSelectHarmonicMag35
3036	0_AnalogInput	mMax5minSelectHarmonicMag35

Object ID	Object Type	Name
3037	0_AnalogInput	mAvg5minSelectHarmonicMag35
3038	0_AnalogInput	mMin5minSelectHarmonicMag36
3039	0_AnalogInput	mMax5minSelectHarmonicMag36
3040	0_AnalogInput	mAvg5minSelectHarmonicMag36
3041	0_AnalogInput	mMin5minSelectHarmonicMag37
3042	0_AnalogInput	mMax5minSelectHarmonicMag37
3043	0_AnalogInput	mAvg5minSelectHarmonicMag37
3044	0_AnalogInput	mMin5minSelectHarmonicMag38
3045	0_AnalogInput	mMax5minSelectHarmonicMag38
3046	0_AnalogInput	mAvg5minSelectHarmonicMag38
3047	0_AnalogInput	mMin5minSelectHarmonicMag39
3048	0_AnalogInput	mMax5minSelectHarmonicMag39
3049	0_AnalogInput	mAvg5minSelectHarmonicMag39
3050	0_AnalogInput	mMin5minSelectHarmonicMag40
3051	0_AnalogInput	mMax5minSelectHarmonicMag40
3052	0_AnalogInput	mAvg5minSelectHarmonicMag40
3053	0_AnalogInput	mMin5minSelectHarmonicMag41
3054	0_AnalogInput	mMax5minSelectHarmonicMag41
3055	0_AnalogInput	mAvg5minSelectHarmonicMag41
3056	0_AnalogInput	mMin5minSelectHarmonicMag42
3057	0_AnalogInput	mMax5minSelectHarmonicMag42
3058	0_AnalogInput	mAvg5minSelectHarmonicMag42
3059	0_AnalogInput	mMin5minSelectHarmonicMag43
3060	0_AnalogInput	mMax5minSelectHarmonicMag43
3061	0_AnalogInput	mAvg5minSelectHarmonicMag43
3062	0_AnalogInput	mMin5minSelectHarmonicMag44
3063	0_AnalogInput	mMax5minSelectHarmonicMag44
3064	0_AnalogInput	mAvg5minSelectHarmonicMag44
3065	0_AnalogInput	mMin5minSelectHarmonicMag45
3066	0_AnalogInput	mMax5minSelectHarmonicMag45
3067	0_AnalogInput	mAvg5minSelectHarmonicMag45
3068	0_AnalogInput	mMin5minSelectHarmonicMag46
3069	0_AnalogInput	mMax5minSelectHarmonicMag46
3070	0_AnalogInput	mAvg5minSelectHarmonicMag46
3071	0_AnalogInput	mMin5minSelectHarmonicMag47
3072	0_AnalogInput	mMax5minSelectHarmonicMag47
3073	0_AnalogInput	mAvg5minSelectHarmonicMag47
3074	0_AnalogInput	mMin5minSelectHarmonicMag48
3075	0_AnalogInput	mMax5minSelectHarmonicMag48
3076	0_AnalogInput	mAvg5minSelectHarmonicMag48
3077	0_AnalogInput	mMin5minSelectHarmonicMag49

Object ID	Object Type	Name
3078	0_AnalogInput	mMax5minSelectHarmonicMag49
3079	0_AnalogInput	mAvg5minSelectHarmonicMag49
3080	0_AnalogInput	mMin5minSelectHarmonicMag50
3081	0_AnalogInput	mMax5minSelectHarmonicMag50
3082	0_AnalogInput	mAvg5minSelectHarmonicMag50
3083	0_AnalogInput	mMin15minPinstVan
3084	0_AnalogInput	mMax15minPinstVan
3085	0_AnalogInput	mAvg15minPinstVan
3086	0_AnalogInput	mMin15minPinstVbn
3087	0_AnalogInput	mMax15minPinstVbn
3088	0_AnalogInput	mAvg15minPinstVbn
3089	0_AnalogInput	mMin15minPinstVcn
3090	0_AnalogInput	mMax15minPinstVcn
3091	0_AnalogInput	mAvg15minPinstVcn
3092	0_AnalogInput	mMin15minPinstVab
3093	0_AnalogInput	mMax15minPinstVab
3094	0_AnalogInput	mAvg15minPinstVab
3095	0_AnalogInput	mMin15minPinstVbc
3096	0_AnalogInput	mMax15minPinstVbc
3097	0_AnalogInput	mAvg15minPinstVbc
3098	0_AnalogInput	mMin15minPinstVca
3099	0_AnalogInput	mMax15minPinstVca
3100	0_AnalogInput	mAvg15minPinstVca
3101	0_AnalogInput	mMin15minPstVan
3102	0_AnalogInput	mMax15minPstVan
3103	0_AnalogInput	mAvg15minPstVan
3104	0_AnalogInput	mMin15minPstVab
3105	0_AnalogInput	mMax15minPstVab
3106	0_AnalogInput	mAvg15minPstVab
3107	0_AnalogInput	mMin15minPstVbn
3108	0_AnalogInput	mMax15minPstVbn
3109	0_AnalogInput	mAvg15minPstVbn
3110	0_AnalogInput	mMin15minPstVbc
3111	0_AnalogInput	mMax15minPstVbc
3112	0_AnalogInput	mAvg15minPstVbc
3113	0_AnalogInput	mMin15minPstVcn
3114	0_AnalogInput	mMax15minPstVcn
3115	0_AnalogInput	mAvg15minPstVcn
3116	0_AnalogInput	mMin15minPstVca
3117	0_AnalogInput	mMax15minPstVca
3118	0_AnalogInput	mAvg15minPstVca

Object ID	Object Type	Name
3119	0_AnalogInput	mMin15minCFa
3120	0_AnalogInput	mMax15minCFa
3121	0_AnalogInput	mAvg15minCFa
3122	0_AnalogInput	mMin15minCFb
3123	0_AnalogInput	mMax15minCFb
3124	0_AnalogInput	mAvg15minCFb
3125	0_AnalogInput	mMin15minCFc
3126	0_AnalogInput	mMax15minCFc
3127	0_AnalogInput	mAvg15minCFc
3128	0_AnalogInput	mMin15minCFsys
3129	0_AnalogInput	mMax15minCFsys
3130	0_AnalogInput	mAvg15minCFsys
3131	0_AnalogInput	mMin15minKFa
3132	0_AnalogInput	mMax15minKFa
3133	0_AnalogInput	mAvg15minKFa
3134	0_AnalogInput	mMin15minKFb
3135	0_AnalogInput	mMax15minKFb
3136	0_AnalogInput	mAvg15minKFb
3137	0_AnalogInput	mMin15minKFc
3138	0_AnalogInput	mMax15minKFc
3139	0_AnalogInput	mAvg15minKFc
3140	0_AnalogInput	mMin15minKFsys
3141	0_AnalogInput	mMax15minKFsys
3142	0_AnalogInput	mAvg15minKFsys
3143	0_AnalogInput	mMin15minTotalHarmonicIa
3144	0_AnalogInput	mMax15minTotalHarmonicIa
3145	0_AnalogInput	mAvg15minTotalHarmonicIa
3146	0_AnalogInput	mMin15minTotalHarmonicIb
3147	0_AnalogInput	mMax15minTotalHarmonicIb
3148	0_AnalogInput	mAvg15minTotalHarmonicIb
3149	0_AnalogInput	mMin15minTotalHarmonicIc
3150	0_AnalogInput	mMax15minTotalHarmonicIc
3151	0_AnalogInput	mAvg15minTotalHarmonicIc
3152	0_AnalogInput	mMin15minTotalHarmonicVab
3153	0_AnalogInput	mMax15minTotalHarmonicVab
3154	0_AnalogInput	mAvg15minTotalHarmonicVab
3155	0_AnalogInput	mMin15minTotalHarmonicVbc
3156	0_AnalogInput	mMax15minTotalHarmonicVbc
3157	0_AnalogInput	mAvg15minTotalHarmonicVbc
3158	0_AnalogInput	mMin15minTotalHarmonicVca
3159	0_AnalogInput	mMax15minTotalHarmonicVca

Object ID	Object Type	Name
3160	0_AnalogInput	mAvg15minTotalHarmonicVca
3161	0_AnalogInput	mMin15minTotalHarmonicVan
3162	0_AnalogInput	mMax15minTotalHarmonicVan
3163	0_AnalogInput	mAvg15minTotalHarmonicVan
3164	0_AnalogInput	mMin15minTotalHarmonicVbn
3165	0_AnalogInput	mMax15minTotalHarmonicVbn
3166	0_AnalogInput	mAvg15minTotalHarmonicVbn
3167	0_AnalogInput	mMin15minTotalHarmonicVcn
3168	0_AnalogInput	mMax15minTotalHarmonicVcn
3169	0_AnalogInput	mAvg15minTotalHarmonicVcn
3170	0_AnalogInput	mMin15minTotalHarmonicIn
3171	0_AnalogInput	mMax15minTotalHarmonicIn
3172	0_AnalogInput	mAvg15minTotalHarmonicIn
3173	0_AnalogInput	mMin15minTotalHarmonicIaPct
3174	0_AnalogInput	mMax15minTotalHarmonicIaPct
3175	0_AnalogInput	mAvg15minTotalHarmonicIaPct
3176	0_AnalogInput	mMin15minTotalHarmonicIbPct
3177	0_AnalogInput	mMax15minTotalHarmonicIbPct
3178	0_AnalogInput	mAvg15minTotalHarmonicIbPct
3179	0_AnalogInput	mMin15minTotalHarmonicIcPct
3180	0_AnalogInput	mMax15minTotalHarmonicIcPct
3181	0_AnalogInput	mAvg15minTotalHarmonicIcPct
3182	0_AnalogInput	mMin15minTotalHarmonicVabPct
3183	0_AnalogInput	mMax15minTotalHarmonicVabPct
3184	0_AnalogInput	mAvg15minTotalHarmonicVabPct
3185	0_AnalogInput	mMin15minTotalHarmonicVbcPct
3186	0_AnalogInput	mMax15minTotalHarmonicVbcPct
3187	0_AnalogInput	mAvg15minTotalHarmonicVbcPct
3188	0_AnalogInput	mMin15minTotalHarmonicVcaPct
3189	0_AnalogInput	mMax15minTotalHarmonicVcaPct
3190	0_AnalogInput	mAvg15minTotalHarmonicVcaPct
3191	0_AnalogInput	mMin15minTotalHarmonicVanPct
3192	0_AnalogInput	mMax15minTotalHarmonicVanPct
3193	0_AnalogInput	mAvg15minTotalHarmonicVanPct
3194	0_AnalogInput	mMin15minTotalHarmonicVbnPct
3195	0_AnalogInput	mMax15minTotalHarmonicVbnPct
3196	0_AnalogInput	mAvg15minTotalHarmonicVbnPct
3197	0_AnalogInput	mMin15minTotalHarmonicVcnPct
3198	0_AnalogInput	mMax15minTotalHarmonicVcnPct
3199	0_AnalogInput	mAvg15minTotalHarmonicVcnPct
3200	0_AnalogInput	mMin15minTotalHarmonicInPct

Object ID	Object Type	Name
3201	0_AnalogInput	mMax15minTotalHarmonicInPct
3202	0_AnalogInput	mAvg15minTotalHarmonicInPct
3203	0_AnalogInput	mMin15minTDDIaPct
3204	0_AnalogInput	mMax15minTDDIaPct
3205	0_AnalogInput	mAvg15minTDDIaPct
3206	0_AnalogInput	mMin15minTDDIbPct
3207	0_AnalogInput	mMax15minTDDIbPct
3208	0_AnalogInput	mAvg15minTDDIbPct
3209	0_AnalogInput	mMin15minTDDIcPct
3210	0_AnalogInput	mMax15minTDDIcPct
3211	0_AnalogInput	mAvg15minTDDIcPct
3212	0_AnalogInput	mMin15minTDDInPct
3213	0_AnalogInput	mMax15minTDDInPct
3214	0_AnalogInput	mAvg15minTDDInPct
3215	0_AnalogInput	mMin15minEvenHarmonicIa
3216	0_AnalogInput	mMax15minEvenHarmonicIa
3217	0_AnalogInput	mAvg15minEvenHarmonicIa
3218	0_AnalogInput	mMin15minEvenHarmonicIb
3219	0_AnalogInput	mMax15minEvenHarmonicIb
3220	0_AnalogInput	mAvg15minEvenHarmonicIb
3221	0_AnalogInput	mMin15minEvenHarmonicIc
3222	0_AnalogInput	mMax15minEvenHarmonicIc
3223	0_AnalogInput	mAvg15minEvenHarmonicIc
3224	0_AnalogInput	mMin15minEvenHarmonicVab
3225	0_AnalogInput	mMax15minEvenHarmonicVab
3226	0_AnalogInput	mAvg15minEvenHarmonicVab
3227	0_AnalogInput	mMin15minEvenHarmonicVbc
3228	0_AnalogInput	mMax15minEvenHarmonicVbc
3229	0_AnalogInput	mAvg15minEvenHarmonicVbc
3230	0_AnalogInput	mMin15minEvenHarmonicVca
3231	0_AnalogInput	mMax15minEvenHarmonicVca
3232	0_AnalogInput	mAvg15minEvenHarmonicVca
3233	0_AnalogInput	mMin15minEvenHarmonicVan
3234	0_AnalogInput	mMax15minEvenHarmonicVan
3235	0_AnalogInput	mAvg15minEvenHarmonicVan
3236	0_AnalogInput	mMin15minEvenHarmonicVbn
3237	0_AnalogInput	mMax15minEvenHarmonicVbn
3238	0_AnalogInput	mAvg15minEvenHarmonicVbn
3239	0_AnalogInput	mMin15minEvenHarmonicVcn
3240	0_AnalogInput	mMax15minEvenHarmonicVcn
3241	0_AnalogInput	mAvg15minEvenHarmonicVcn

Object ID	Object Type	Name
3242	0_AnalogInput	mMin15minEvenHarmonicIn
3243	0_AnalogInput	mMax15minEvenHarmonicIn
3244	0_AnalogInput	mAvg15minEvenHarmonicIn
3245	0_AnalogInput	mMin15minEvenHarmonicIaPct
3246	0_AnalogInput	mMax15minEvenHarmonicIaPct
3247	0_AnalogInput	mAvg15minEvenHarmonicIaPct
3248	0_AnalogInput	mMin15minEvenHarmonicIbPct
3249	0_AnalogInput	mMax15minEvenHarmonicIbPct
3250	0_AnalogInput	mAvg15minEvenHarmonicIbPct
3251	0_AnalogInput	mMin15minEvenHarmonicIcPct
3252	0_AnalogInput	mMax15minEvenHarmonicIcPct
3253	0_AnalogInput	mAvg15minEvenHarmonicIcPct
3254	0_AnalogInput	mMin15minEvenHarmonicVabPct
3255	0_AnalogInput	mMax15minEvenHarmonicVabPct
3256	0_AnalogInput	mAvg15minEvenHarmonicVabPct
3257	0_AnalogInput	mMin15minEvenHarmonicVbcPct
3258	0_AnalogInput	mMax15minEvenHarmonicVbcPct
3259	0_AnalogInput	mAvg15minEvenHarmonicVbcPct
3260	0_AnalogInput	mMin15minEvenHarmonicVcaPct
3261	0_AnalogInput	mMax15minEvenHarmonicVcaPct
3262	0_AnalogInput	mAvg15minEvenHarmonicVcaPct
3263	0_AnalogInput	mMin15minEvenHarmonicVanPct
3264	0_AnalogInput	mMax15minEvenHarmonicVanPct
3265	0_AnalogInput	mAvg15minEvenHarmonicVanPct
3266	0_AnalogInput	mMin15minEvenHarmonicVbnPct
3267	0_AnalogInput	mMax15minEvenHarmonicVbnPct
3268	0_AnalogInput	mAvg15minEvenHarmonicVbnPct
3269	0_AnalogInput	mMin15minEvenHarmonicVcnPct
3270	0_AnalogInput	mMax15minEvenHarmonicVcnPct
3271	0_AnalogInput	mAvg15minEvenHarmonicVcnPct
3272	0_AnalogInput	mMin15minEvenHarmonicInPct
3273	0_AnalogInput	mMax15minEvenHarmonicInPct
3274	0_AnalogInput	mAvg15minEvenHarmonicInPct
3275	0_AnalogInput	mMin15minEvenTDDIaPct
3276	0_AnalogInput	mMax15minEvenTDDIaPct
3277	0_AnalogInput	mAvg15minEvenTDDIaPct
3278	0_AnalogInput	mMin15minEvenTDDIbPct
3279	0_AnalogInput	mMax15minEvenTDDIbPct
3280	0_AnalogInput	mAvg15minEvenTDDIbPct
3281	0_AnalogInput	mMin15minEvenTDDIcPct
3282	0_AnalogInput	mMax15minEvenTDDIcPct

Object ID	Object Type	Name
3283	0_AnalogInput	mAvg15minEvenTDDIcPct
3284	0_AnalogInput	mMin15minEvenTDDInPct
3285	0_AnalogInput	mMax15minEvenTDDInPct
3286	0_AnalogInput	mAvg15minEvenTDDInPct
3287	0_AnalogInput	mMin15minOddHarmonicIa
3288	0_AnalogInput	mMax15minOddHarmonicIa
3289	0_AnalogInput	mAvg15minOddHarmonicIa
3290	0_AnalogInput	mMin15minOddHarmonicIb
3291	0_AnalogInput	mMax15minOddHarmonicIb
3292	0_AnalogInput	mAvg15minOddHarmonicIb
3293	0_AnalogInput	mMin15minOddHarmonicIc
3294	0_AnalogInput	mMax15minOddHarmonicIc
3295	0_AnalogInput	mAvg15minOddHarmonicIc
3296	0_AnalogInput	mMin15minOddHarmonicVab
3297	0_AnalogInput	mMax15minOddHarmonicVab
3298	0_AnalogInput	mAvg15minOddHarmonicVab
3299	0_AnalogInput	mMin15minOddHarmonicVbc
3300	0_AnalogInput	mMax15minOddHarmonicVbc
3301	0_AnalogInput	mAvg15minOddHarmonicVbc
3302	0_AnalogInput	mMin15minOddHarmonicVca
3303	0_AnalogInput	mMax15minOddHarmonicVca
3304	0_AnalogInput	mAvg15minOddHarmonicVca
3305	0_AnalogInput	mMin15minOddHarmonicVan
3306	0_AnalogInput	mMax15minOddHarmonicVan
3307	0_AnalogInput	mAvg15minOddHarmonicVan
3308	0_AnalogInput	mMin15minOddHarmonicVbn
3309	0_AnalogInput	mMax15minOddHarmonicVbn
3310	0_AnalogInput	mAvg15minOddHarmonicVbn
3311	0_AnalogInput	mMin15minOddHarmonicVcn
3312	0_AnalogInput	mMax15minOddHarmonicVcn
3313	0_AnalogInput	mAvg15minOddHarmonicVcn
3314	0_AnalogInput	mMin15minOddHarmonicIn
3315	0_AnalogInput	mMax15minOddHarmonicIn
3316	0_AnalogInput	mAvg15minOddHarmonicIn
3317	0_AnalogInput	mMin15minOddHarmonicIaPct
3318	0_AnalogInput	mMax15minOddHarmonicIaPct
3319	0_AnalogInput	mAvg15minOddHarmonicIaPct
3320	0_AnalogInput	mMin15minOddHarmonicIbPct
3321	0_AnalogInput	mMax15minOddHarmonicIbPct
3322	0_AnalogInput	mAvg15minOddHarmonicIbPct
3323	0_AnalogInput	mMin15minOddHarmonicIcPct

Object ID	Object Type	Name
3324	0_AnalogInput	mMax15minOddHarmonicIcPct
3325	0_AnalogInput	mAvg15minOddHarmonicIcPct
3326	0_AnalogInput	mMin15minOddHarmonicVabPct
3327	0_AnalogInput	mMax15minOddHarmonicVabPct
3328	0_AnalogInput	mAvg15minOddHarmonicVabPct
3329	0_AnalogInput	mMin15minOddHarmonicVbcPct
3330	0_AnalogInput	mMax15minOddHarmonicVbcPct
3331	0_AnalogInput	mAvg15minOddHarmonicVbcPct
3332	0_AnalogInput	mMin15minOddHarmonicVcaPct
3333	0_AnalogInput	mMax15minOddHarmonicVcaPct
3334	0_AnalogInput	mAvg15minOddHarmonicVcaPct
3335	0_AnalogInput	mMin15minOddHarmonicVanPct
3336	0_AnalogInput	mMax15minOddHarmonicVanPct
3337	0_AnalogInput	mAvg15minOddHarmonicVanPct
3338	0_AnalogInput	mMin15minOddHarmonicVbnPct
3339	0_AnalogInput	mMax15minOddHarmonicVbnPct
3340	0_AnalogInput	mAvg15minOddHarmonicVbnPct
3341	0_AnalogInput	mMin15minOddHarmonicVcnPct
3342	0_AnalogInput	mMax15minOddHarmonicVcnPct
3343	0_AnalogInput	mAvg15minOddHarmonicVcnPct
3344	0_AnalogInput	mMin15minOddHarmonicInPct
3345	0_AnalogInput	mMax15minOddHarmonicInPct
3346	0_AnalogInput	mAvg15minOddHarmonicInPct
3347	0_AnalogInput	mMin15minOddTDDIaPct
3348	0_AnalogInput	mMax15minOddTDDIaPct
3349	0_AnalogInput	mAvg15minOddTDDIaPct
3350	0_AnalogInput	mMin15minOddTDDIbPct
3351	0_AnalogInput	mMax15minOddTDDIbPct
3352	0_AnalogInput	mAvg15minOddTDDIbPct
3353	0_AnalogInput	mMin15minOddTDDIcPct
3354	0_AnalogInput	mMax15minOddTDDIcPct
3355	0_AnalogInput	mAvg15minOddTDDIcPct
3356	0_AnalogInput	mMin15minOddTDDInPct
3357	0_AnalogInput	mMax15minOddTDDInPct
3358	0_AnalogInput	mAvg15minOddTDDInPct
3359	0_AnalogInput	mMin15minInterHarmonicIa
3360	0_AnalogInput	mMax15minInterHarmonicIa
3361	0_AnalogInput	mAvg15minInterHarmonicIa
3362	0_AnalogInput	mMin15minInterHarmonicIb
3363	0_AnalogInput	mMax15minInterHarmonicIb
3364	0_AnalogInput	mAvg15minInterHarmonicIb

Object ID	Object Type	Name
3365	0_AnalogInput	mMin15minInterHarmonicIc
3366	0_AnalogInput	mMax15minInterHarmonicIc
3367	0_AnalogInput	mAvg15minInterHarmonicIc
3368	0_AnalogInput	mMin15minInterHarmonicVab
3369	0_AnalogInput	mMax15minInterHarmonicVab
3370	0_AnalogInput	mAvg15minInterHarmonicVab
3371	0_AnalogInput	mMin15minInterHarmonicVbc
3372	0_AnalogInput	mMax15minInterHarmonicVbc
3373	0_AnalogInput	mAvg15minInterHarmonicVbc
3374	0_AnalogInput	mMin15minInterHarmonicVca
3375	0_AnalogInput	mMax15minInterHarmonicVca
3376	0_AnalogInput	mAvg15minInterHarmonicVca
3377	0_AnalogInput	mMin15minInterHarmonicVan
3378	0_AnalogInput	mMax15minInterHarmonicVan
3379	0_AnalogInput	mAvg15minInterHarmonicVan
3380	0_AnalogInput	mMin15minInterHarmonicVbn
3381	0_AnalogInput	mMax15minInterHarmonicVbn
3382	0_AnalogInput	mAvg15minInterHarmonicVbn
3383	0_AnalogInput	mMin15minInterHarmonicVcn
3384	0_AnalogInput	mMax15minInterHarmonicVcn
3385	0_AnalogInput	mAvg15minInterHarmonicVcn
3386	0_AnalogInput	mMin15minInterHarmonicIn
3387	0_AnalogInput	mMax15minInterHarmonicIn
3388	0_AnalogInput	mAvg15minInterHarmonicIn
3389	0_AnalogInput	mMin15minInterHarmonicIaPct
3390	0_AnalogInput	mMax15minInterHarmonicIaPct
3391	0_AnalogInput	mAvg15minInterHarmonicIaPct
3392	0_AnalogInput	mMin15minInterHarmonicIbPct
3393	0_AnalogInput	mMax15minInterHarmonicIbPct
3394	0_AnalogInput	mAvg15minInterHarmonicIbPct
3395	0_AnalogInput	mMin15minInterHarmonicIcPct
3396	0_AnalogInput	mMax15minInterHarmonicIcPct
3397	0_AnalogInput	mAvg15minInterHarmonicIcPct
3398	0_AnalogInput	mMin15minInterHarmonicVabPct
3399	0_AnalogInput	mMax15minInterHarmonicVabPct
3400	0_AnalogInput	mAvg15minInterHarmonicVabPct
3401	0_AnalogInput	mMin15minInterHarmonicVbcPct
3402	0_AnalogInput	mMax15minInterHarmonicVbcPct
3403	0_AnalogInput	mAvg15minInterHarmonicVbcPct
3404	0_AnalogInput	mMin15minInterHarmonicVcaPct
3405	0_AnalogInput	mMax15minInterHarmonicVcaPct

Object ID	Object Type	Name
3406	0_AnalogInput	mAvg15minInterHarmonicVcaPct
3407	0_AnalogInput	mMin15minInterHarmonicVanPct
3408	0_AnalogInput	mMax15minInterHarmonicVanPct
3409	0_AnalogInput	mAvg15minInterHarmonicVanPct
3410	0_AnalogInput	mMin15minInterHarmonicVbnPct
3411	0_AnalogInput	mMax15minInterHarmonicVbnPct
3412	0_AnalogInput	mAvg15minInterHarmonicVbnPct
3413	0_AnalogInput	mMin15minInterHarmonicVcnPct
3414	0_AnalogInput	mMax15minInterHarmonicVcnPct
3415	0_AnalogInput	mAvg15minInterHarmonicVcnPct
3416	0_AnalogInput	mMin15minInterHarmonicInPct
3417	0_AnalogInput	mMax15minInterHarmonicInPct
3418	0_AnalogInput	mAvg15minInterHarmonicInPct
3419	0_AnalogInput	mMin15minInterTDDIaPct
3420	0_AnalogInput	mMax15minInterTDDIaPct
3421	0_AnalogInput	mAvg15minInterTDDIaPct
3422	0_AnalogInput	mMin15minInterTDDIbPct
3423	0_AnalogInput	mMax15minInterTDDIbPct
3424	0_AnalogInput	mAvg15minInterTDDIbPct
3425	0_AnalogInput	mMin15minInterTDDIcPct
3426	0_AnalogInput	mMax15minInterTDDIcPct
3427	0_AnalogInput	mAvg15minInterTDDIcPct
3428	0_AnalogInput	mMin15minInterTDDInPct
3429	0_AnalogInput	mMax15minInterTDDInPct
3430	0_AnalogInput	mAvg15minInterTDDInPct
3431	0_AnalogInput	mMin15minSelectHarmonicMag1
3432	0_AnalogInput	mMax15minSelectHarmonicMag1
3433	0_AnalogInput	mAvg15minSelectHarmonicMag1
3434	0_AnalogInput	mMin15minSelectHarmonicMag2
3435	0_AnalogInput	mMax15minSelectHarmonicMag2
3436	0_AnalogInput	mAvg15minSelectHarmonicMag2
3437	0_AnalogInput	mMin15minSelectHarmonicMag3
3438	0_AnalogInput	mMax15minSelectHarmonicMag3
3439	0_AnalogInput	mAvg15minSelectHarmonicMag3
3440	0_AnalogInput	mMin15minSelectHarmonicMag4
3441	0_AnalogInput	mMax15minSelectHarmonicMag4
3442	0_AnalogInput	mAvg15minSelectHarmonicMag4
3443	0_AnalogInput	mMin15minSelectHarmonicMag5
3444	0_AnalogInput	mMax15minSelectHarmonicMag5
3445	0_AnalogInput	mAvg15minSelectHarmonicMag5
3446	0_AnalogInput	mMin15minSelectHarmonicMag6

Object ID	Object Type	Name
3447	0_AnalogInput	mMax15minSelectHarmonicMag6
3448	0_AnalogInput	mAvg15minSelectHarmonicMag6
3449	0_AnalogInput	mMin15minSelectHarmonicMag7
3450	0_AnalogInput	mMax15minSelectHarmonicMag7
3451	0_AnalogInput	mAvg15minSelectHarmonicMag7
3452	0_AnalogInput	mMin15minSelectHarmonicMag8
3453	0_AnalogInput	mMax15minSelectHarmonicMag8
3454	0_AnalogInput	mAvg15minSelectHarmonicMag8
3455	0_AnalogInput	mMin15minSelectHarmonicMag9
3456	0_AnalogInput	mMax15minSelectHarmonicMag9
3457	0_AnalogInput	mAvg15minSelectHarmonicMag9
3458	0_AnalogInput	mMin15minSelectHarmonicMag10
3459	0_AnalogInput	mMax15minSelectHarmonicMag10
3460	0_AnalogInput	mAvg15minSelectHarmonicMag10
3461	0_AnalogInput	mMin15minSelectHarmonicMag11
3462	0_AnalogInput	mMax15minSelectHarmonicMag11
3463	0_AnalogInput	mAvg15minSelectHarmonicMag11
3464	0_AnalogInput	mMin15minSelectHarmonicMag12
3465	0_AnalogInput	mMax15minSelectHarmonicMag12
3466	0_AnalogInput	mAvg15minSelectHarmonicMag12
3467	0_AnalogInput	mMin15minSelectHarmonicMag13
3468	0_AnalogInput	mMax15minSelectHarmonicMag13
3469	0_AnalogInput	mAvg15minSelectHarmonicMag13
3470	0_AnalogInput	mMin15minSelectHarmonicMag14
3471	0_AnalogInput	mMax15minSelectHarmonicMag14
3472	0_AnalogInput	mAvg15minSelectHarmonicMag14
3473	0_AnalogInput	mMin15minSelectHarmonicMag15
3474	0_AnalogInput	mMax15minSelectHarmonicMag15
3475	0_AnalogInput	mAvg15minSelectHarmonicMag15
3476	0_AnalogInput	mMin15minSelectHarmonicMag16
3477	0_AnalogInput	mMax15minSelectHarmonicMag16
3478	0_AnalogInput	mAvg15minSelectHarmonicMag16
3479	0_AnalogInput	mMin15minSelectHarmonicMag17
3480	0_AnalogInput	mMax15minSelectHarmonicMag17
3481	0_AnalogInput	mAvg15minSelectHarmonicMag17
3482	0_AnalogInput	mMin15minSelectHarmonicMag18
3483	0_AnalogInput	mMax15minSelectHarmonicMag18
3484	0_AnalogInput	mAvg15minSelectHarmonicMag18
3485	0_AnalogInput	mMin15minSelectHarmonicMag19
3486	0_AnalogInput	mMax15minSelectHarmonicMag19
3487	0_AnalogInput	mAvg15minSelectHarmonicMag19

Object ID	Object Type	Name
3488	0_AnalogInput	mMin15minSelectHarmonicMag20
3489	0_AnalogInput	mMax15minSelectHarmonicMag20
3490	0_AnalogInput	mAvg15minSelectHarmonicMag20
3491	0_AnalogInput	mMin15minSelectHarmonicMag21
3492	0_AnalogInput	mMax15minSelectHarmonicMag21
3493	0_AnalogInput	mAvg15minSelectHarmonicMag21
3494	0_AnalogInput	mMin15minSelectHarmonicMag22
3495	0_AnalogInput	mMax15minSelectHarmonicMag22
3496	0_AnalogInput	mAvg15minSelectHarmonicMag22
3497	0_AnalogInput	mMin15minSelectHarmonicMag23
3498	0_AnalogInput	mMax15minSelectHarmonicMag23
3499	0_AnalogInput	mAvg15minSelectHarmonicMag23
3500	0_AnalogInput	mMin15minSelectHarmonicMag24
3501	0_AnalogInput	mMax15minSelectHarmonicMag24
3502	0_AnalogInput	mAvg15minSelectHarmonicMag24
3503	0_AnalogInput	mMin15minSelectHarmonicMag25
3504	0_AnalogInput	mMax15minSelectHarmonicMag25
3505	0_AnalogInput	mAvg15minSelectHarmonicMag25
3506	0_AnalogInput	mMin15minSelectHarmonicMag26
3507	0_AnalogInput	mMax15minSelectHarmonicMag26
3508	0_AnalogInput	mAvg15minSelectHarmonicMag26
3509	0_AnalogInput	mMin15minSelectHarmonicMag27
3510	0_AnalogInput	mMax15minSelectHarmonicMag27
3511	0_AnalogInput	mAvg15minSelectHarmonicMag27
3512	0_AnalogInput	mMin15minSelectHarmonicMag28
3513	0_AnalogInput	mMax15minSelectHarmonicMag28
3514	0_AnalogInput	mAvg15minSelectHarmonicMag28
3515	0_AnalogInput	mMin15minSelectHarmonicMag29
3516	0_AnalogInput	mMax15minSelectHarmonicMag29
3517	0_AnalogInput	mAvg15minSelectHarmonicMag29
3518	0_AnalogInput	mMin15minSelectHarmonicMag30
3519	0_AnalogInput	mMax15minSelectHarmonicMag30
3520	0_AnalogInput	mAvg15minSelectHarmonicMag30
3521	0_AnalogInput	mMin15minSelectHarmonicMag31
3522	0_AnalogInput	mMax15minSelectHarmonicMag31
3523	0_AnalogInput	mAvg15minSelectHarmonicMag31
3524	0_AnalogInput	mMin15minSelectHarmonicMag32
3525	0_AnalogInput	mMax15minSelectHarmonicMag32
3526	0_AnalogInput	mAvg15minSelectHarmonicMag32
3527	0_AnalogInput	mMin15minSelectHarmonicMag33
3528	0_AnalogInput	mMax15minSelectHarmonicMag33

Object ID	Object Type	Name
3529	0_AnalogInput	mAvg15minSelectHarmonicMag33
3530	0_AnalogInput	mMin15minSelectHarmonicMag34
3531	0_AnalogInput	mMax15minSelectHarmonicMag34
3532	0_AnalogInput	mAvg15minSelectHarmonicMag34
3533	0_AnalogInput	mMin15minSelectHarmonicMag35
3534	0_AnalogInput	mMax15minSelectHarmonicMag35
3535	0_AnalogInput	mAvg15minSelectHarmonicMag35
3536	0_AnalogInput	mMin15minSelectHarmonicMag36
3537	0_AnalogInput	mMax15minSelectHarmonicMag36
3538	0_AnalogInput	mAvg15minSelectHarmonicMag36
3539	0_AnalogInput	mMin15minSelectHarmonicMag37
3540	0_AnalogInput	mMax15minSelectHarmonicMag37
3541	0_AnalogInput	mAvg15minSelectHarmonicMag37
3542	0_AnalogInput	mMin15minSelectHarmonicMag38
3543	0_AnalogInput	mMax15minSelectHarmonicMag38
3544	0_AnalogInput	mAvg15minSelectHarmonicMag38
3545	0_AnalogInput	mMin15minSelectHarmonicMag39
3546	0_AnalogInput	mMax15minSelectHarmonicMag39
3547	0_AnalogInput	mAvg15minSelectHarmonicMag39
3548	0_AnalogInput	mMin15minSelectHarmonicMag40
3549	0_AnalogInput	mMax15minSelectHarmonicMag40
3550	0_AnalogInput	mAvg15minSelectHarmonicMag40
3551	0_AnalogInput	mMin15minSelectHarmonicMag41
3552	0_AnalogInput	mMax15minSelectHarmonicMag41
3553	0_AnalogInput	mAvg15minSelectHarmonicMag41
3554	0_AnalogInput	mMin15minSelectHarmonicMag42
3555	0_AnalogInput	mMax15minSelectHarmonicMag42
3556	0_AnalogInput	mAvg15minSelectHarmonicMag42
3557	0_AnalogInput	mMin15minSelectHarmonicMag43
3558	0_AnalogInput	mMax15minSelectHarmonicMag43
3559	0_AnalogInput	mAvg15minSelectHarmonicMag43
3560	0_AnalogInput	mMin15minSelectHarmonicMag44
3561	0_AnalogInput	mMax15minSelectHarmonicMag44
3562	0_AnalogInput	mAvg15minSelectHarmonicMag44
3563	0_AnalogInput	mMin15minSelectHarmonicMag45
3564	0_AnalogInput	mMax15minSelectHarmonicMag45
3565	0_AnalogInput	mAvg15minSelectHarmonicMag45
3566	0_AnalogInput	mMin15minSelectHarmonicMag46
3567	0_AnalogInput	mMax15minSelectHarmonicMag46
3568	0_AnalogInput	mAvg15minSelectHarmonicMag46
3569	0_AnalogInput	mMin15minSelectHarmonicMag47

Object ID	Object Type	Name
3570	0_AnalogInput	mMax15minSelectHarmonicMag47
3571	0_AnalogInput	mAvg15minSelectHarmonicMag47
3572	0_AnalogInput	mMin15minSelectHarmonicMag48
3573	0_AnalogInput	mMax15minSelectHarmonicMag48
3574	0_AnalogInput	mAvg15minSelectHarmonicMag48
3575	0_AnalogInput	mMin15minSelectHarmonicMag49
3576	0_AnalogInput	mMax15minSelectHarmonicMag49
3577	0_AnalogInput	mAvg15minSelectHarmonicMag49
3578	0_AnalogInput	mMin15minSelectHarmonicMag50
3579	0_AnalogInput	mMax15minSelectHarmonicMag50
3580	0_AnalogInput	mAvg15minSelectHarmonicMag50

Eaton
1000 Eaton Boulevard
Cleveland, OH 44122
United States
Eaton.com

© 2026 Eaton
All Rights Reserved
Printed in USA
Publication No. MN150016EN/GC
January 2026

Eaton is a registered trademark.

All trademarks are property
of their respective owners.

Powering Business Worldwide